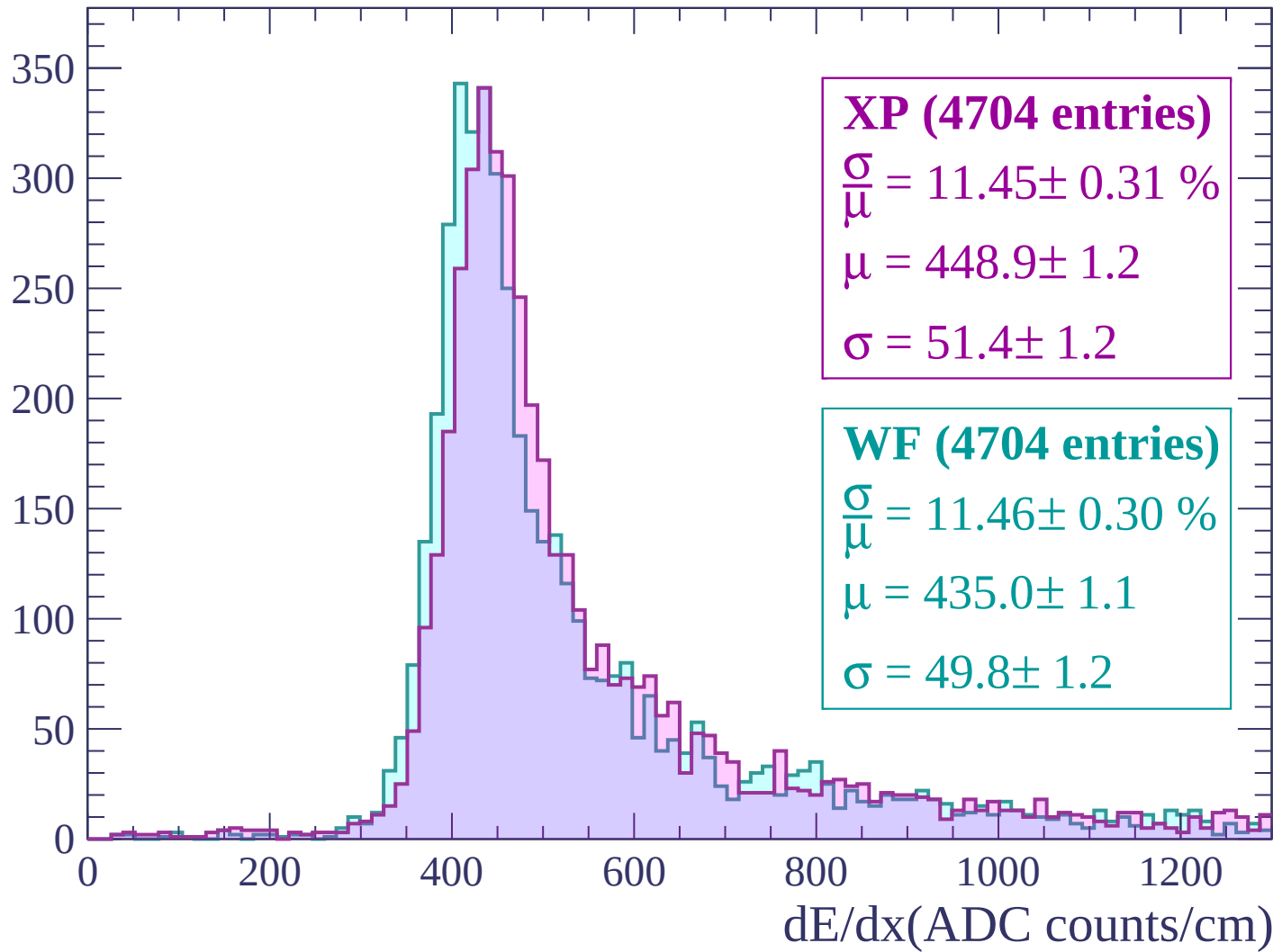
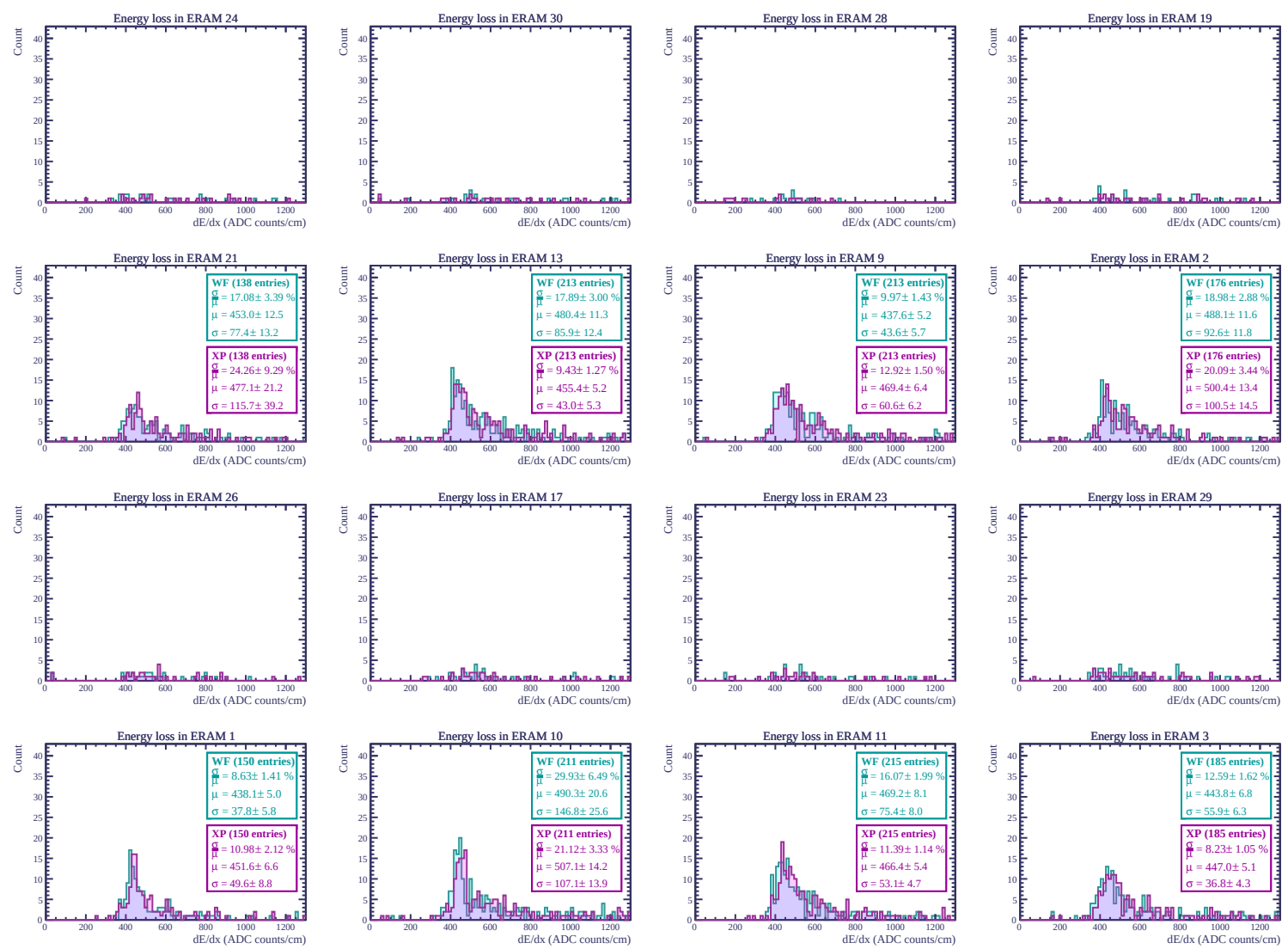
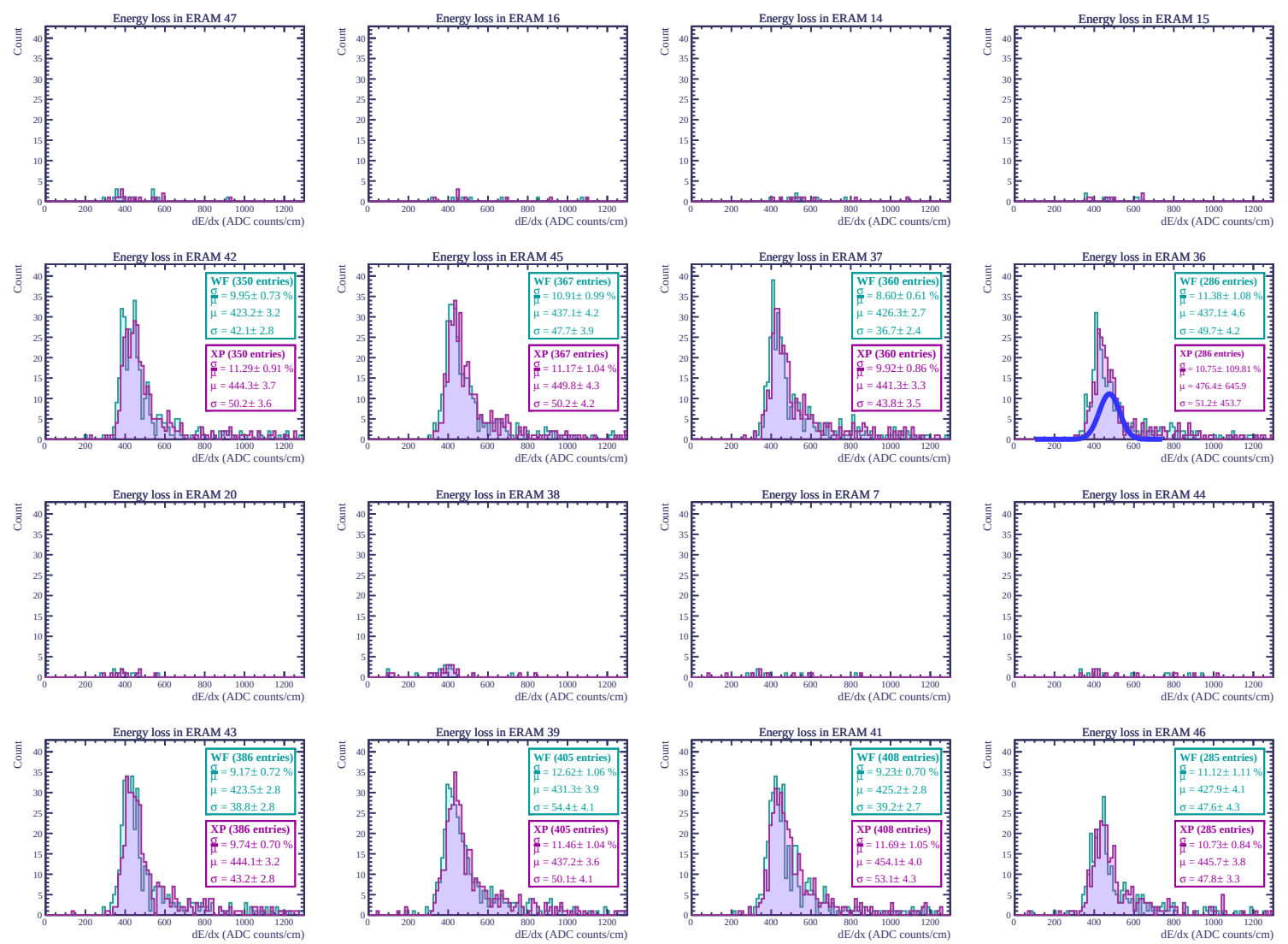
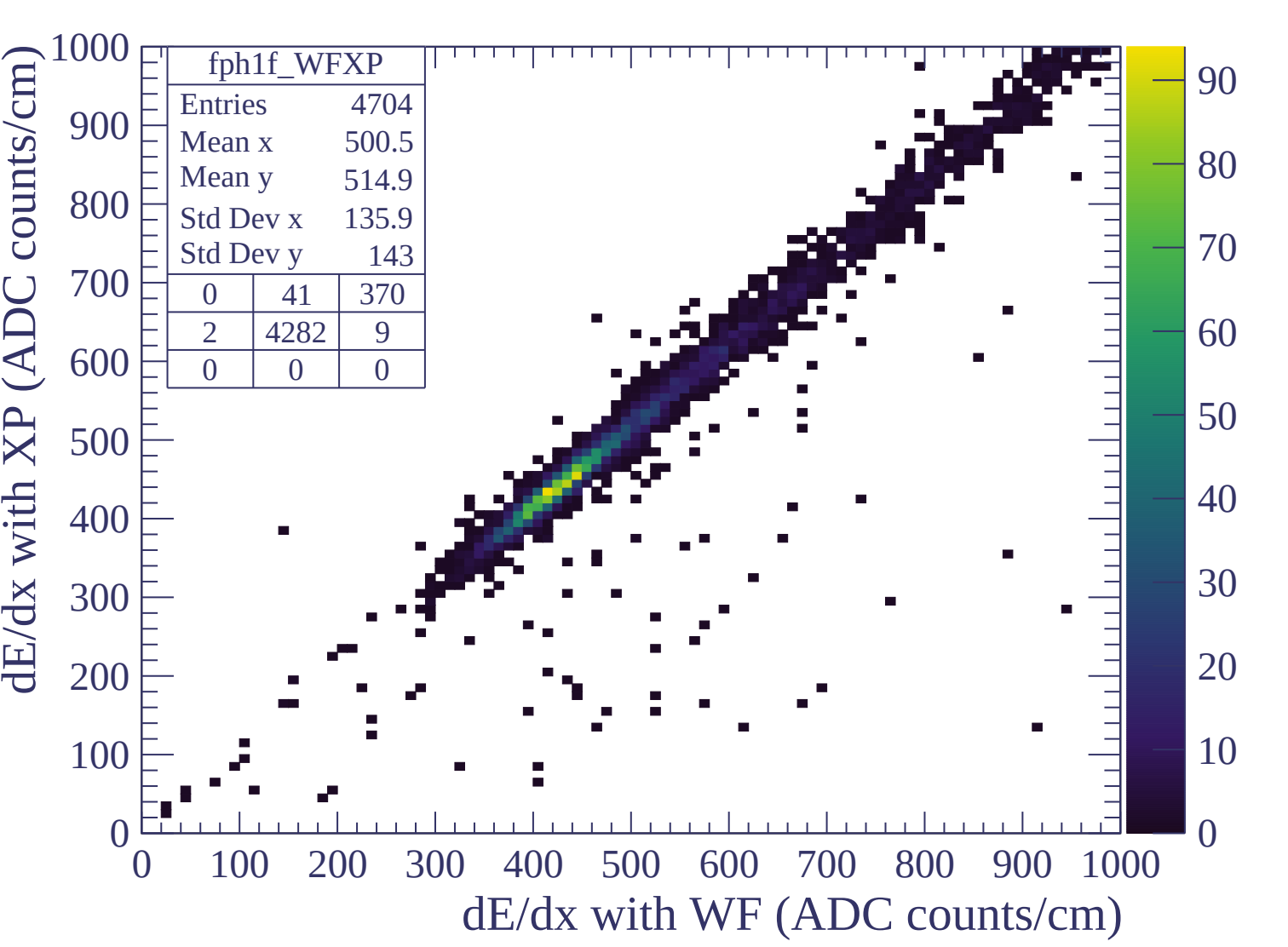


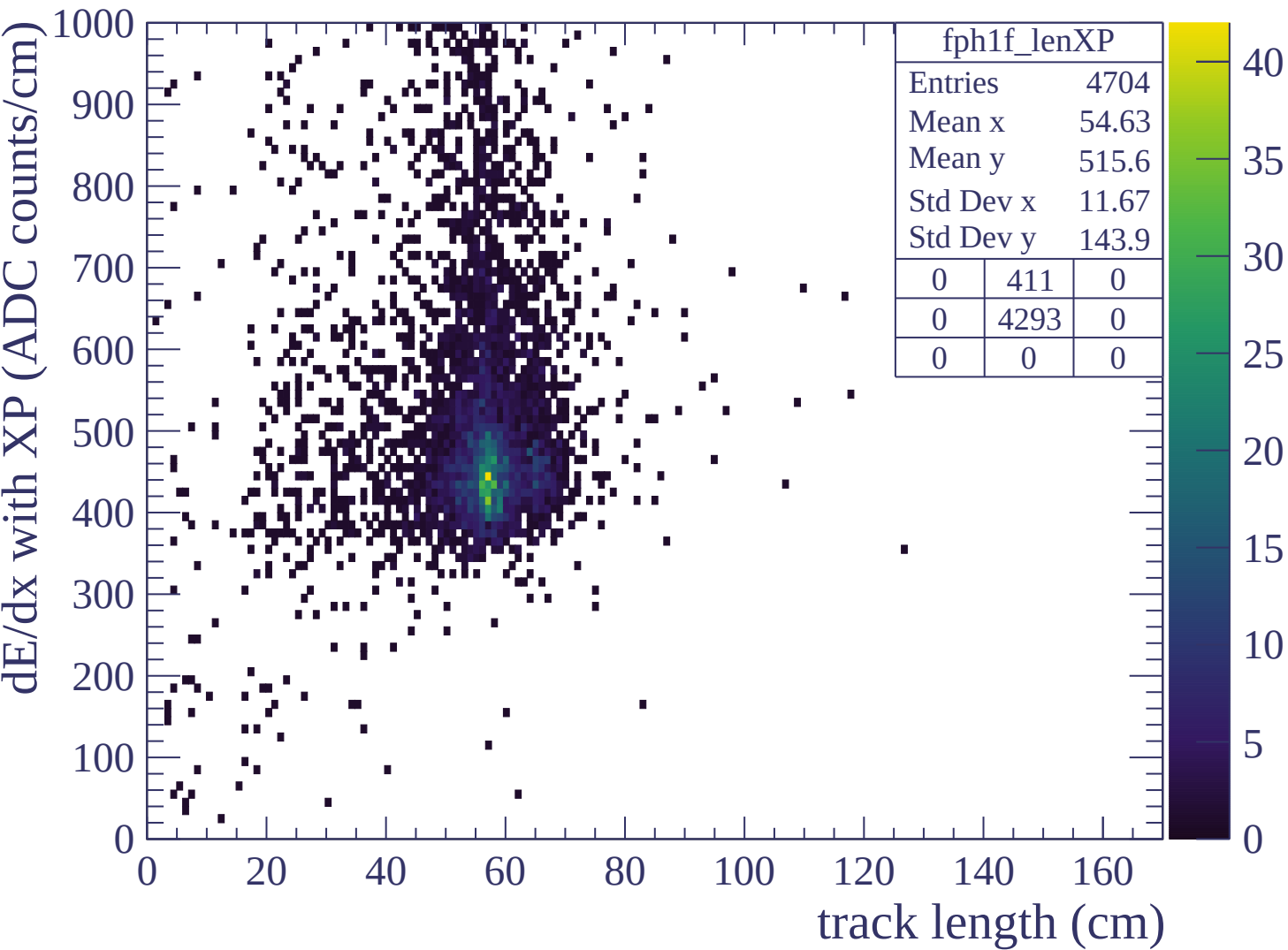
Count

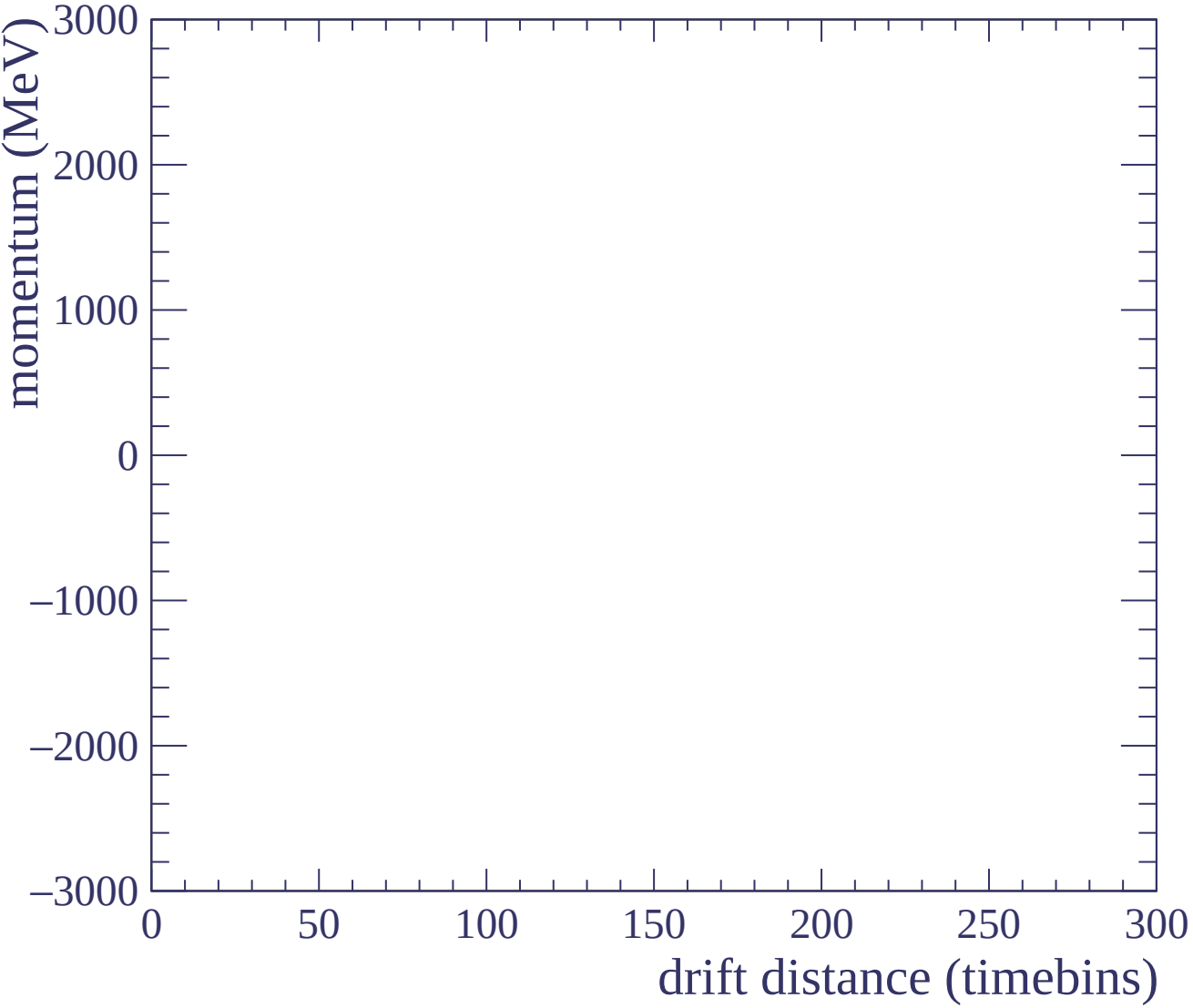


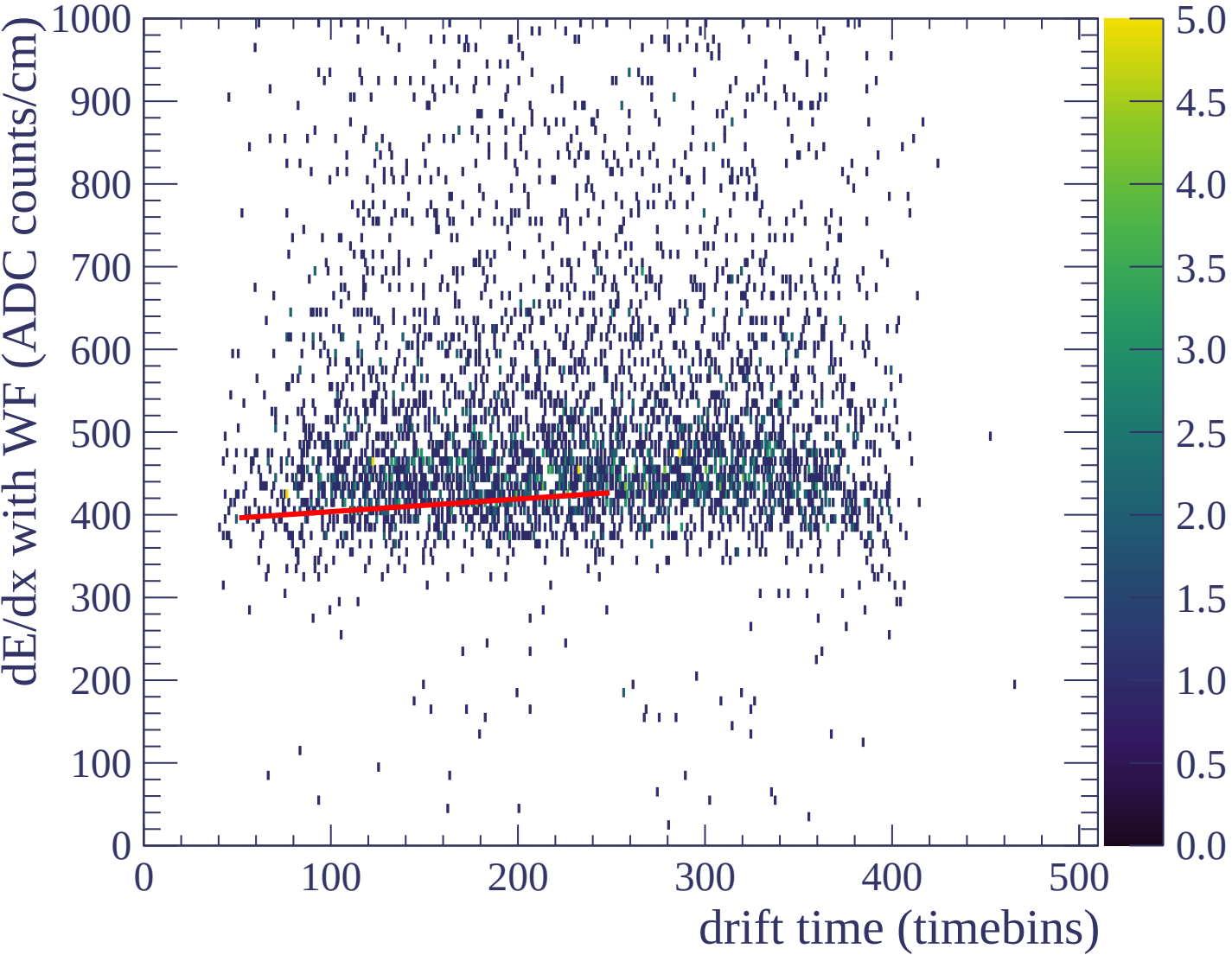


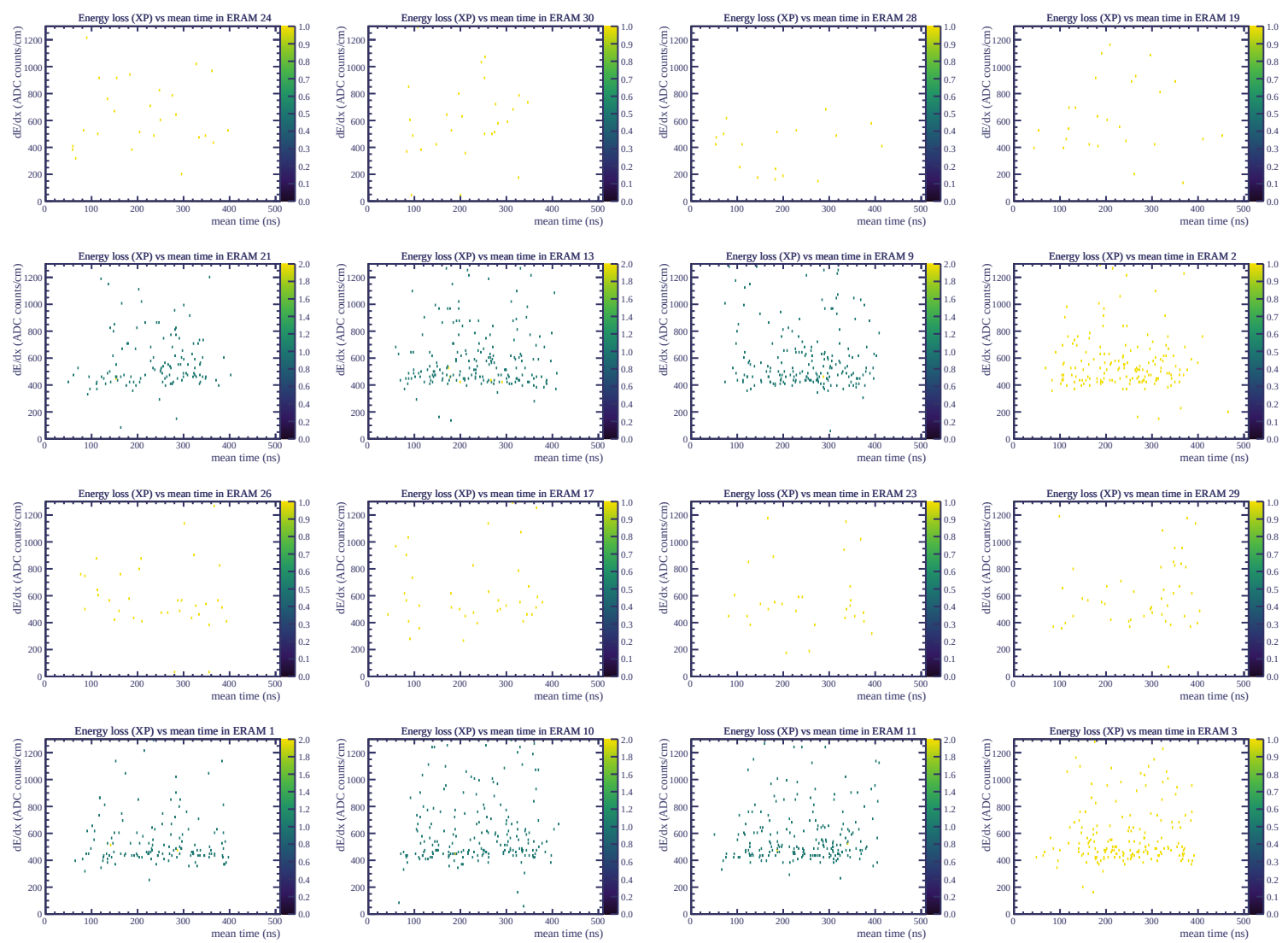


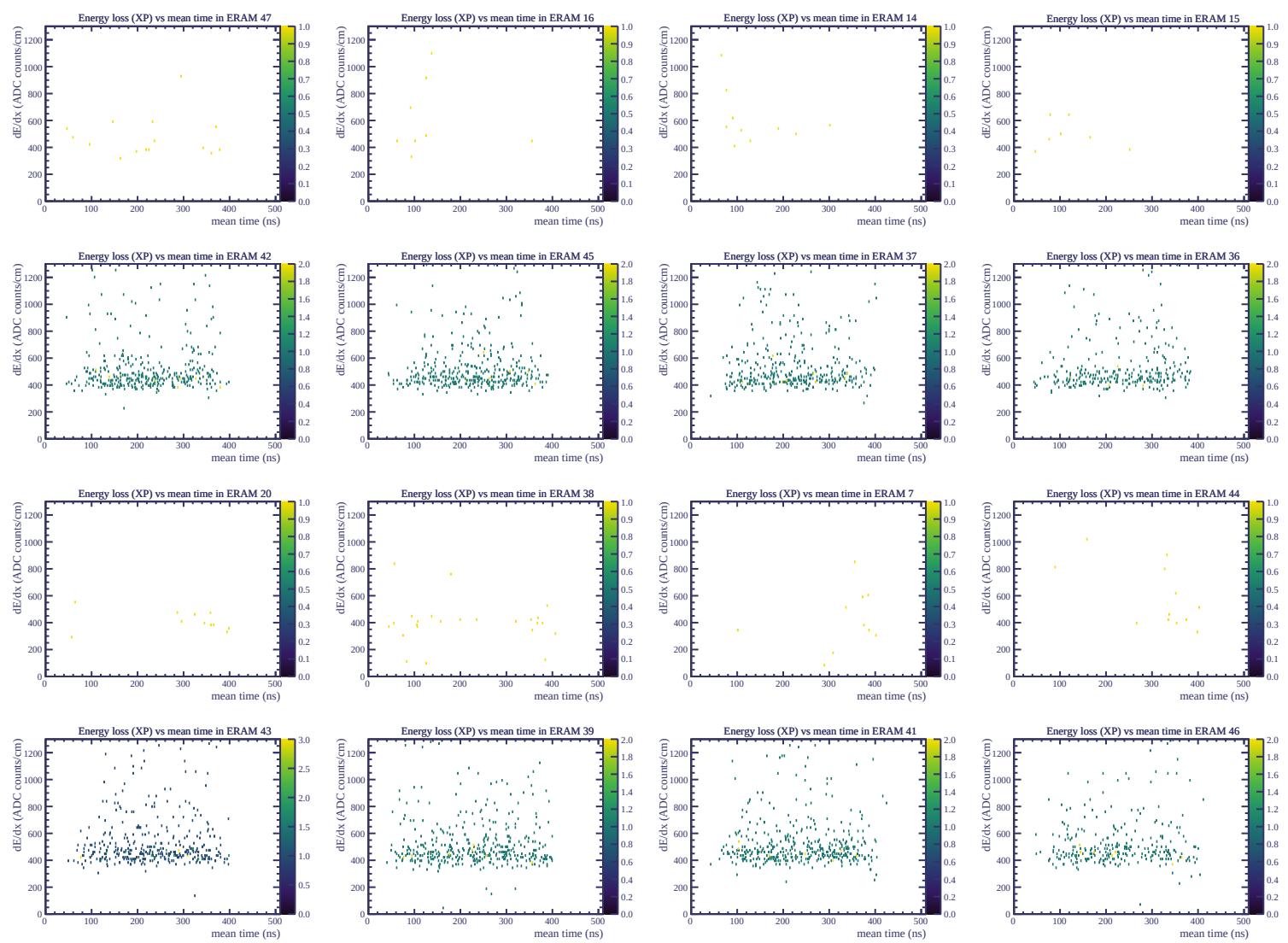


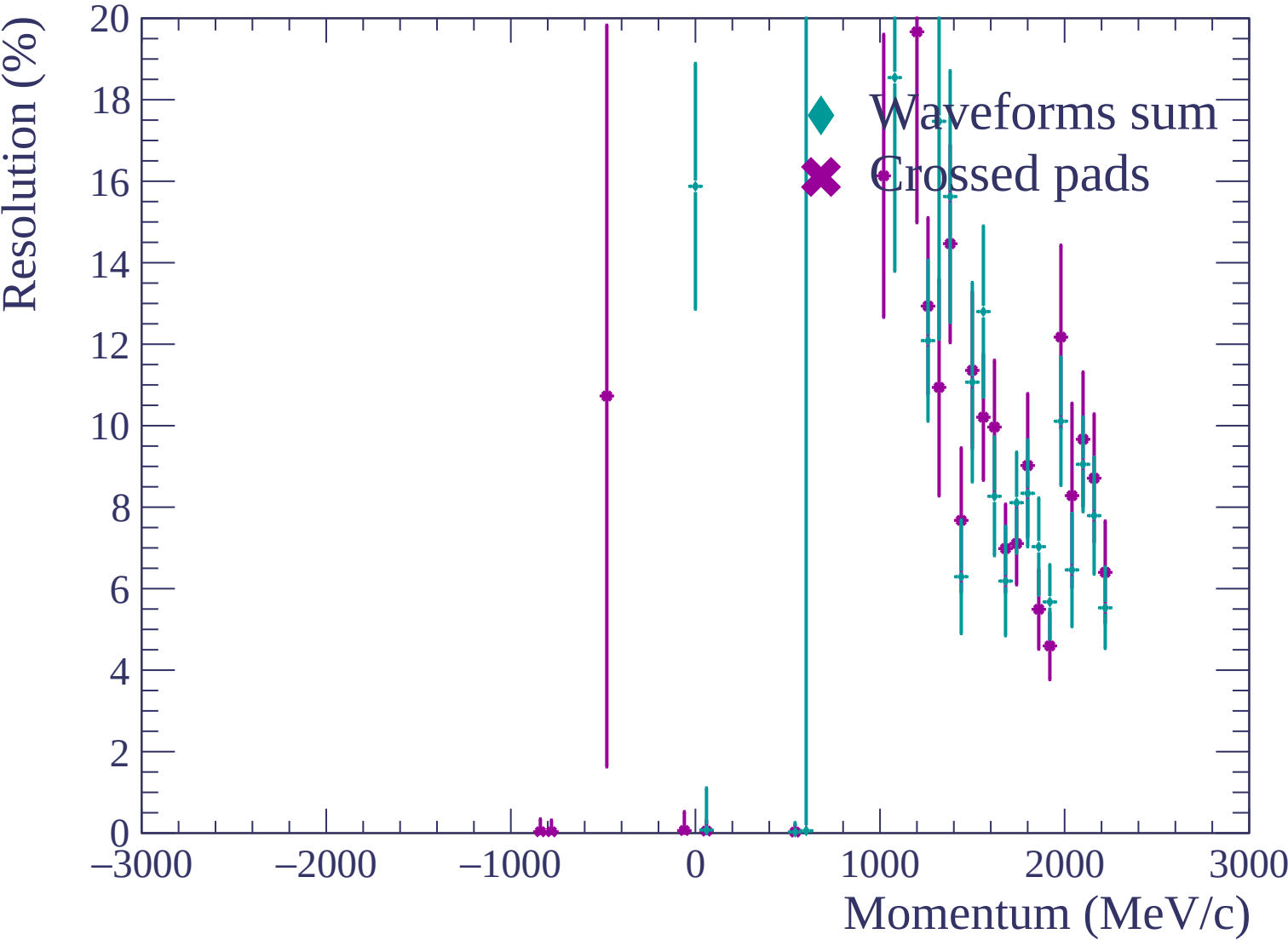


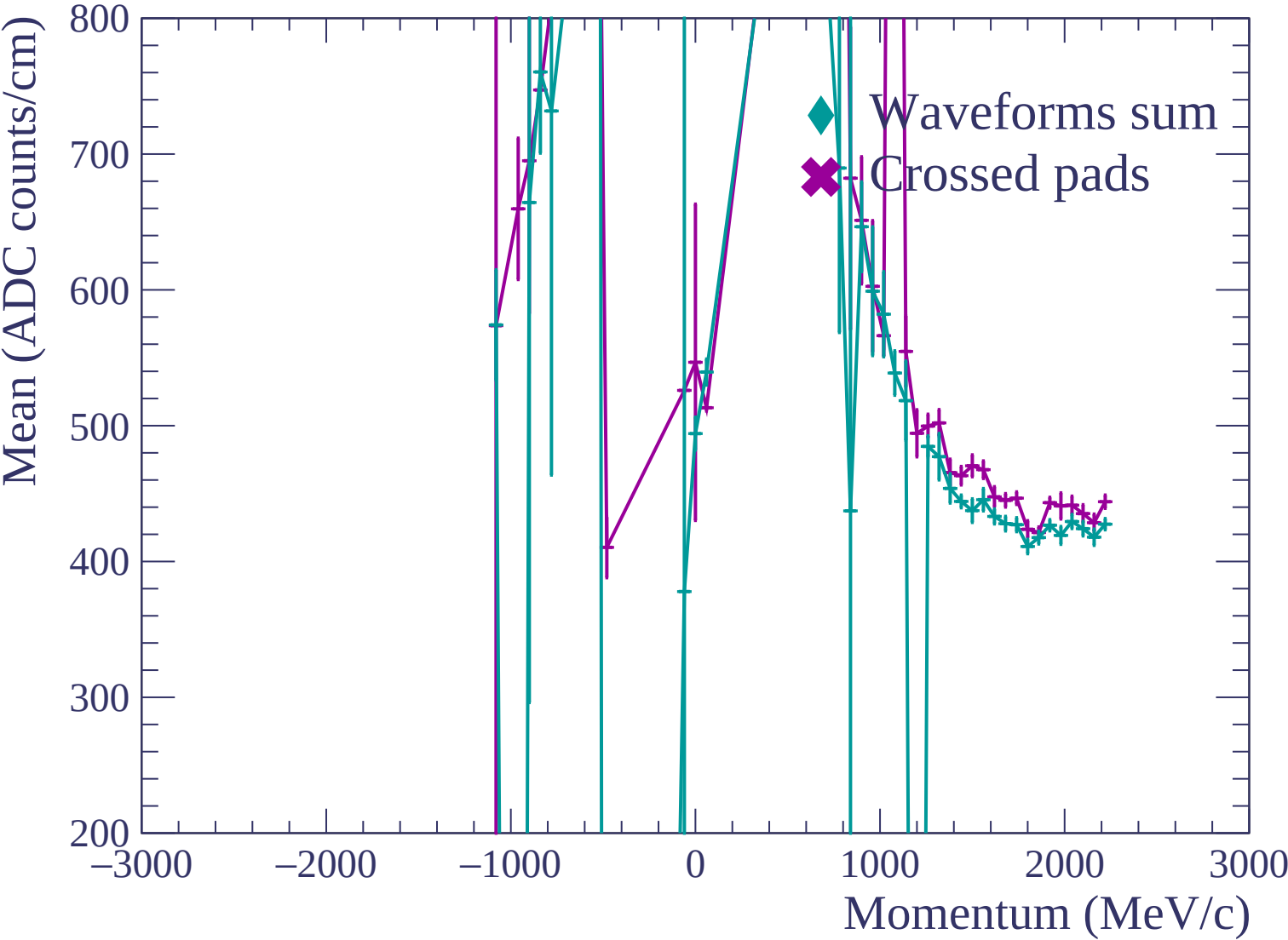


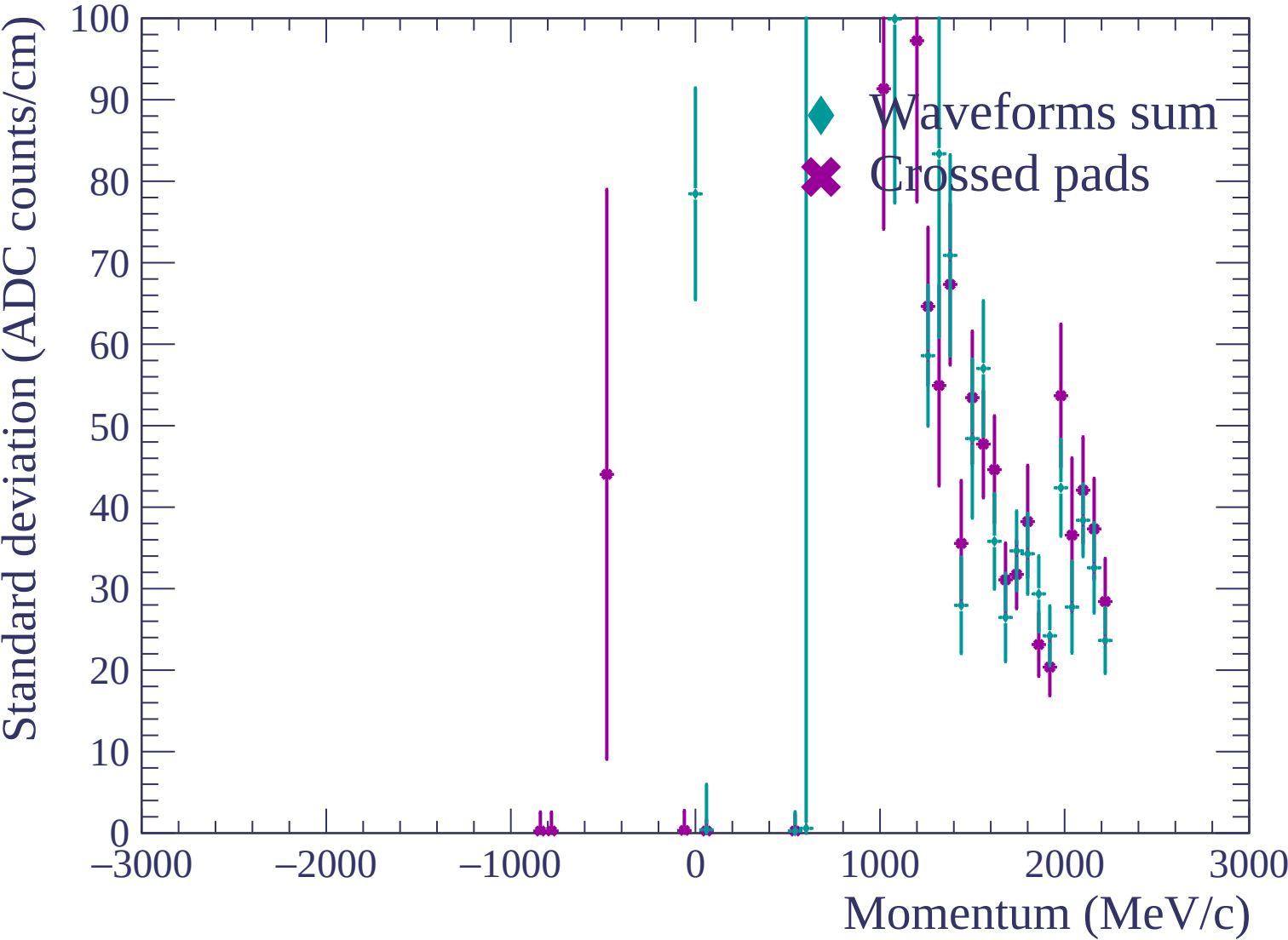


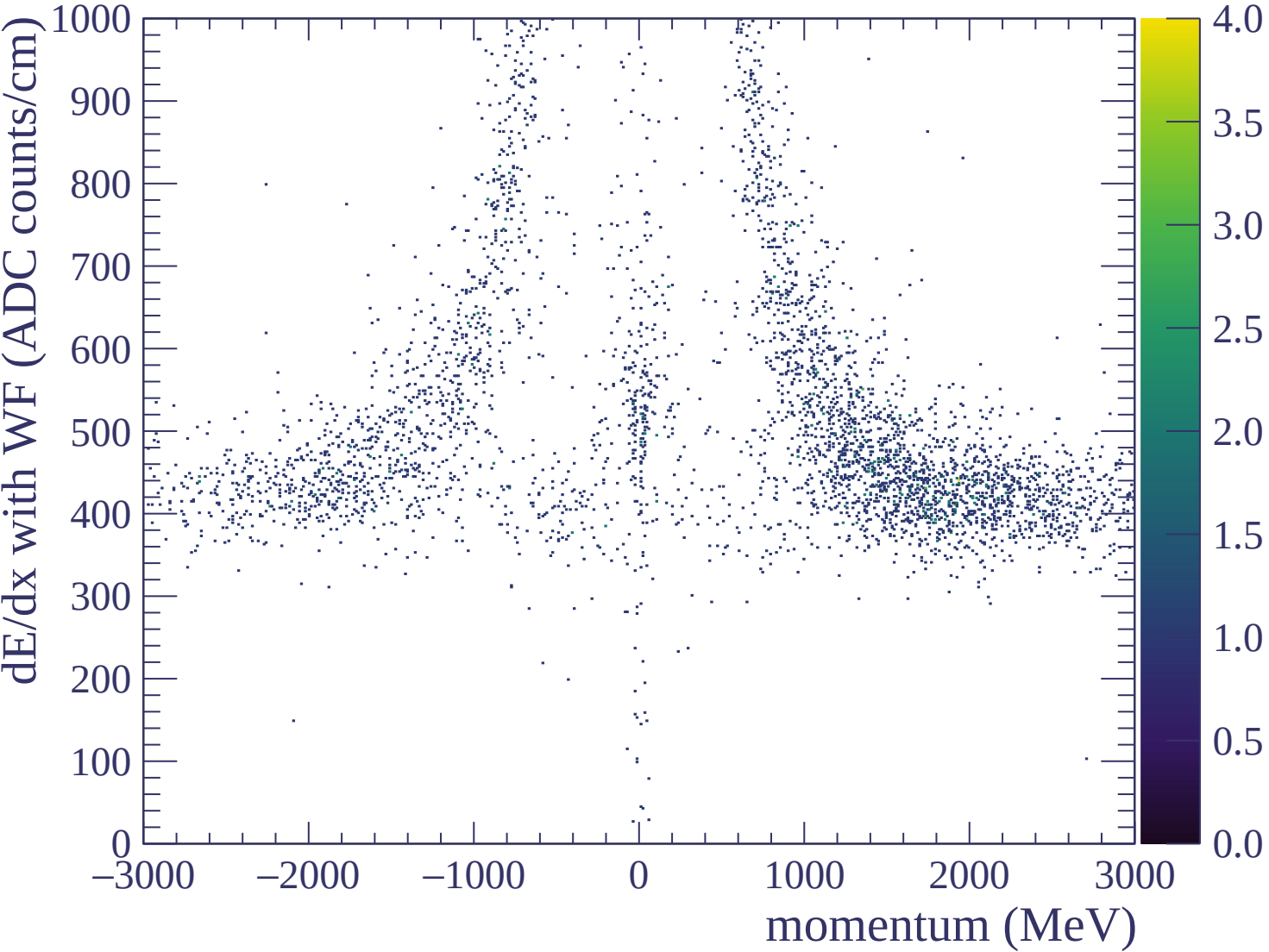


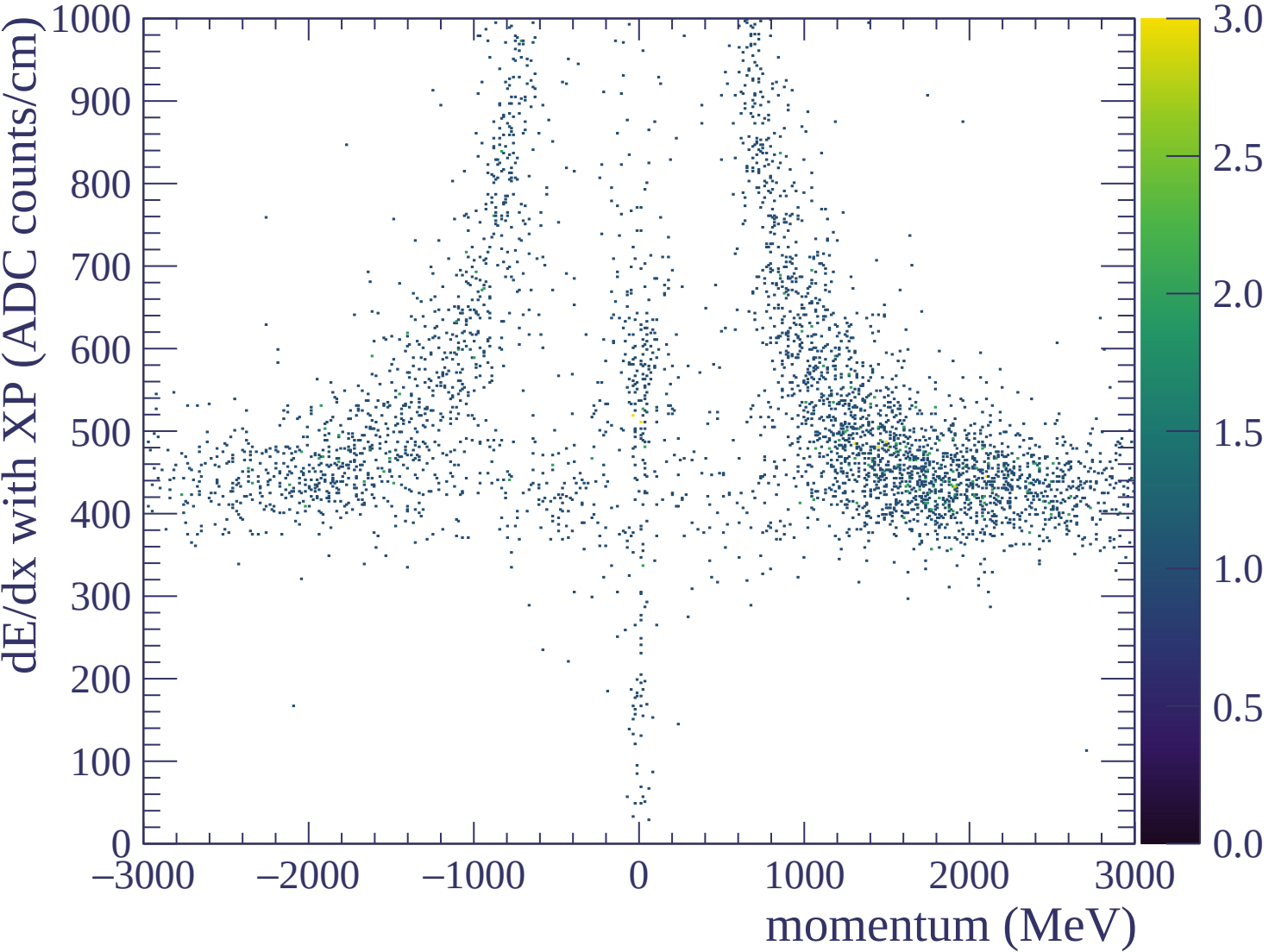


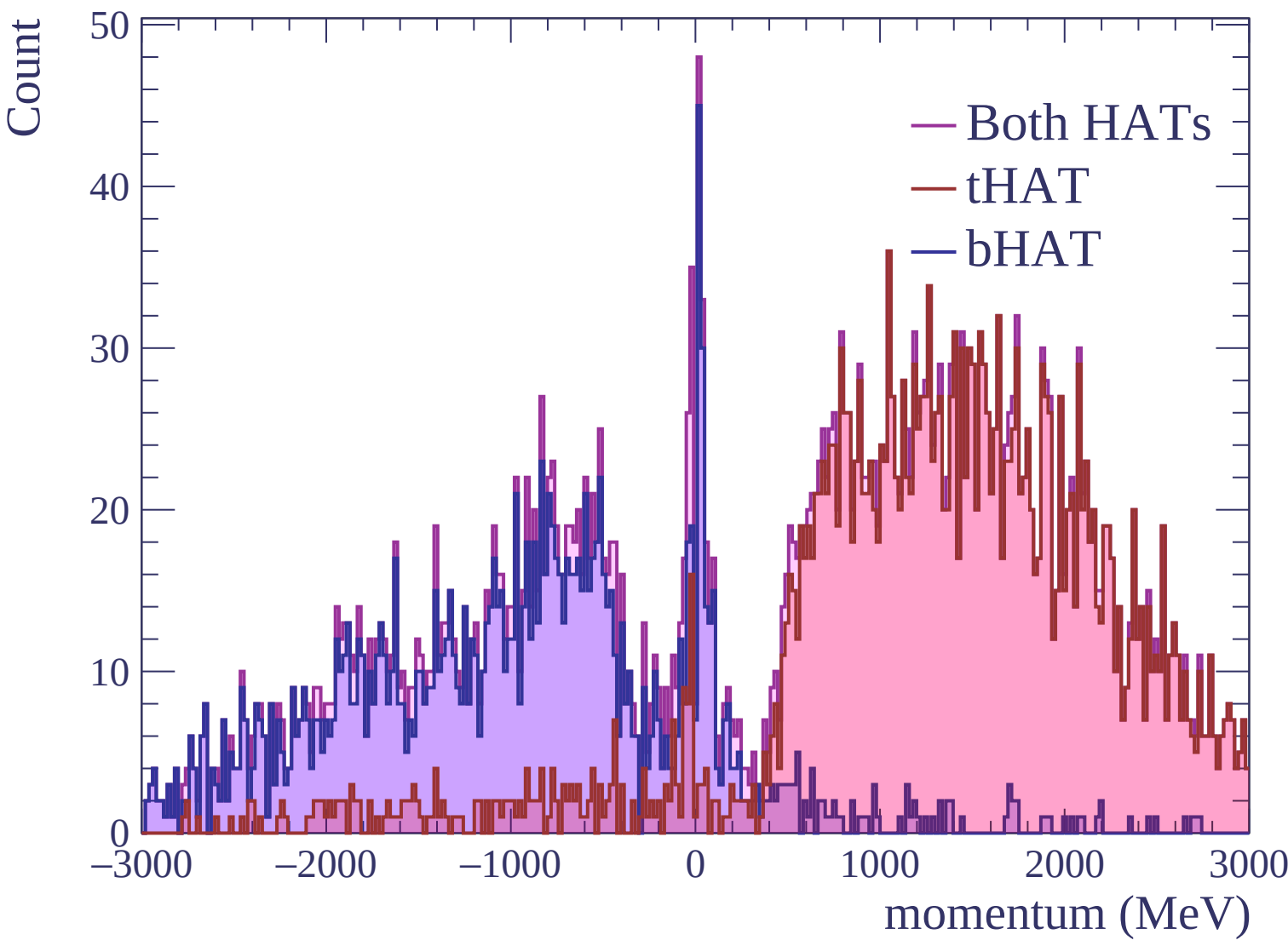


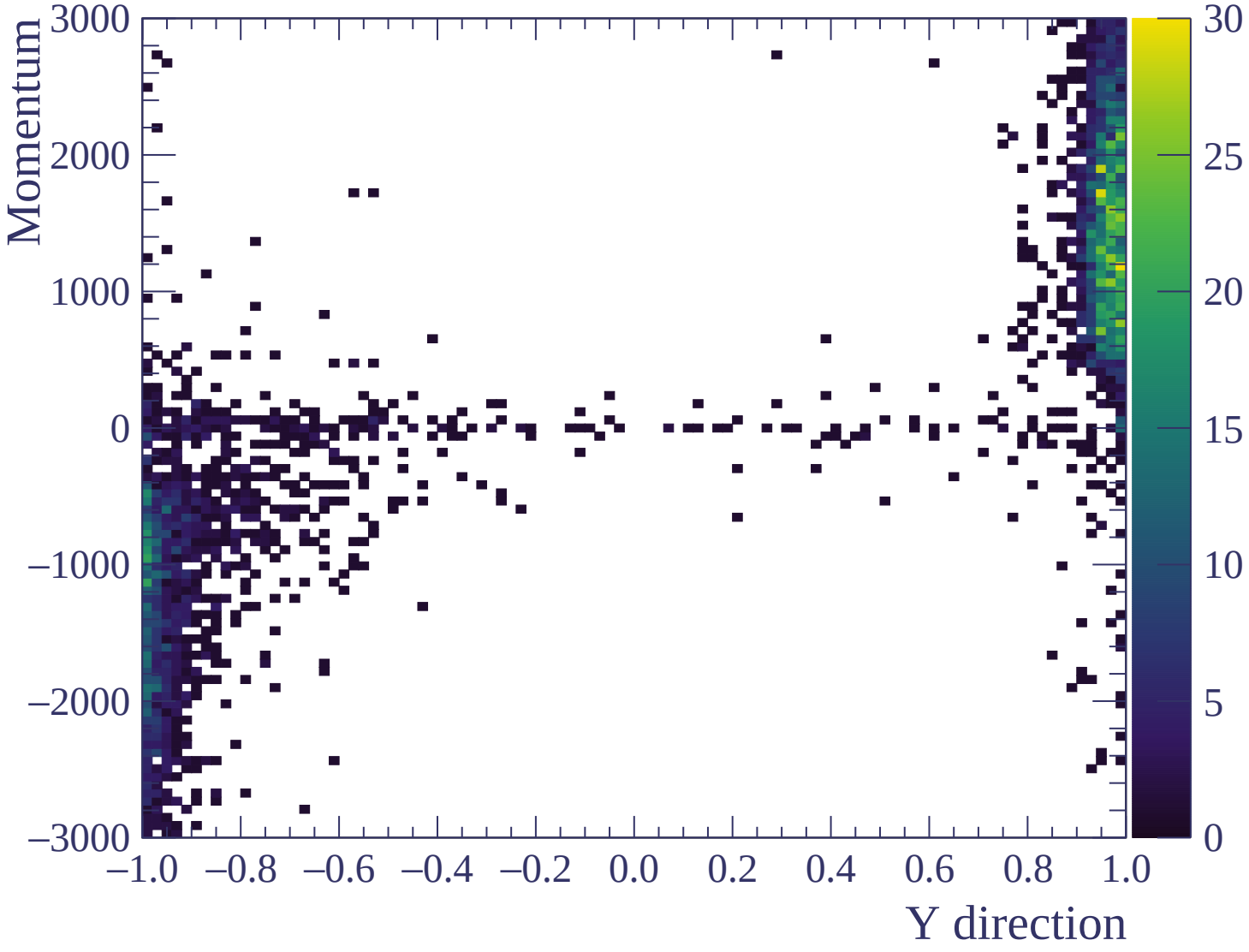


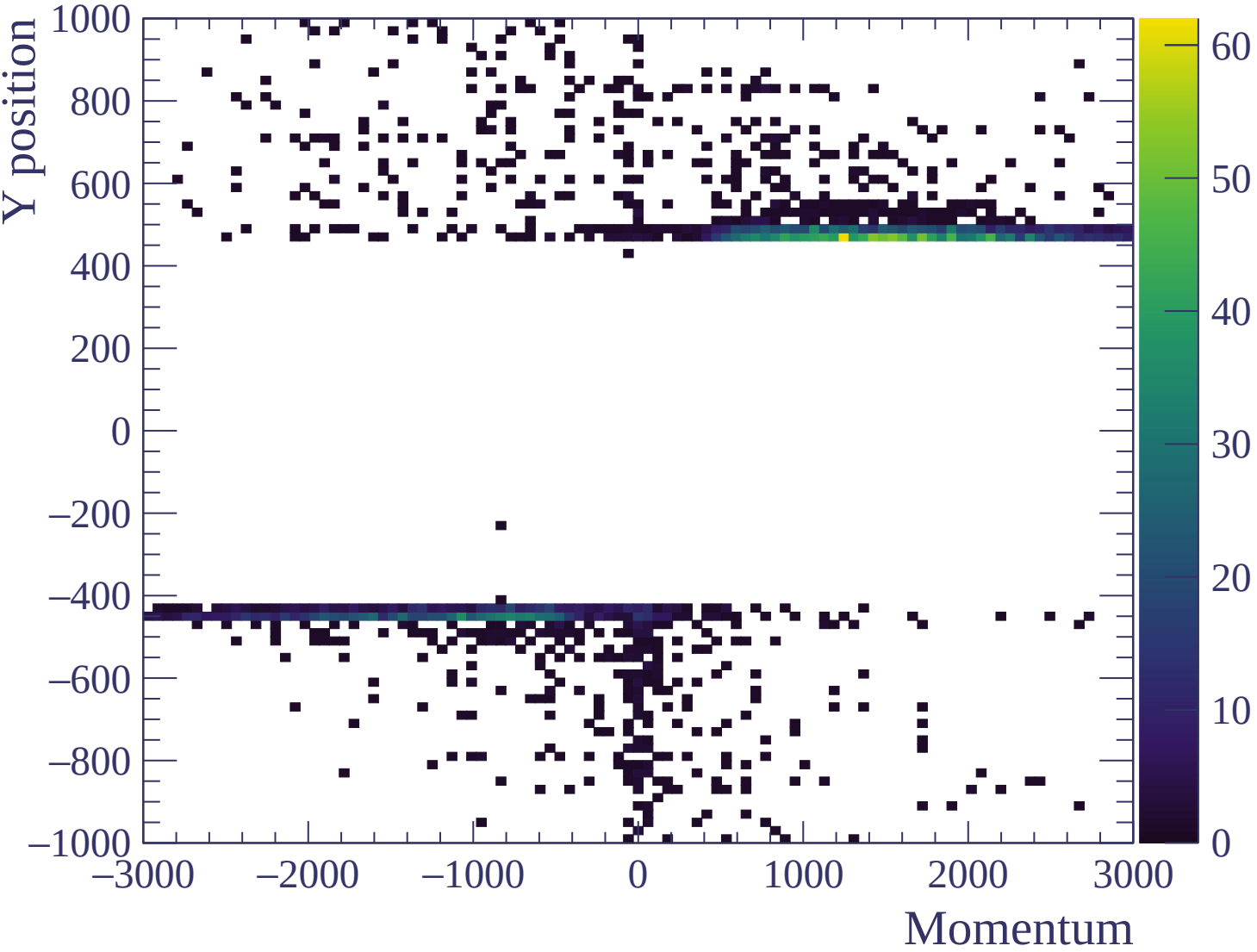


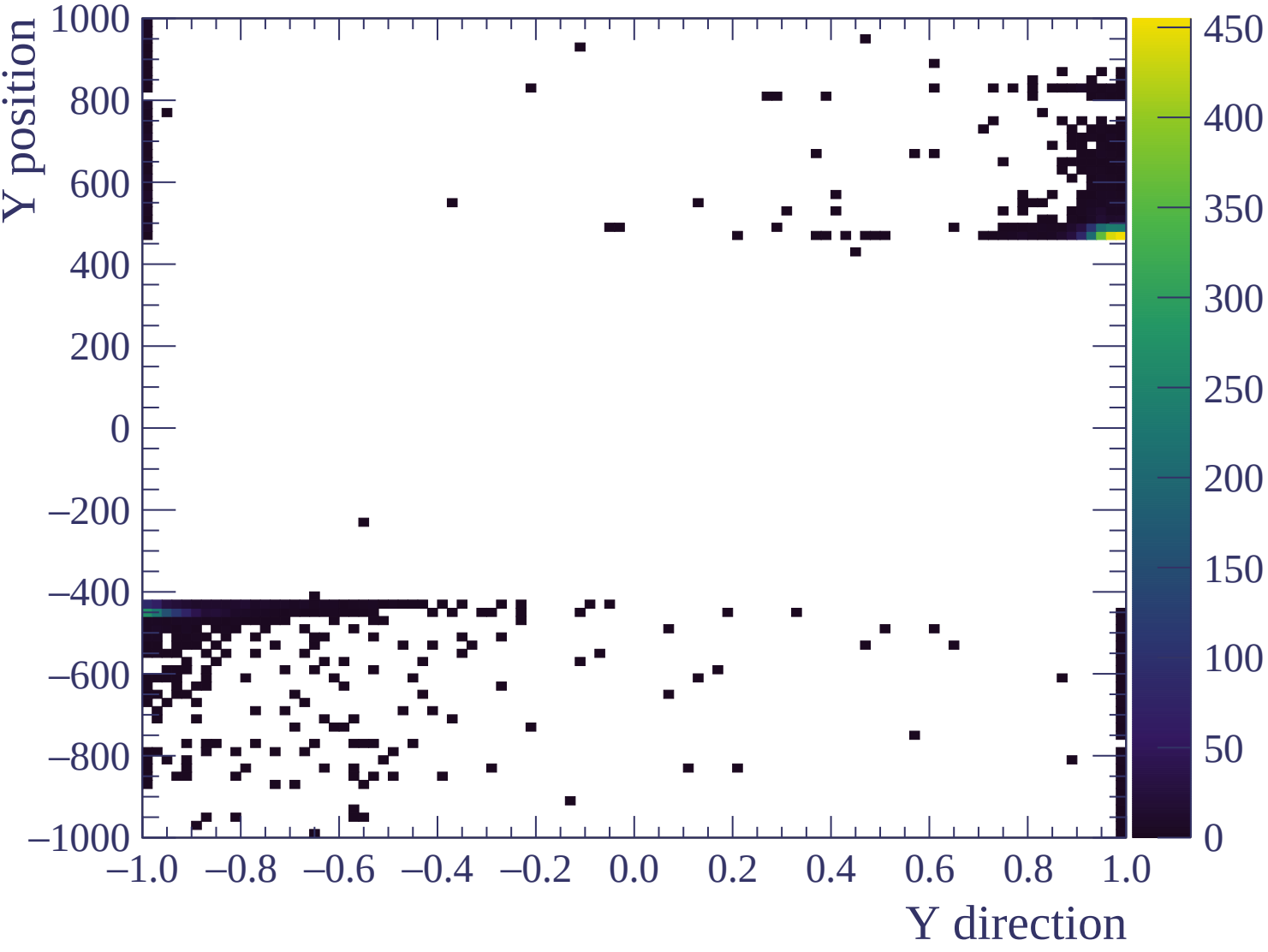


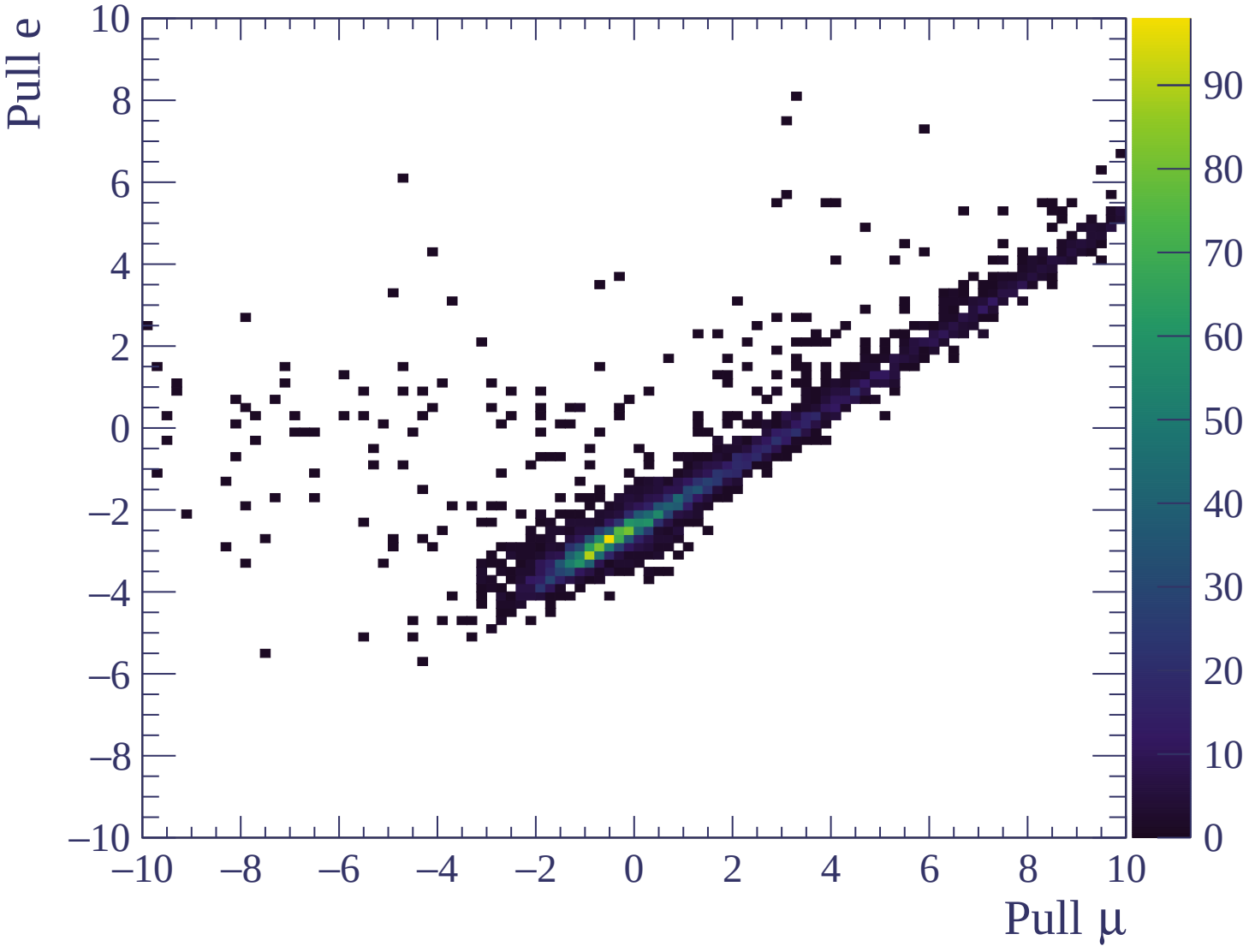












Energy loss | $-3000 < p < -2940$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

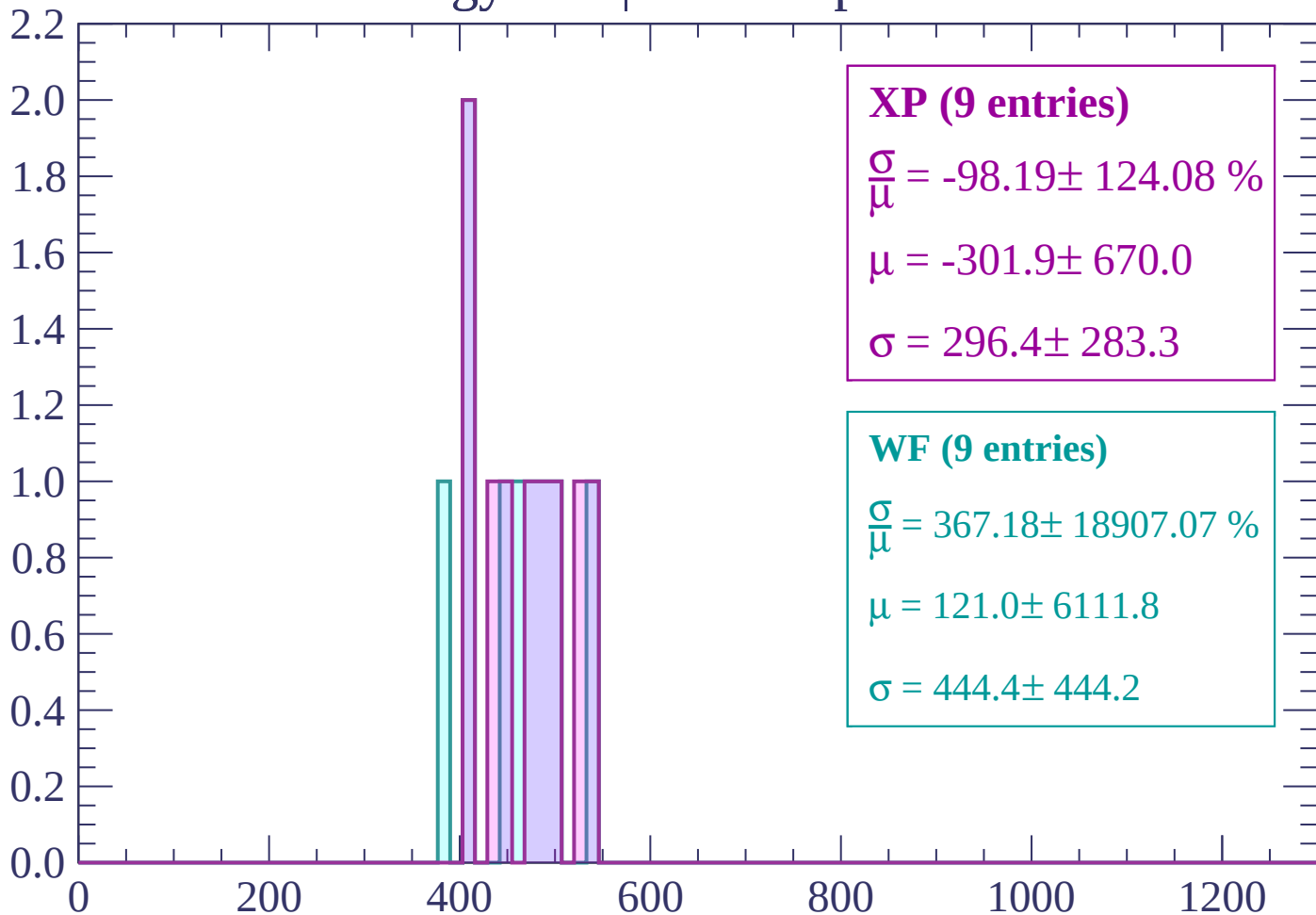
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2940 < p < -2880

Count



XP (9 entries)

$$\frac{\sigma}{\mu} = -98.19 \pm 124.08 \%$$

$$\mu = -301.9 \pm 670.0$$

$$\sigma = 296.4 \pm 283.3$$

WF (9 entries)

$$\frac{\sigma}{\mu} = 367.18 \pm 18907.07 \%$$

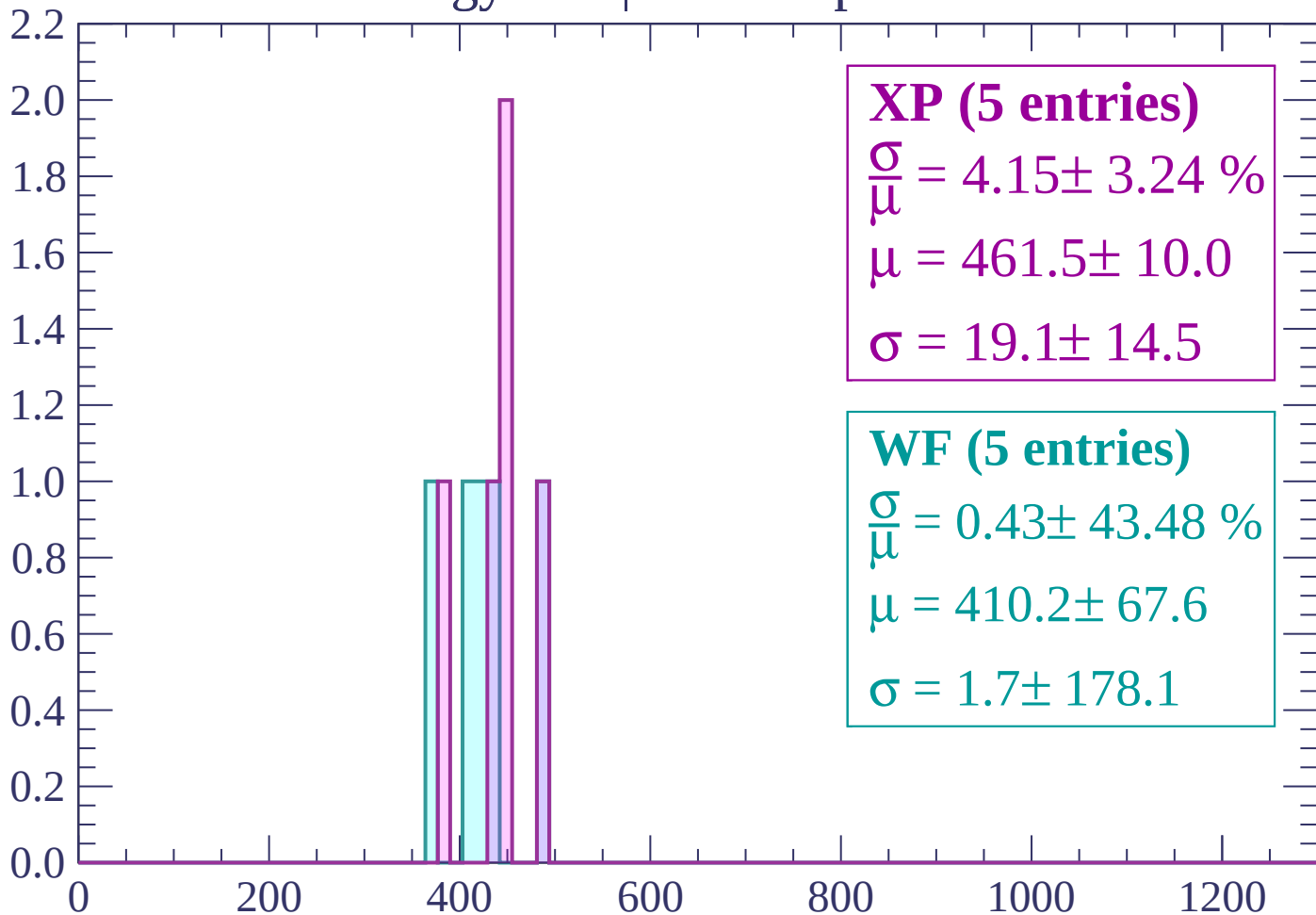
$$\mu = 121.0 \pm 6111.8$$

$$\sigma = 444.4 \pm 444.2$$

dE/dx (ADC counts/cm)

Energy loss | $-2880 < p < -2820$

Count



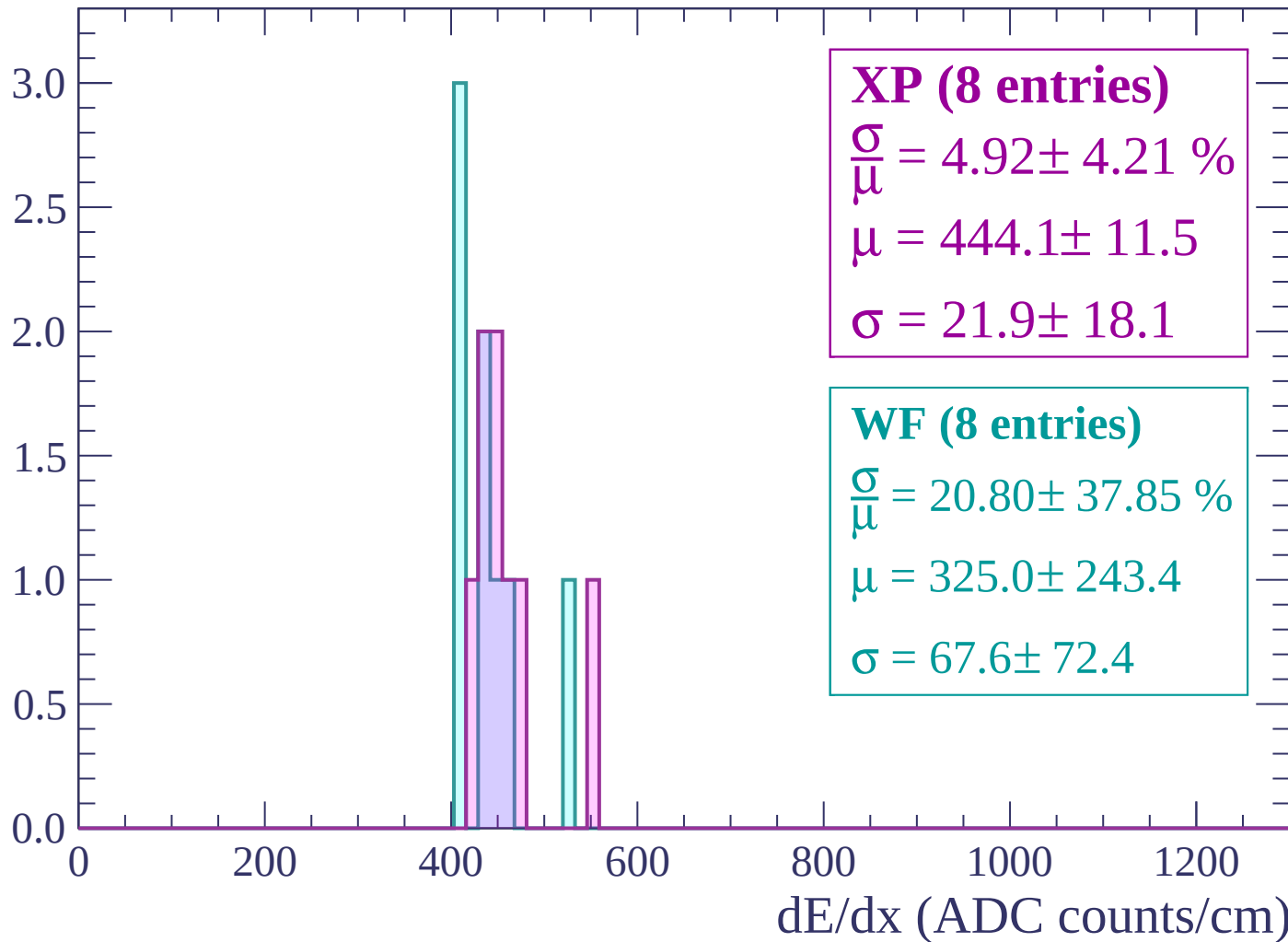
XP (5 entries)
 $\frac{\sigma}{\mu} = 4.15 \pm 3.24 \%$
 $\mu = 461.5 \pm 10.0$
 $\sigma = 19.1 \pm 14.5$

WF (5 entries)
 $\frac{\sigma}{\mu} = 0.43 \pm 43.48 \%$
 $\mu = 410.2 \pm 67.6$
 $\sigma = 1.7 \pm 178.1$

dE/dx (ADC counts/cm)

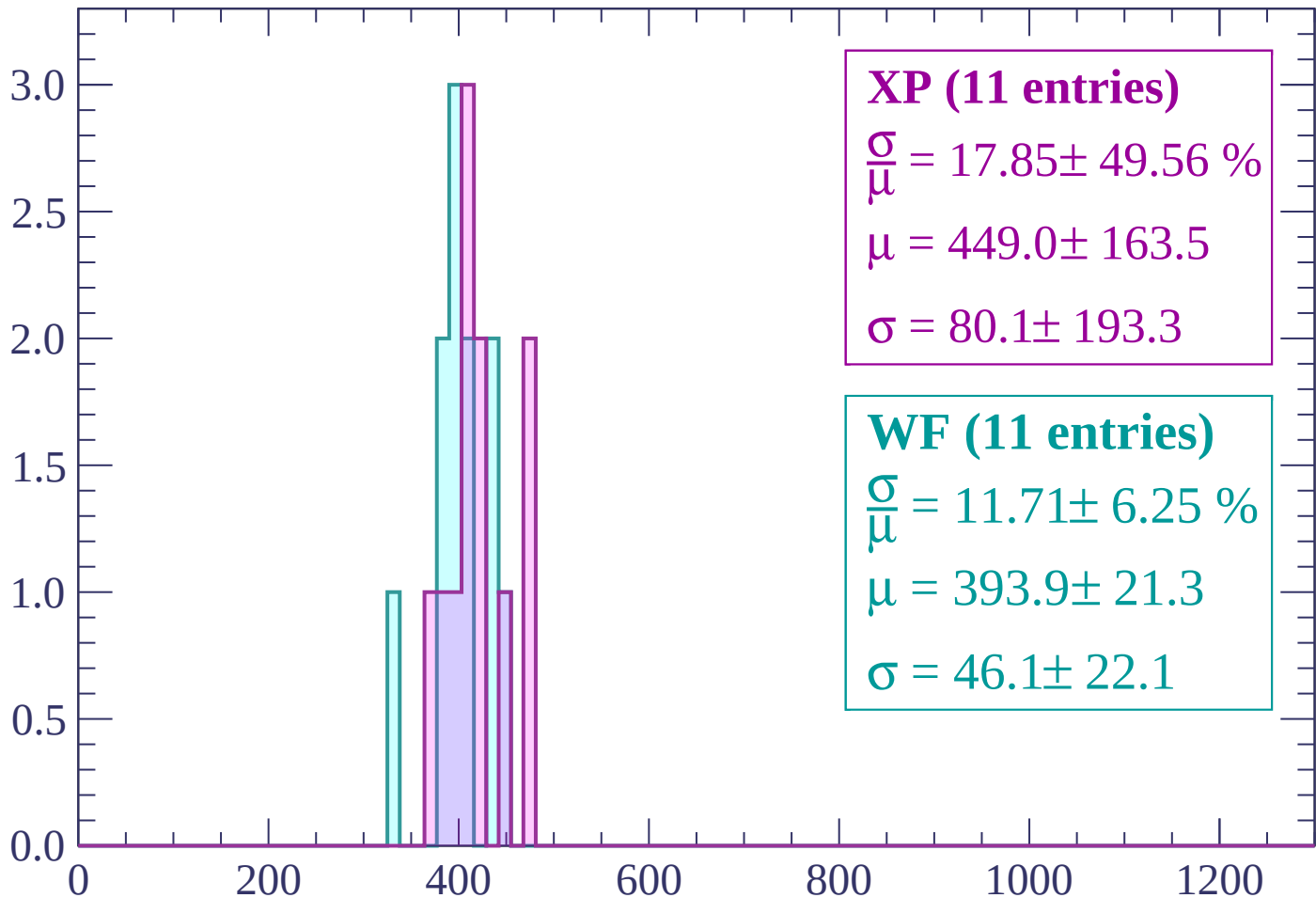
Energy loss | $-2820 < p < -2760$

Count



Energy loss | -2760 < p < -2700

Count



XP (11 entries)

$$\frac{\sigma}{\mu} = 17.85 \pm 49.56 \%$$

$$\mu = 449.0 \pm 163.5$$

$$\sigma = 80.1 \pm 193.3$$

WF (11 entries)

$$\frac{\sigma}{\mu} = 11.71 \pm 6.25 \%$$

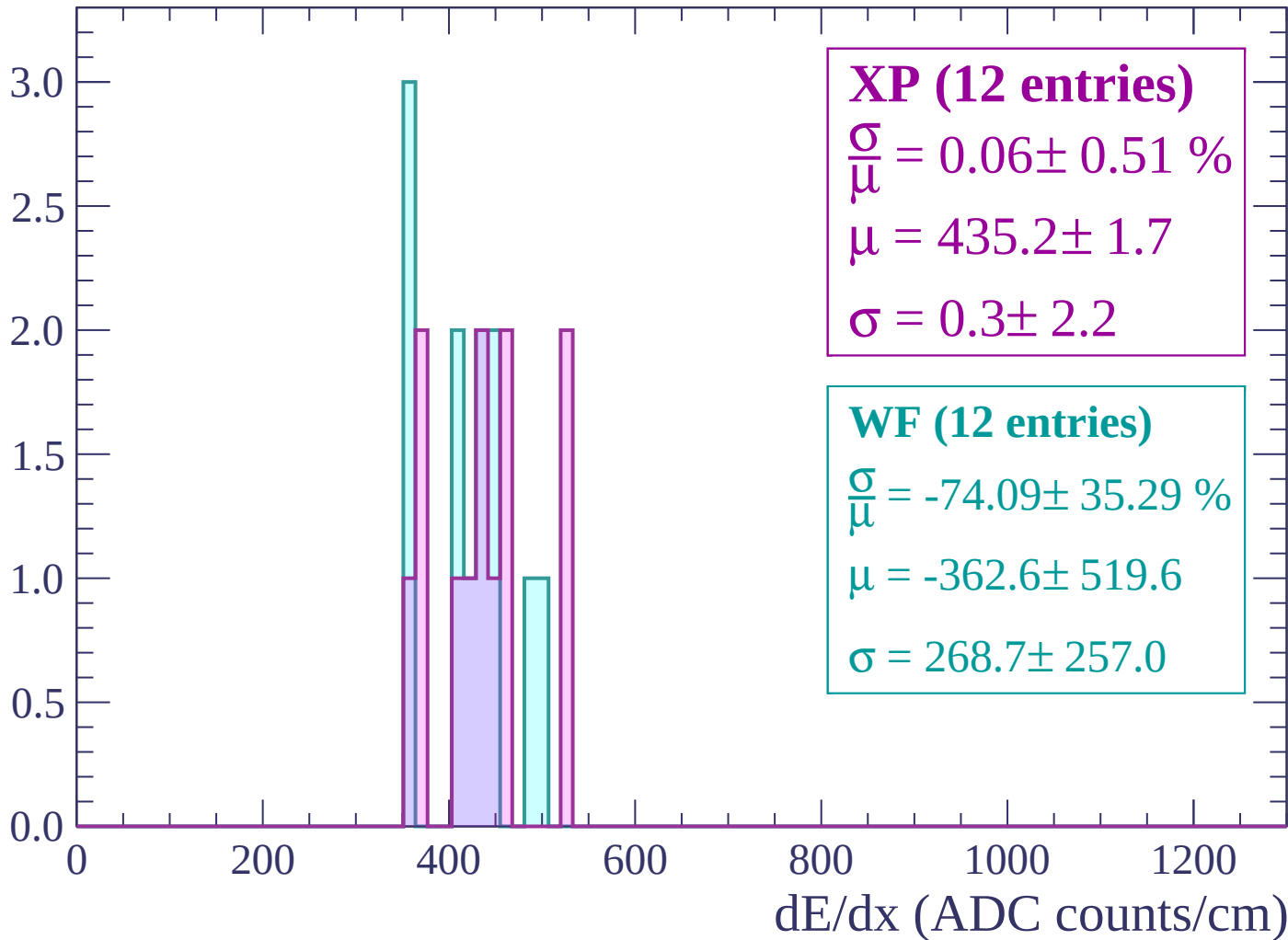
$$\mu = 393.9 \pm 21.3$$

$$\sigma = 46.1 \pm 22.1$$

dE/dx (ADC counts/cm)

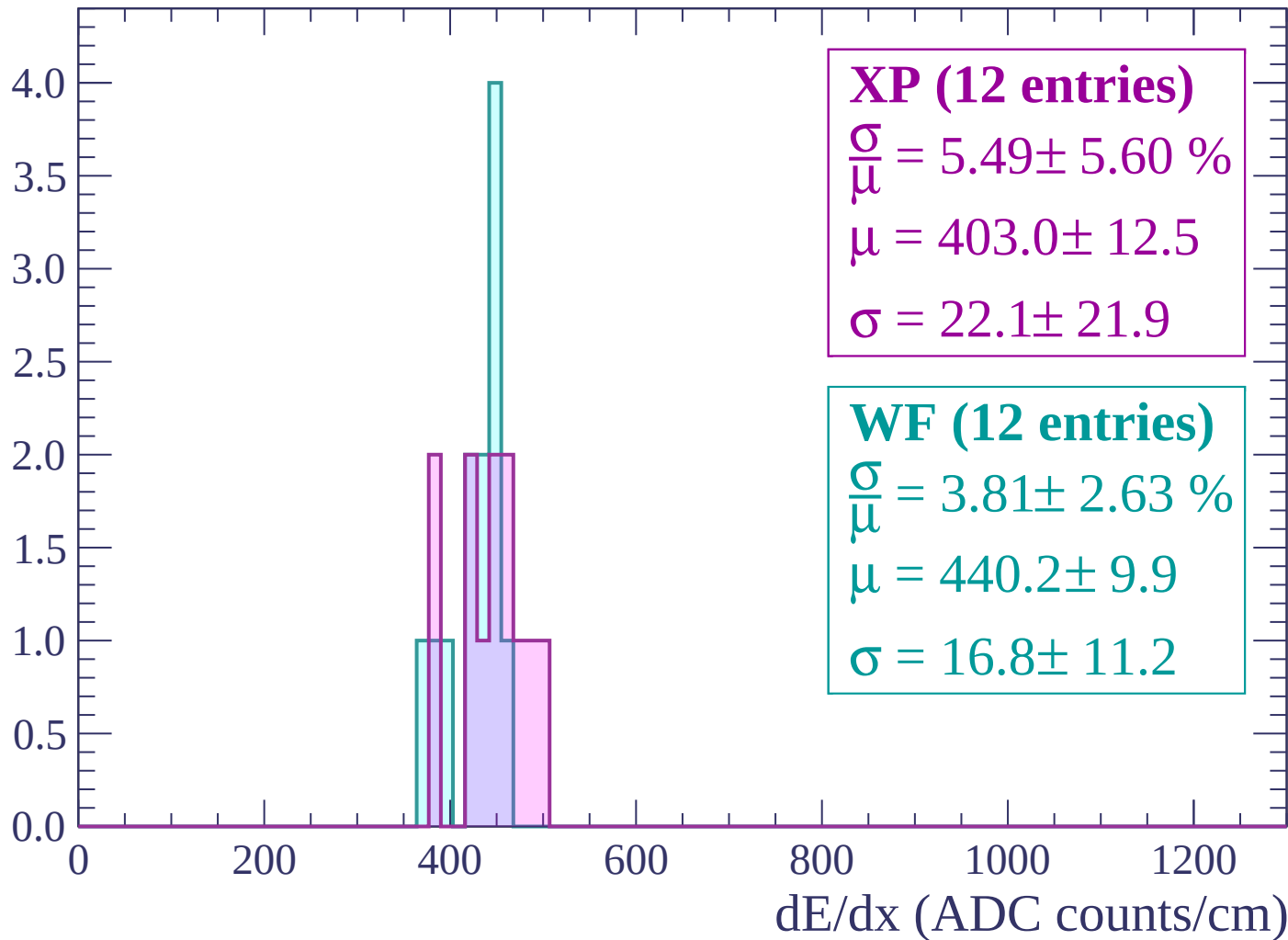
Energy loss | -2700 < p < -2640

Count



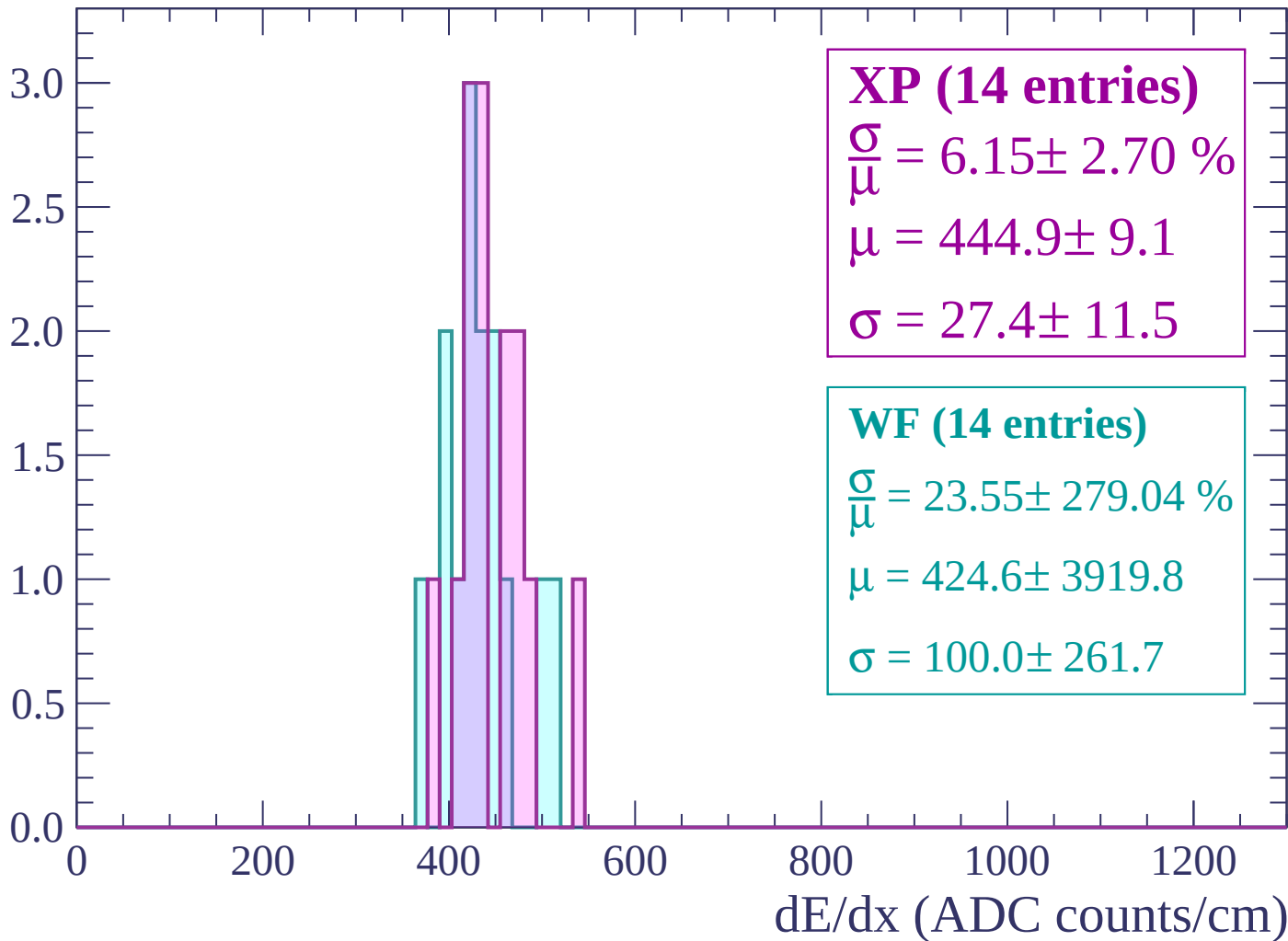
Energy loss | $-2640 < p < -2580$

Count



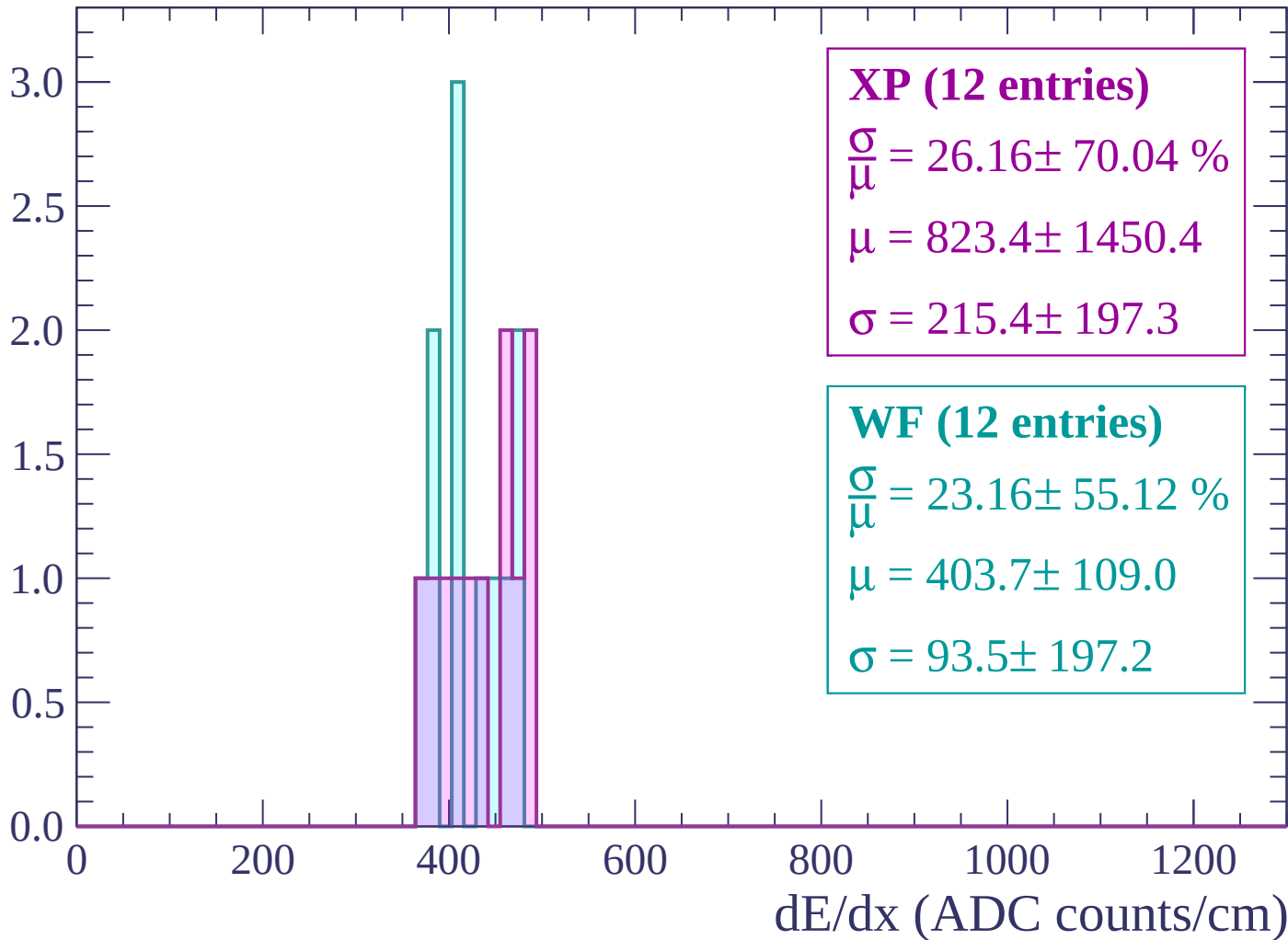
Energy loss | -2580 < p < -2520

Count



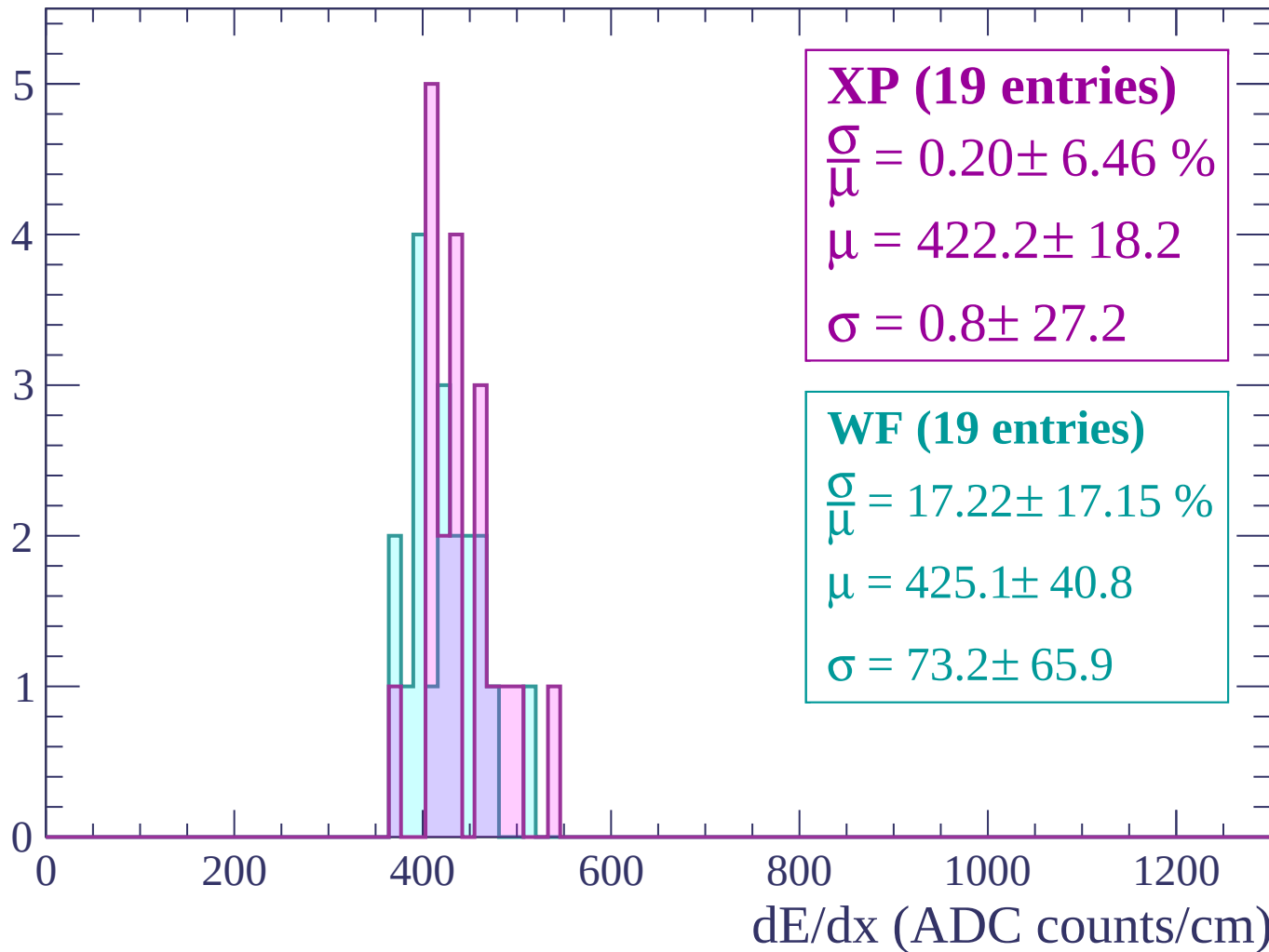
Energy loss | -2520 < p < -2460

Count



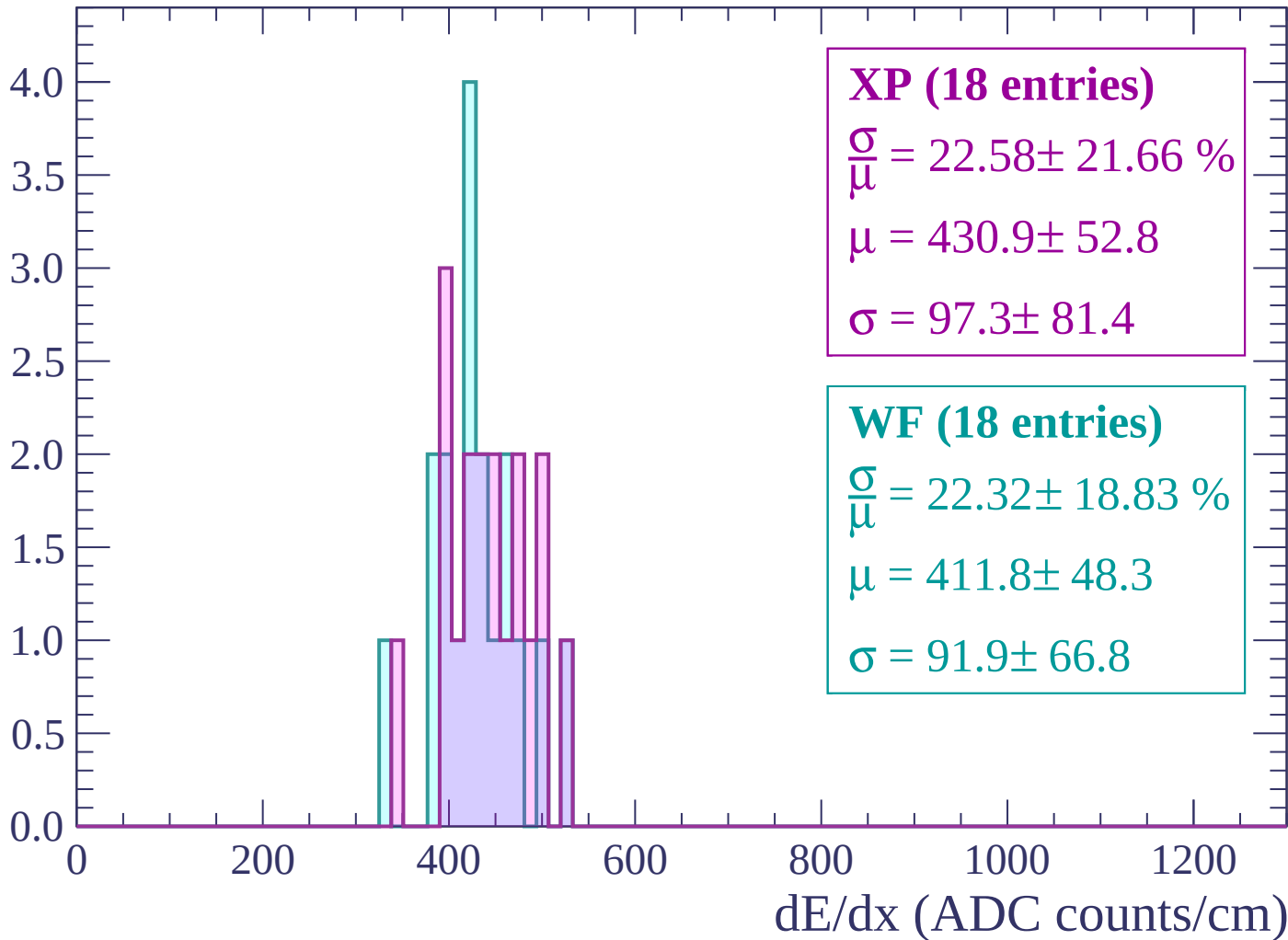
Energy loss | $-2460 < p < -2400$

Count



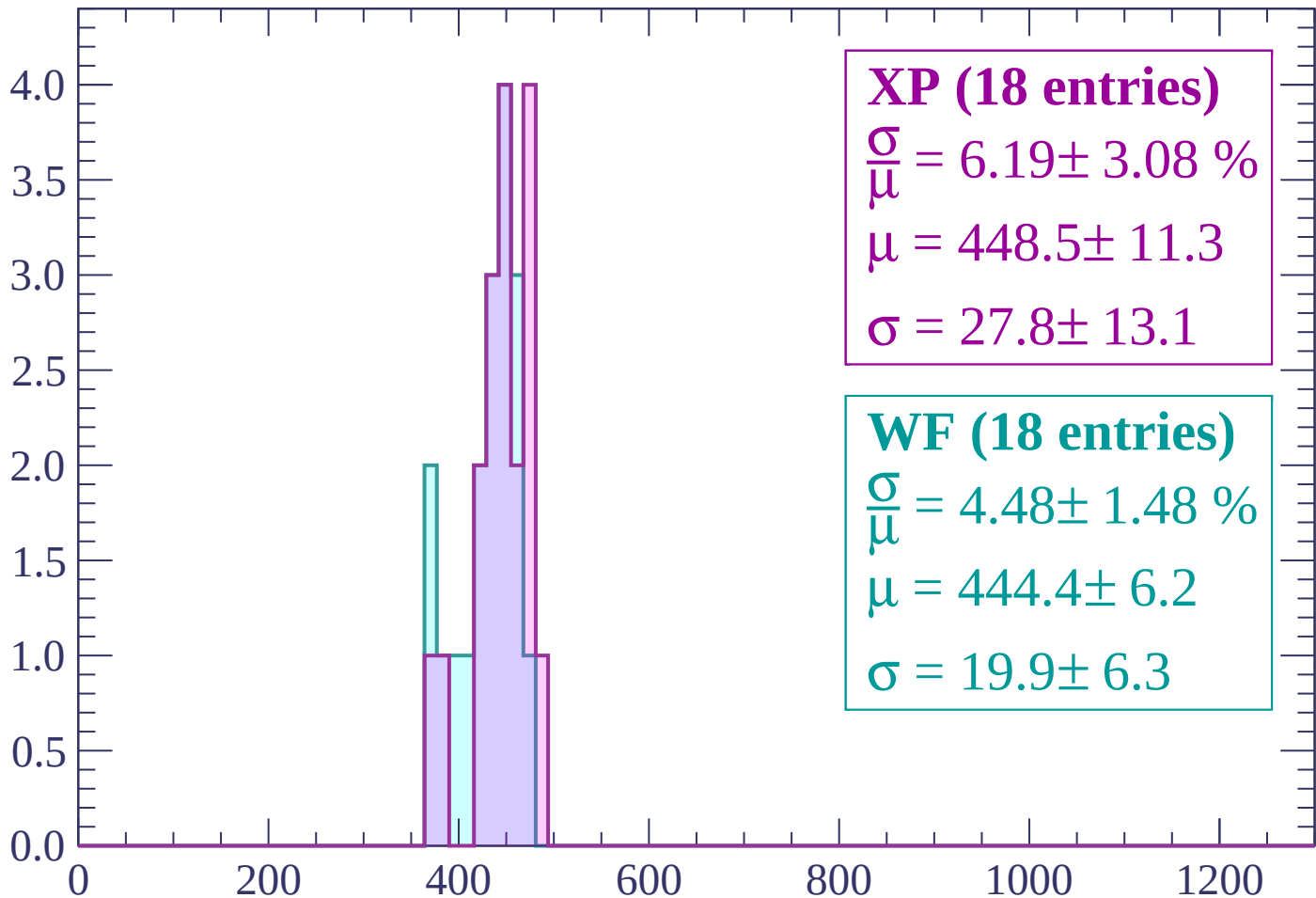
Energy loss | $-2400 < p < -2340$

Count



Energy loss | $-2340 < p < -2280$

Count



XP (18 entries)

$$\frac{\sigma}{\mu} = 6.19 \pm 3.08 \%$$

$$\mu = 448.5 \pm 11.3$$

$$\sigma = 27.8 \pm 13.1$$

WF (18 entries)

$$\frac{\sigma}{\mu} = 4.48 \pm 1.48 \%$$

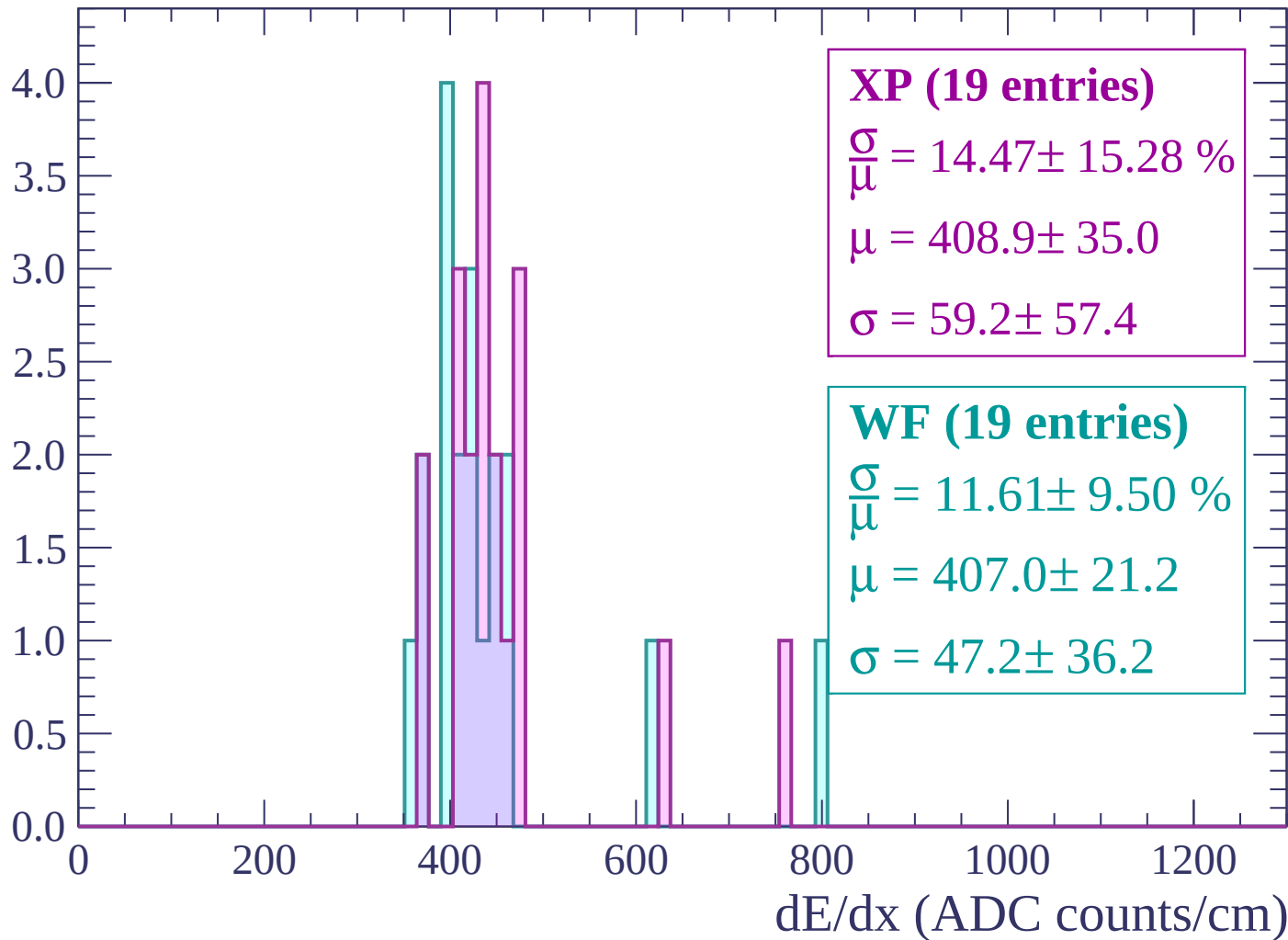
$$\mu = 444.4 \pm 6.2$$

$$\sigma = 19.9 \pm 6.3$$

dE/dx (ADC counts/cm)

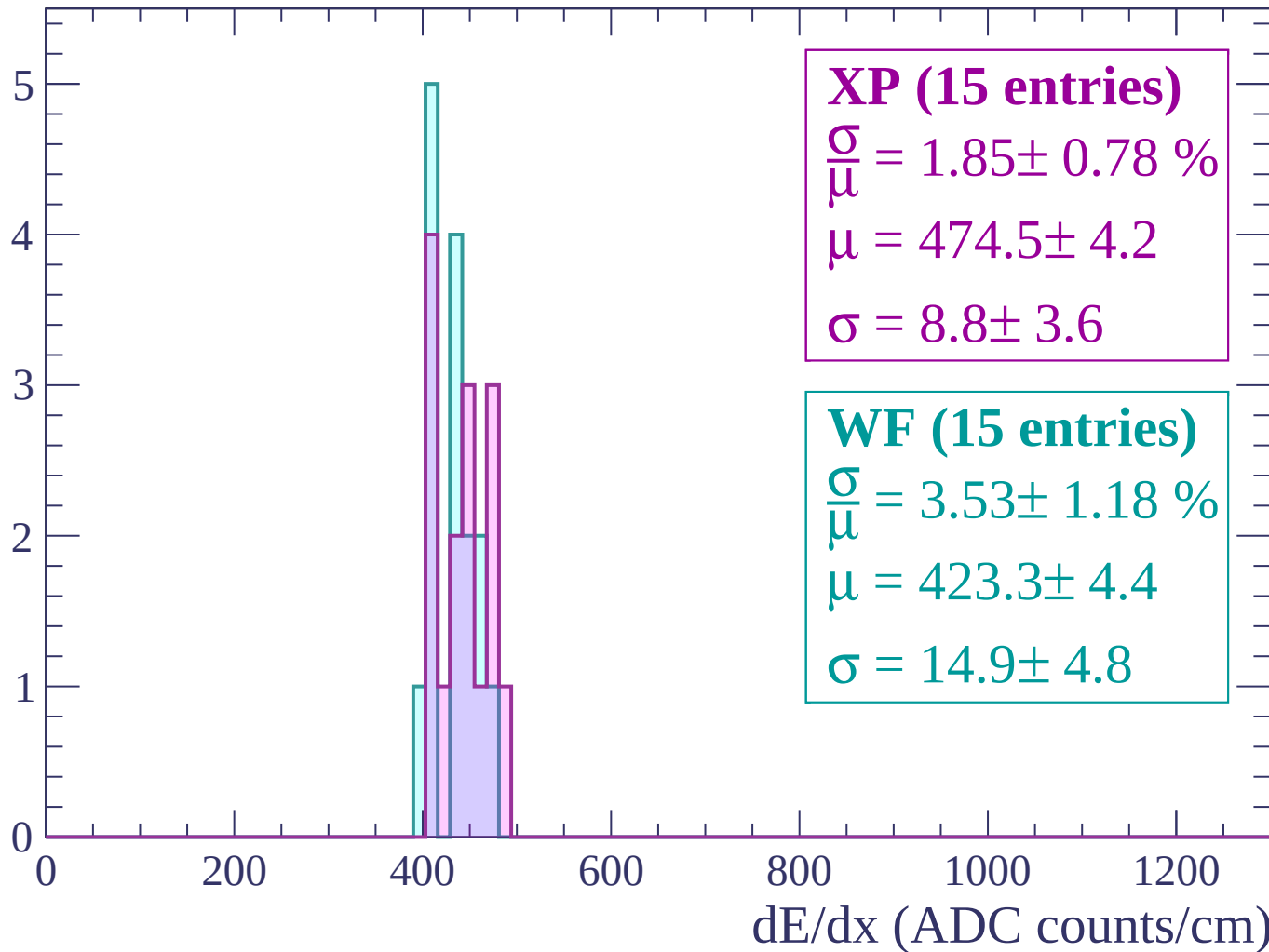
Energy loss | -2280 < p < -2220

Count



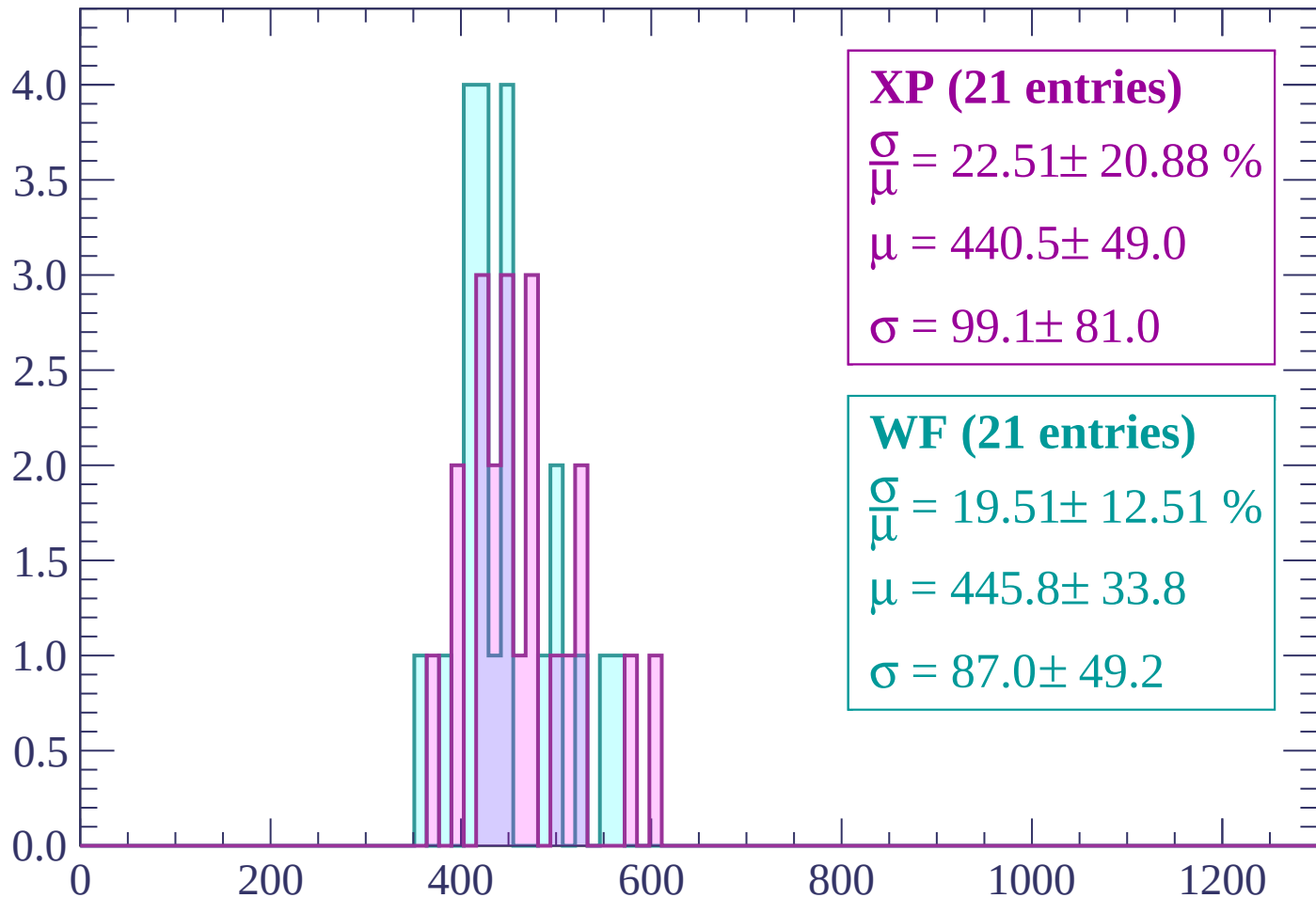
Energy loss | -2220 < p < -2160

Count



Energy loss | -2160 < p < -2100

Count



XP (21 entries)

$$\frac{\sigma}{\mu} = 22.51 \pm 20.88 \%$$

$$\mu = 440.5 \pm 49.0$$

$$\sigma = 99.1 \pm 81.0$$

WF (21 entries)

$$\frac{\sigma}{\mu} = 19.51 \pm 12.51 \%$$

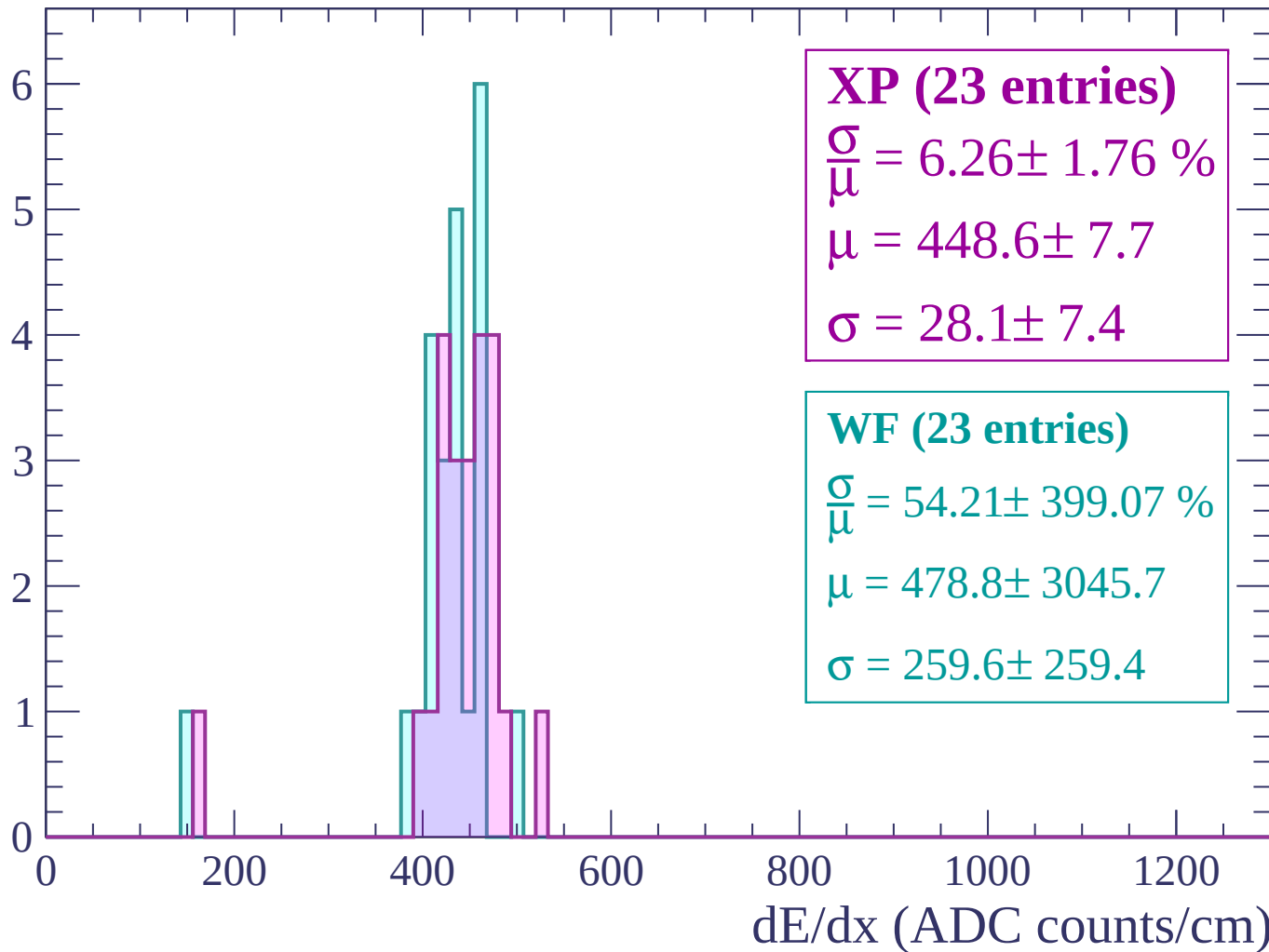
$$\mu = 445.8 \pm 33.8$$

$$\sigma = 87.0 \pm 49.2$$

dE/dx (ADC counts/cm)

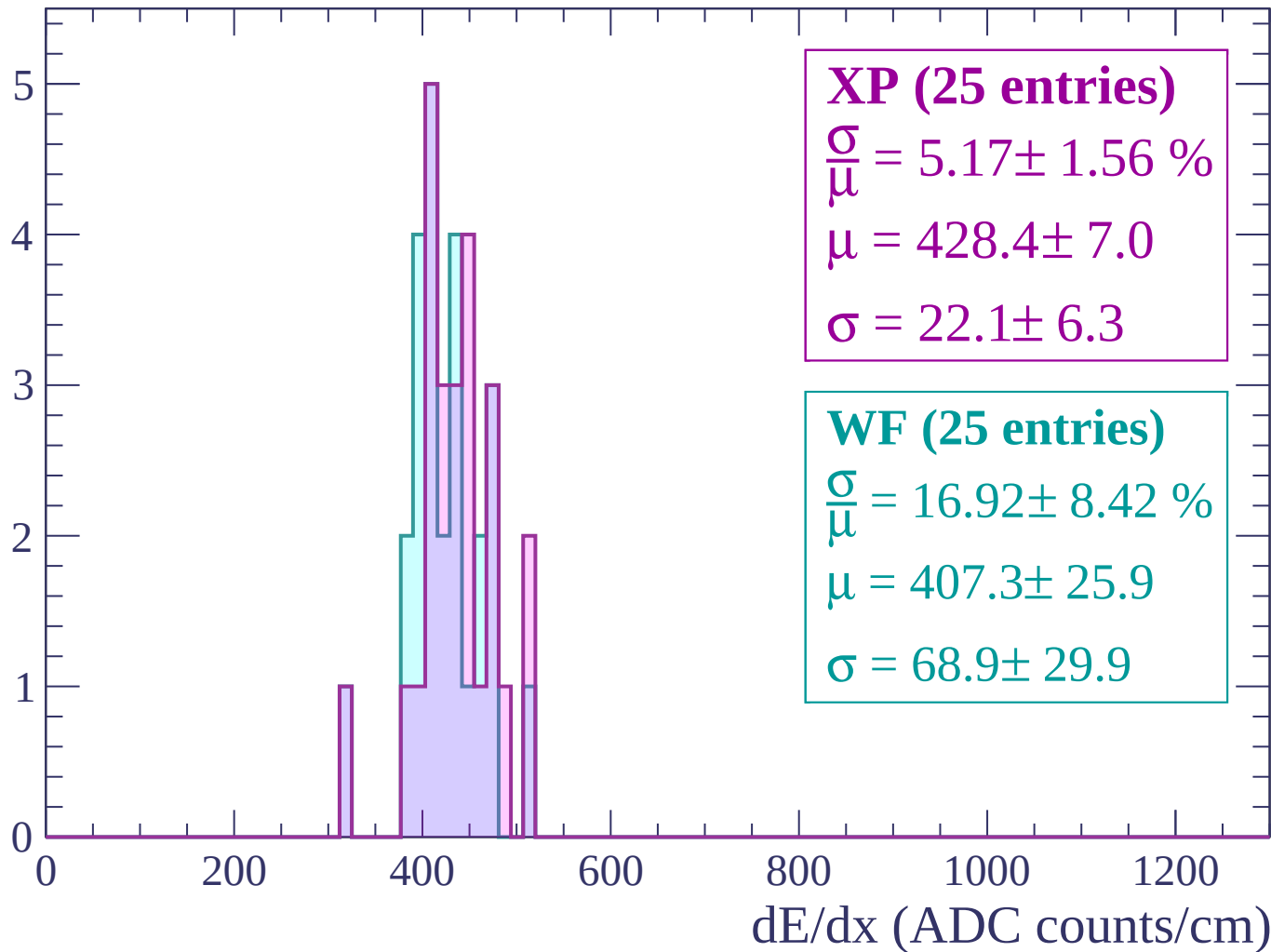
Energy loss | -2100 < p < -2040

Count



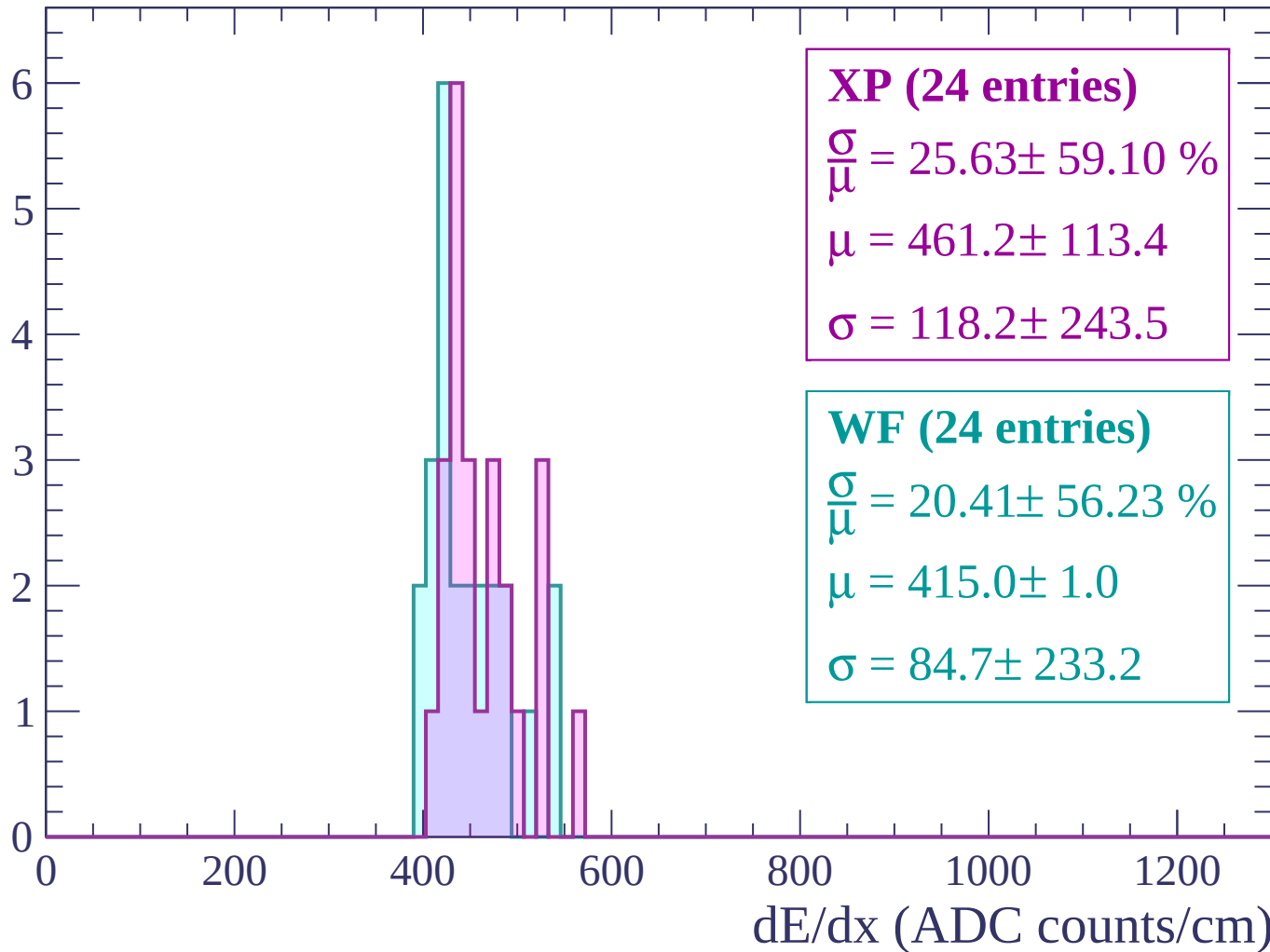
Energy loss | -2040 < p < -1980

Count



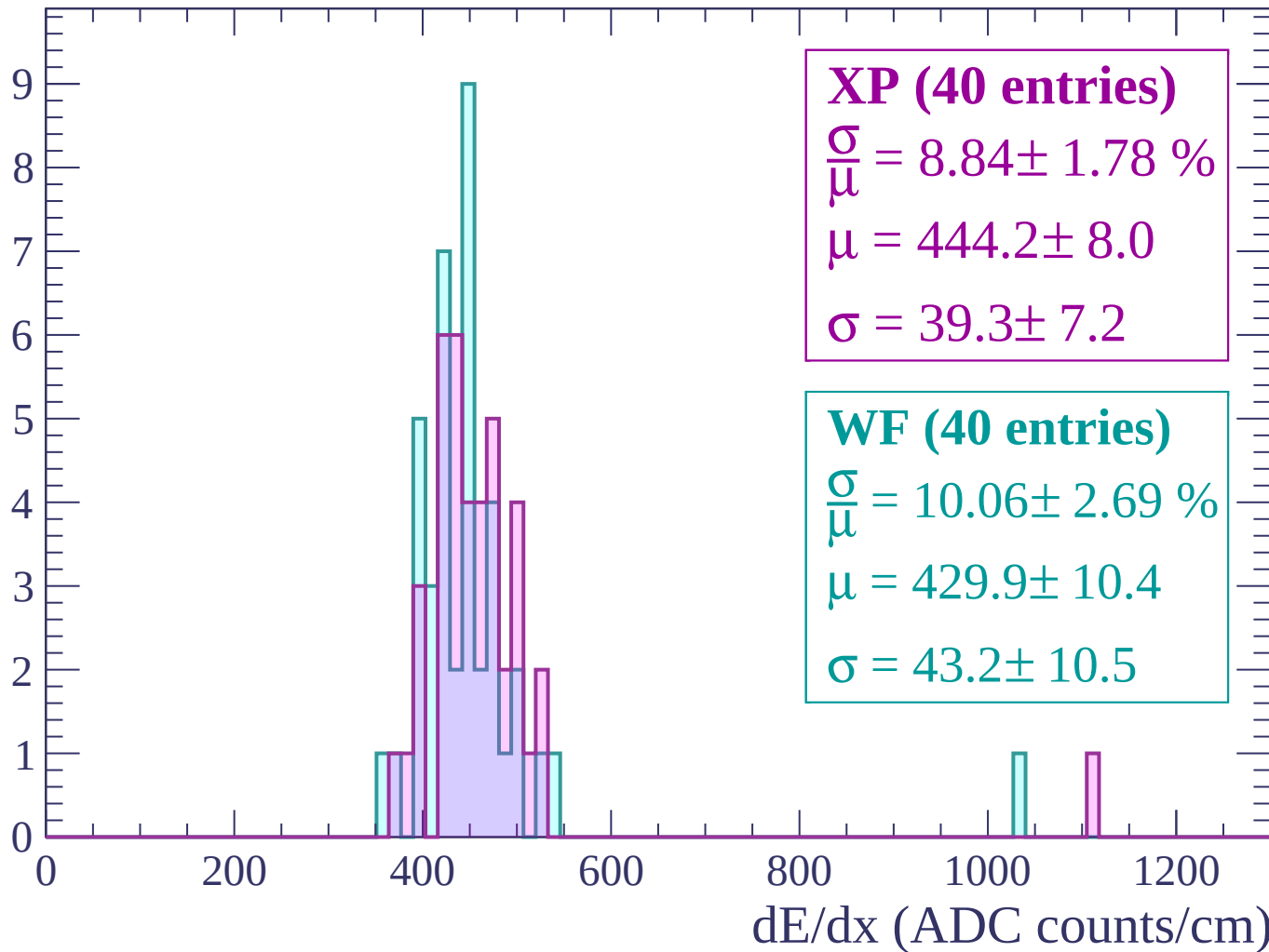
Energy loss | -1980 < p < -1920

Count



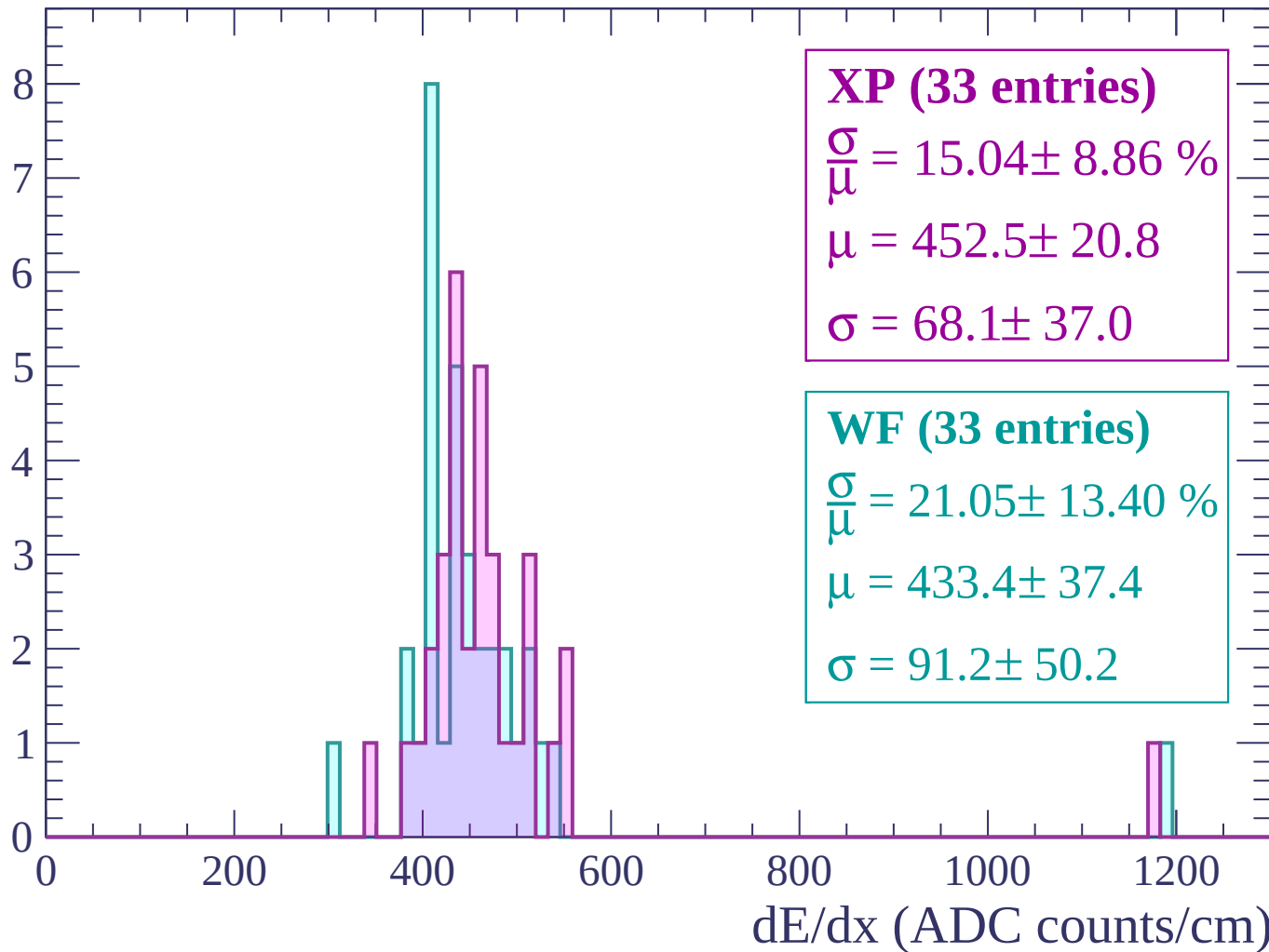
Energy loss | $-1920 < p < -1860$

Count



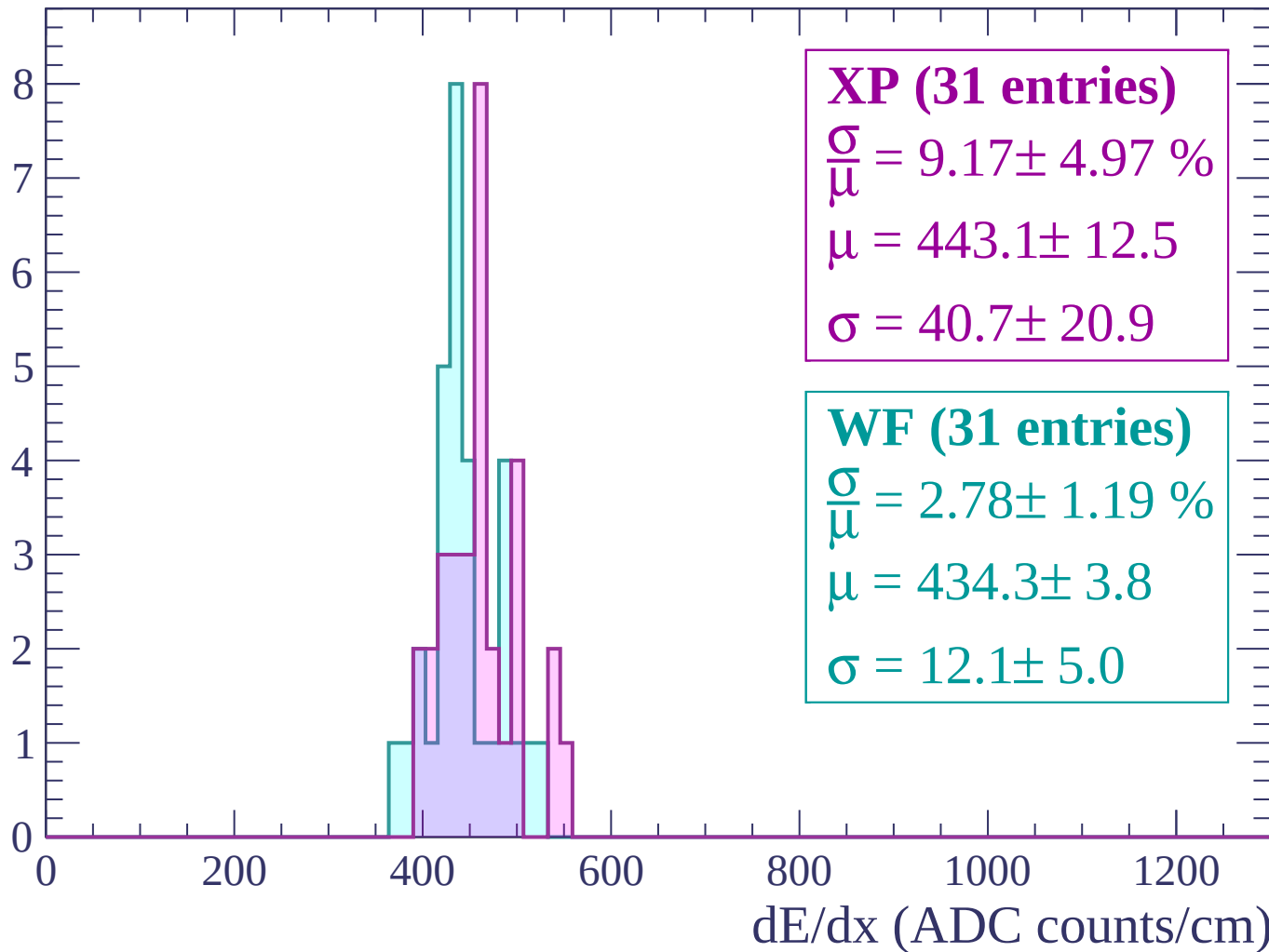
Energy loss | -1860 < p < -1800

Count



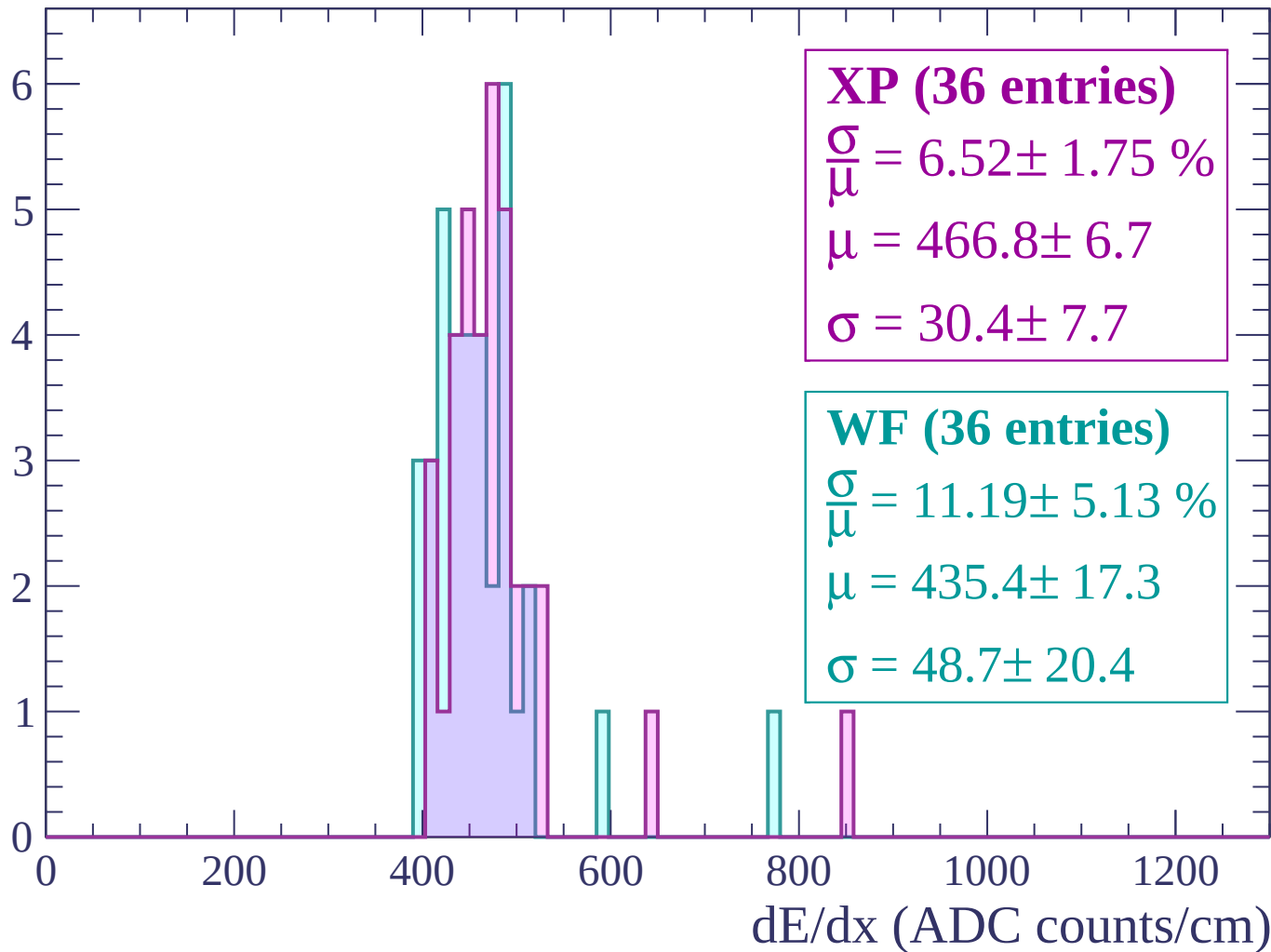
Energy loss | $-1800 < p < -1740$

Count



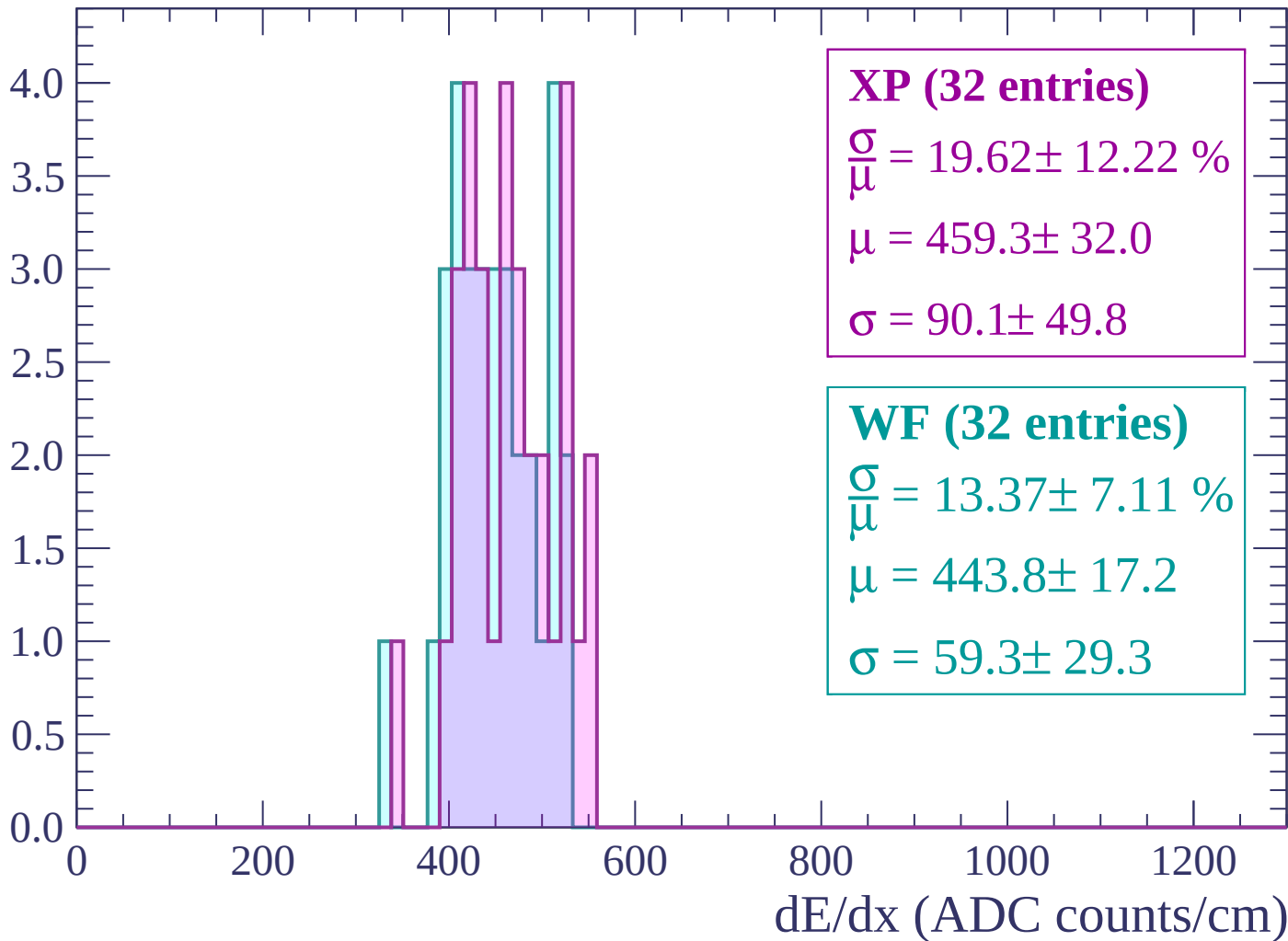
Energy loss | $-1740 < p < -1680$

Count



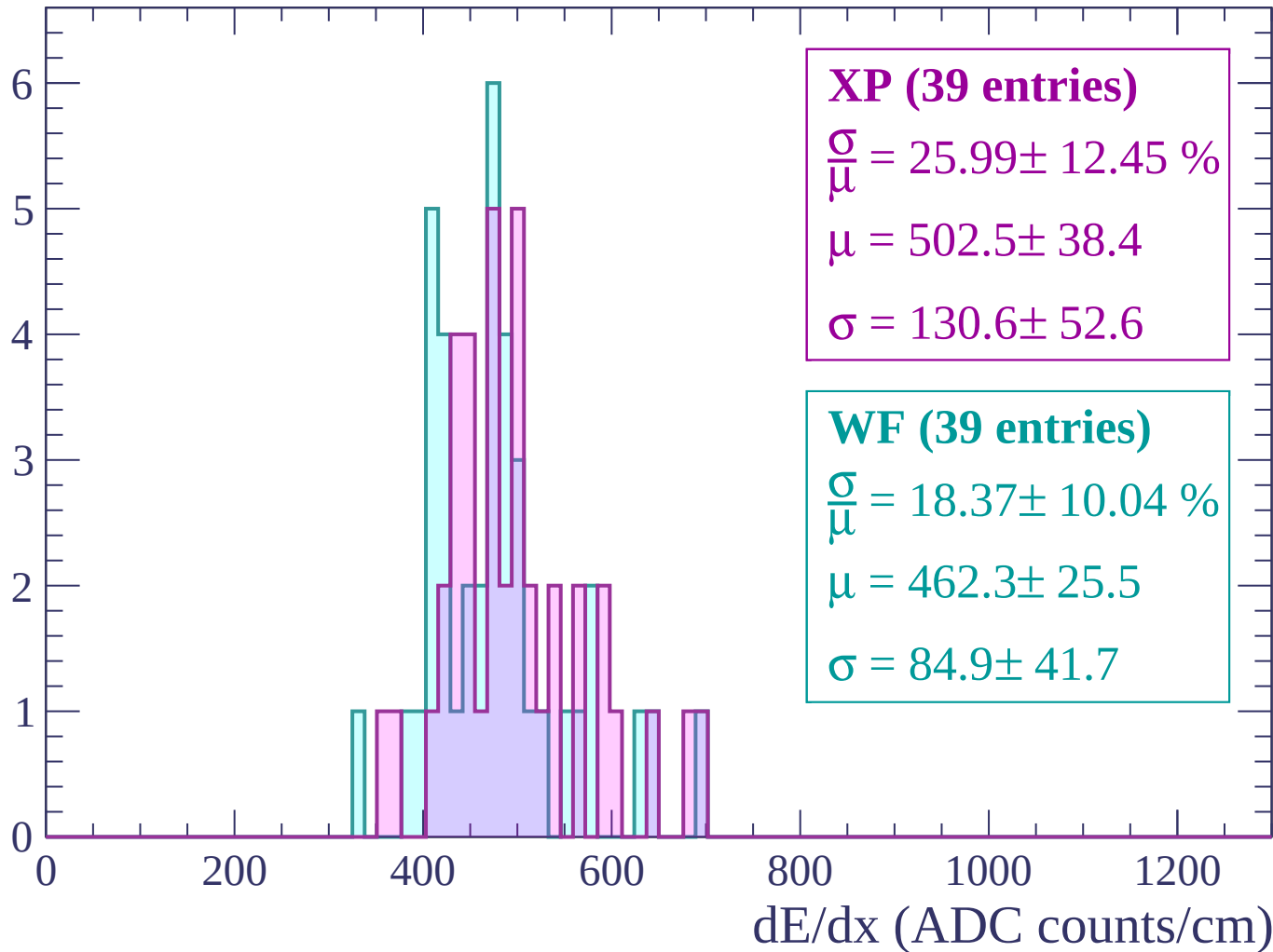
Energy loss | -1680 < p < -1620

Count



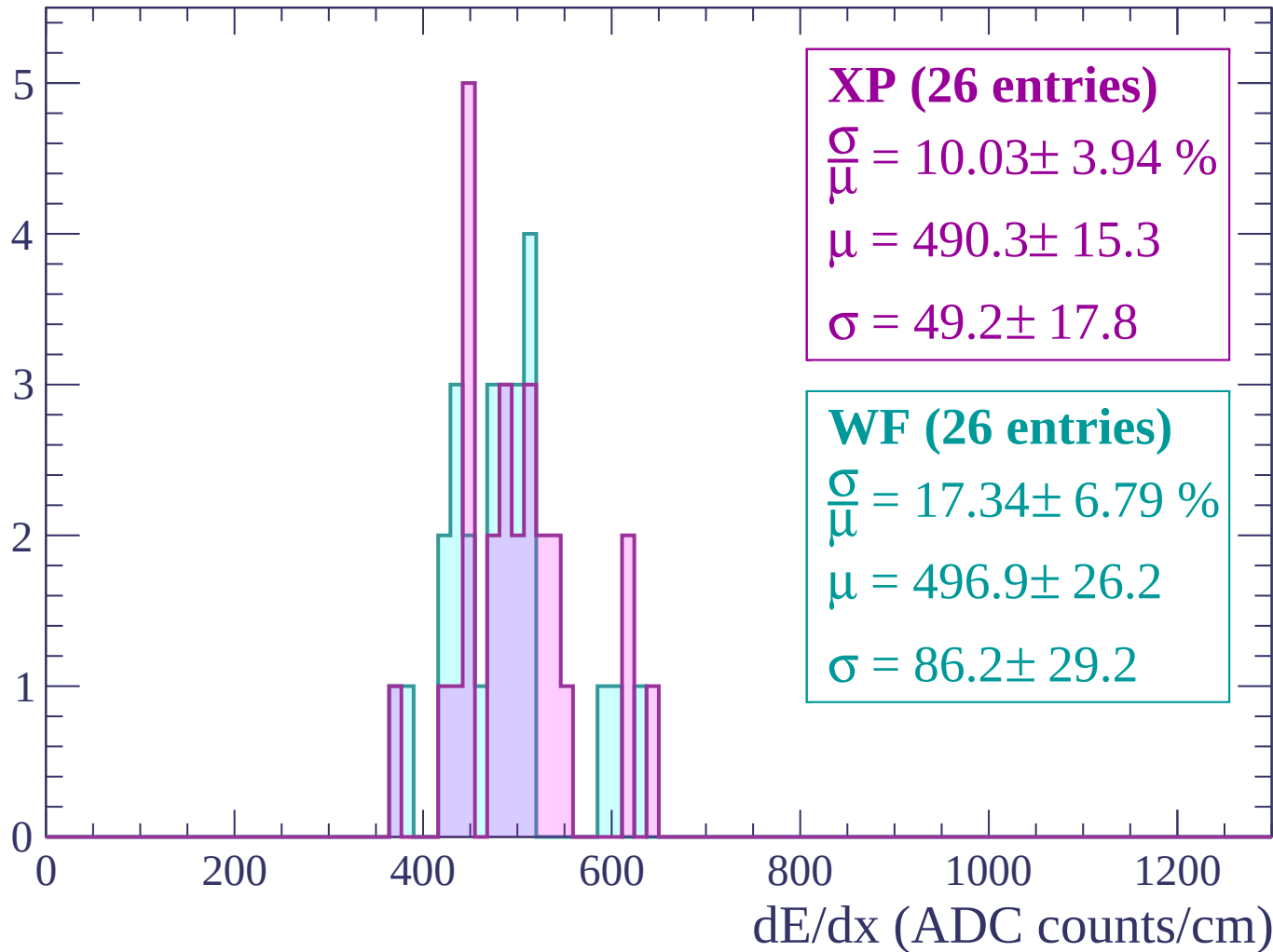
Energy loss | -1620 < p < -1560

Count



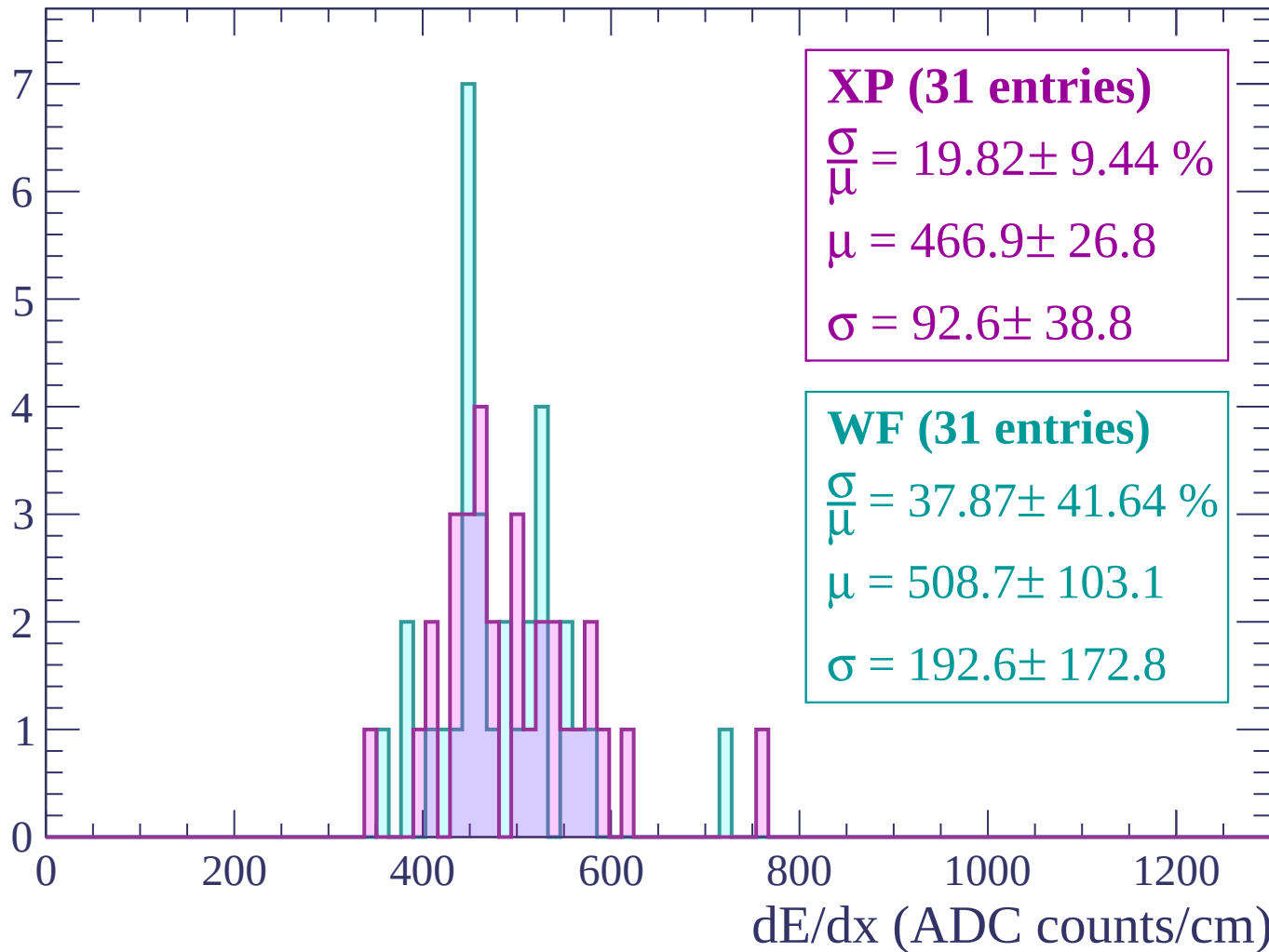
Energy loss | -1560 < p < -1500

Count



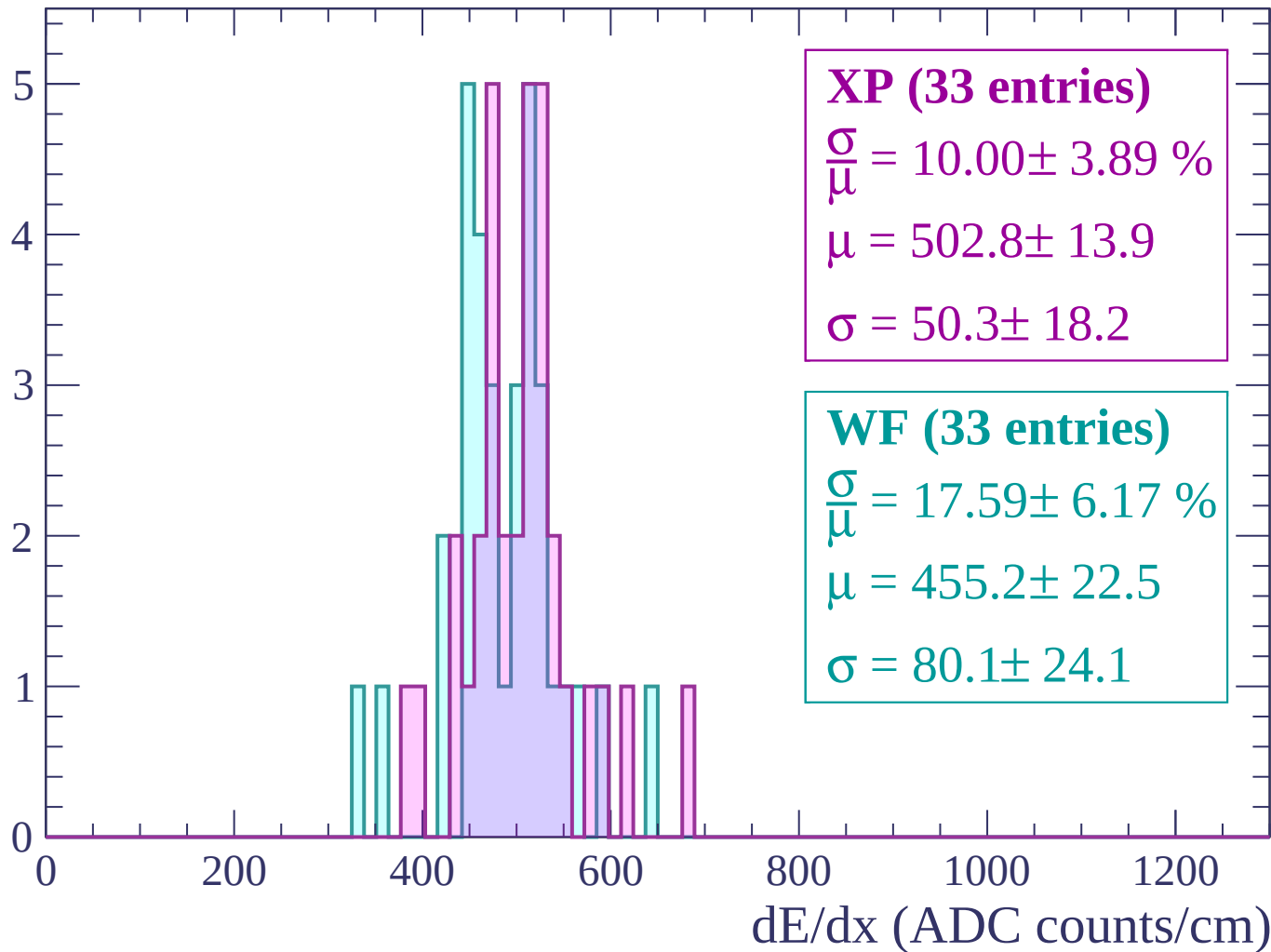
Energy loss | -1500 < p < -1440

Count



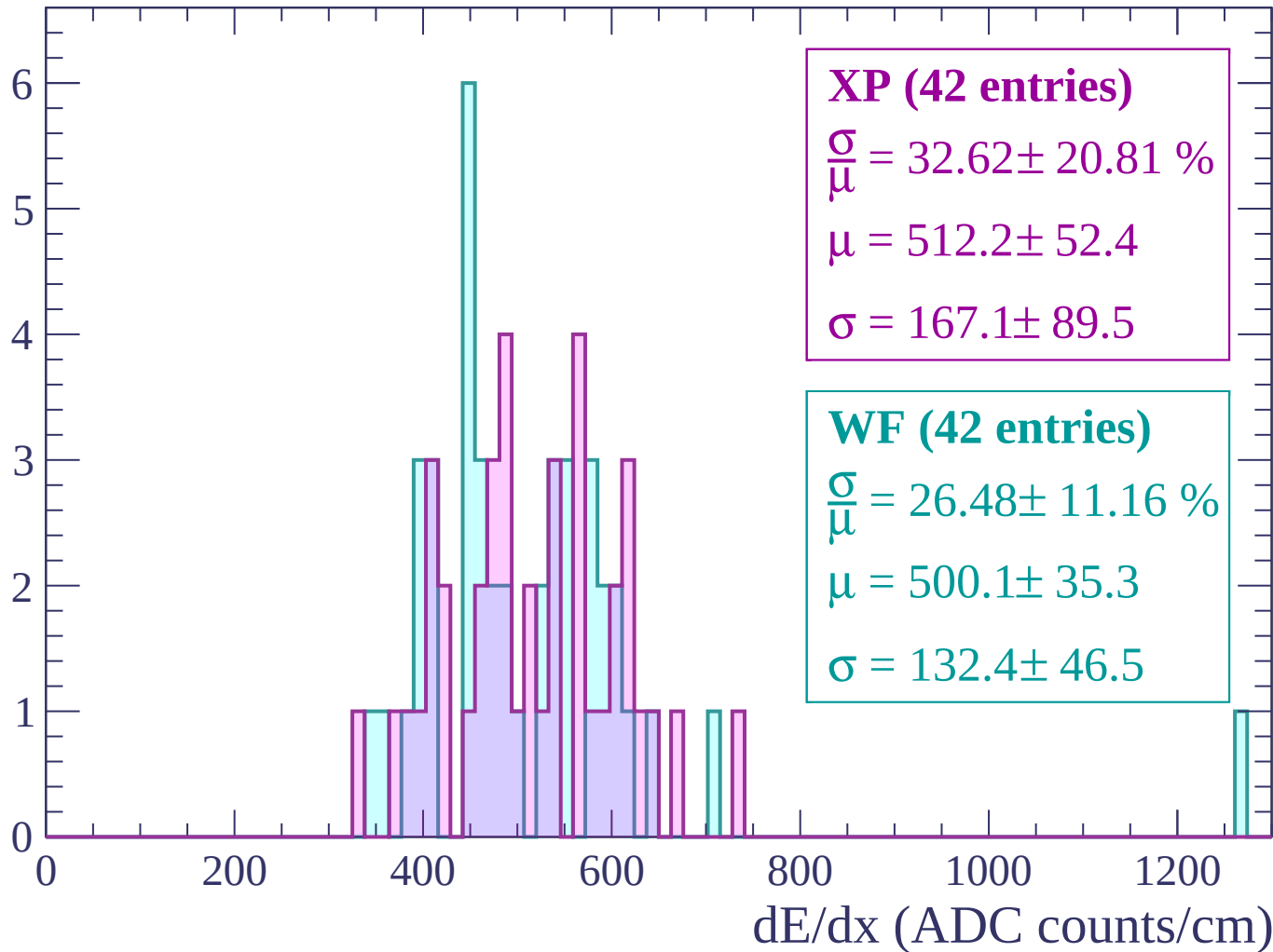
Energy loss | $-1440 < p < -1380$

Count



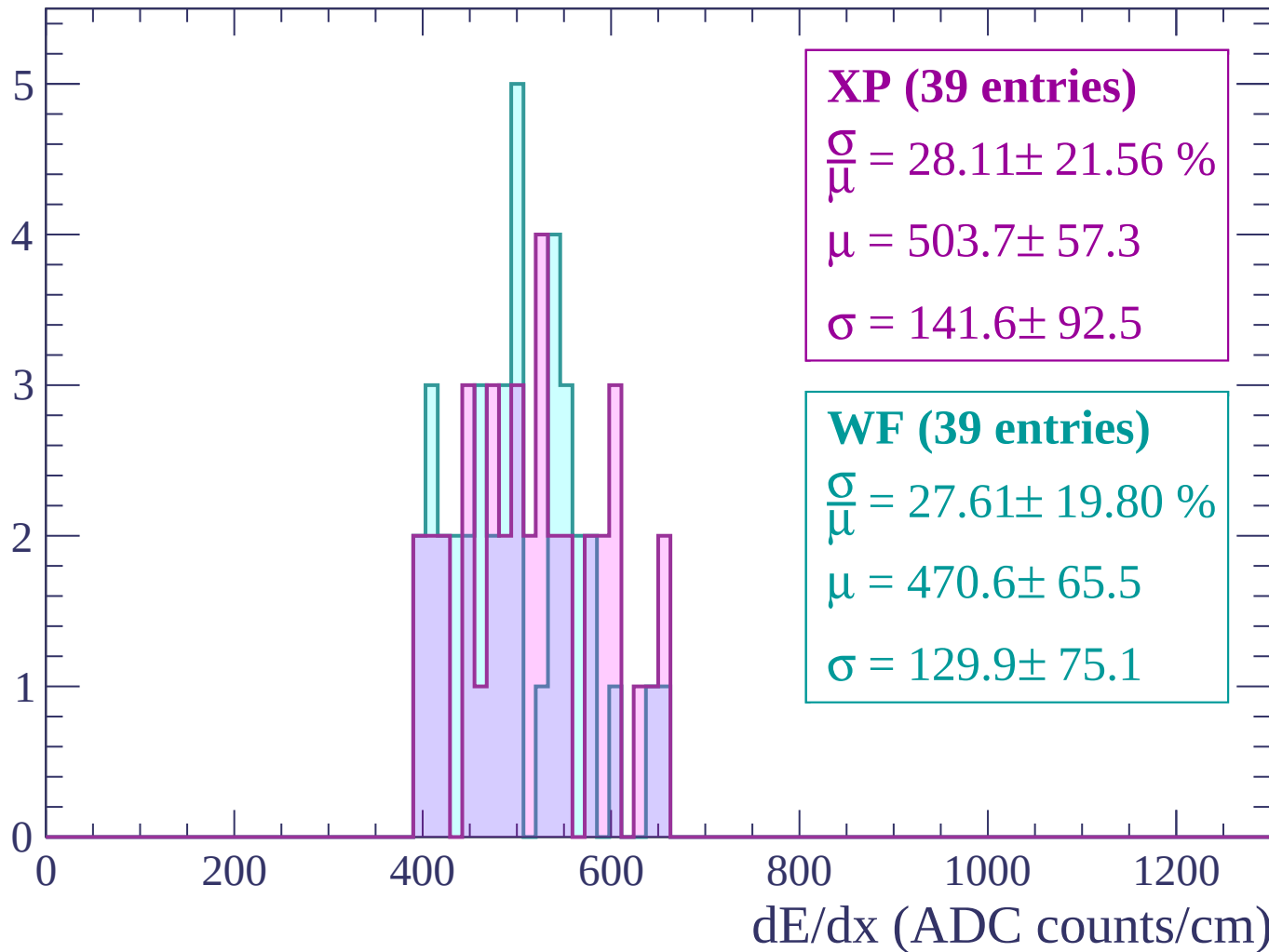
Energy loss | -1380 < p < -1320

Count



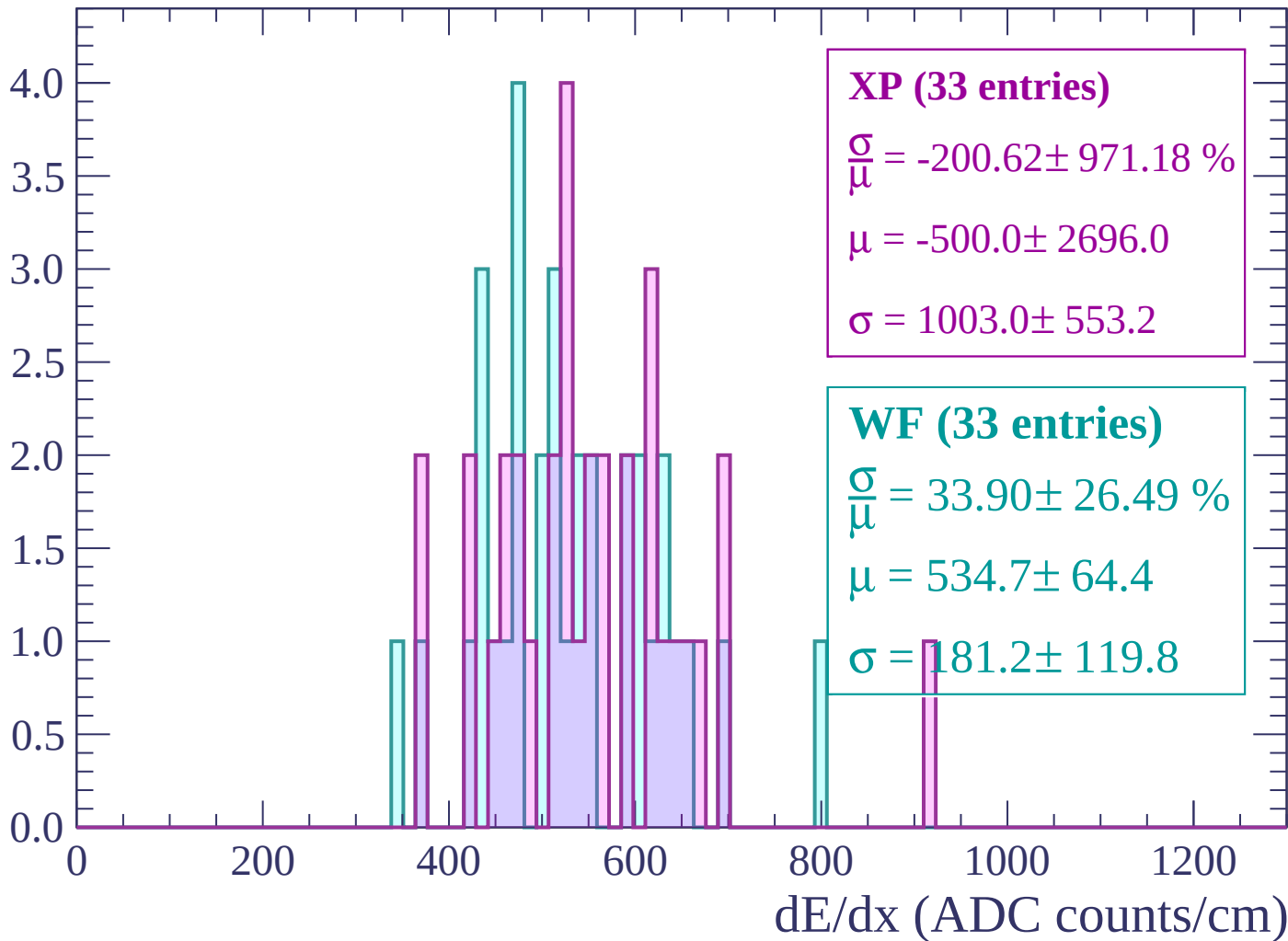
Energy loss | -1320 < p < -1260

Count



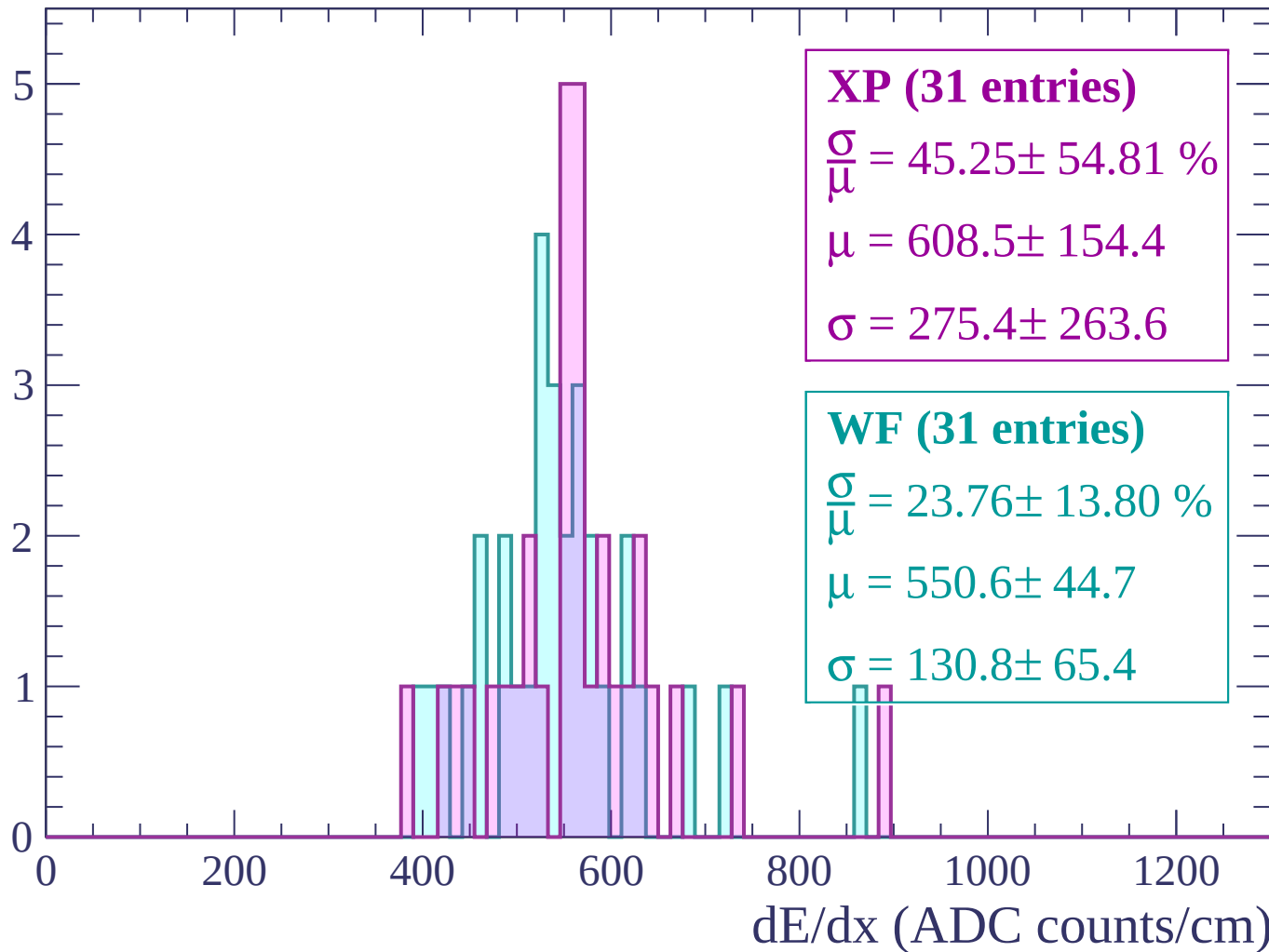
Energy loss | -1260 < p < -1200

Count



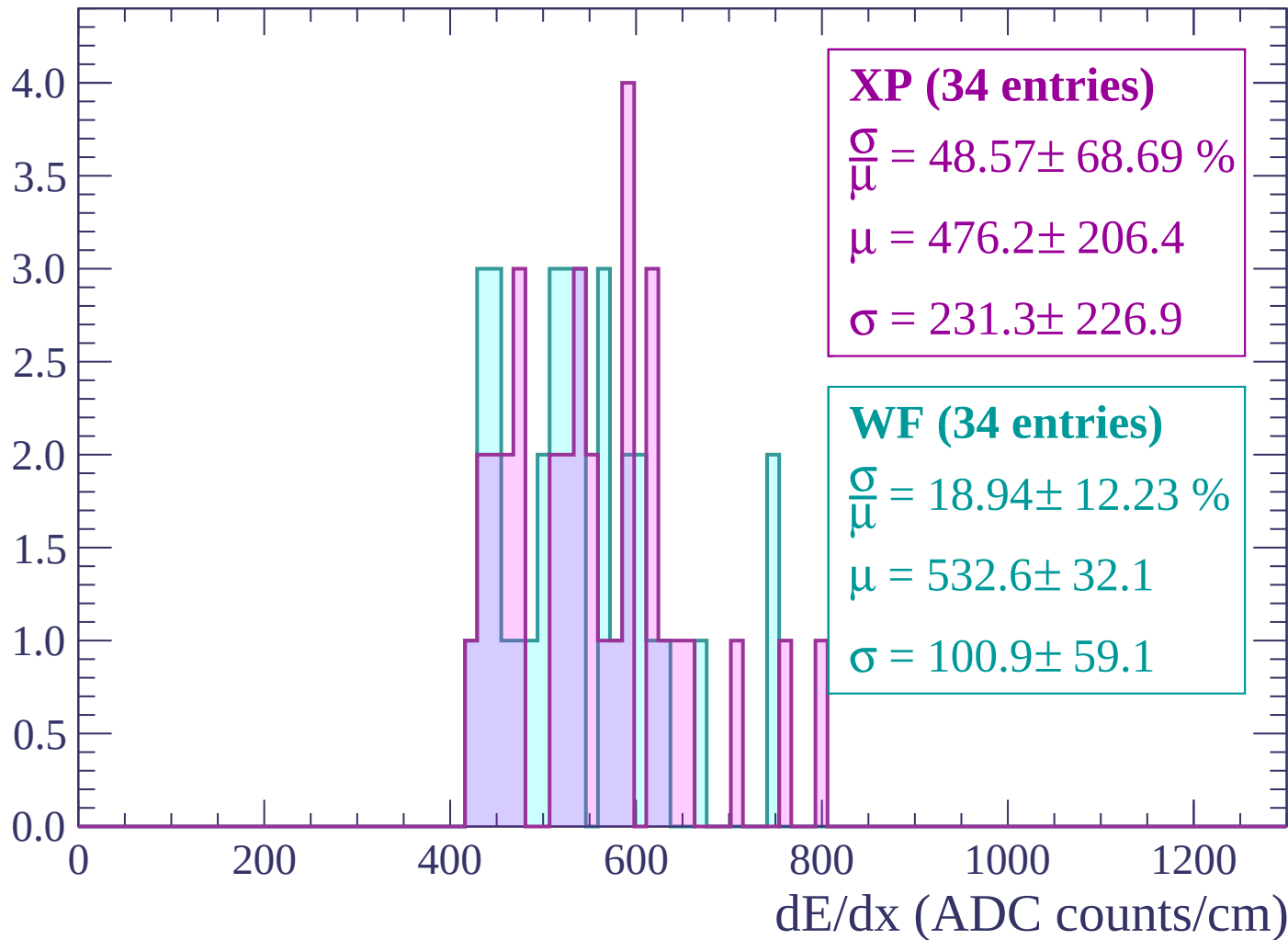
Energy loss | -1200 < p < -1140

Count



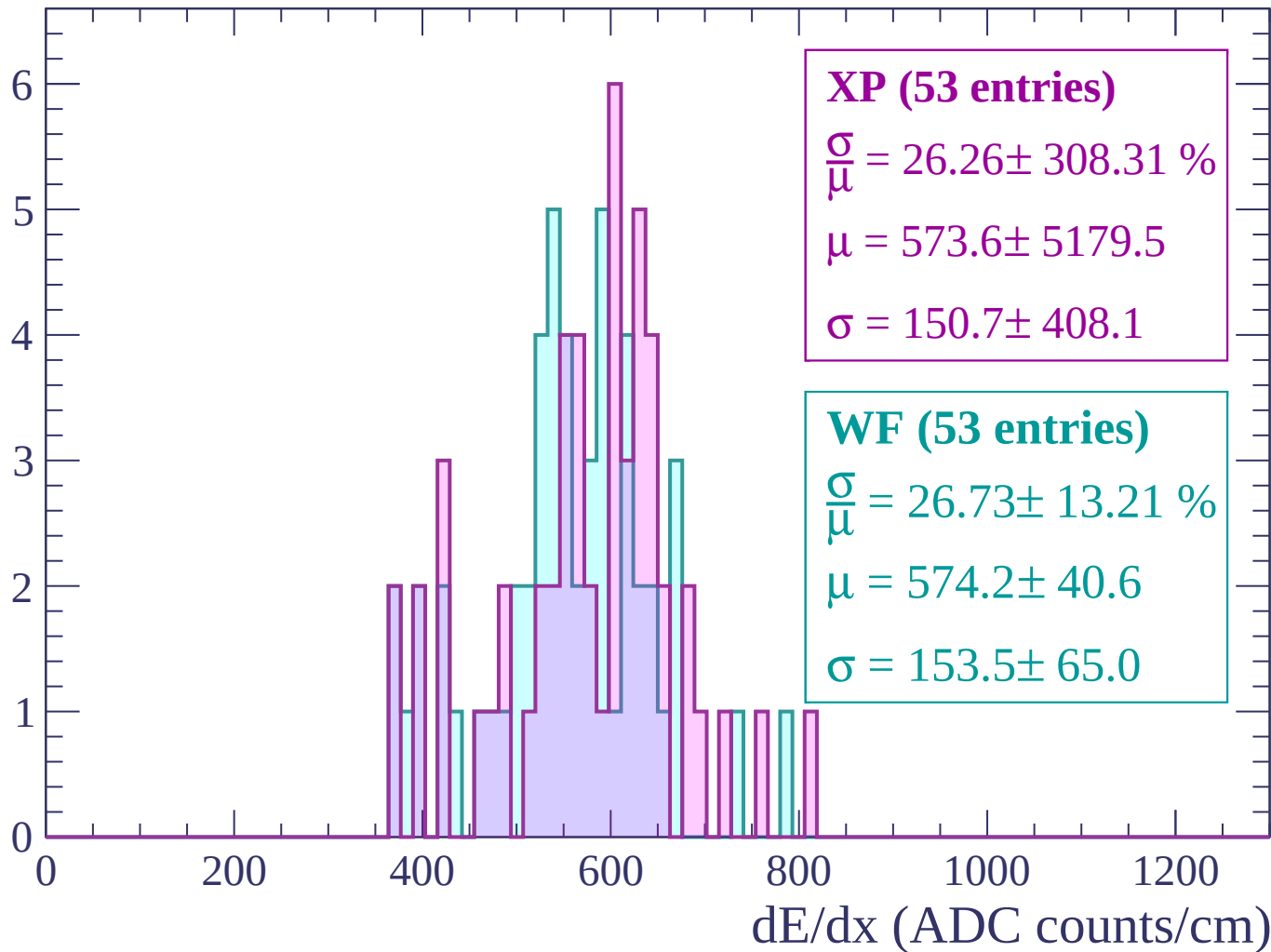
Energy loss | -1140 < p < -1080

Count



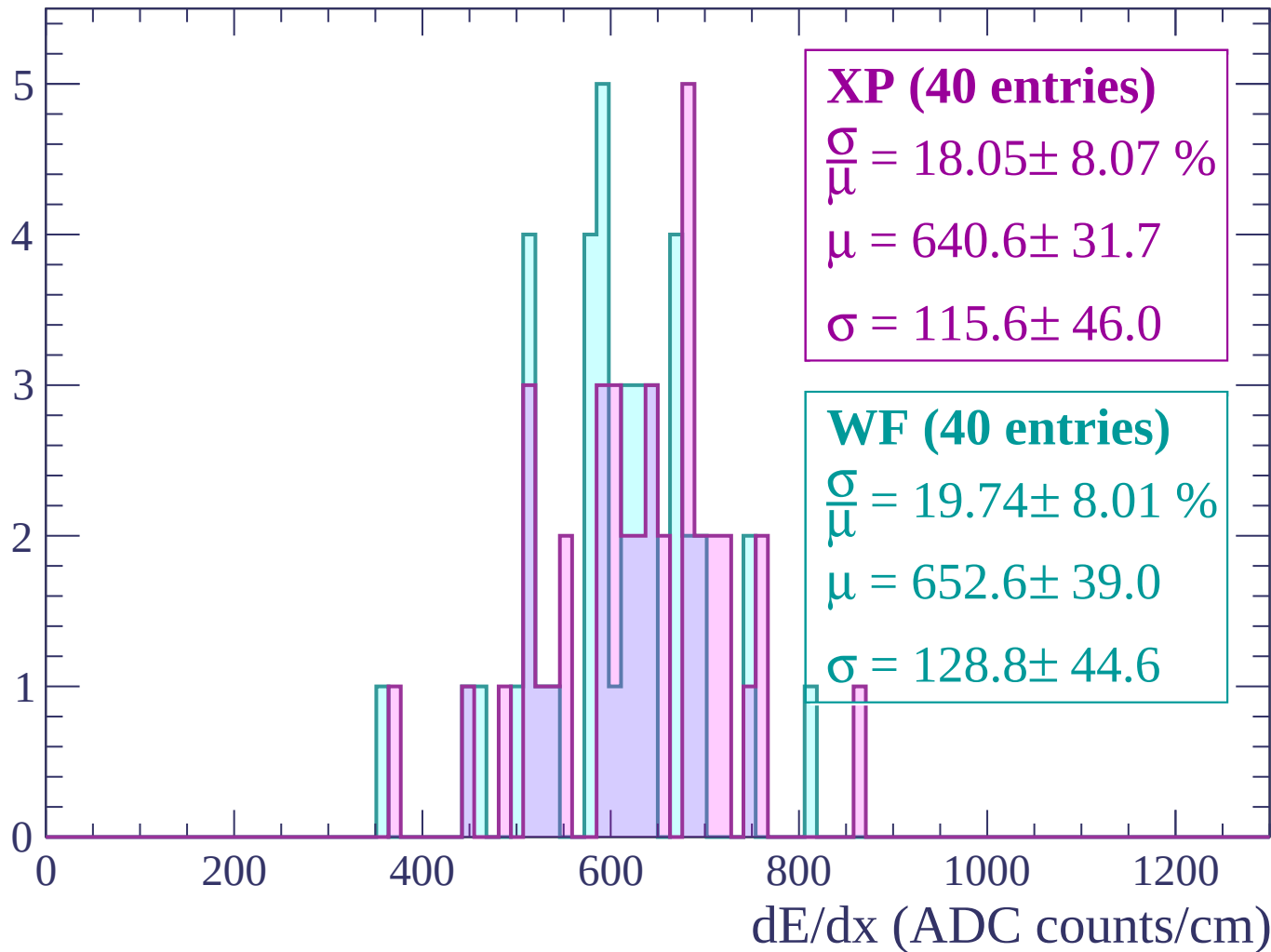
Energy loss | -1080 < p < -1020

Count



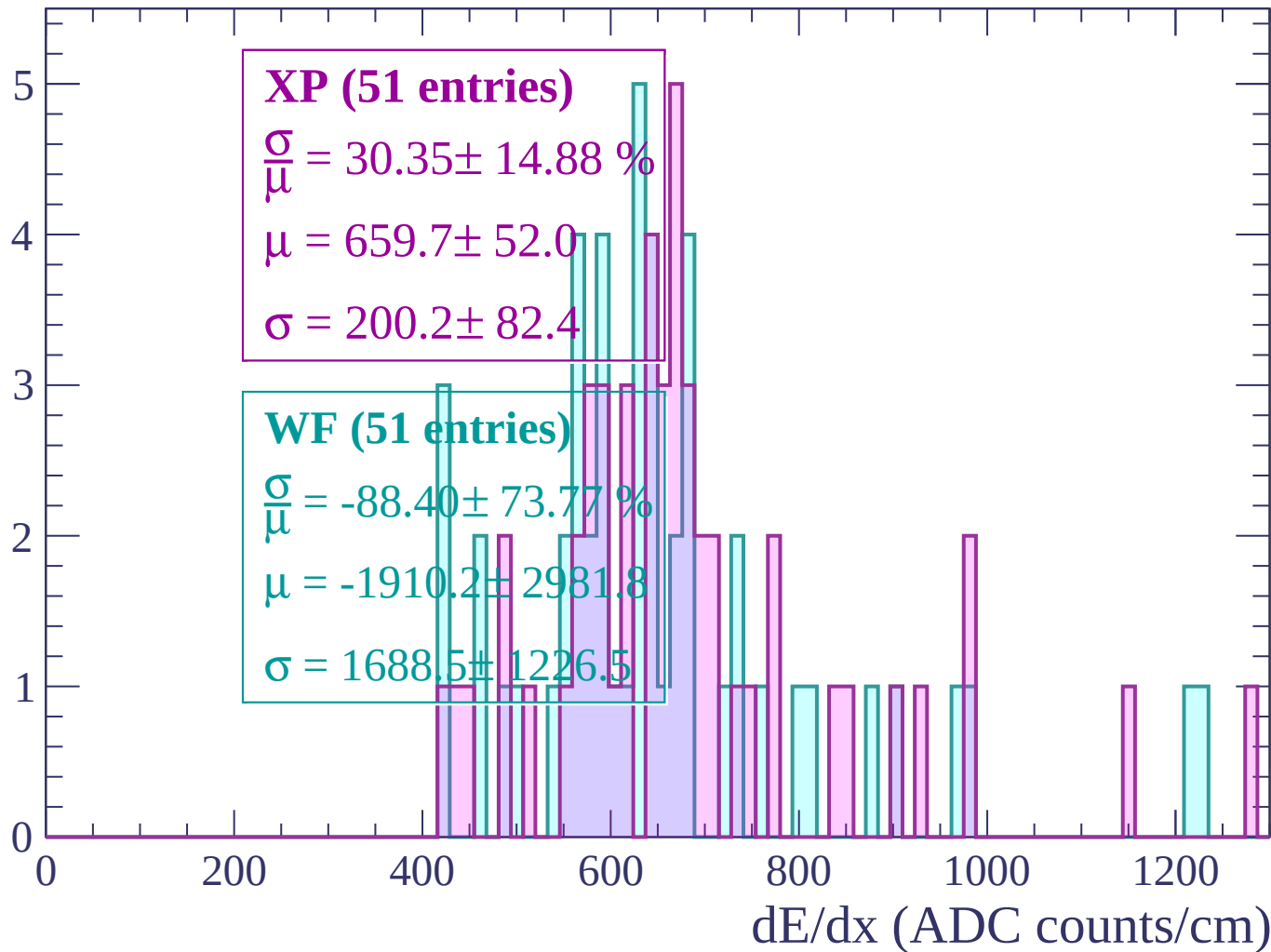
Energy loss | $-1020 < p < -960$

Count



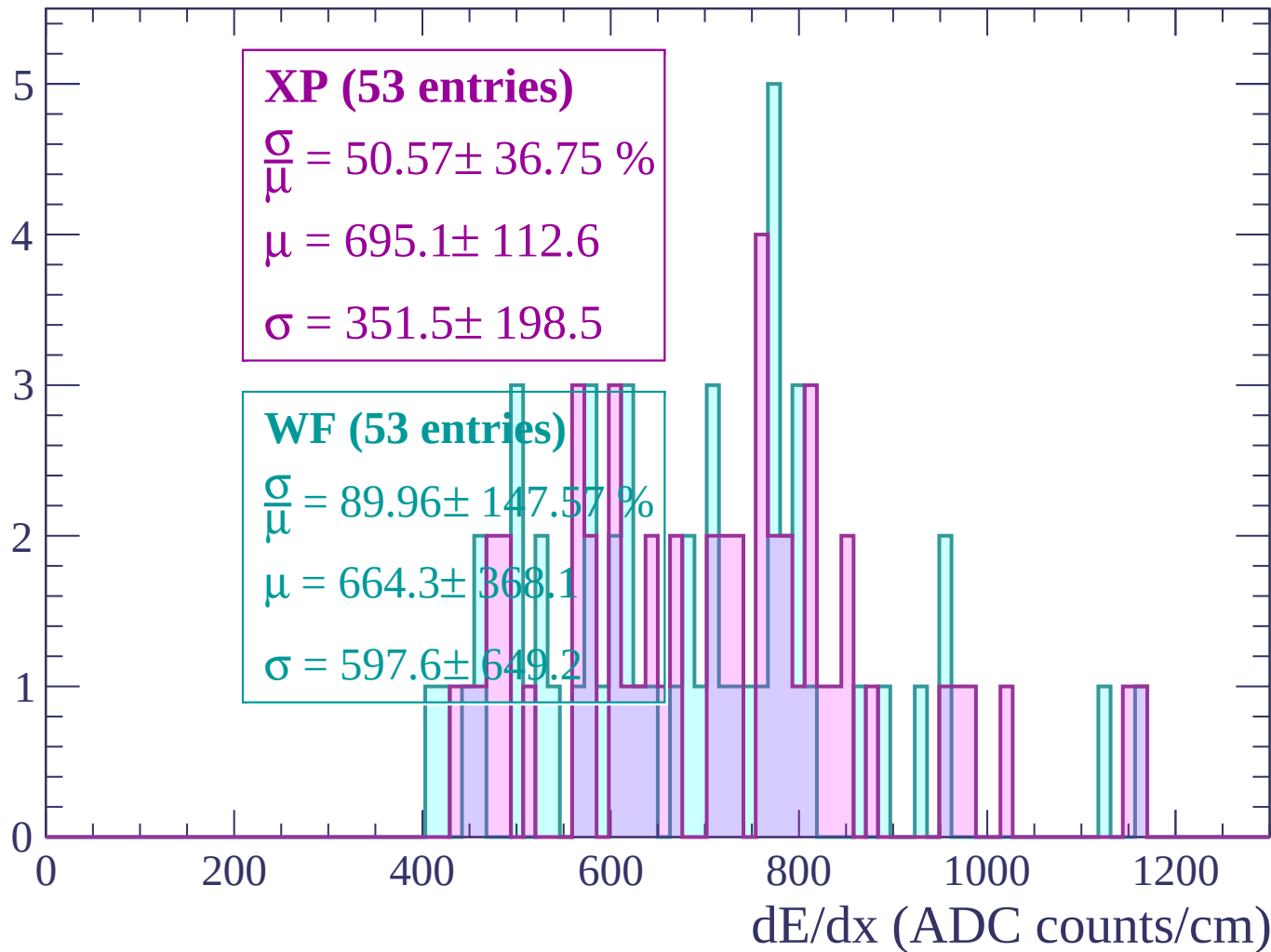
Energy loss | -960 < p < -900

Count



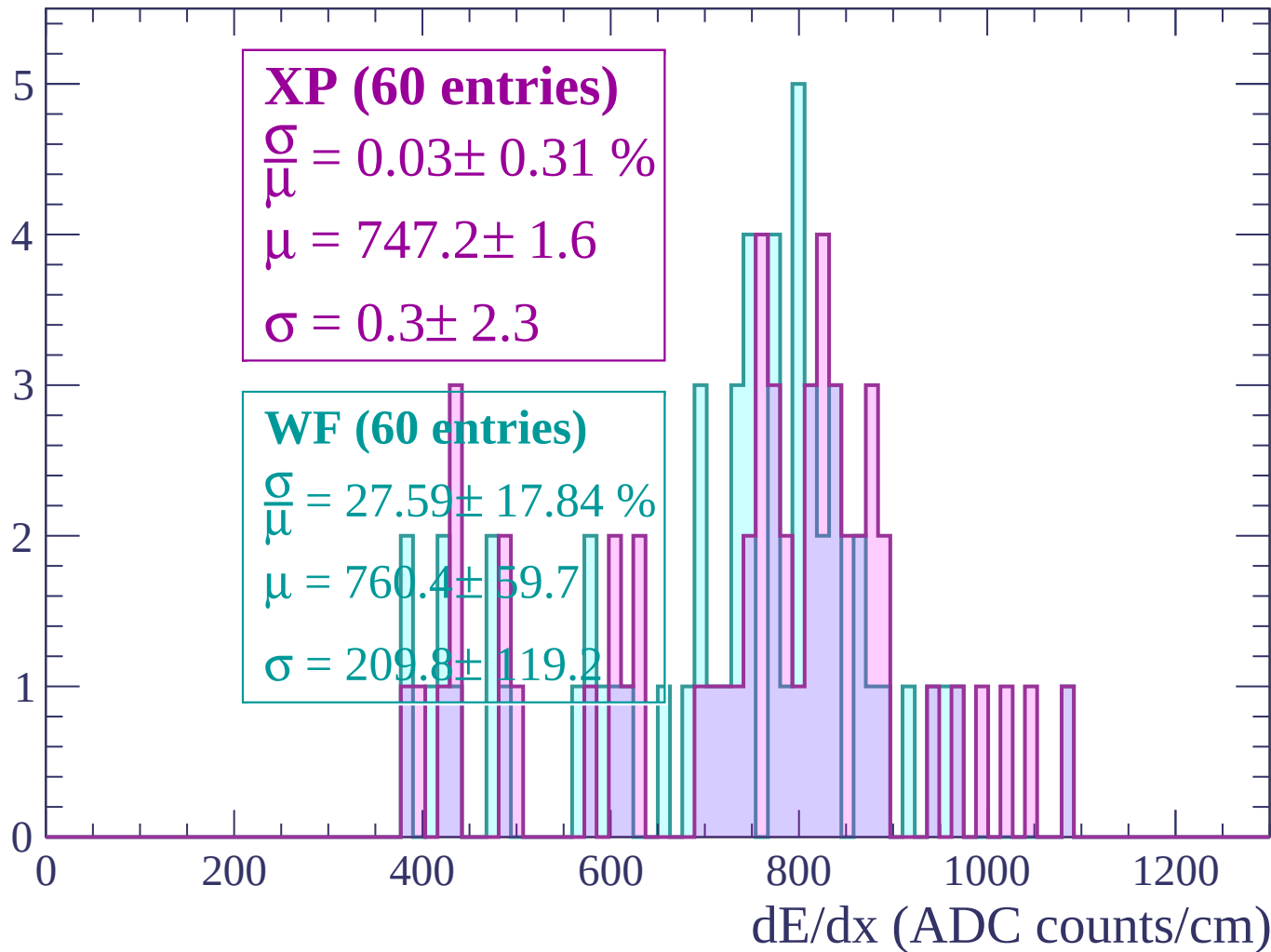
Energy loss | $-900 < p < -840$

Count



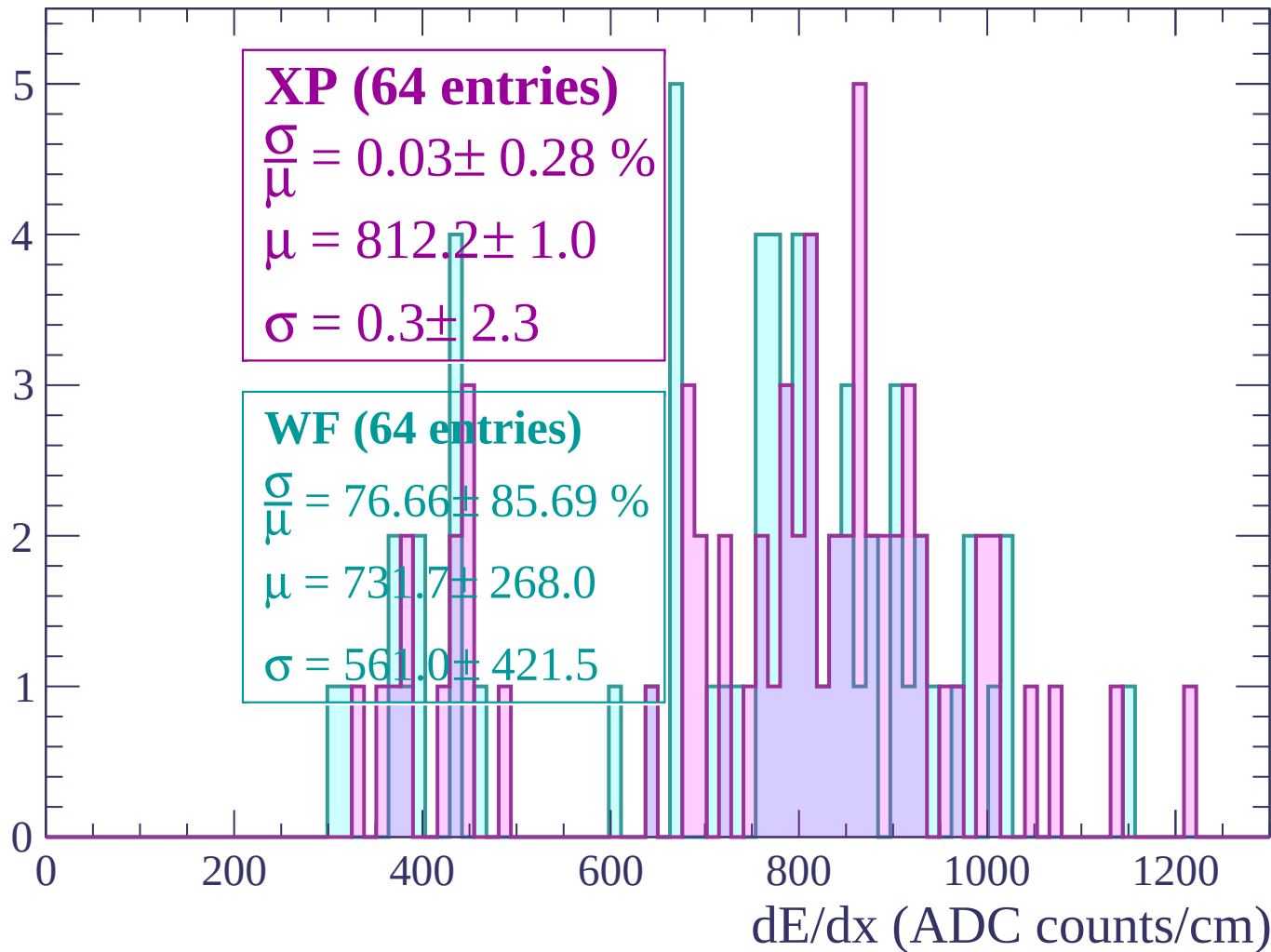
Energy loss | $-840 < p < -780$

Count



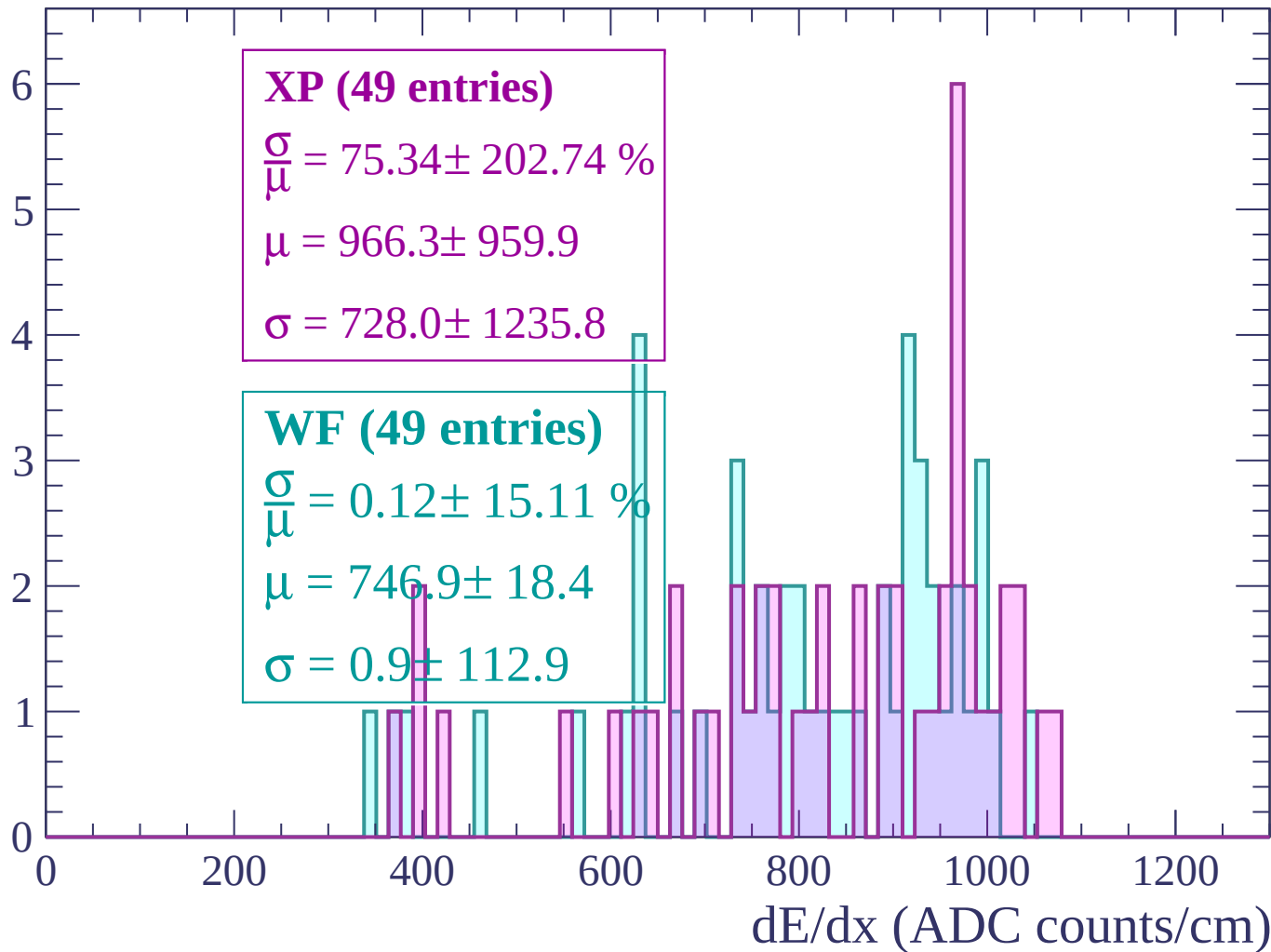
Energy loss | $-780 < p < -720$

Count



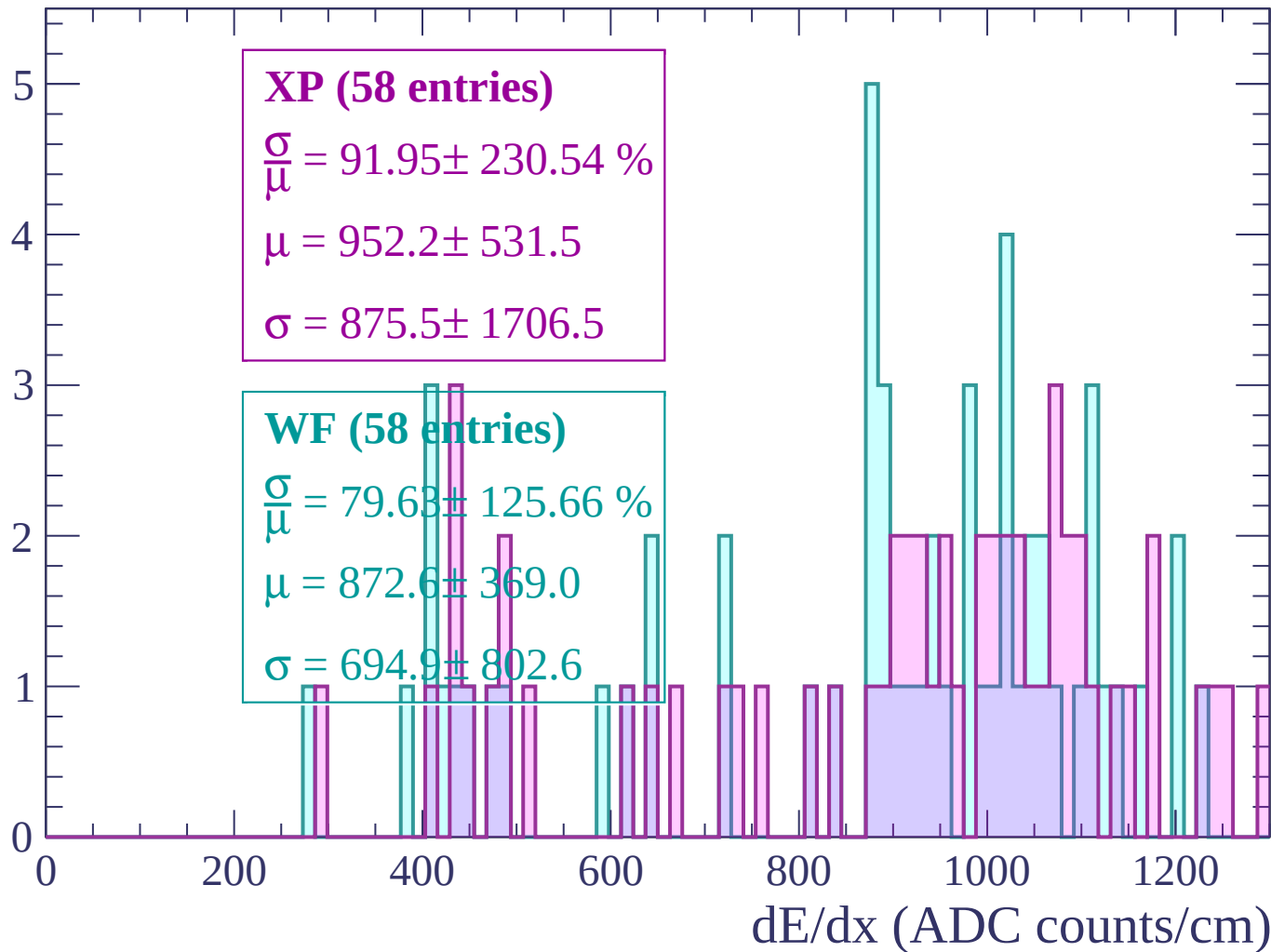
Energy loss | $-720 < p < -660$

Count



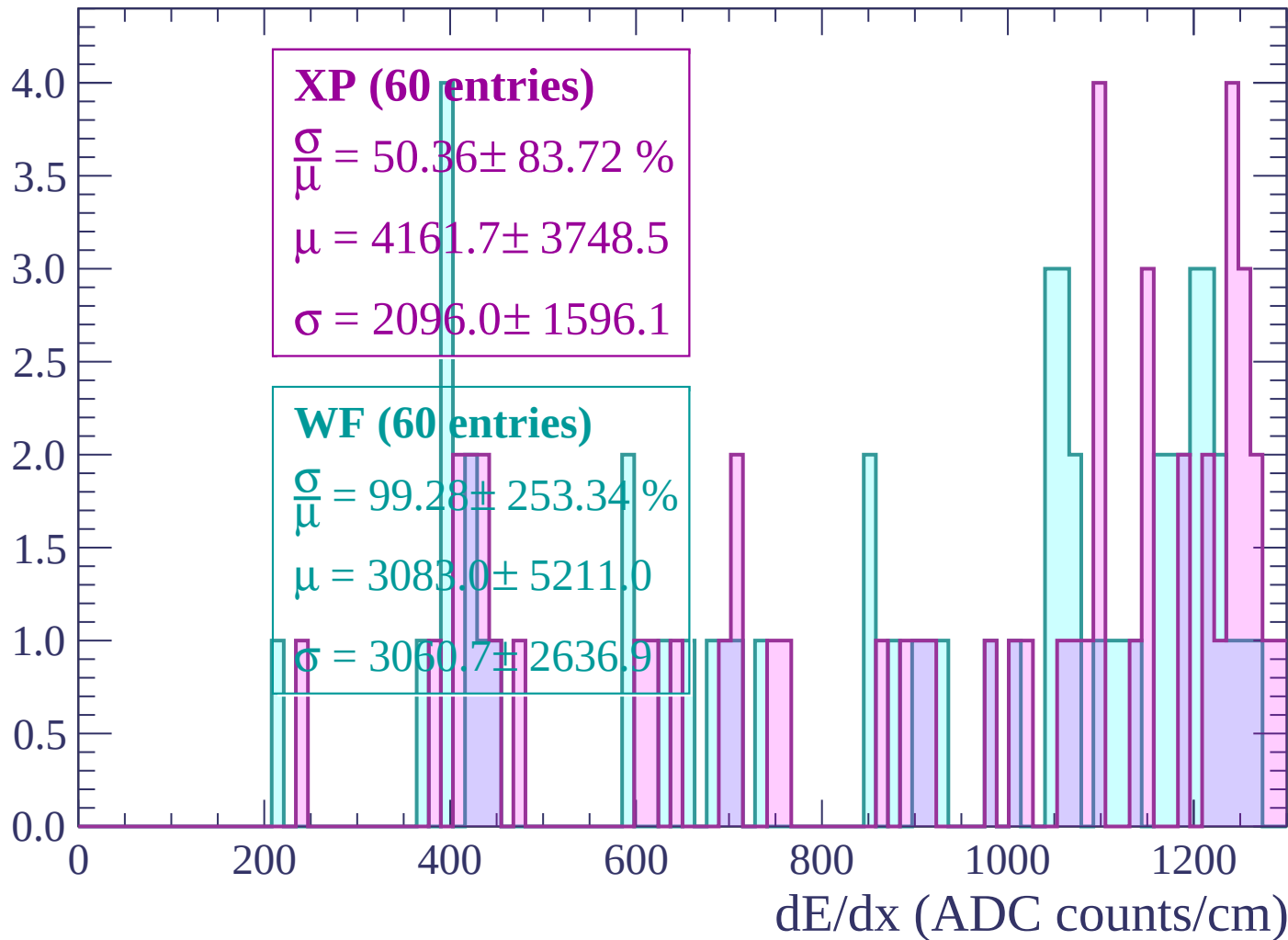
Energy loss | -660 < p < -600

Count



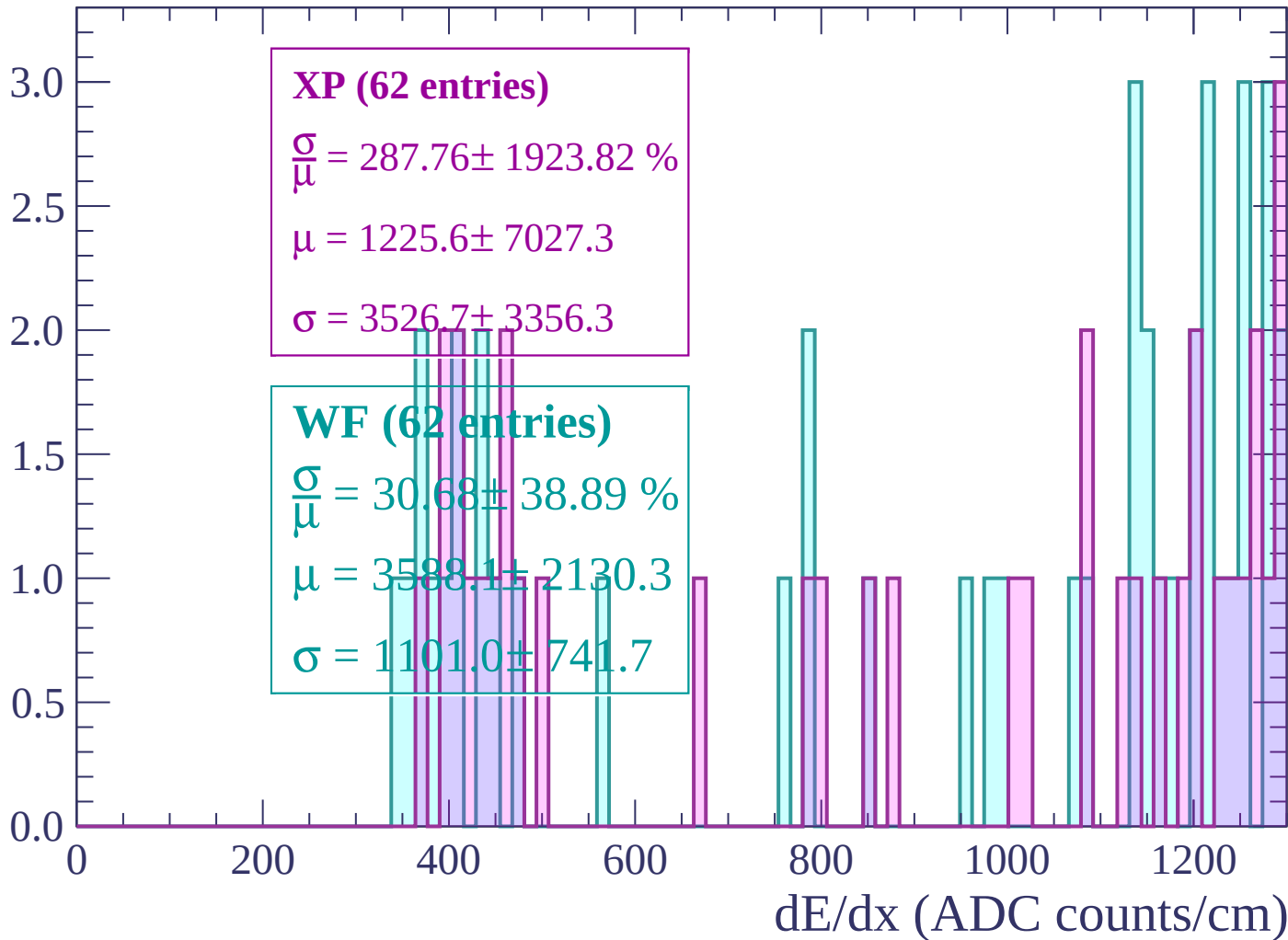
Energy loss | -600 < p < -540

Count



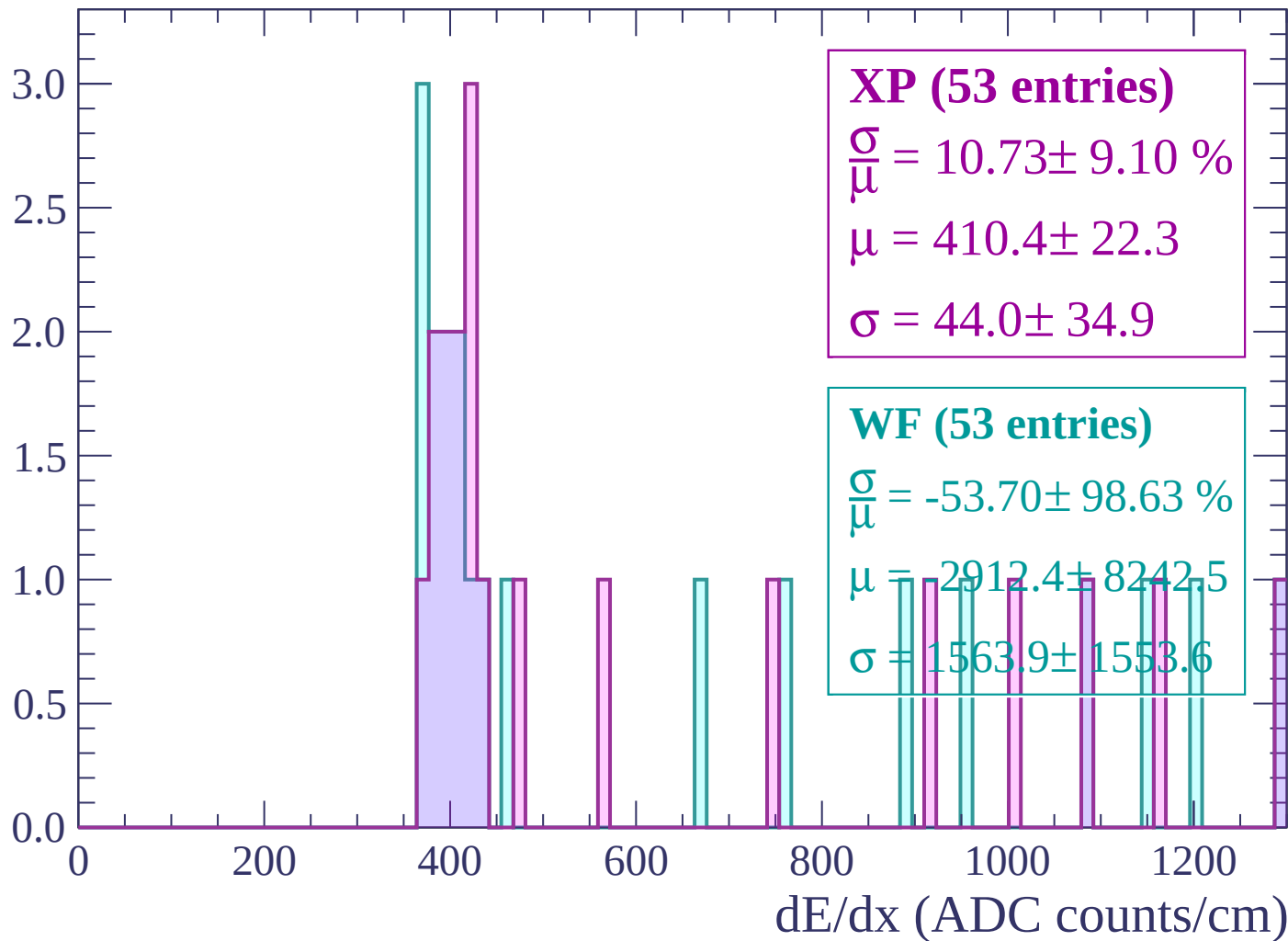
Energy loss | -540 < p < -480

Count



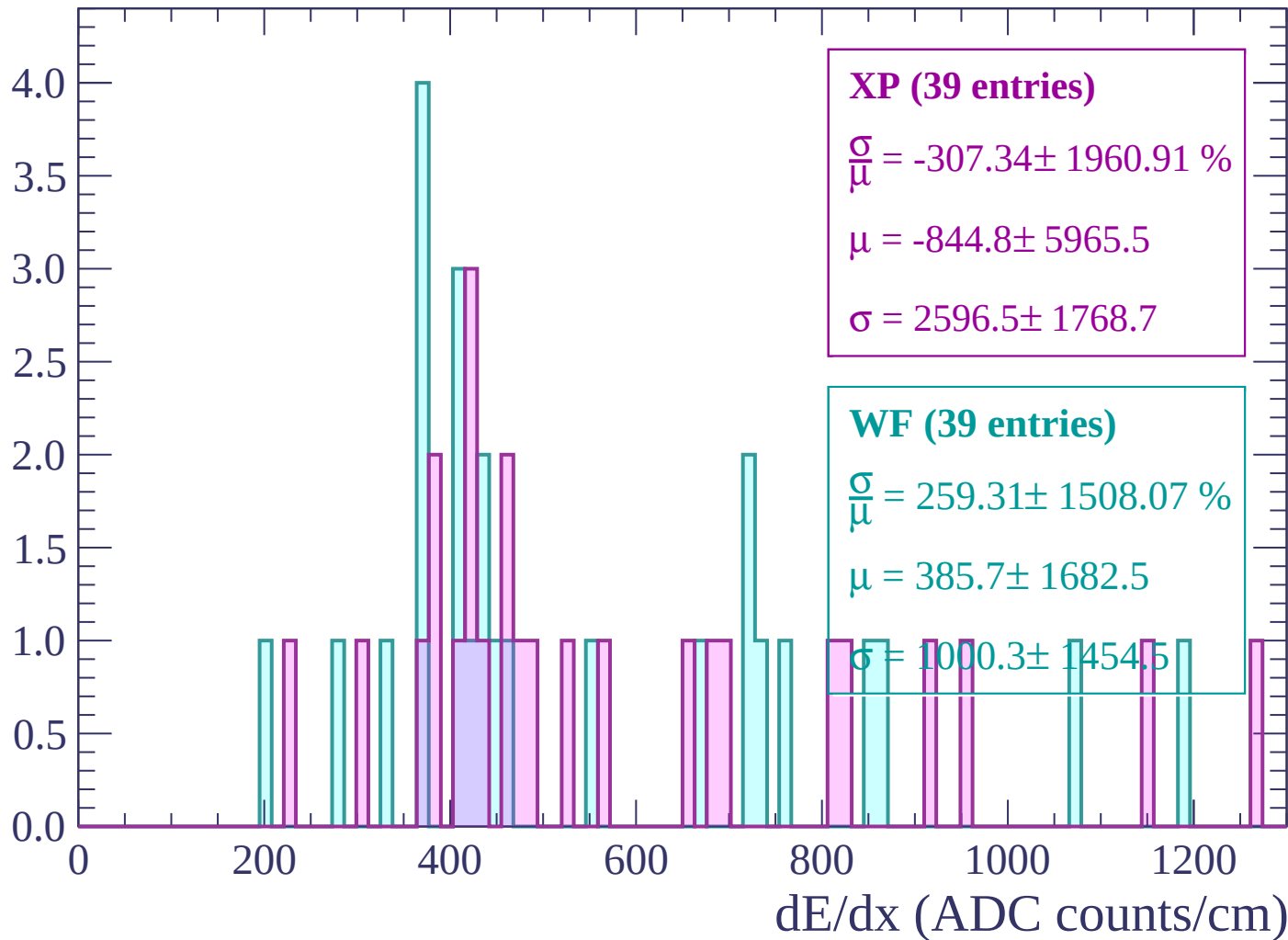
Energy loss | $-480 < p < -420$

Count



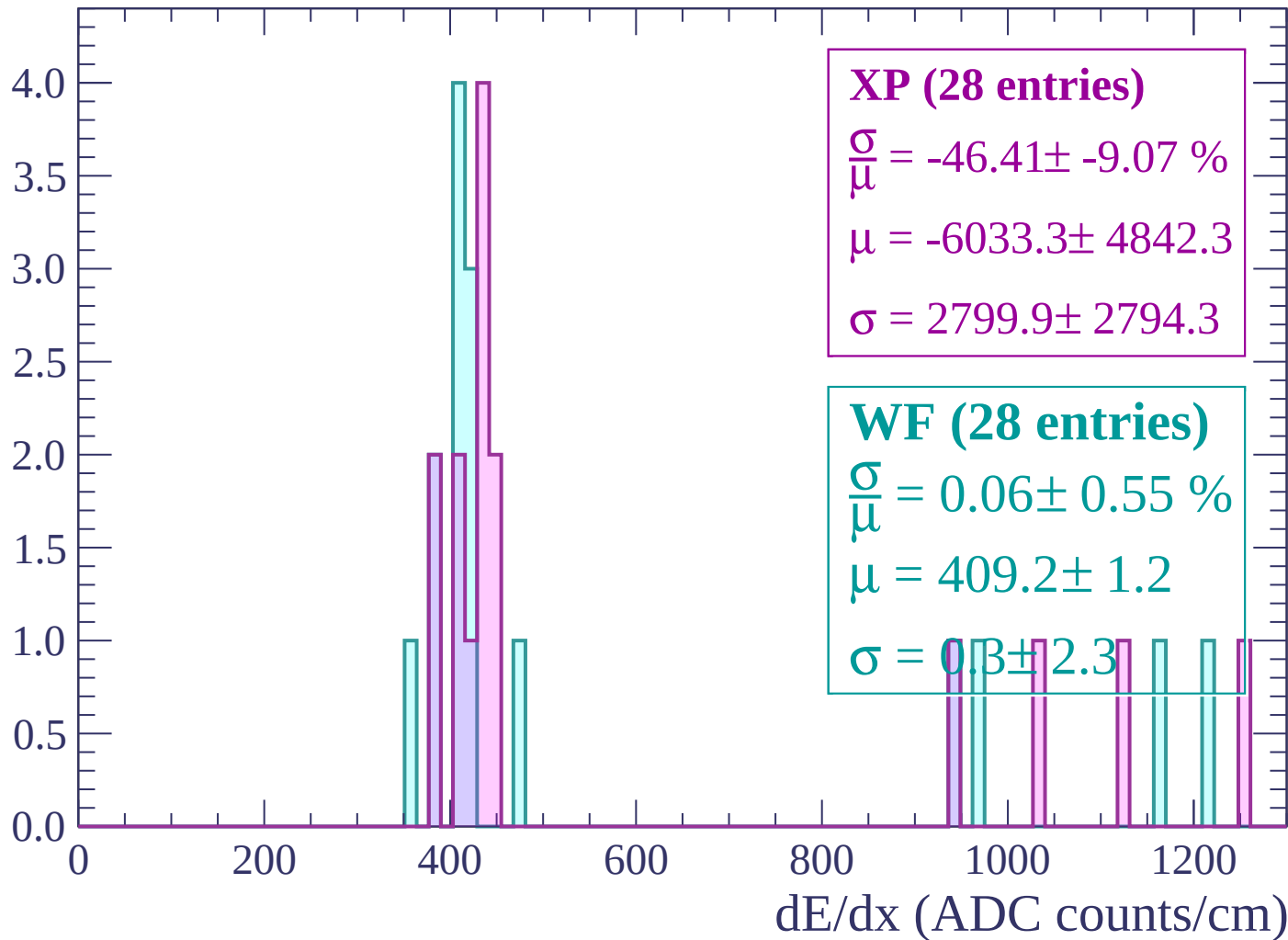
Energy loss | -420 < p < -360

Count



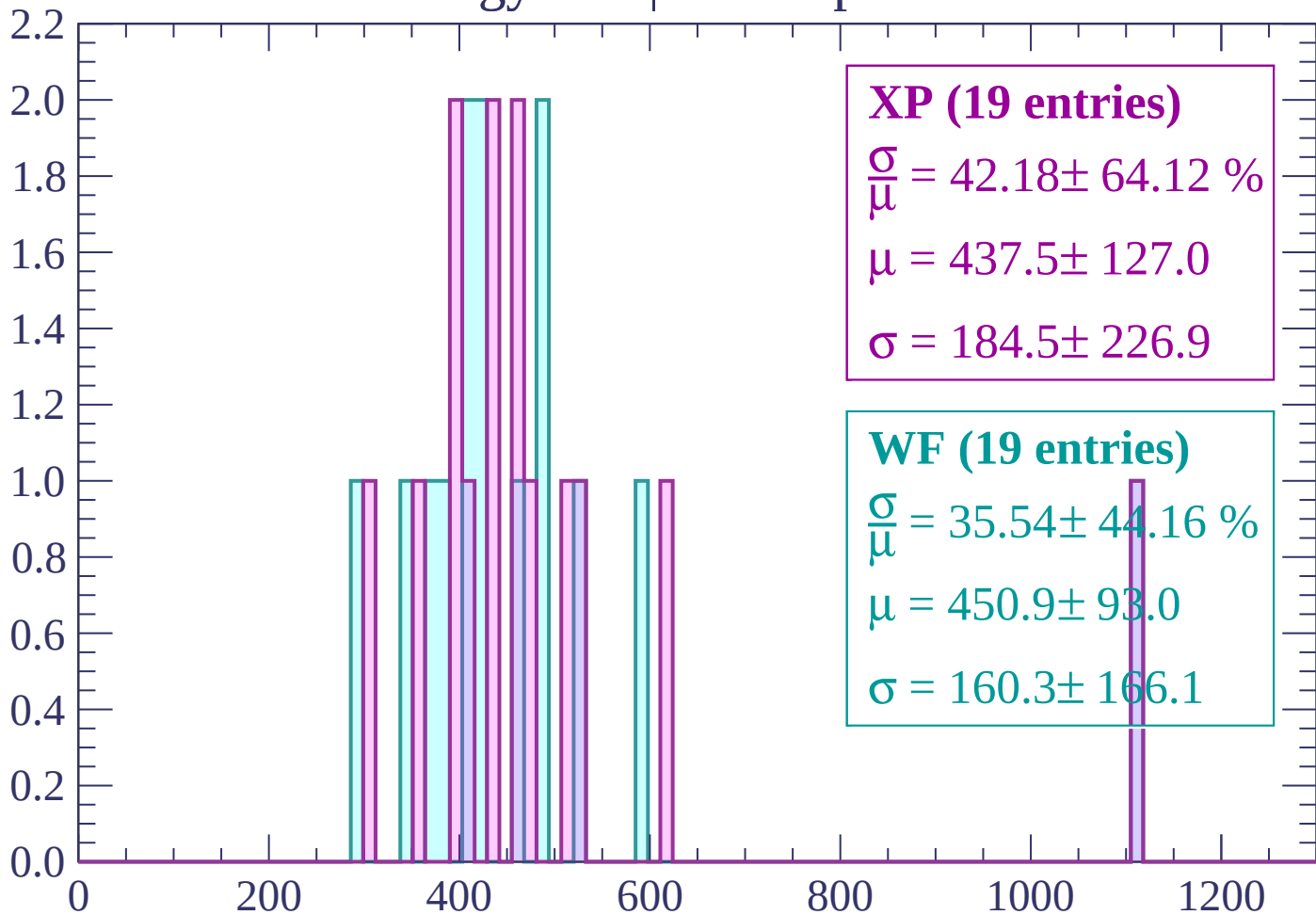
Energy loss | -360 < p < -300

Count



Energy loss | -300 < p < -240

Count



XP (19 entries)

$$\frac{\sigma}{\mu} = 42.18 \pm 64.12 \%$$

$$\mu = 437.5 \pm 127.0$$

$$\sigma = 184.5 \pm 226.9$$

WF (19 entries)

$$\frac{\sigma}{\mu} = 35.54 \pm 44.16 \%$$

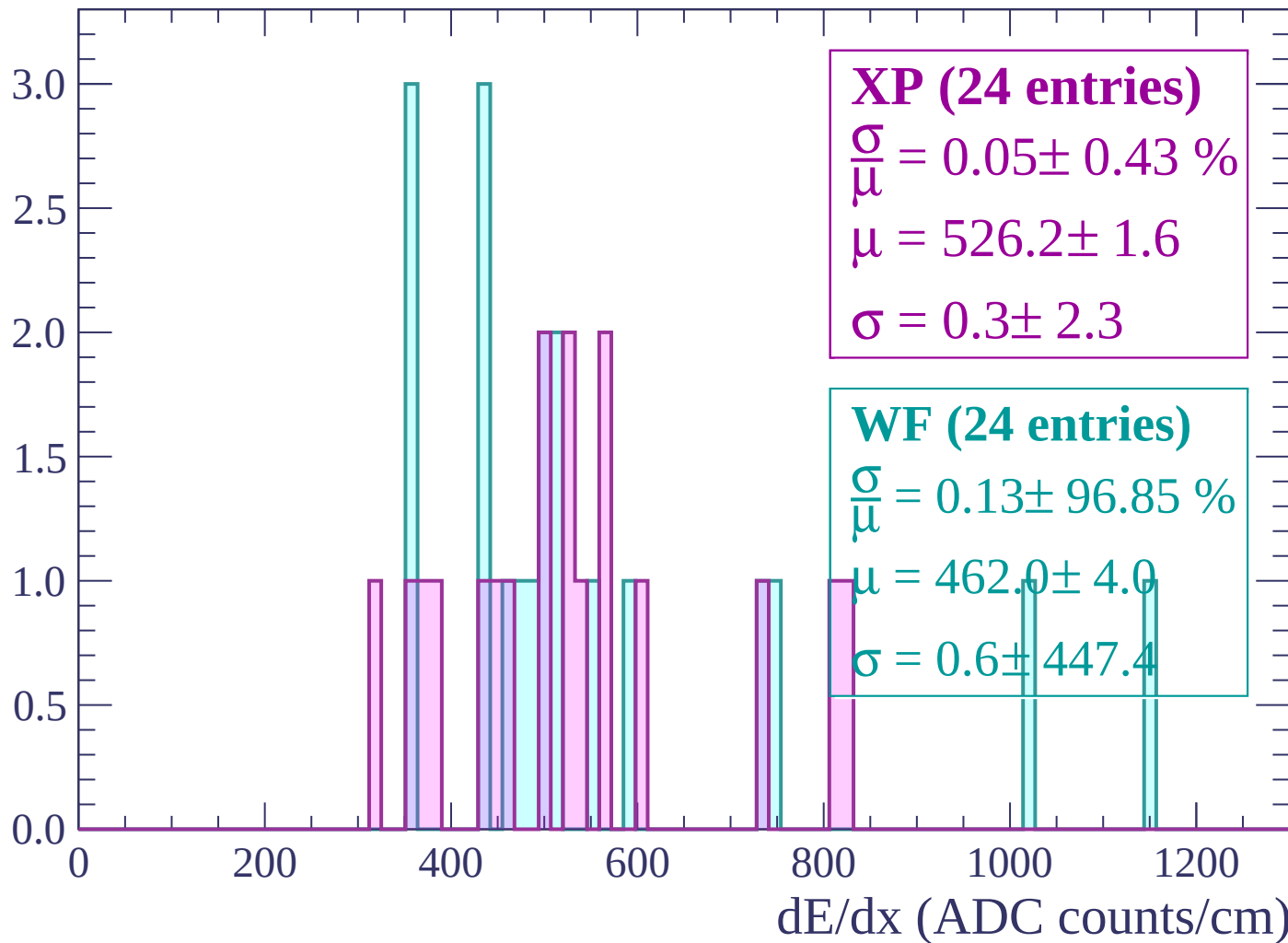
$$\mu = 450.9 \pm 93.0$$

$$\sigma = 160.3 \pm 166.1$$

dE/dx (ADC counts/cm)

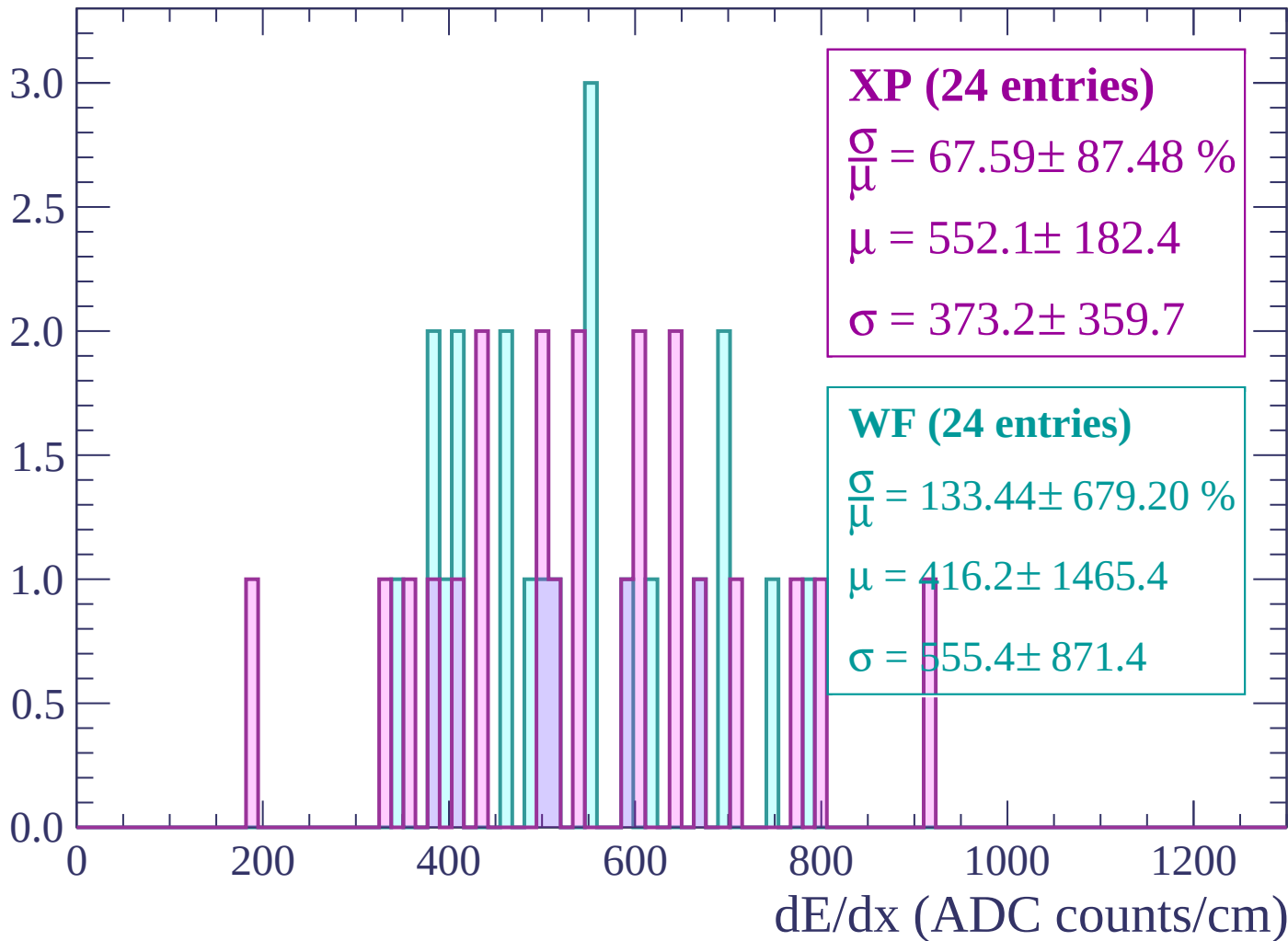
Energy loss | $-240 < p < -180$

Count



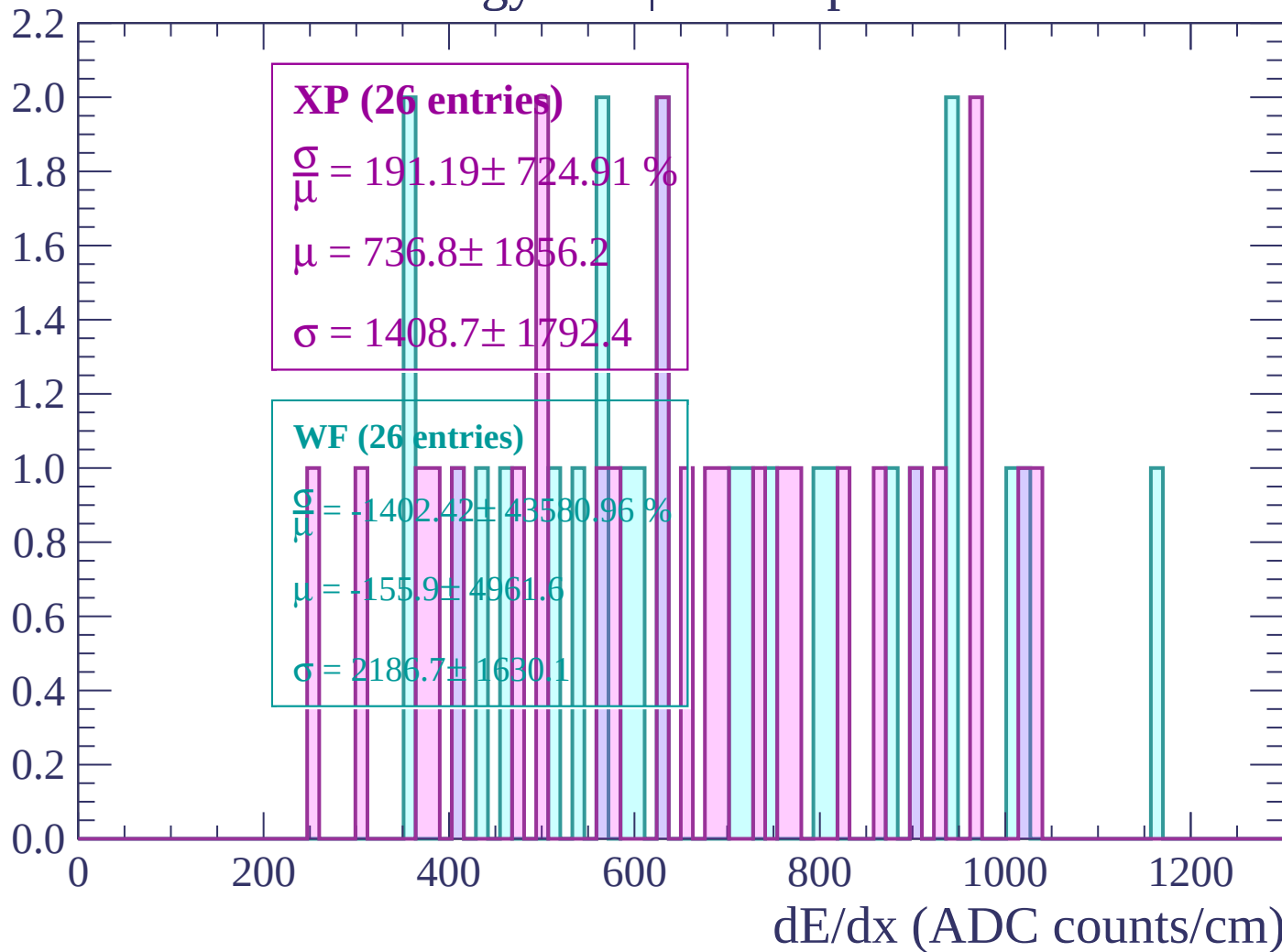
Energy loss | -180 < p < -120

Count



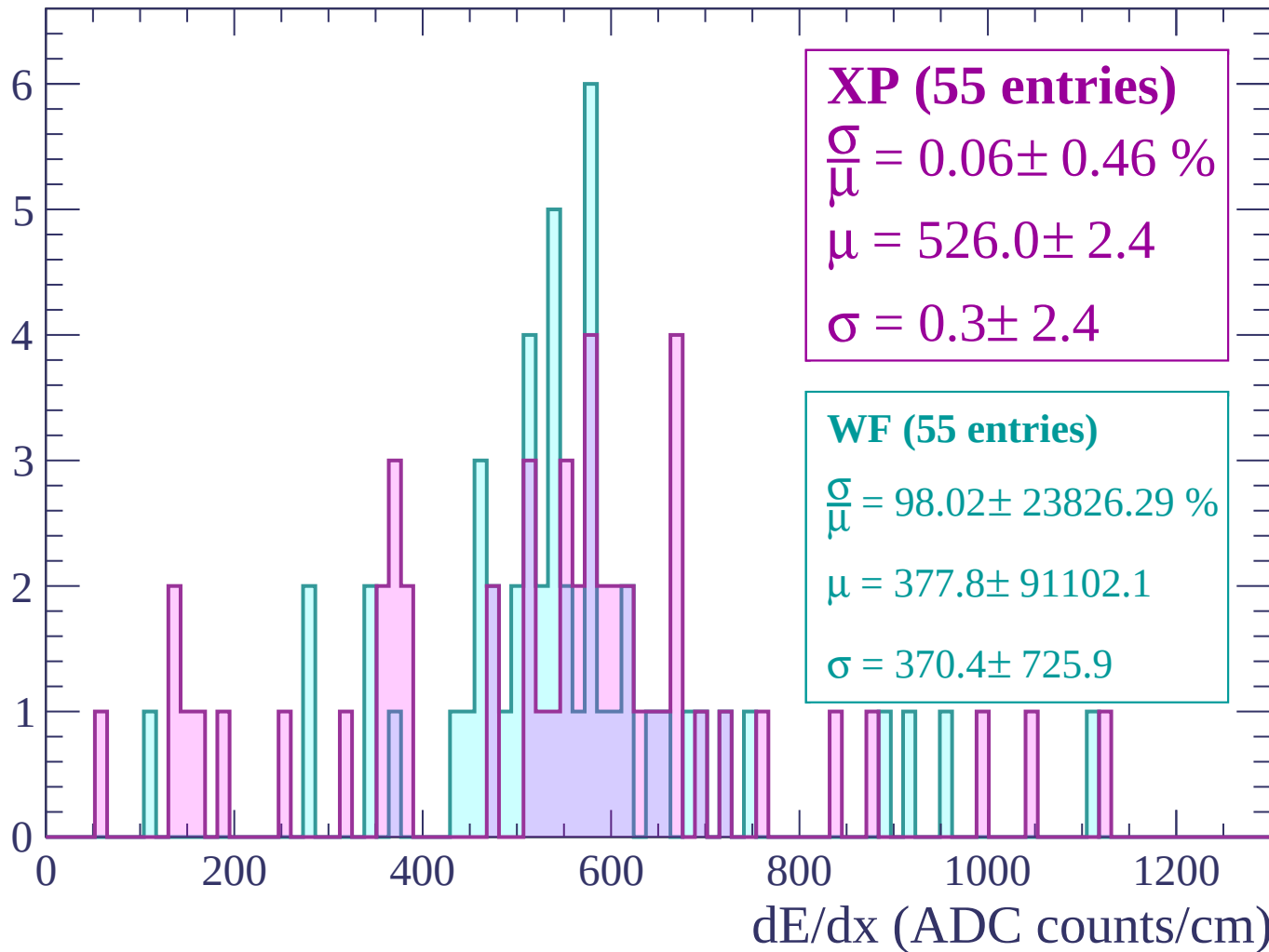
Energy loss | $-120 < p < -60$

Count



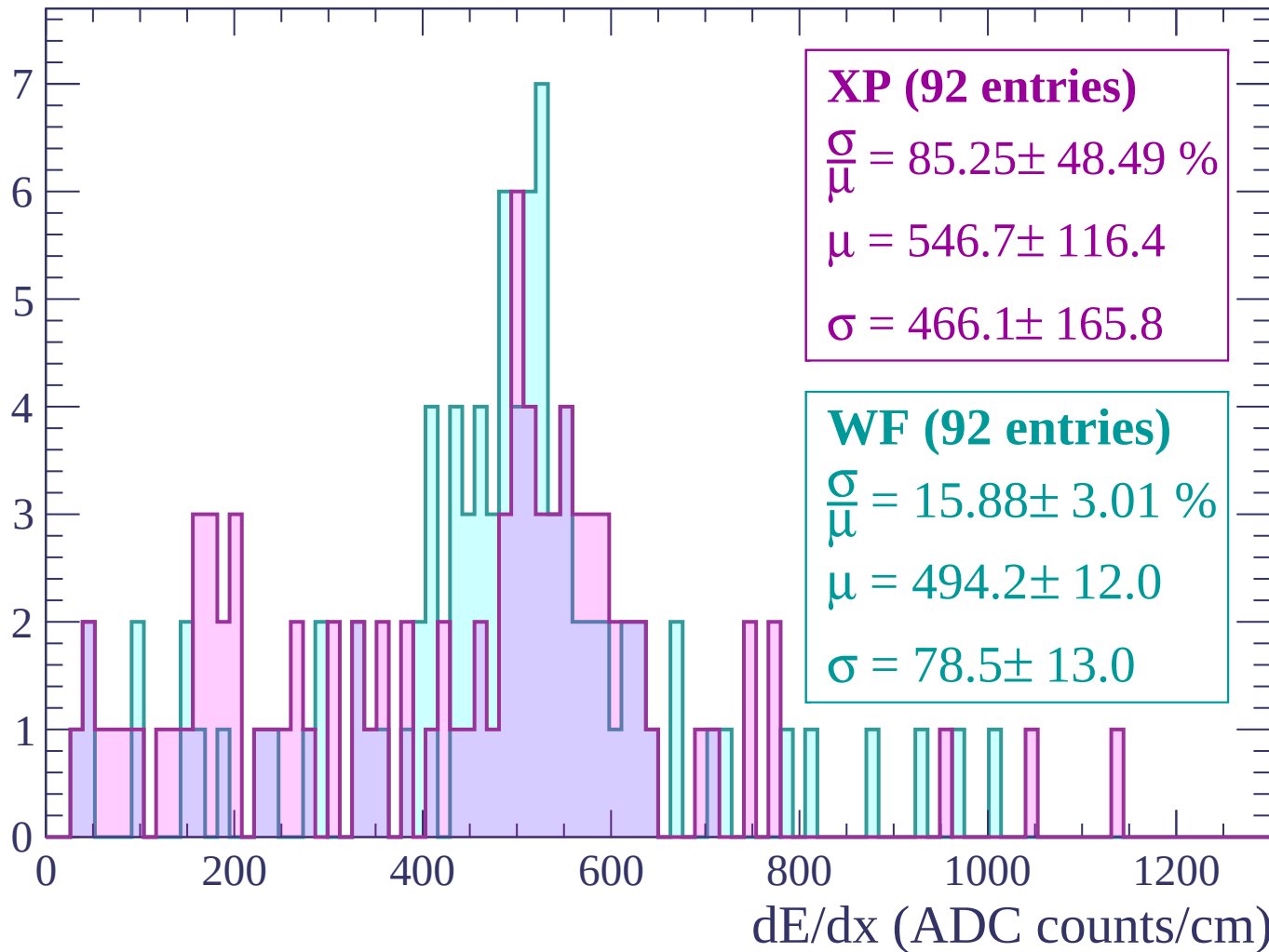
Energy loss | $-60 < p < 0$

Count



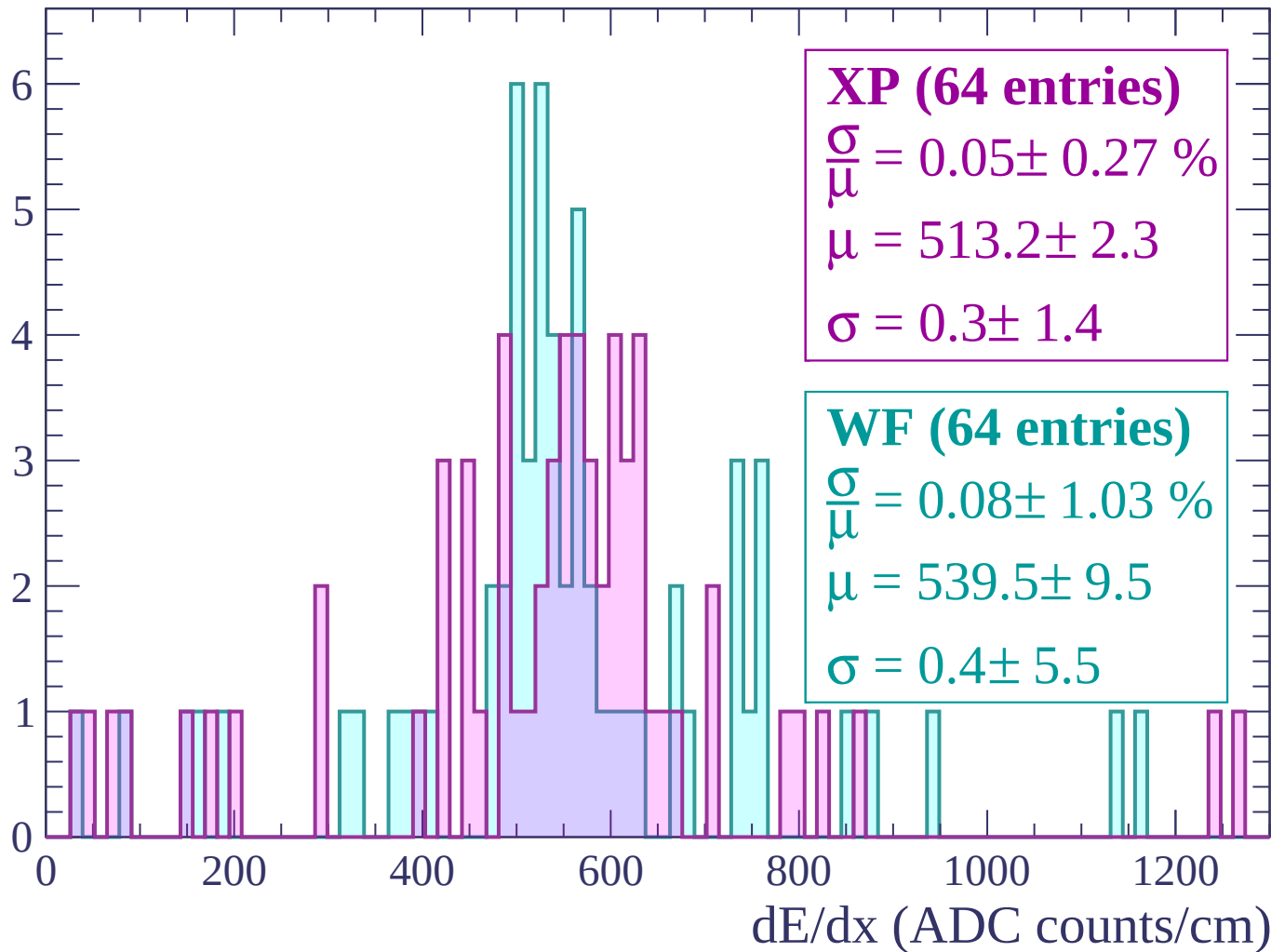
Energy loss | $0 < p < 60$

Count



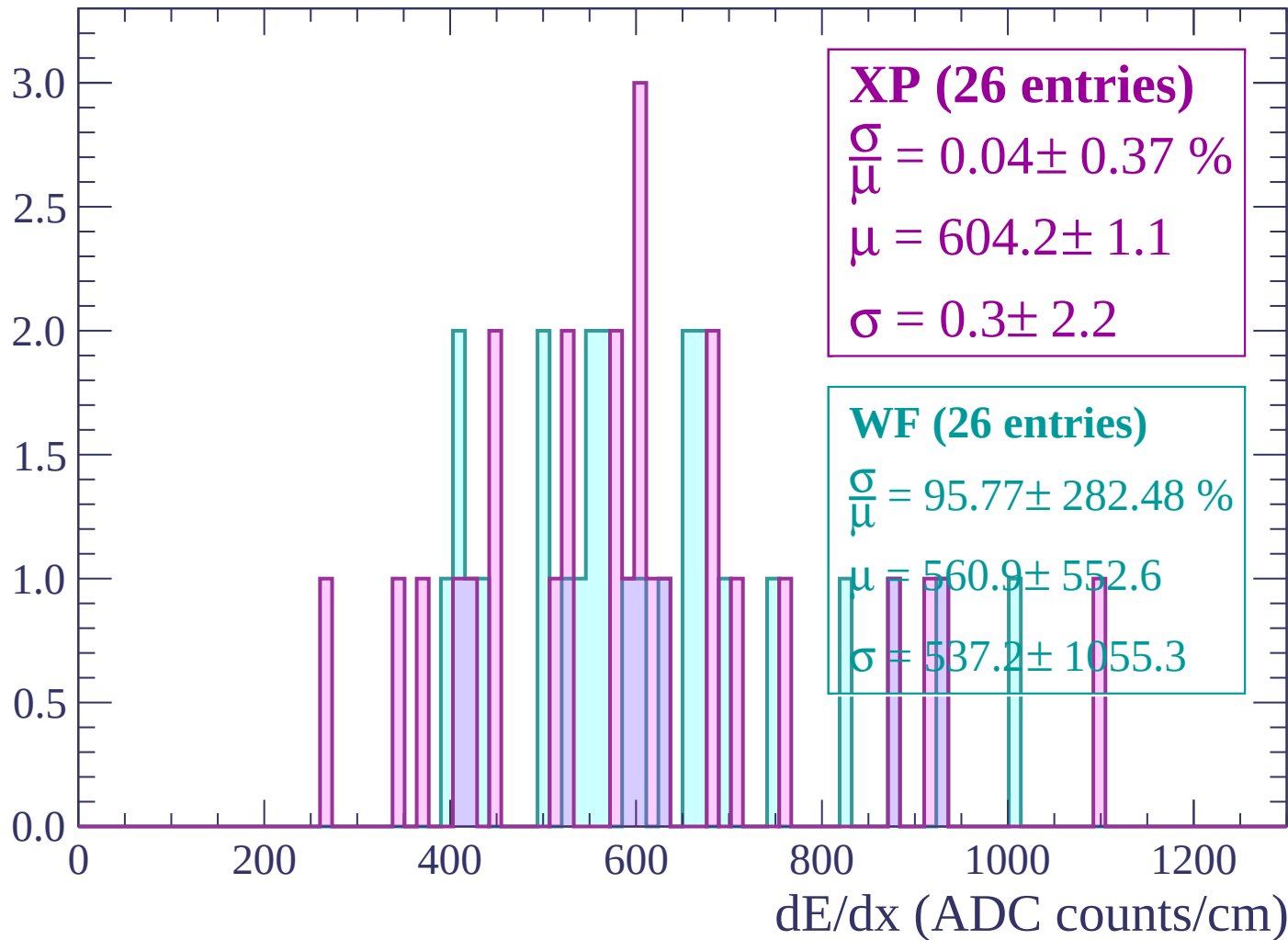
Energy loss | $60 < p < 120$

Count



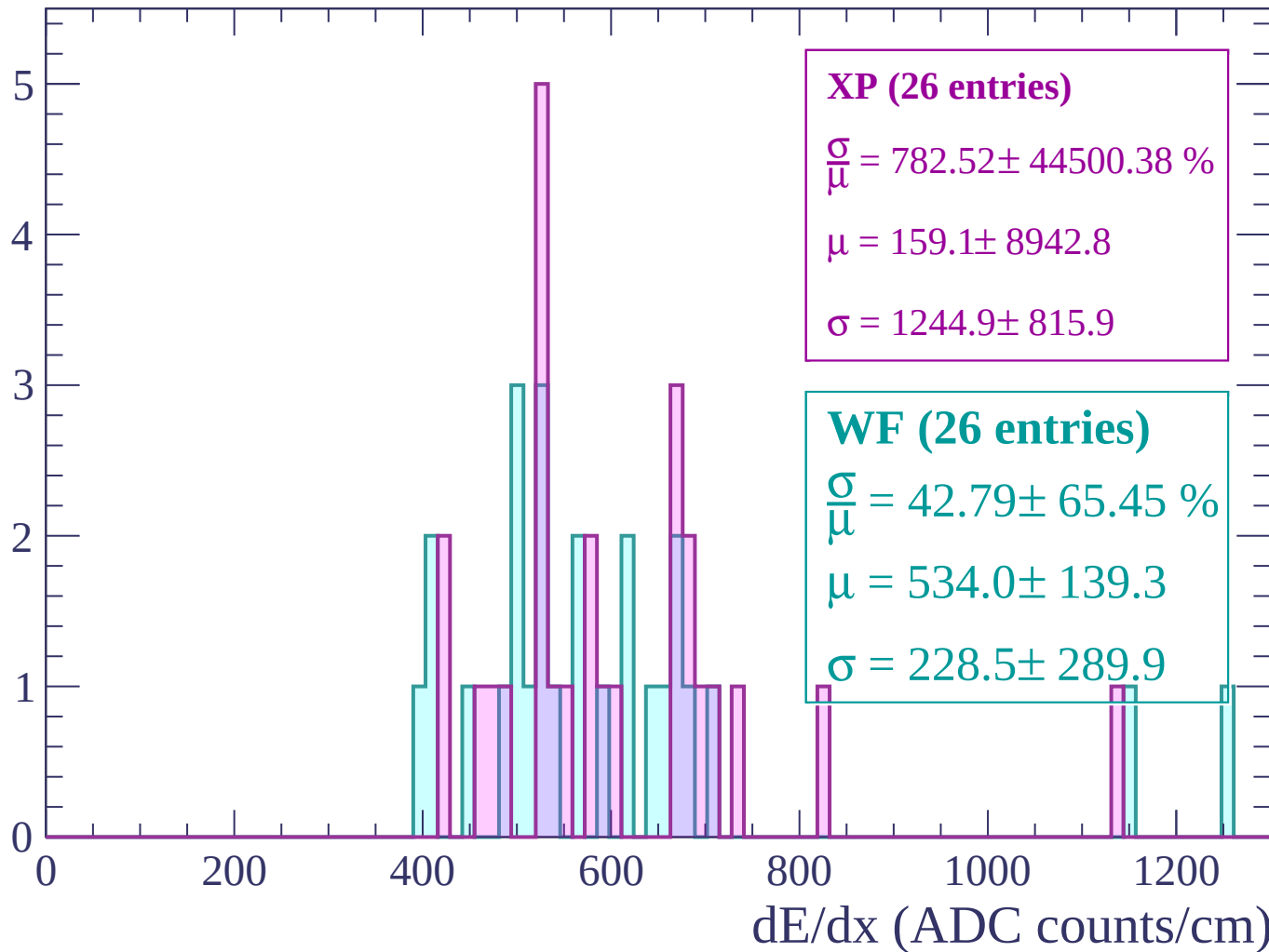
Energy loss | $120 < p < 180$

Count



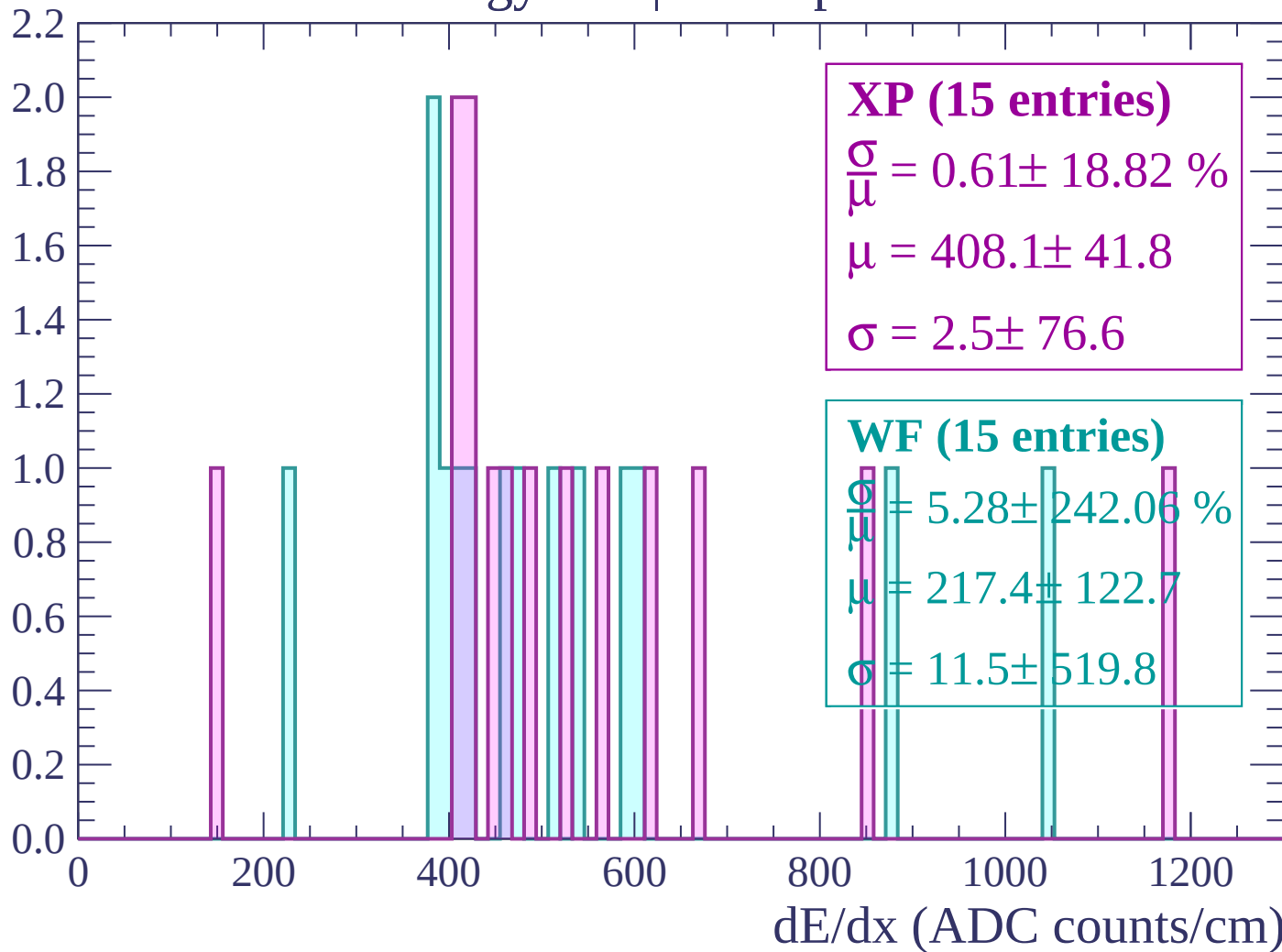
Energy loss | $180 < p < 240$

Count



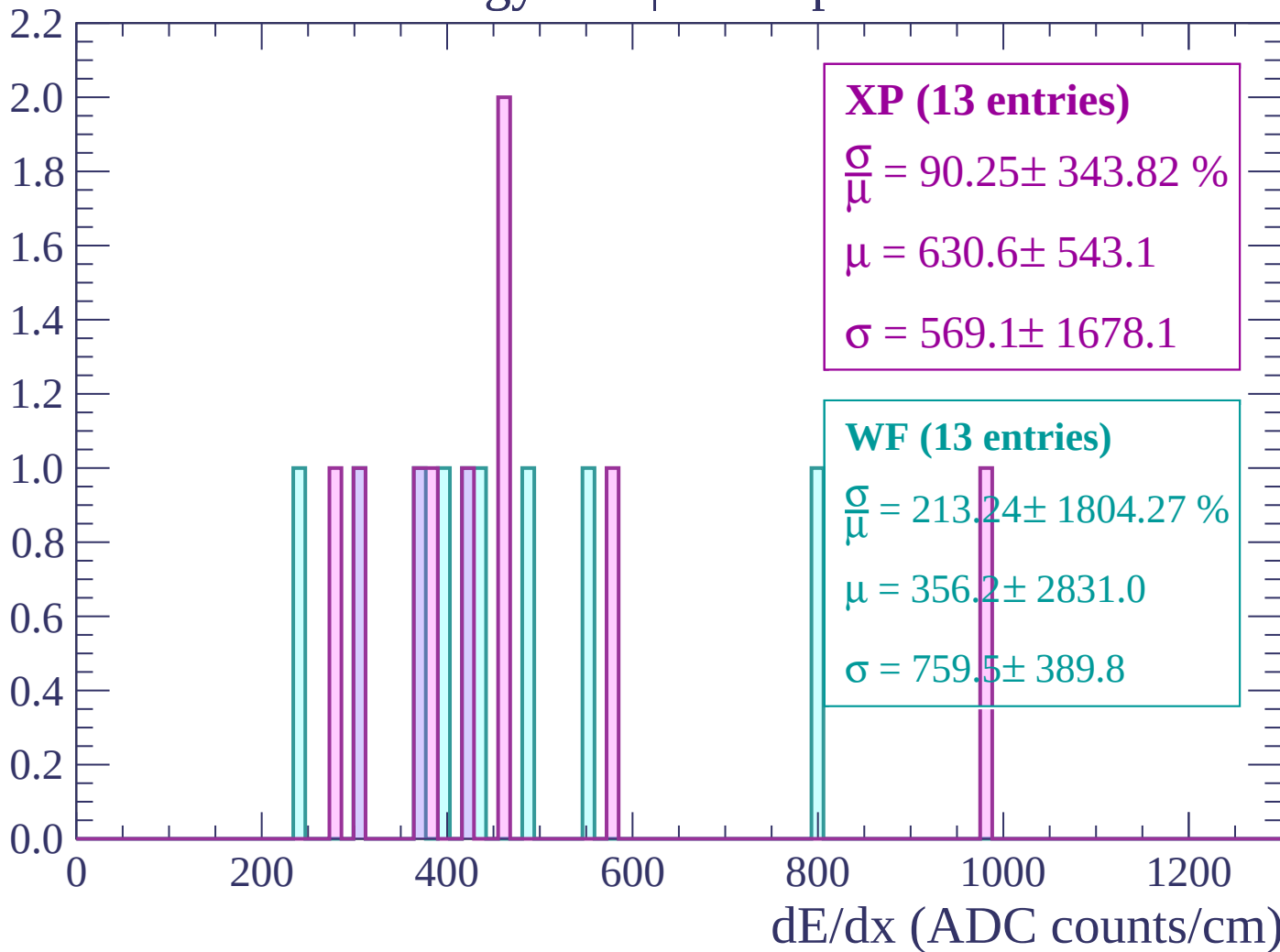
Energy loss | $240 < p < 300$

Count



Energy loss | $300 < p < 360$

Count



Energy loss | $360 < p < 420$

Count

1.0
0.8
0.6
0.4
0.2
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (11 entries)

$\frac{\sigma}{\mu} = 238.09 \pm 2012.87 \%$

$\mu = 656.2 \pm 5037.4$

$\sigma = 1562.3 \pm 1214.3$

WF (11 entries)

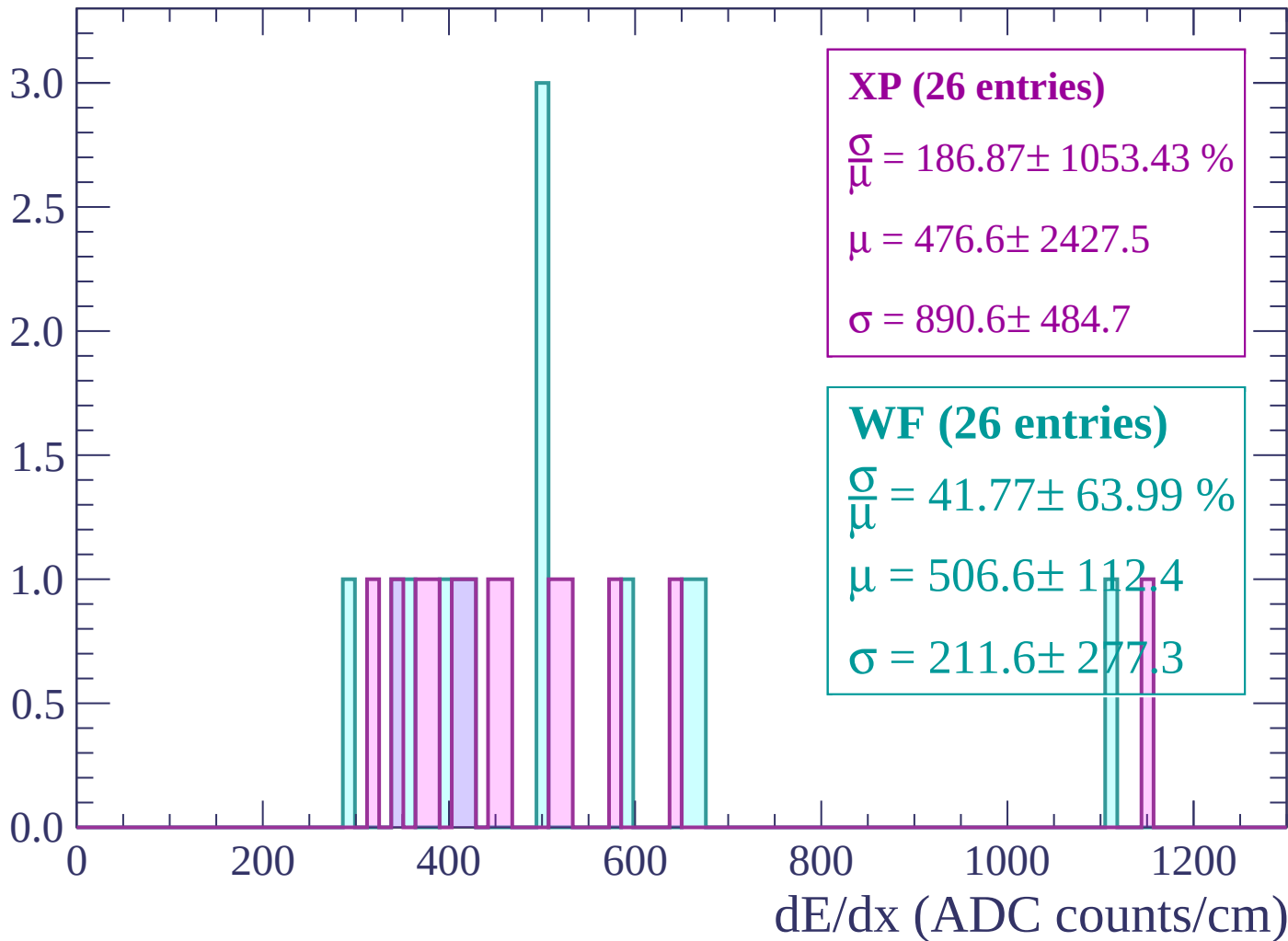
$\frac{\sigma}{\mu} = 104.34 \pm 542.33 \%$

$\mu = 465.8 \pm 2064.5$

$\sigma = 486.0 \pm 372.1$

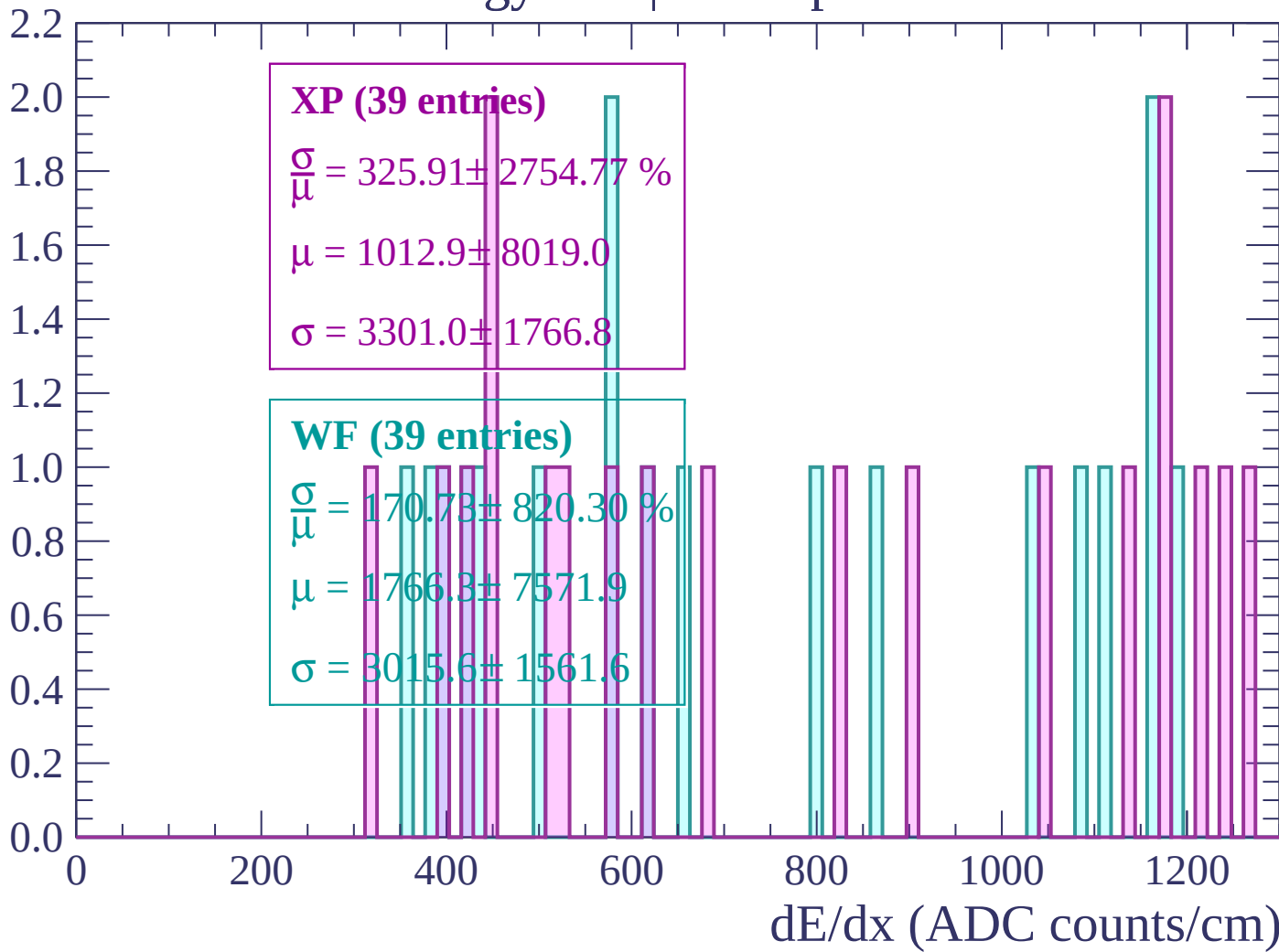
Energy loss | $420 < p < 480$

Count



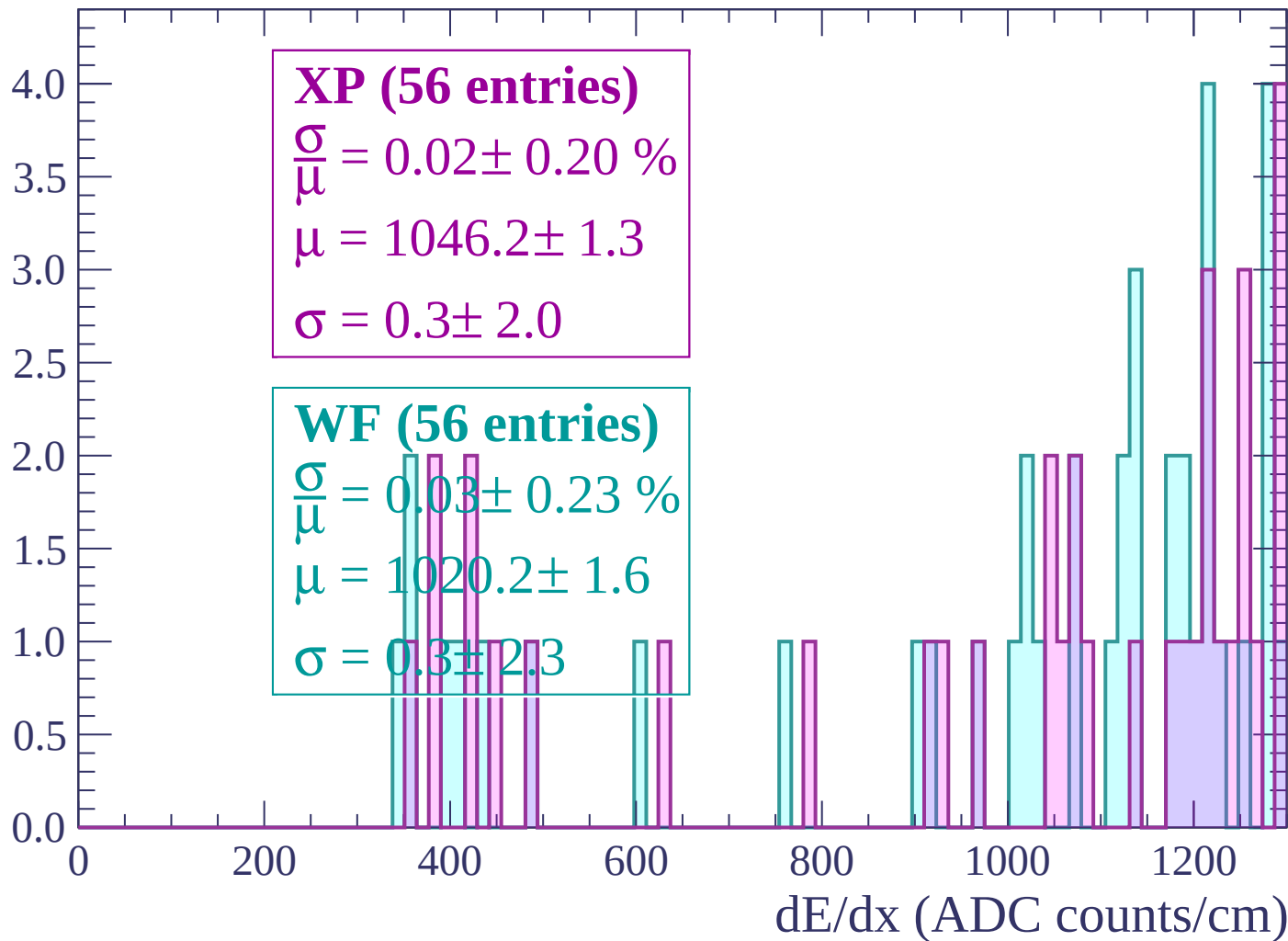
Energy loss | 480 < p < 540

Count



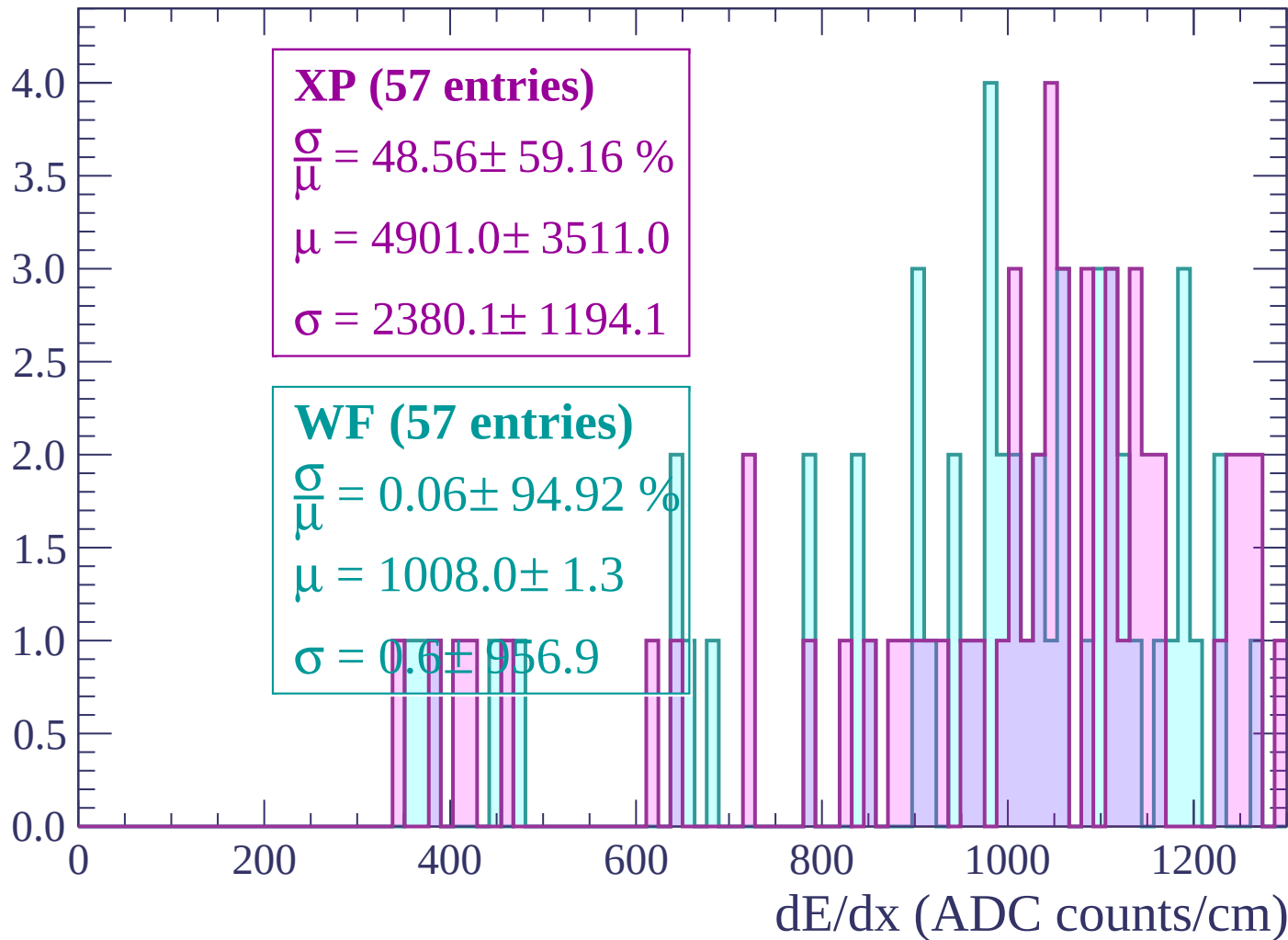
Energy loss | $540 < p < 600$

Count



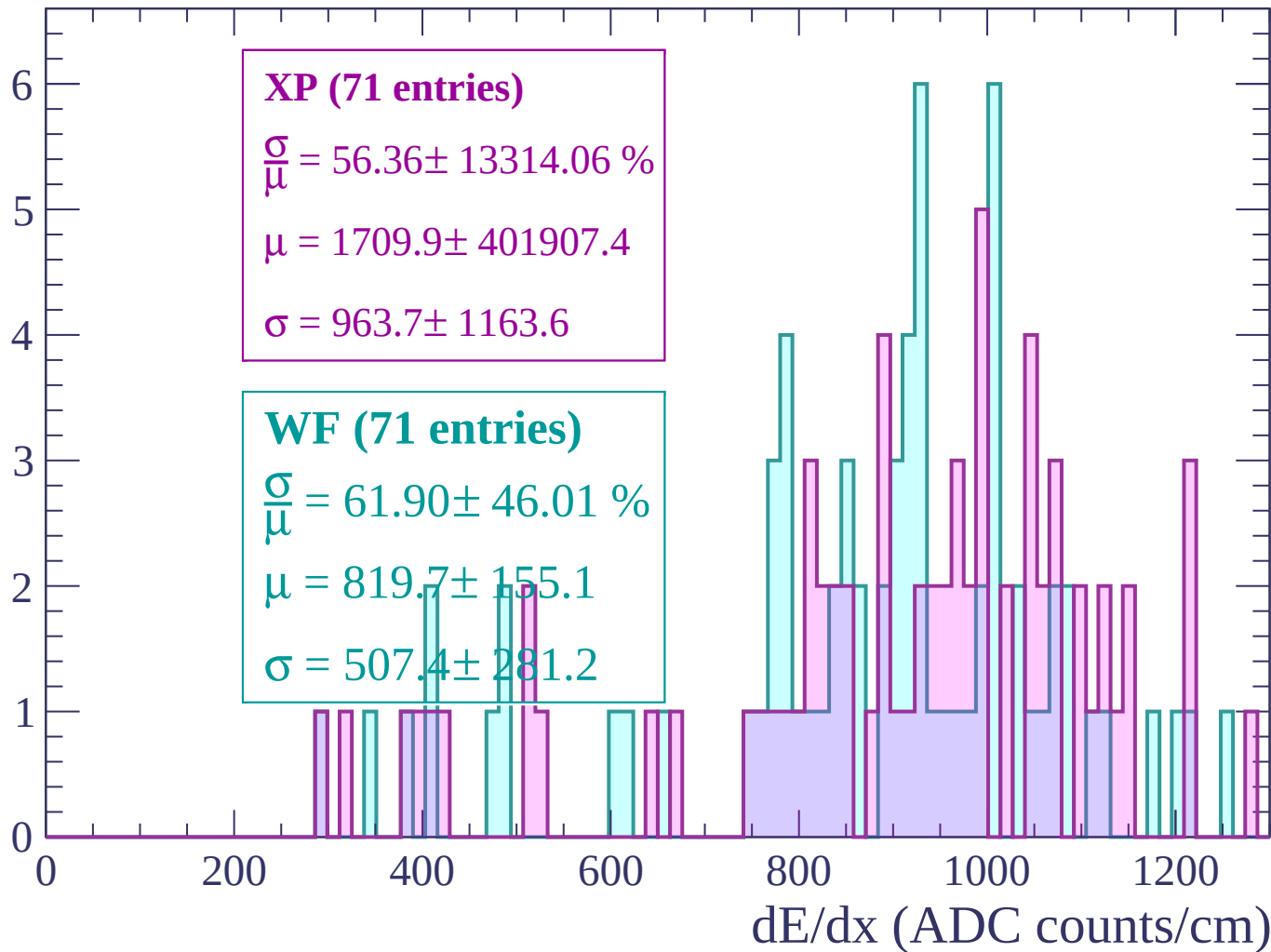
Energy loss | $600 < p < 660$

Count



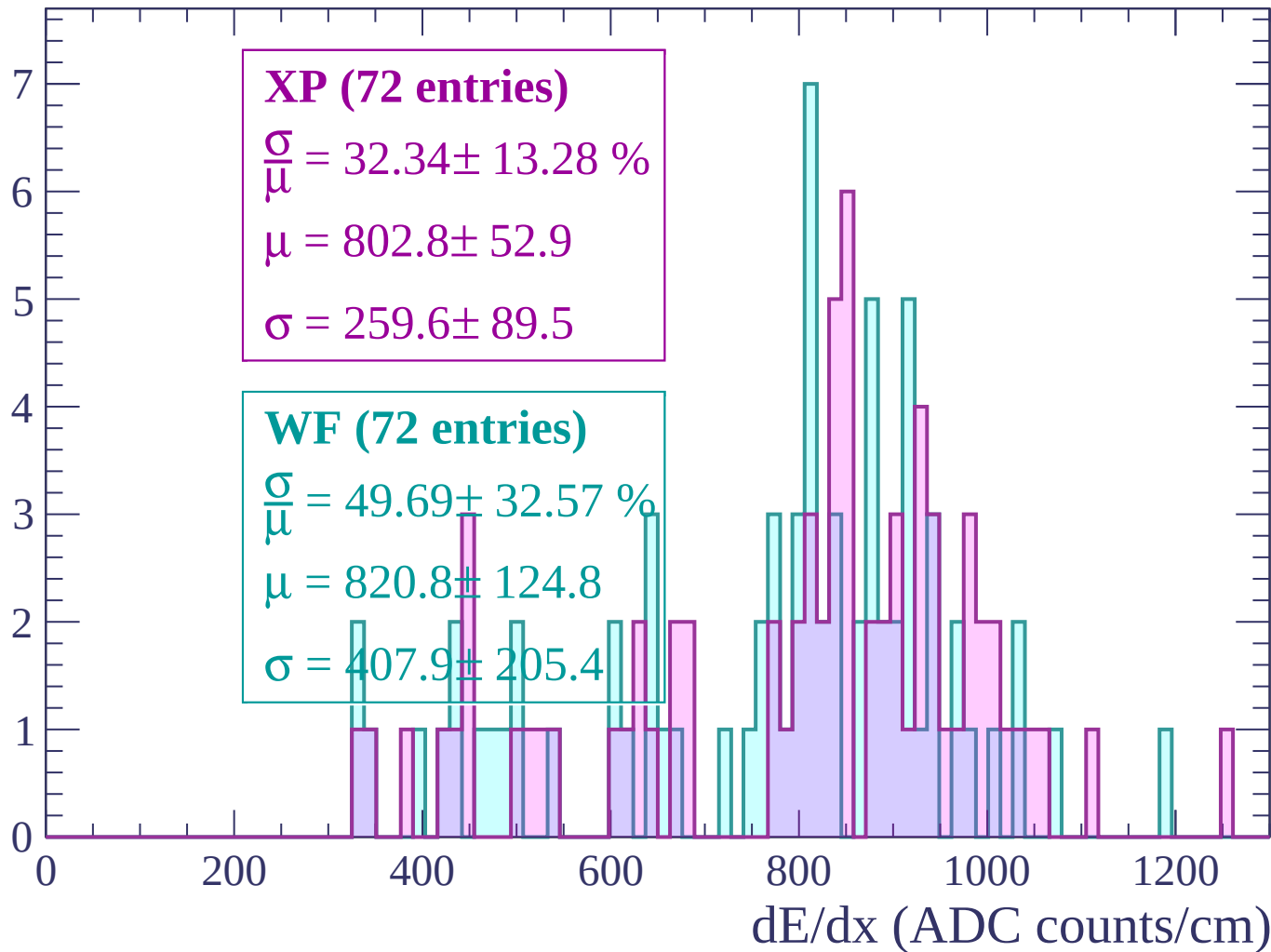
Energy loss | $660 < p < 720$

Count



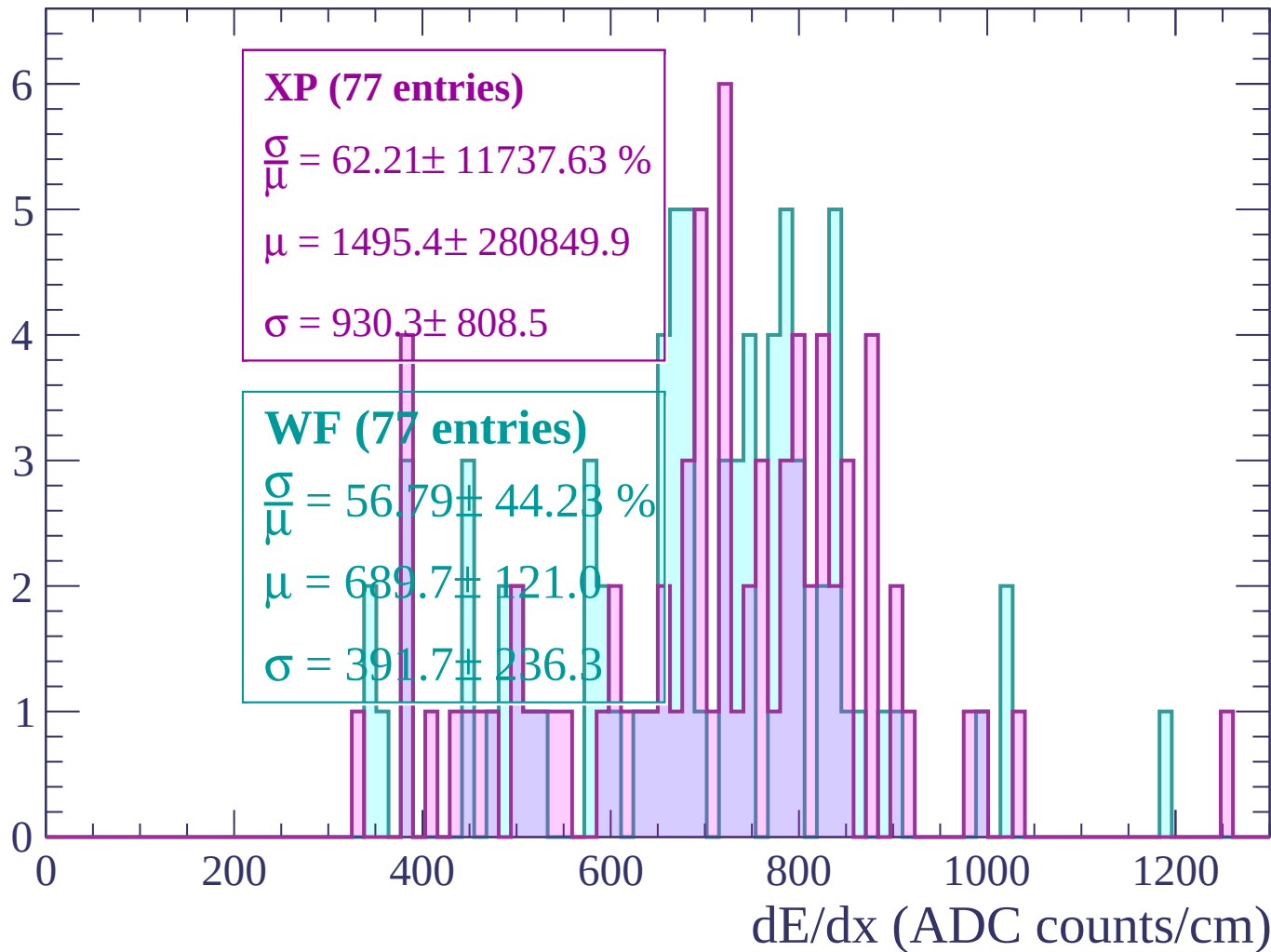
Energy loss | $720 < p < 780$

Count



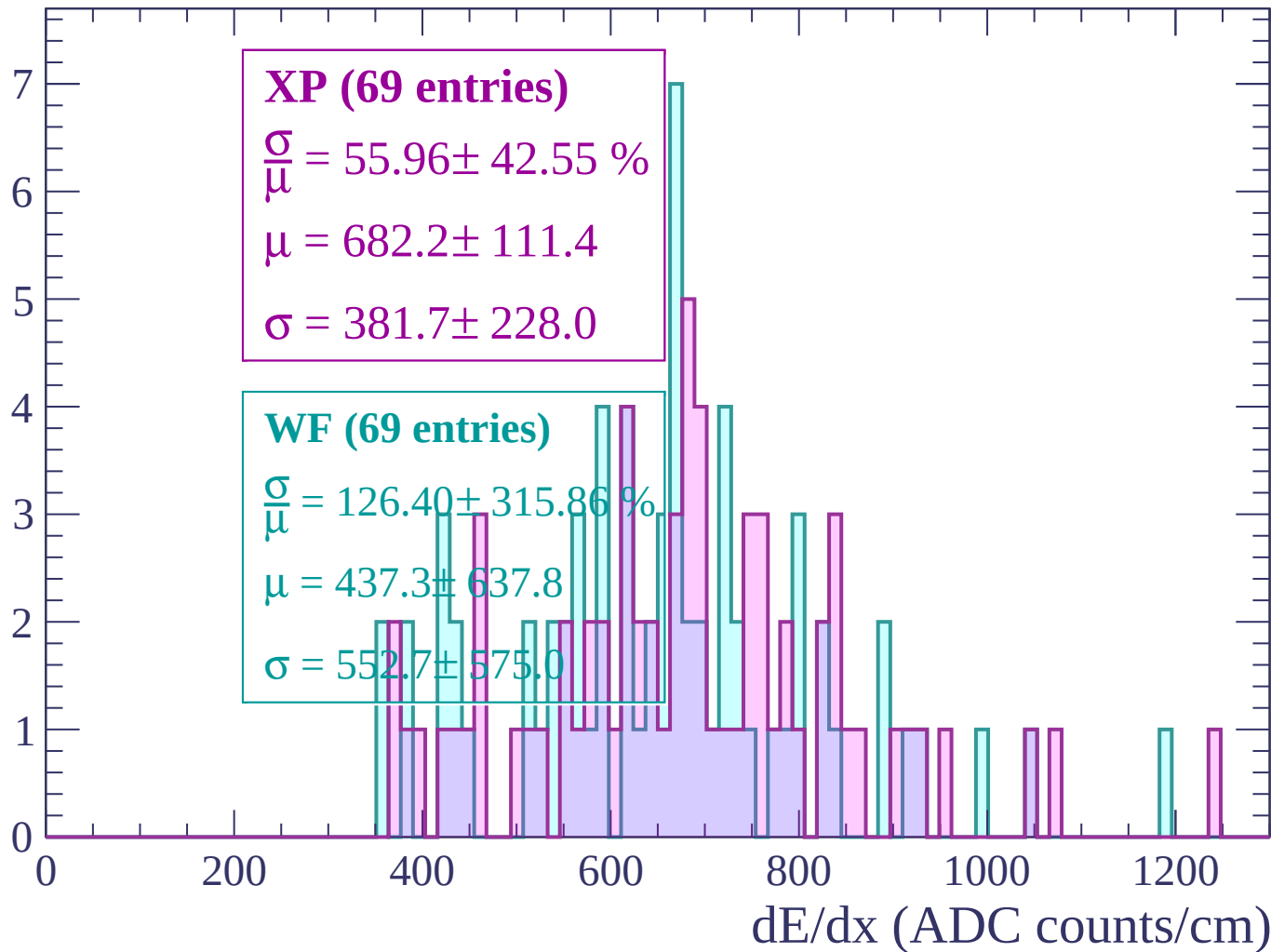
Energy loss | $780 < p < 840$

Count



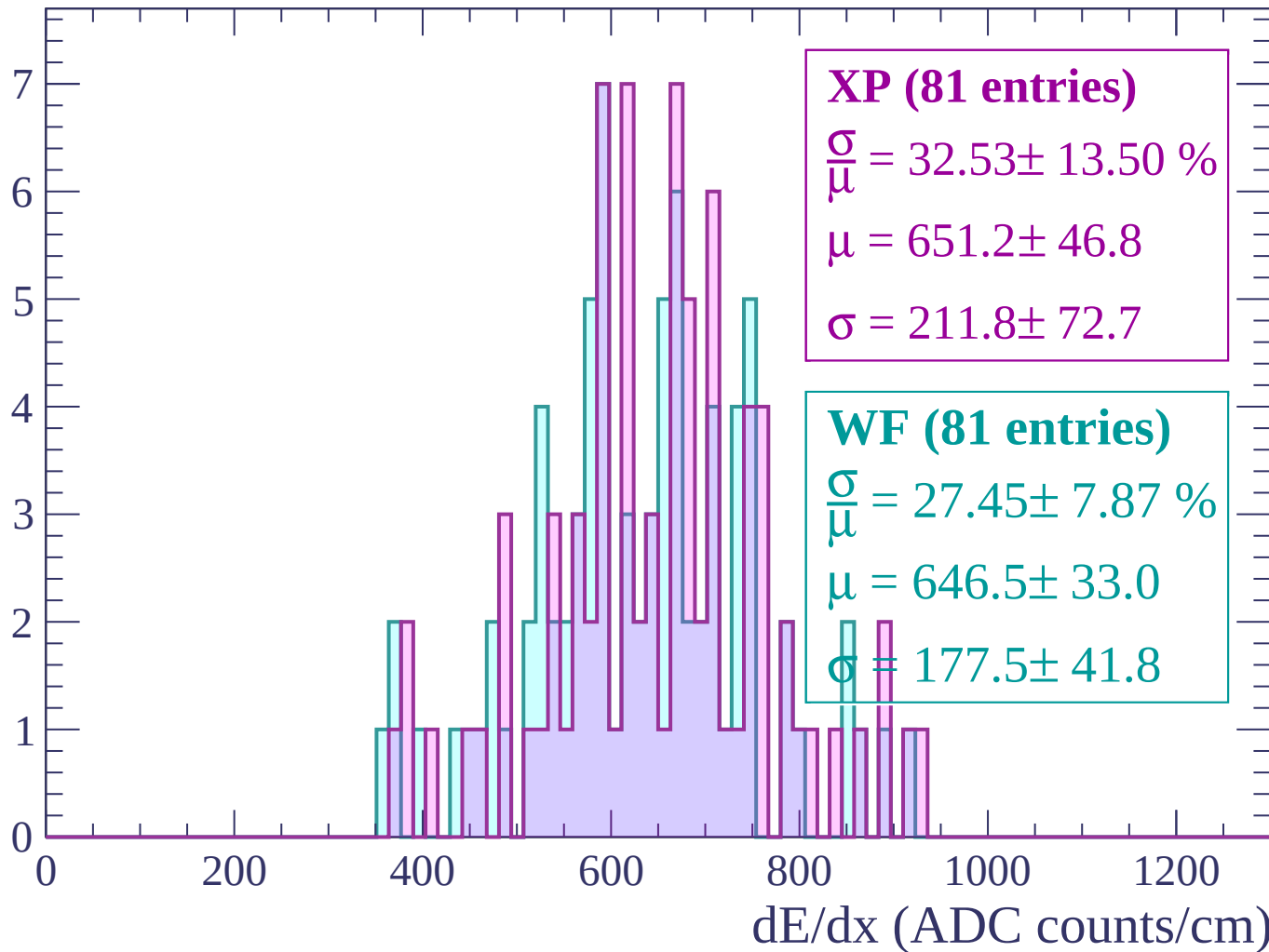
Energy loss | $840 < p < 900$

Count



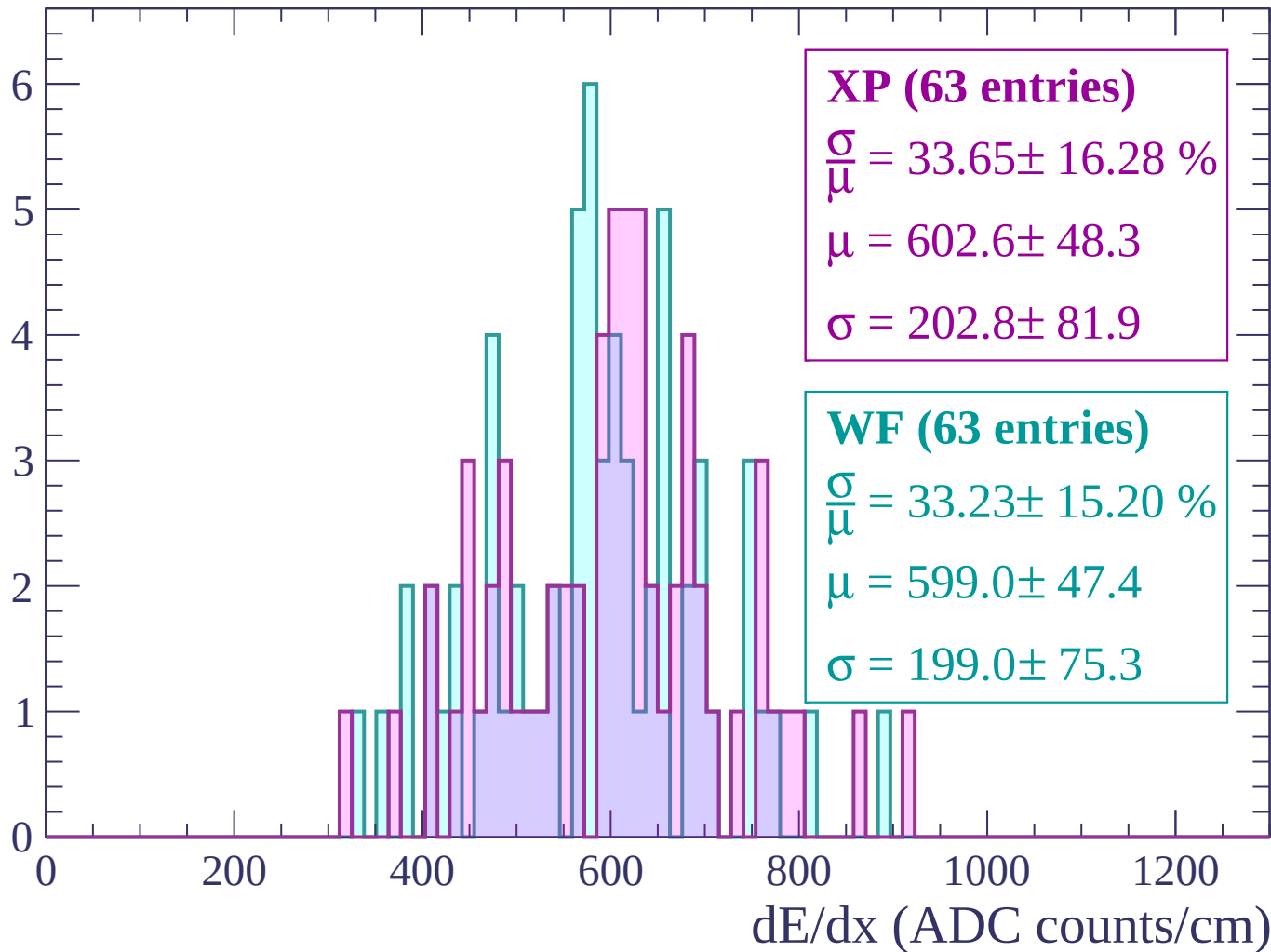
Energy loss | $900 < p < 960$

Count



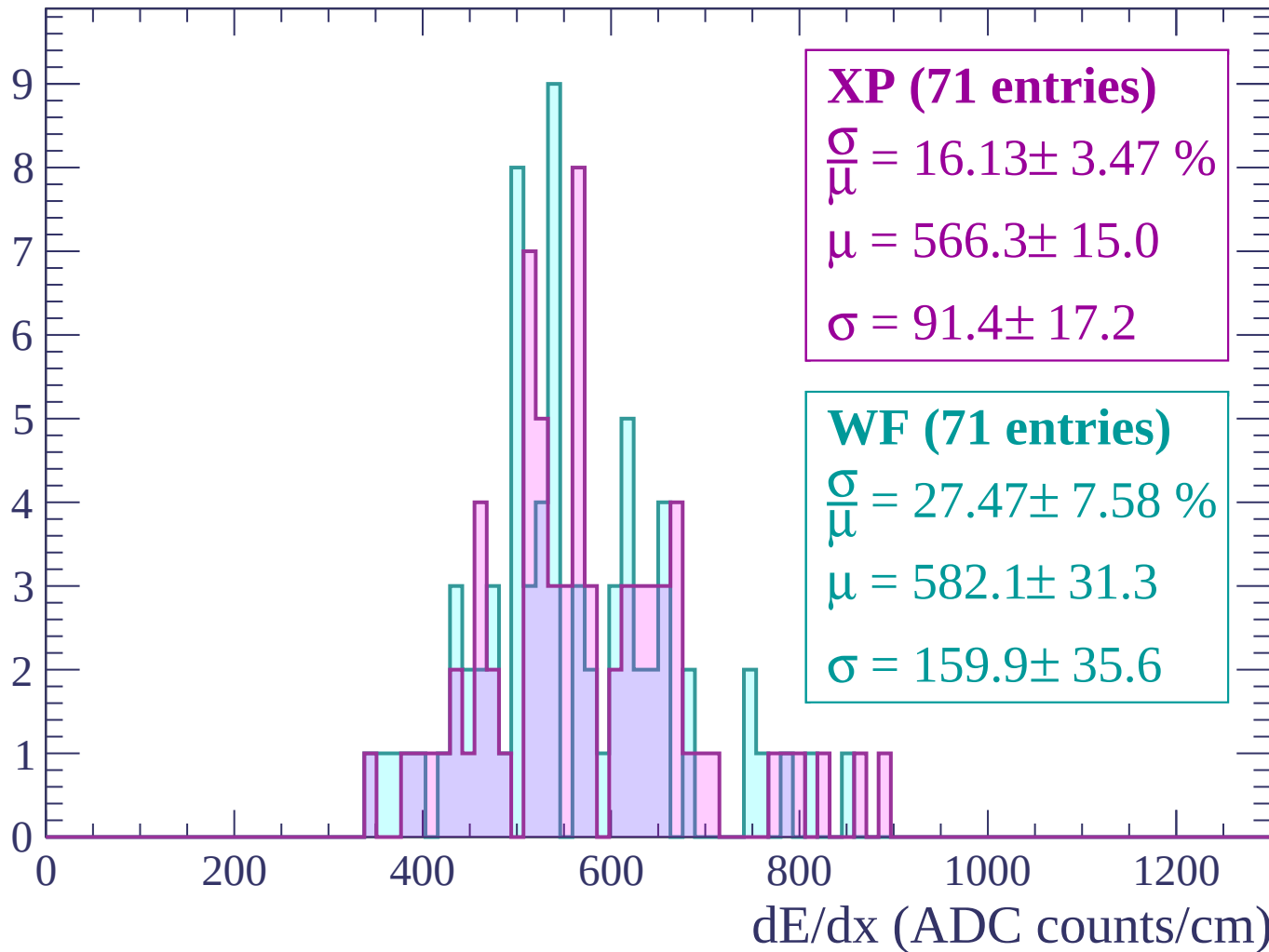
Energy loss | $960 < p < 1020$

Count



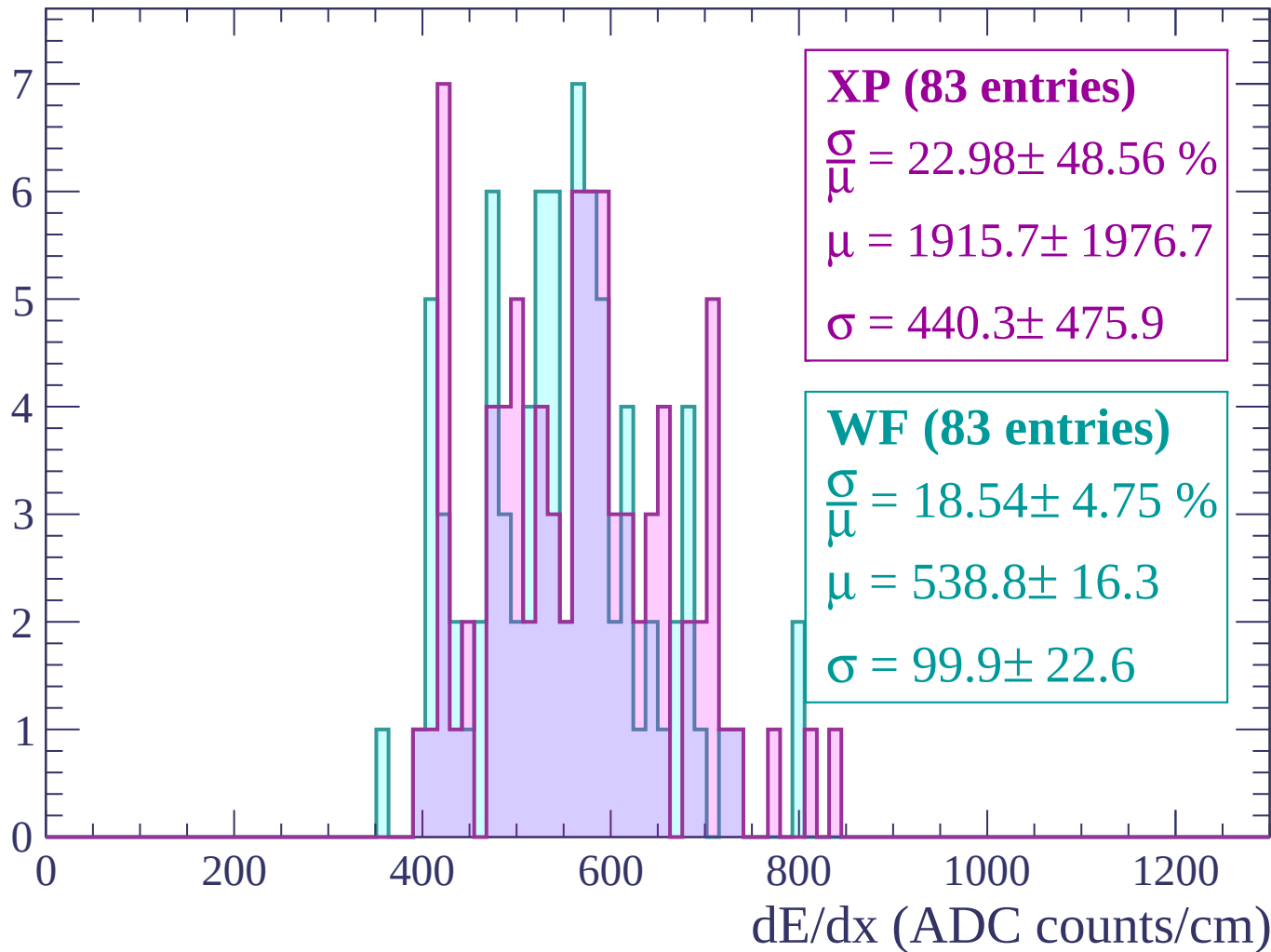
Energy loss | $1020 < p < 1080$

Count



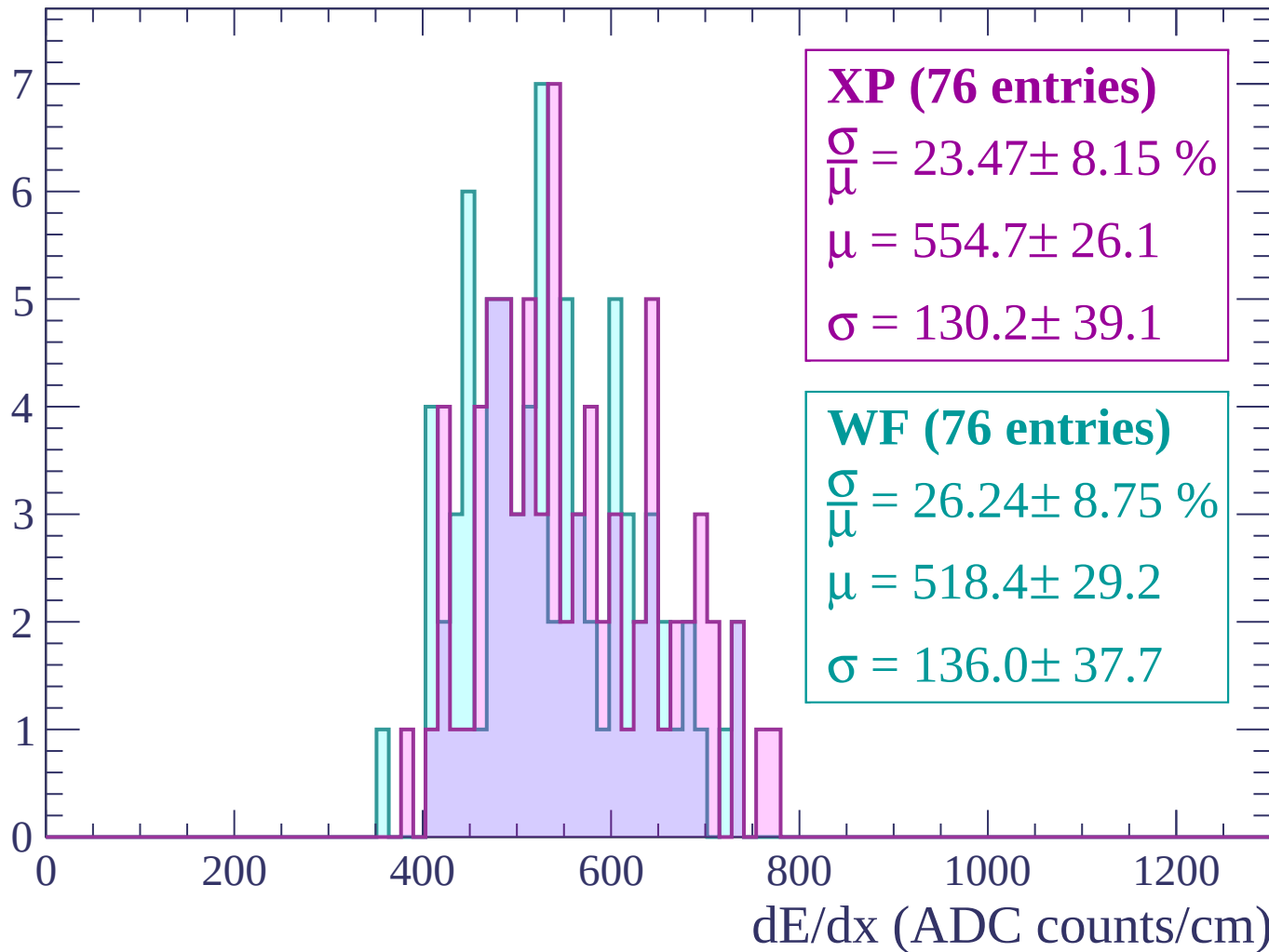
Energy loss | $1080 < p < 1140$

Count



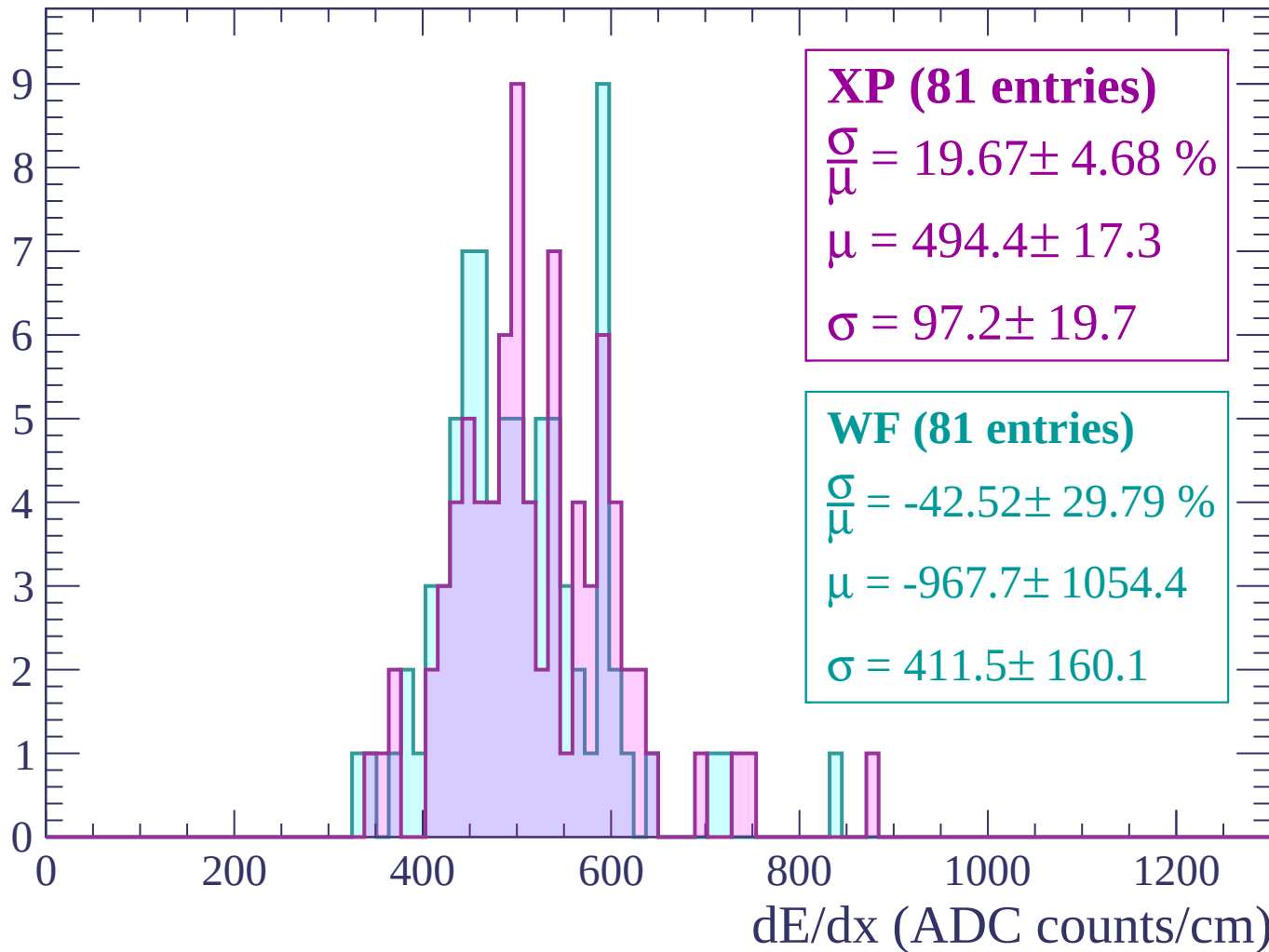
Energy loss | $1140 < p < 1200$

Count



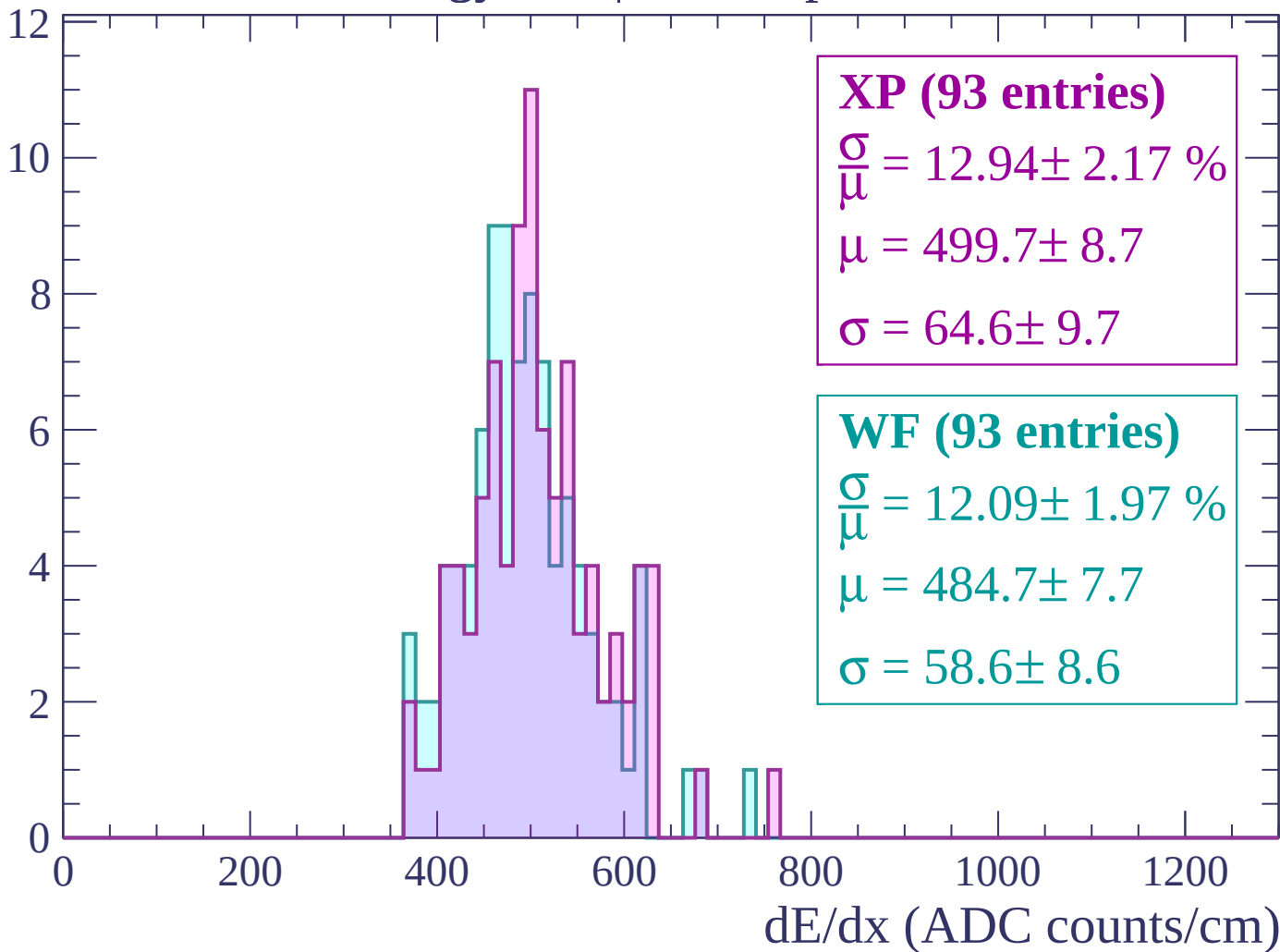
Energy loss | $1200 < p < 1260$

Count



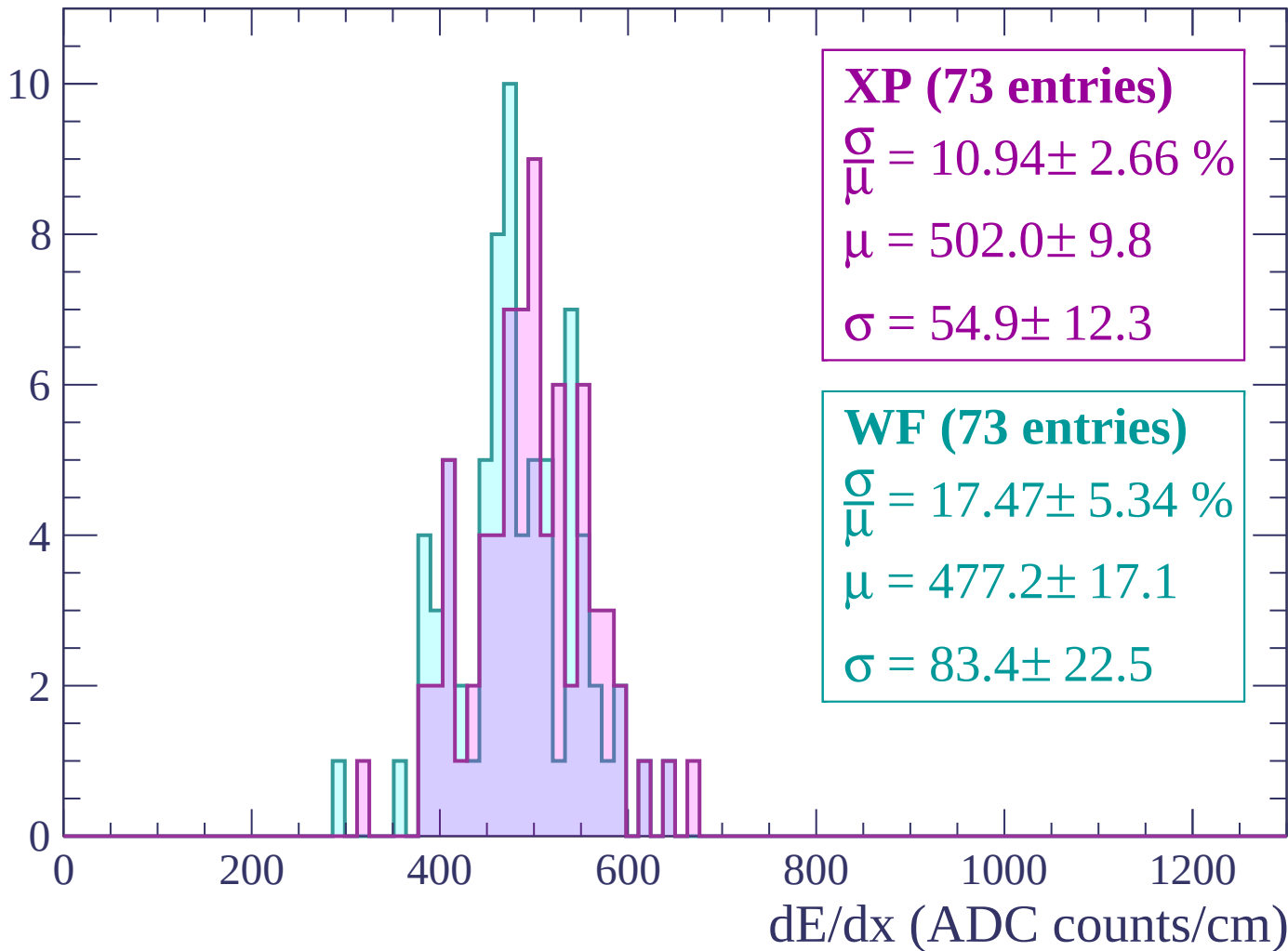
Energy loss | $1260 < p < 1320$

Count



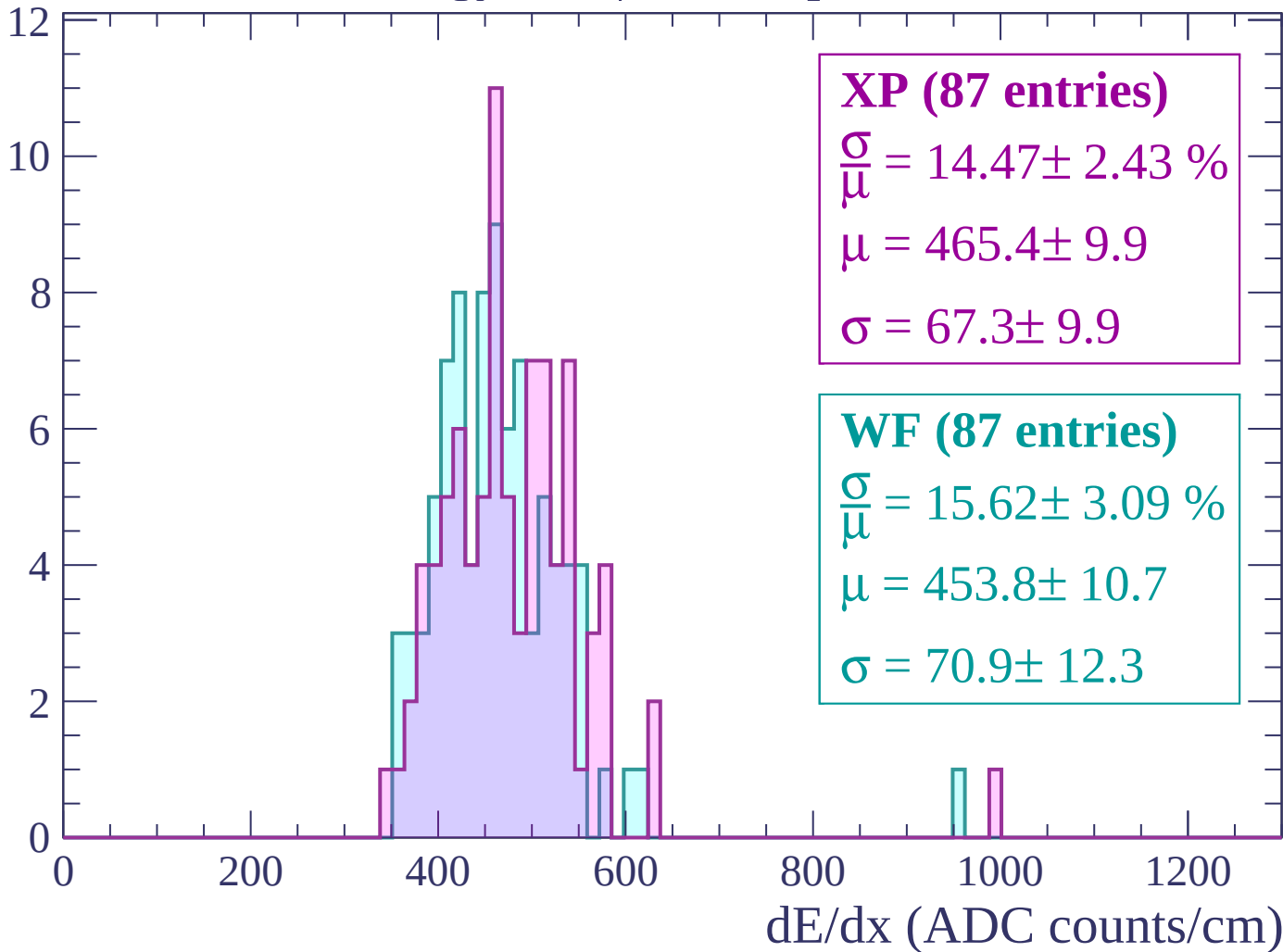
Energy loss | $1320 < p < 1380$

Count



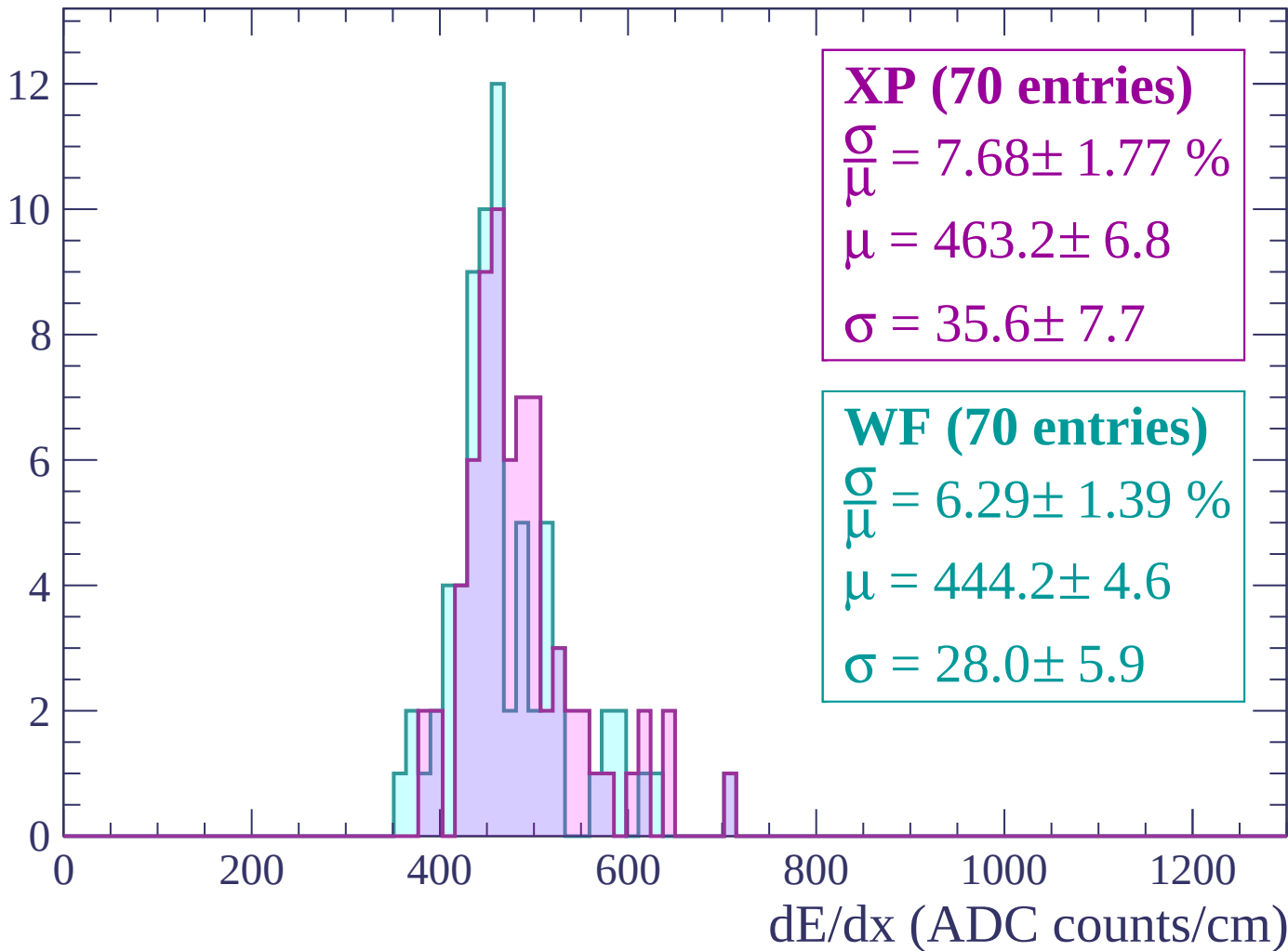
Energy loss | $1380 < p < 1440$

Count



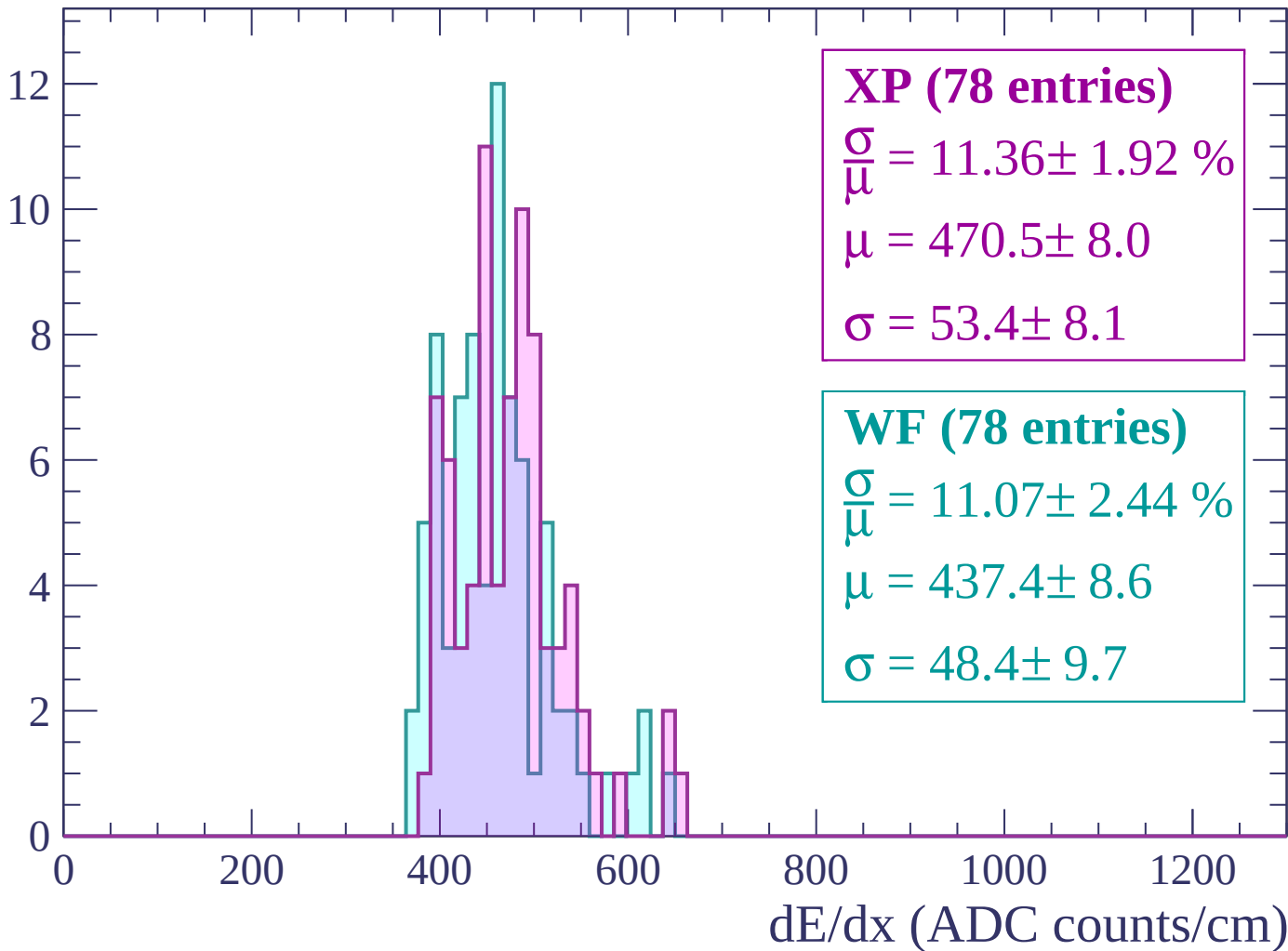
Energy loss | $1440 < p < 1500$

Count



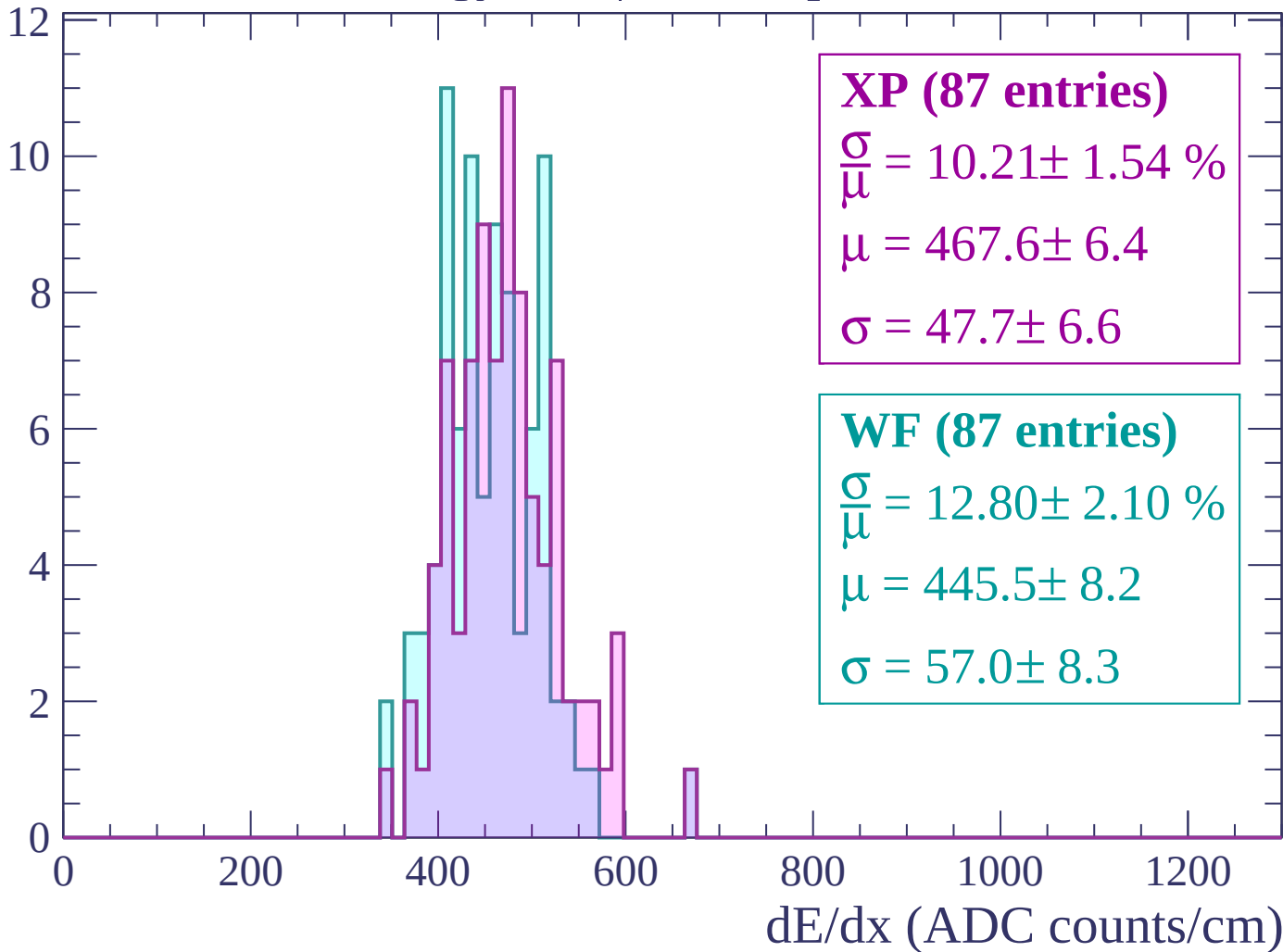
Energy loss | $1500 < p < 1560$

Count



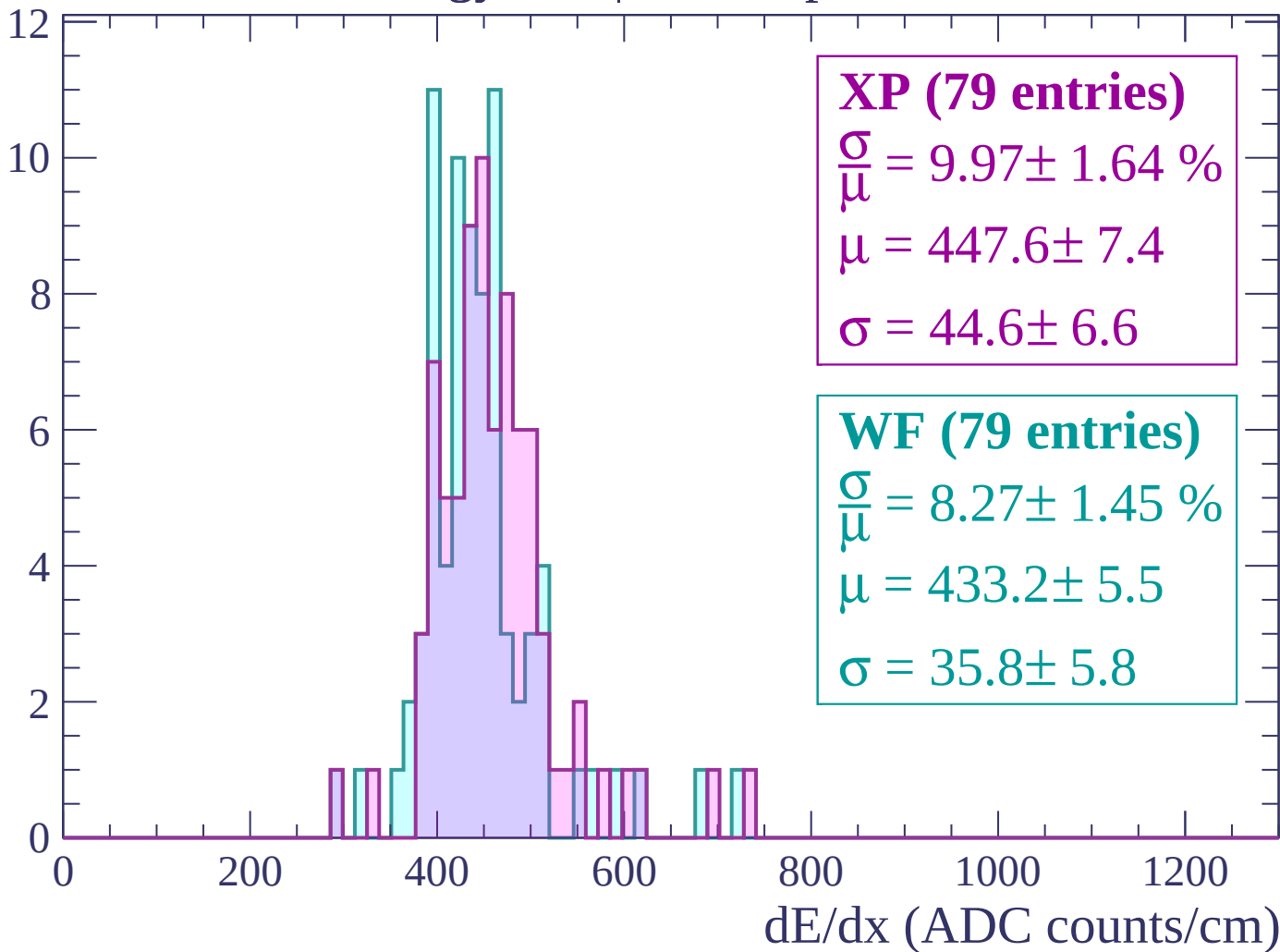
Energy loss | $1560 < p < 1620$

Count



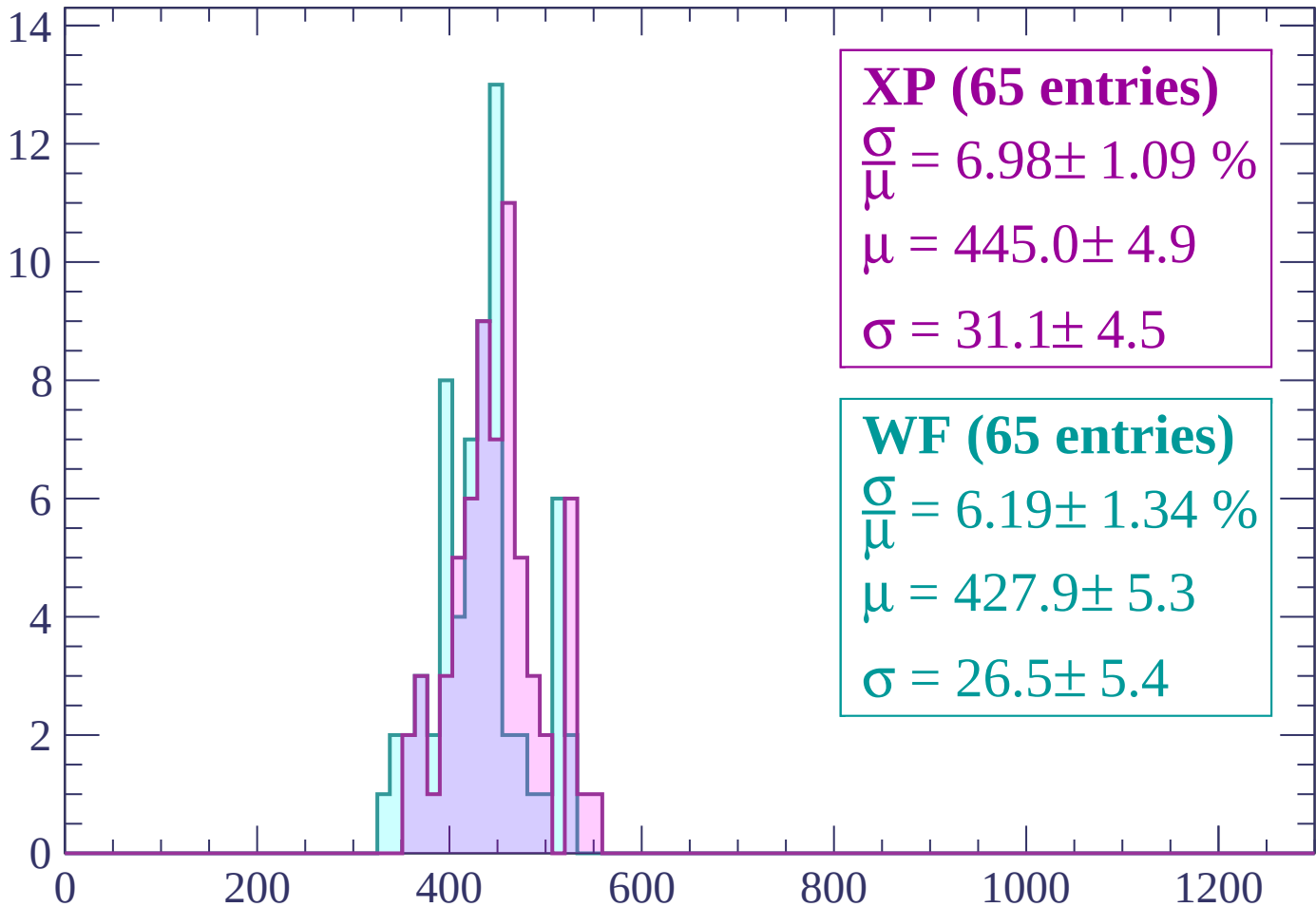
Energy loss | $1620 < p < 1680$

Count



Energy loss | $1680 < p < 1740$

Count



XP (65 entries)

$\frac{\sigma}{\mu} = 6.98 \pm 1.09 \%$

$\mu = 445.0 \pm 4.9$

$\sigma = 31.1 \pm 4.5$

WF (65 entries)

$\frac{\sigma}{\mu} = 6.19 \pm 1.34 \%$

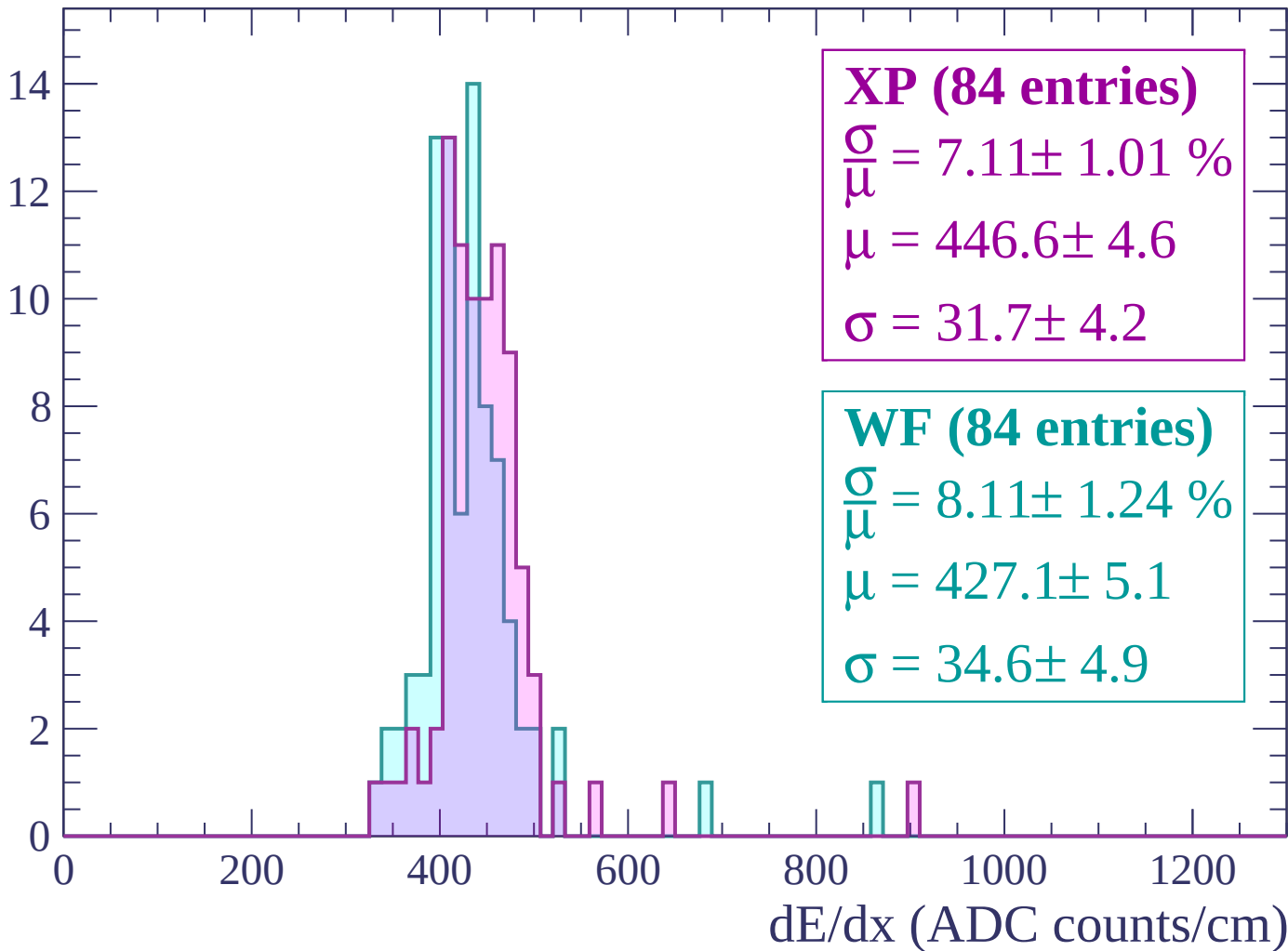
$\mu = 427.9 \pm 5.3$

$\sigma = 26.5 \pm 5.4$

dE/dx (ADC counts/cm)

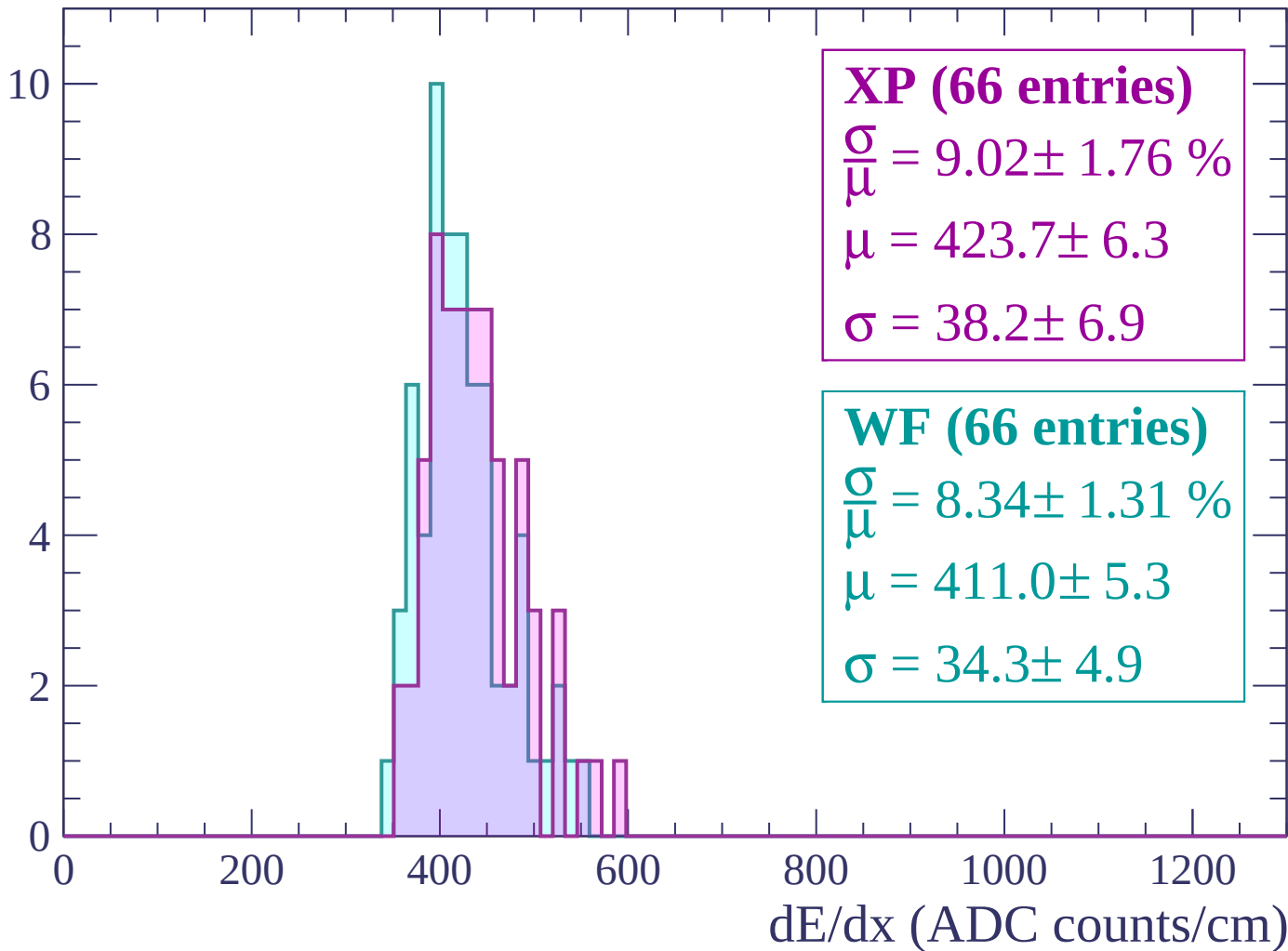
Energy loss | $1740 < p < 1800$

Count



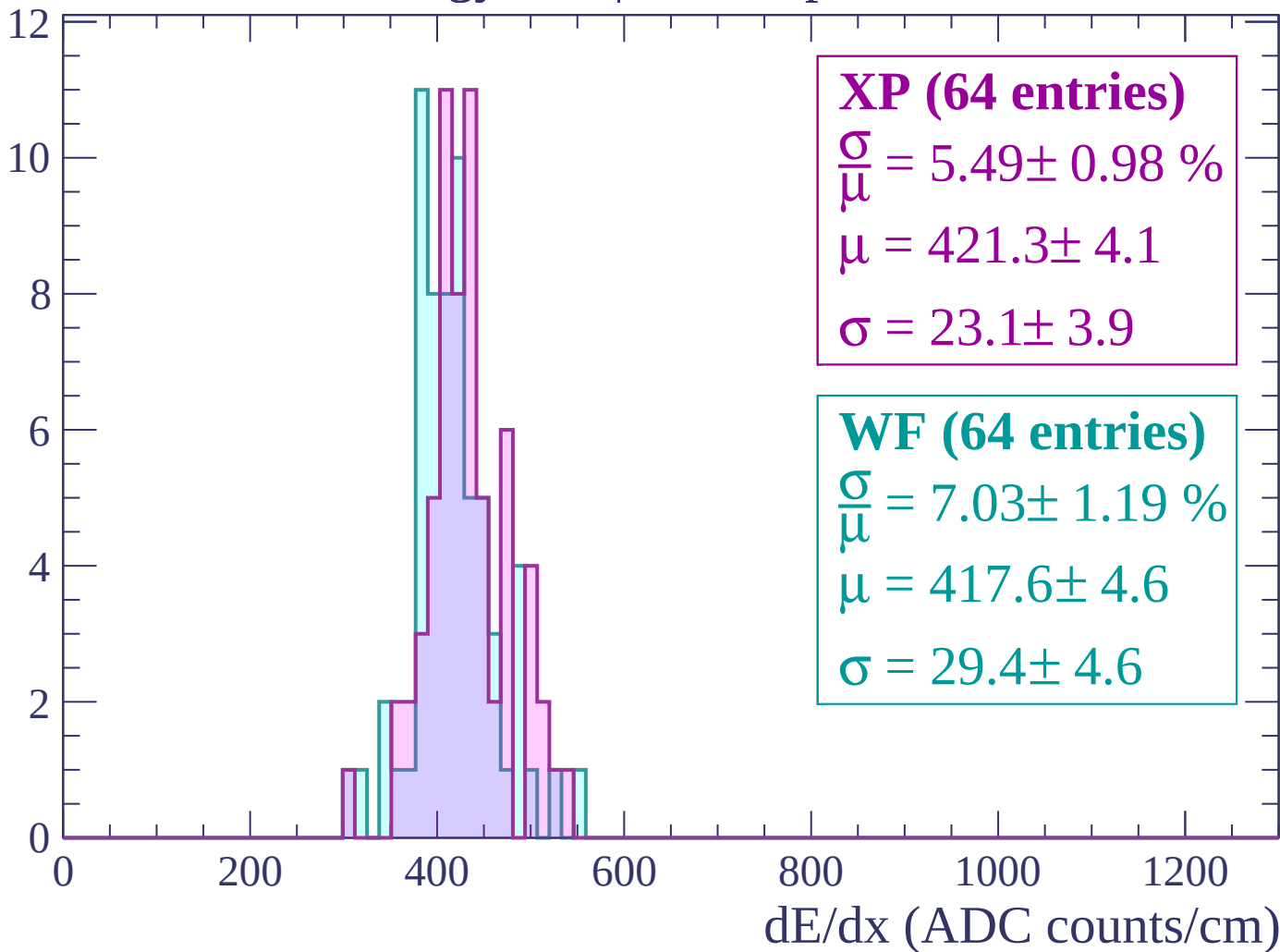
Energy loss | $1800 < p < 1860$

Count



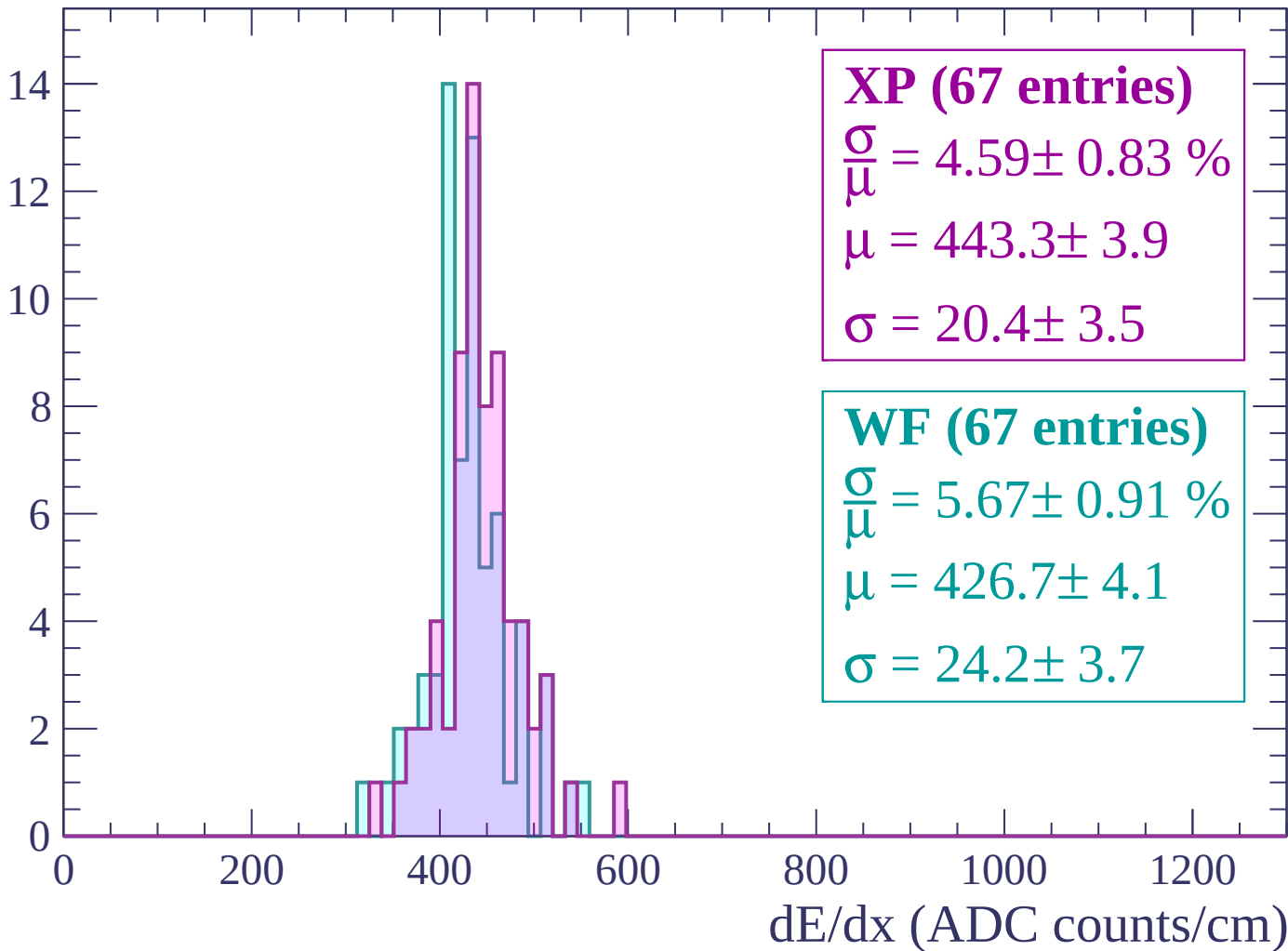
Energy loss | $1860 < p < 1920$

Count



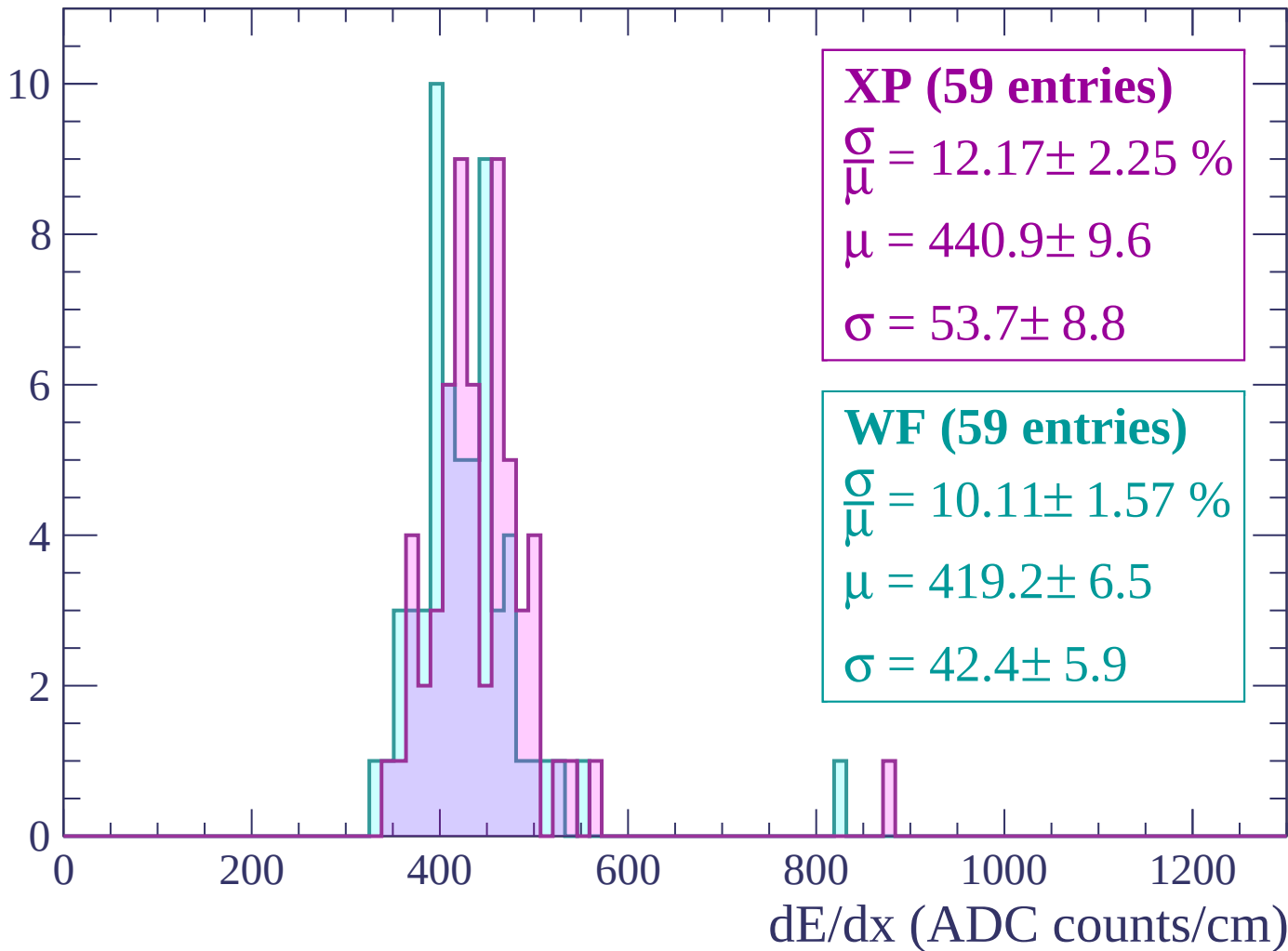
Energy loss | $1920 < p < 1980$

Count



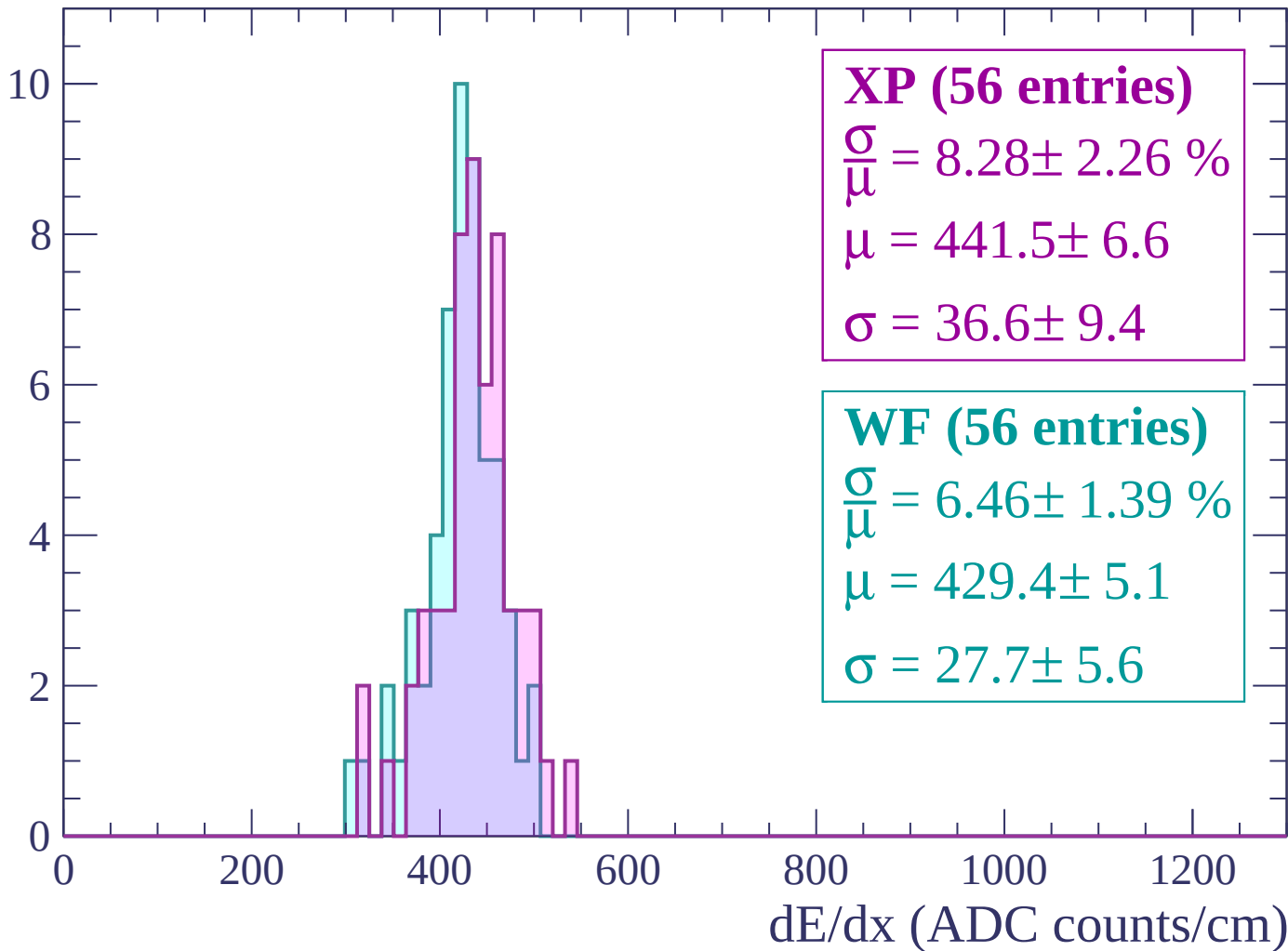
Energy loss | $1980 < p < 2040$

Count



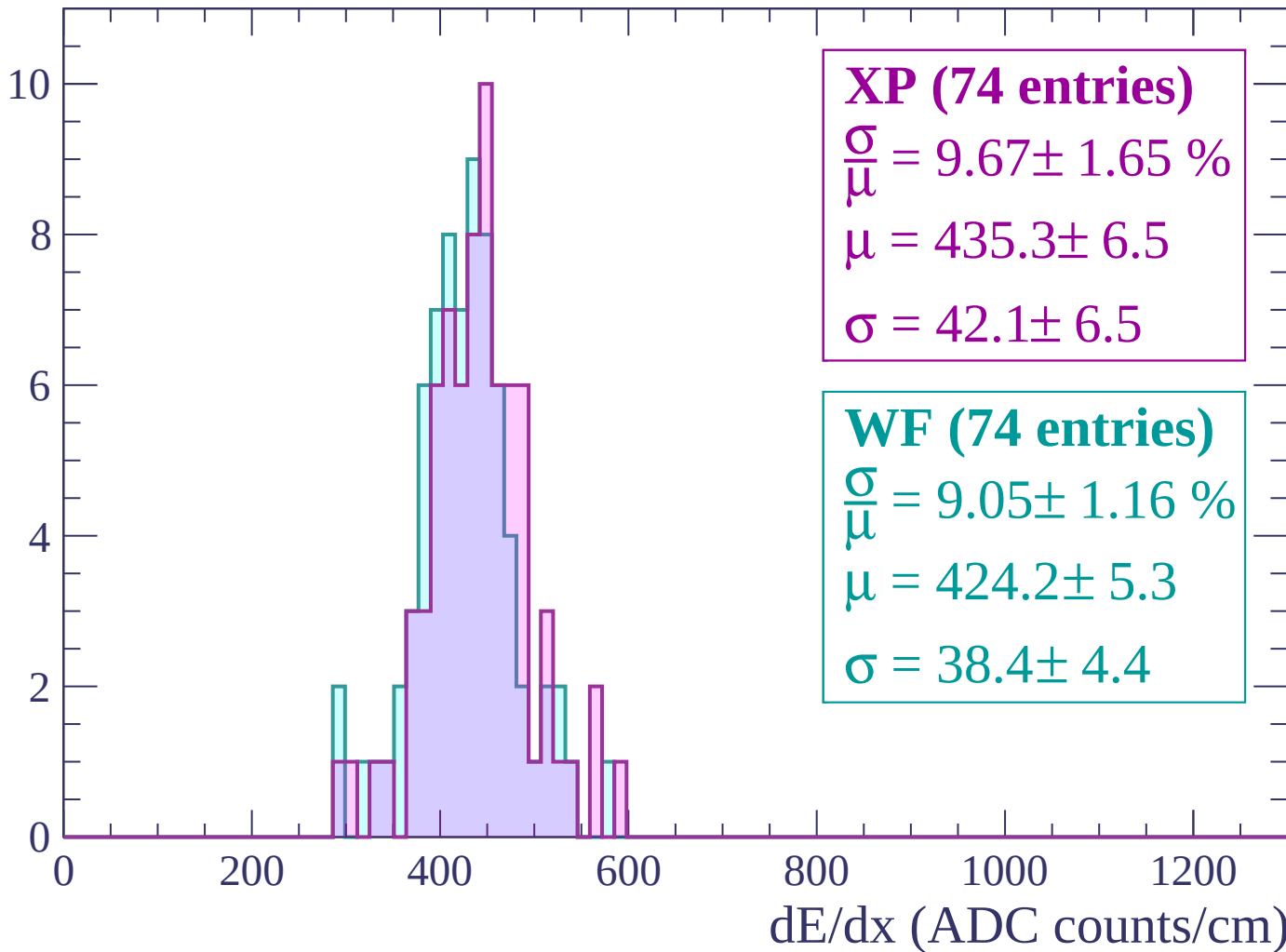
Energy loss | $2040 < p < 2100$

Count



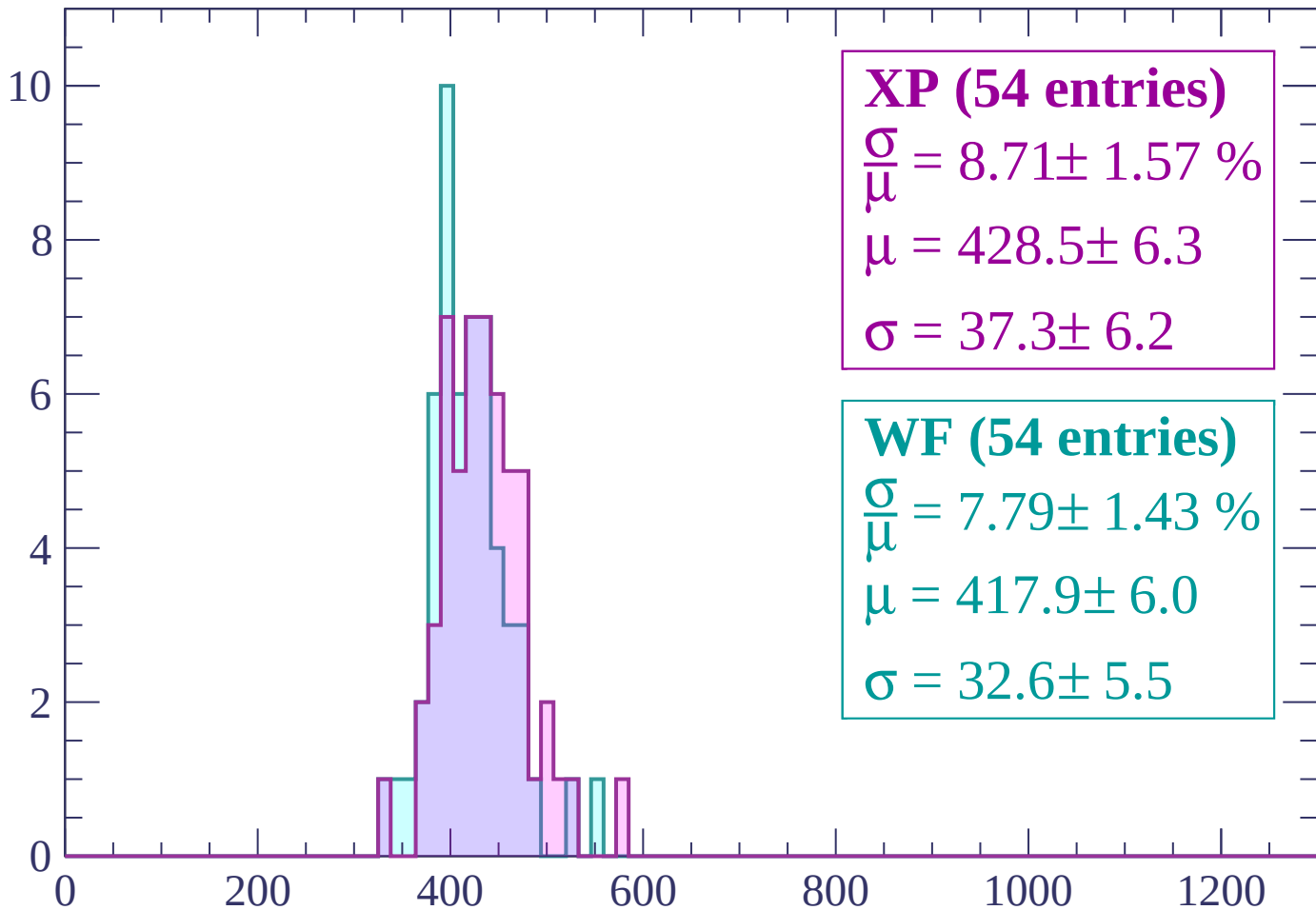
Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count



XP (54 entries)

$\frac{\sigma}{\mu} = 8.71 \pm 1.57 \%$

$\mu = 428.5 \pm 6.3$

$\sigma = 37.3 \pm 6.2$

WF (54 entries)

$\frac{\sigma}{\mu} = 7.79 \pm 1.43 \%$

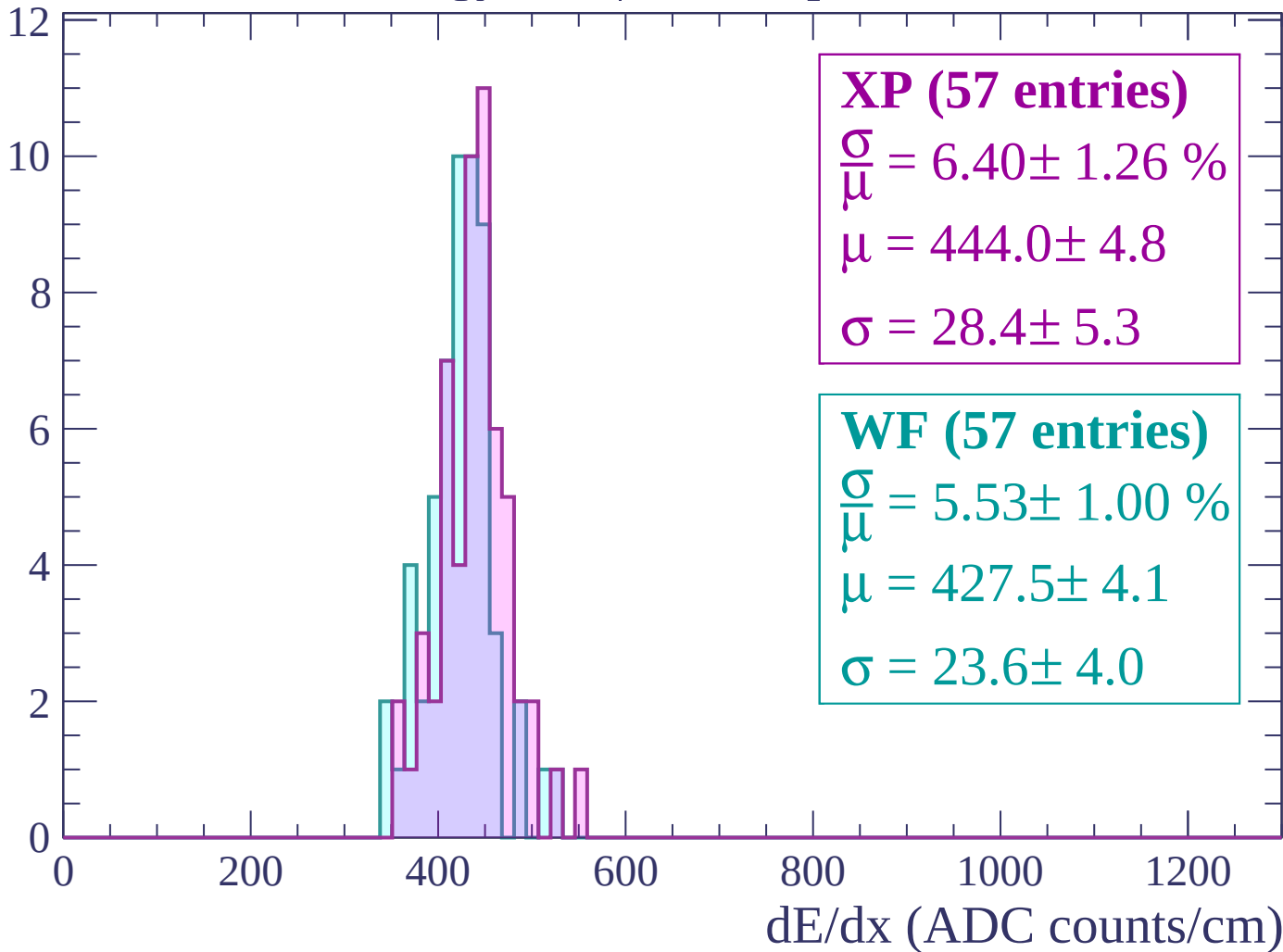
$\mu = 417.9 \pm 6.0$

$\sigma = 32.6 \pm 5.5$

dE/dx (ADC counts/cm)

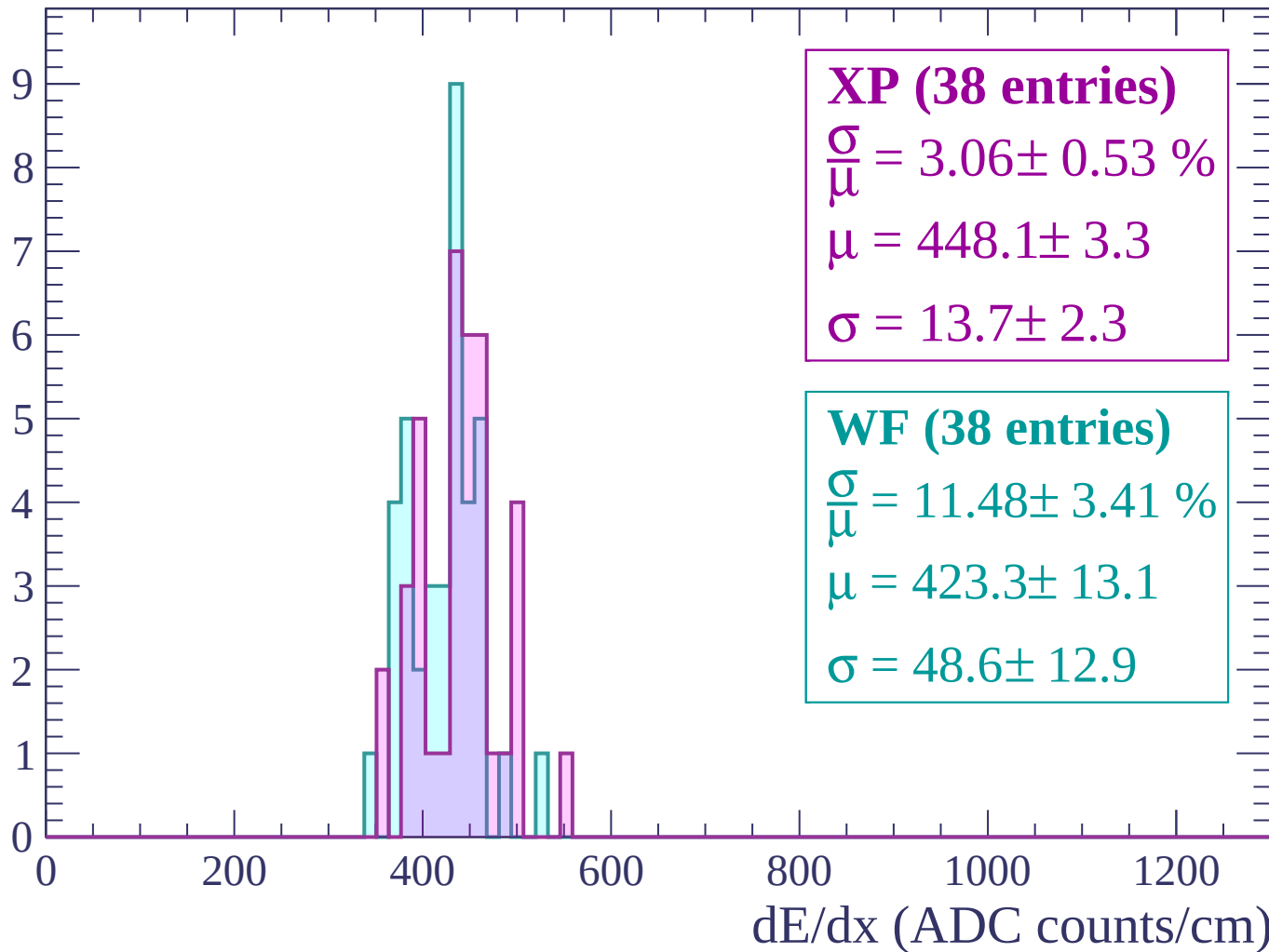
Energy loss | $2220 < p < 2280$

Count



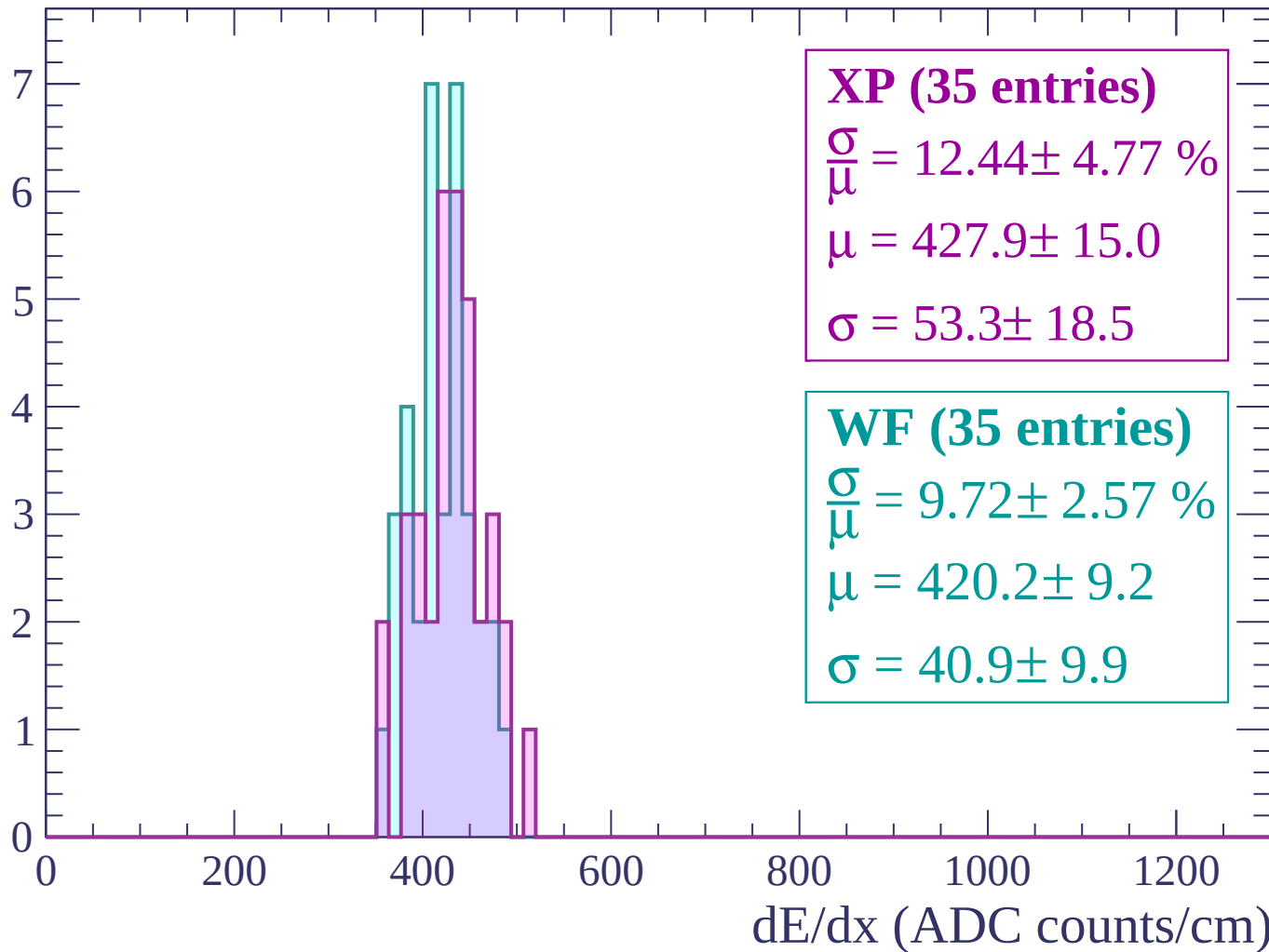
Energy loss | $2280 < p < 2340$

Count



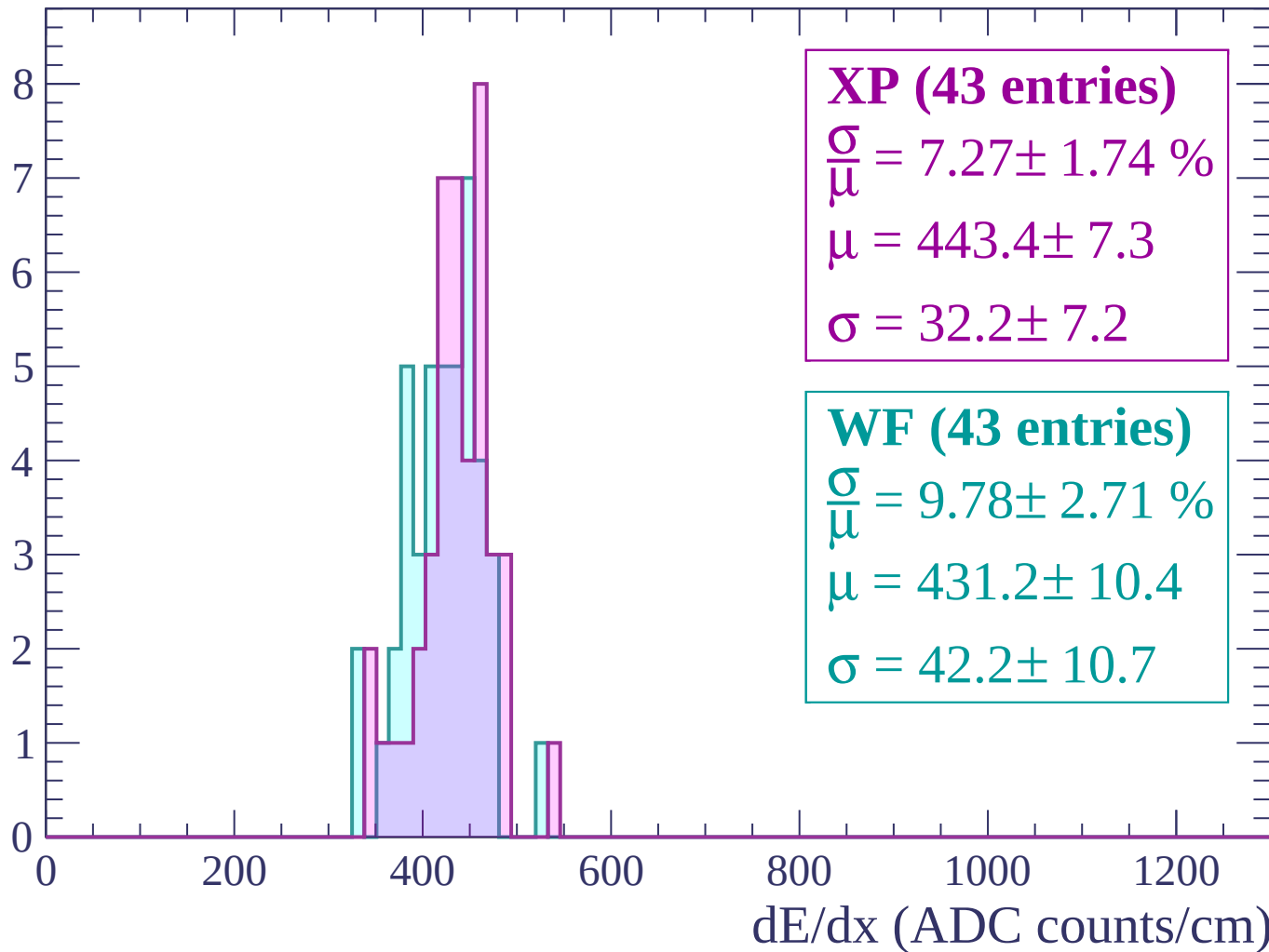
Energy loss | $2340 < p < 2400$

Count



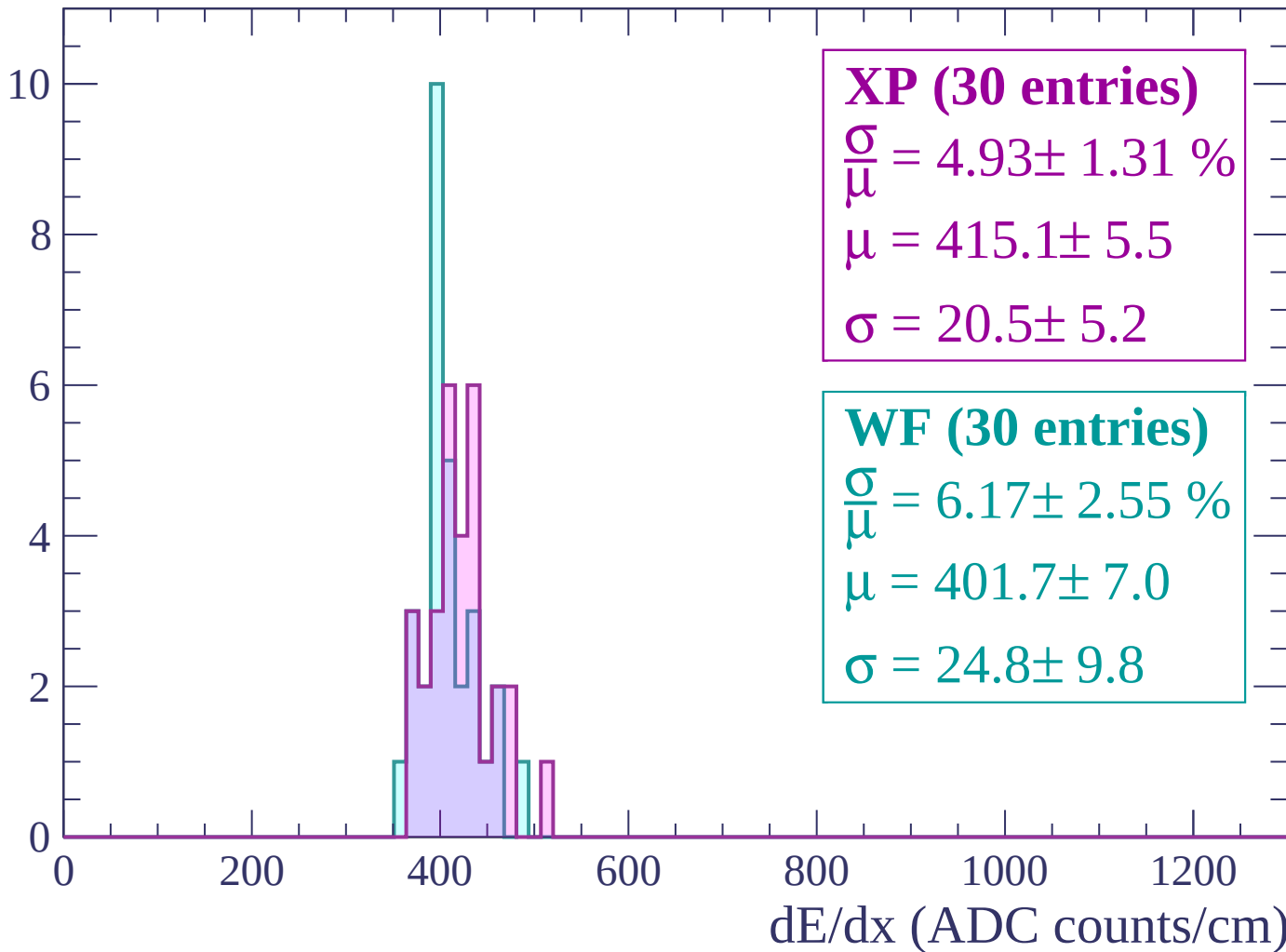
Energy loss | $2400 < p < 2460$

Count



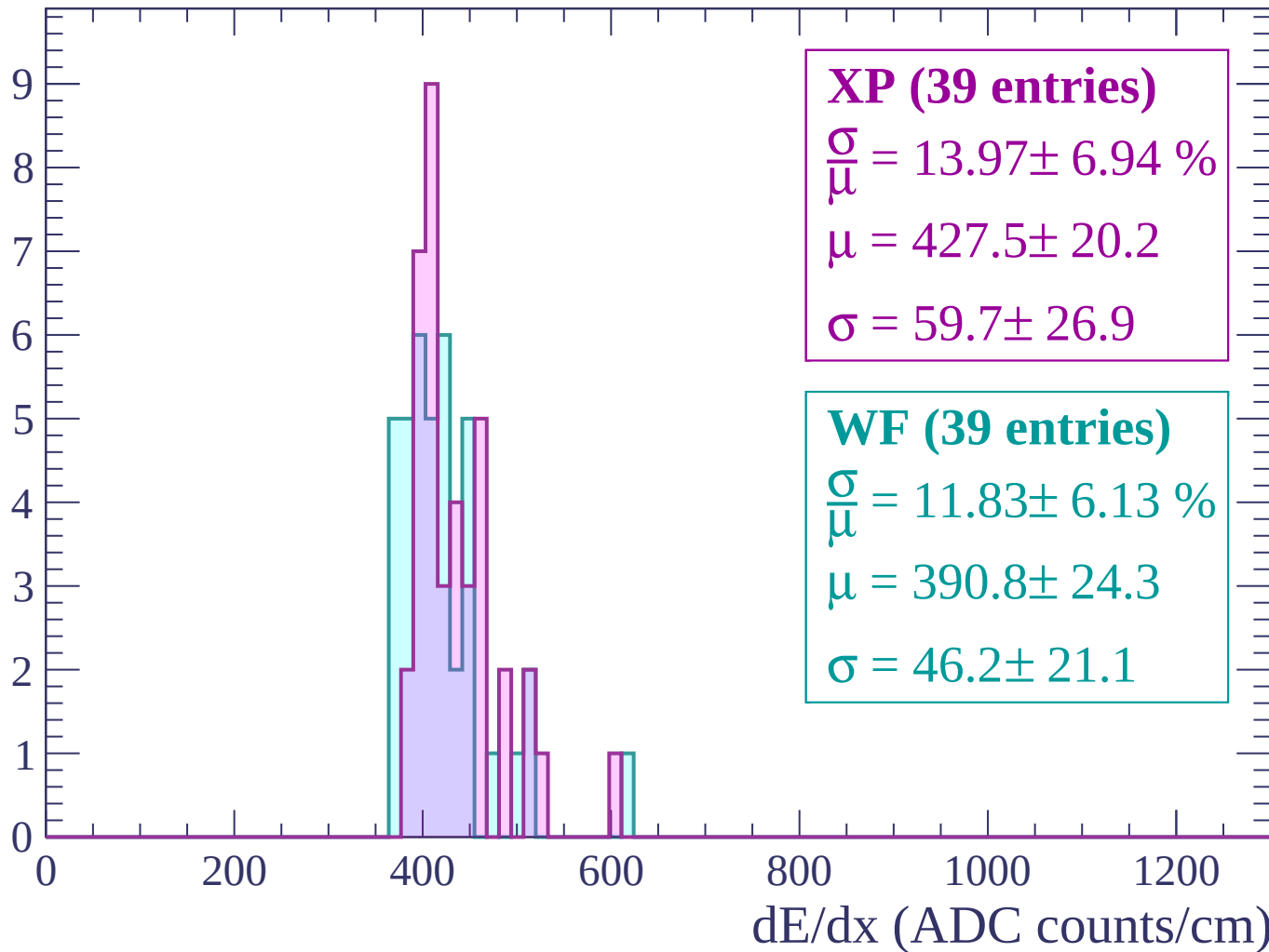
Energy loss | $2460 < p < 2520$

Count



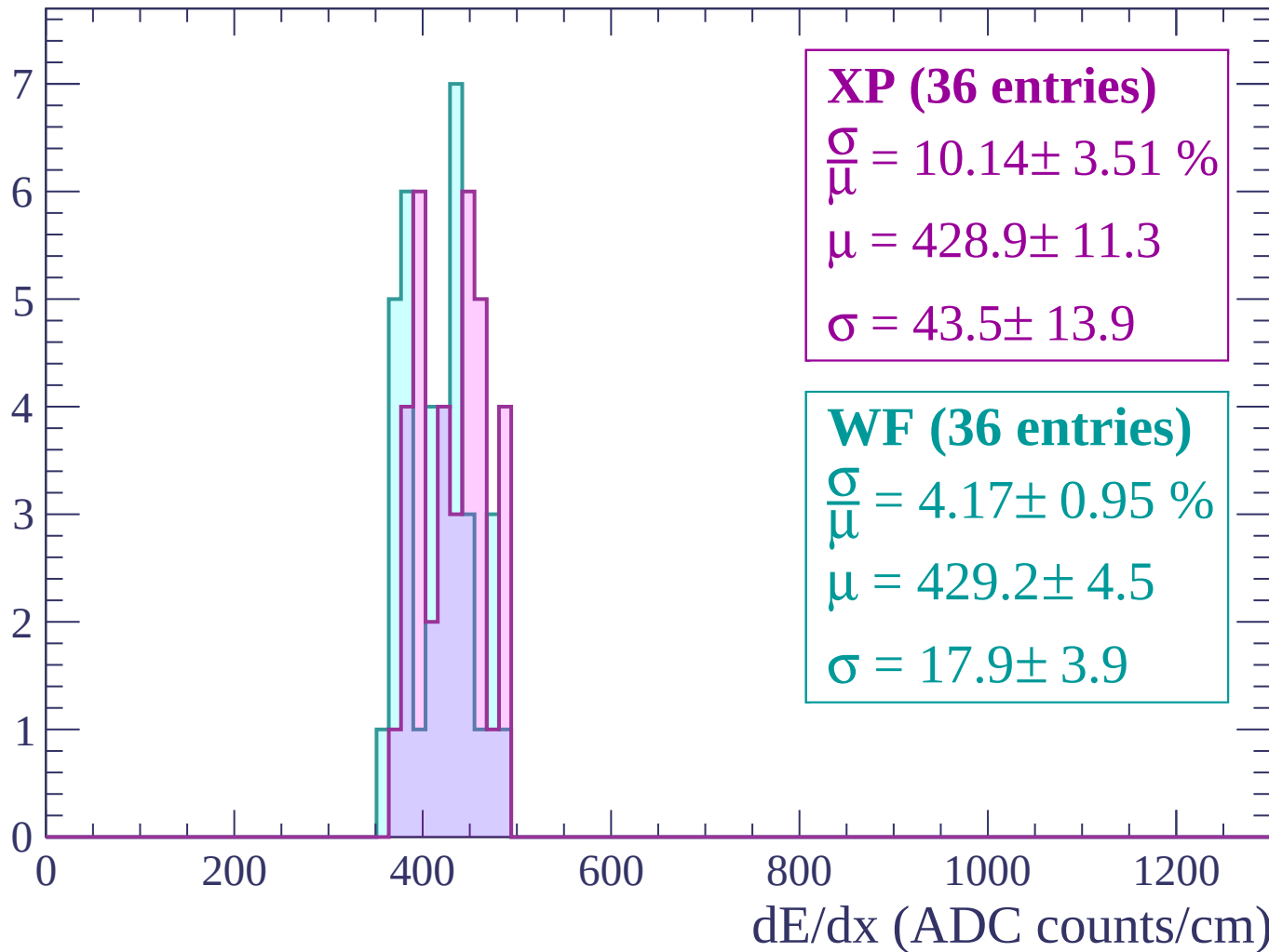
Energy loss | $2520 < p < 2580$

Count



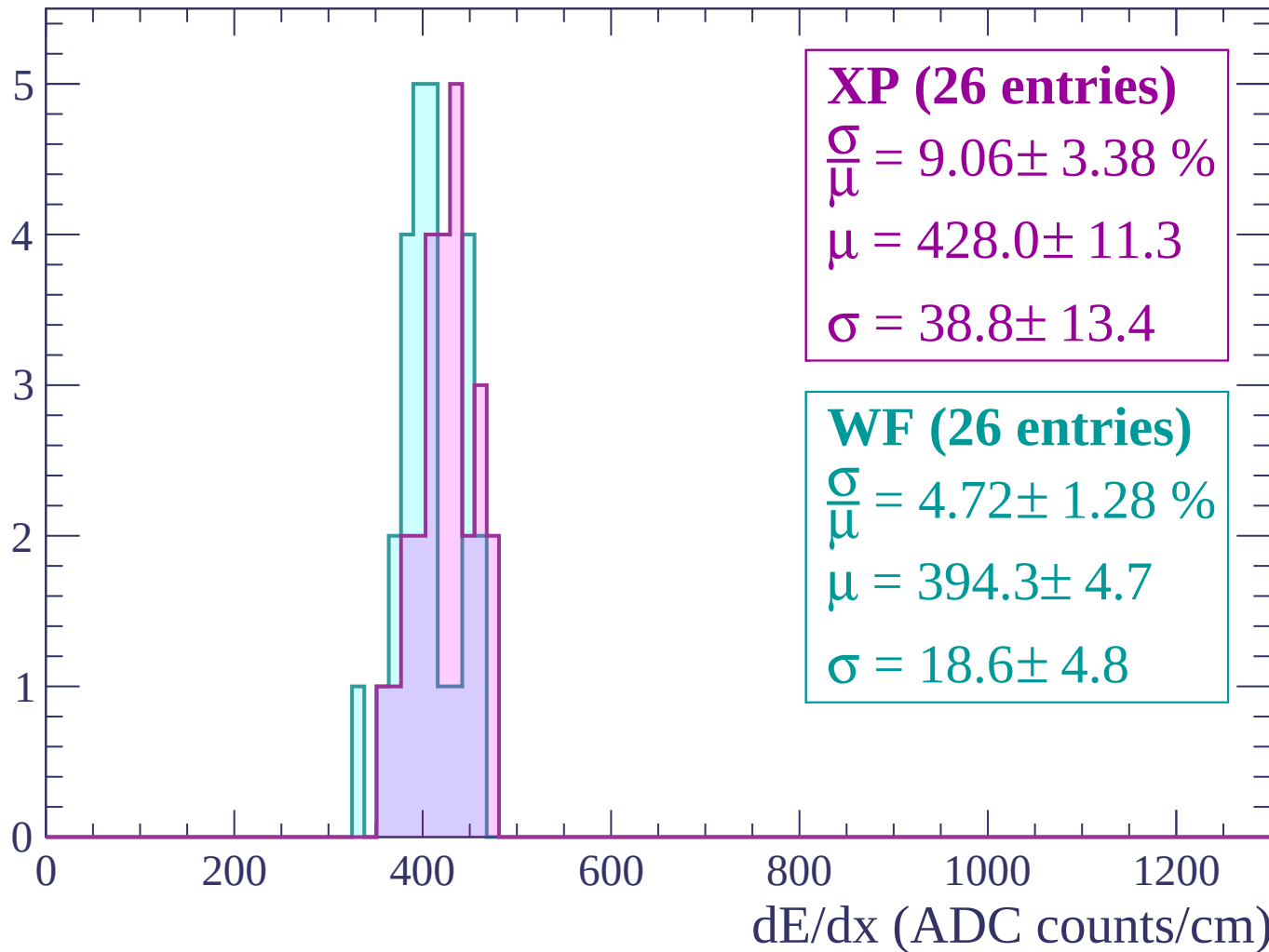
Energy loss | $2580 < p < 2640$

Count



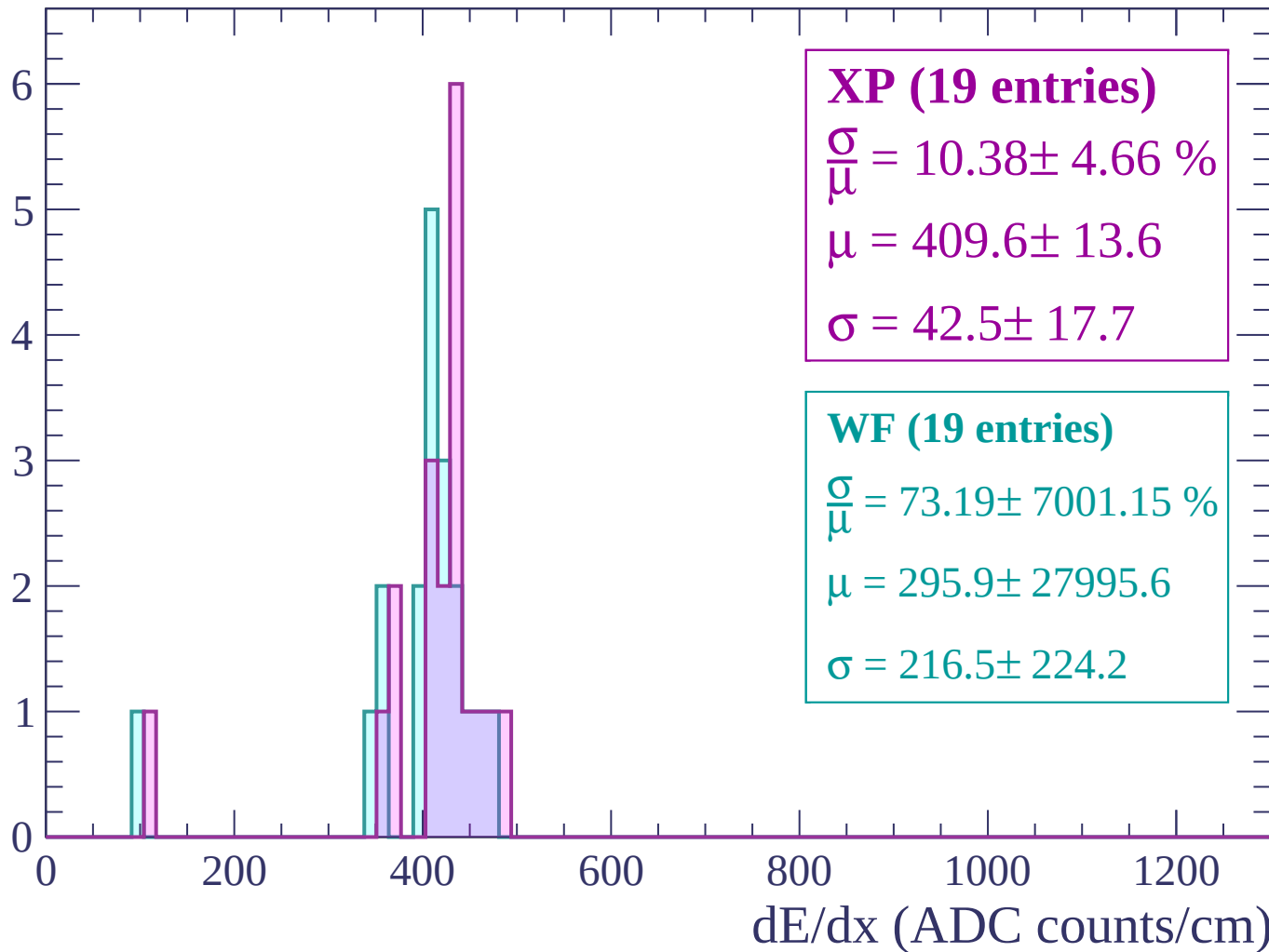
Energy loss | $2640 < p < 2700$

Count



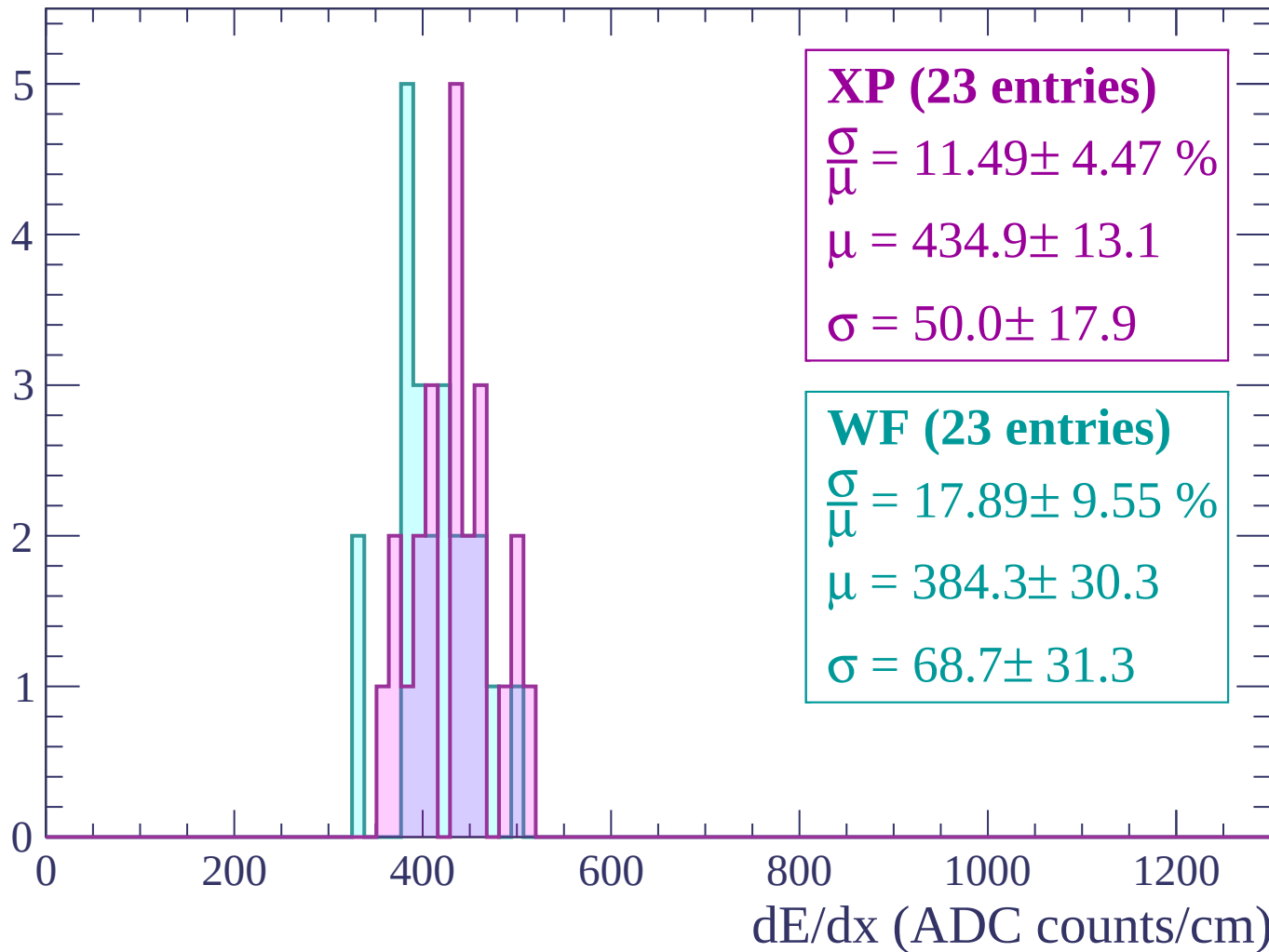
Energy loss | $2700 < p < 2760$

Count



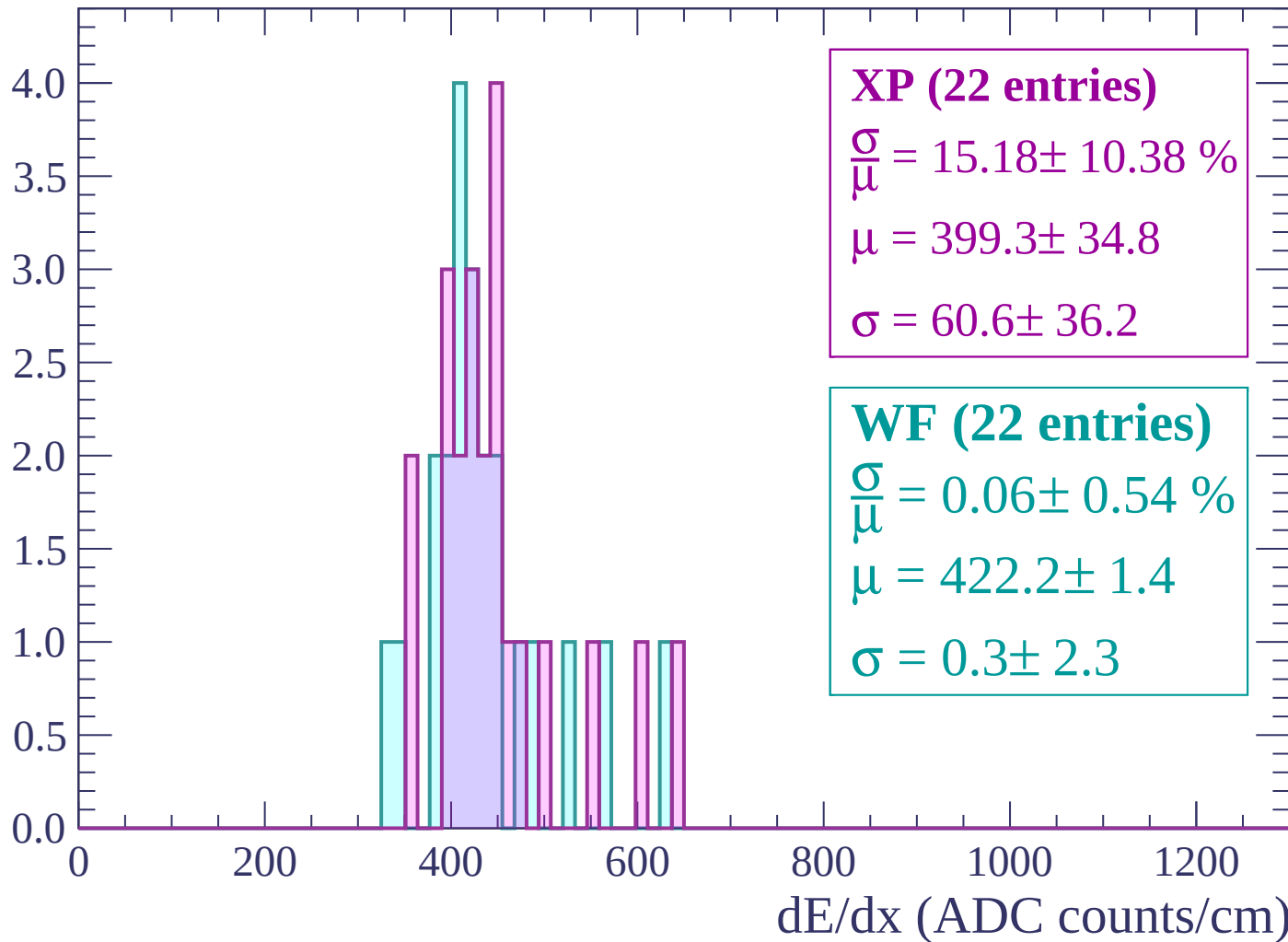
Energy loss | $2760 < p < 2820$

Count



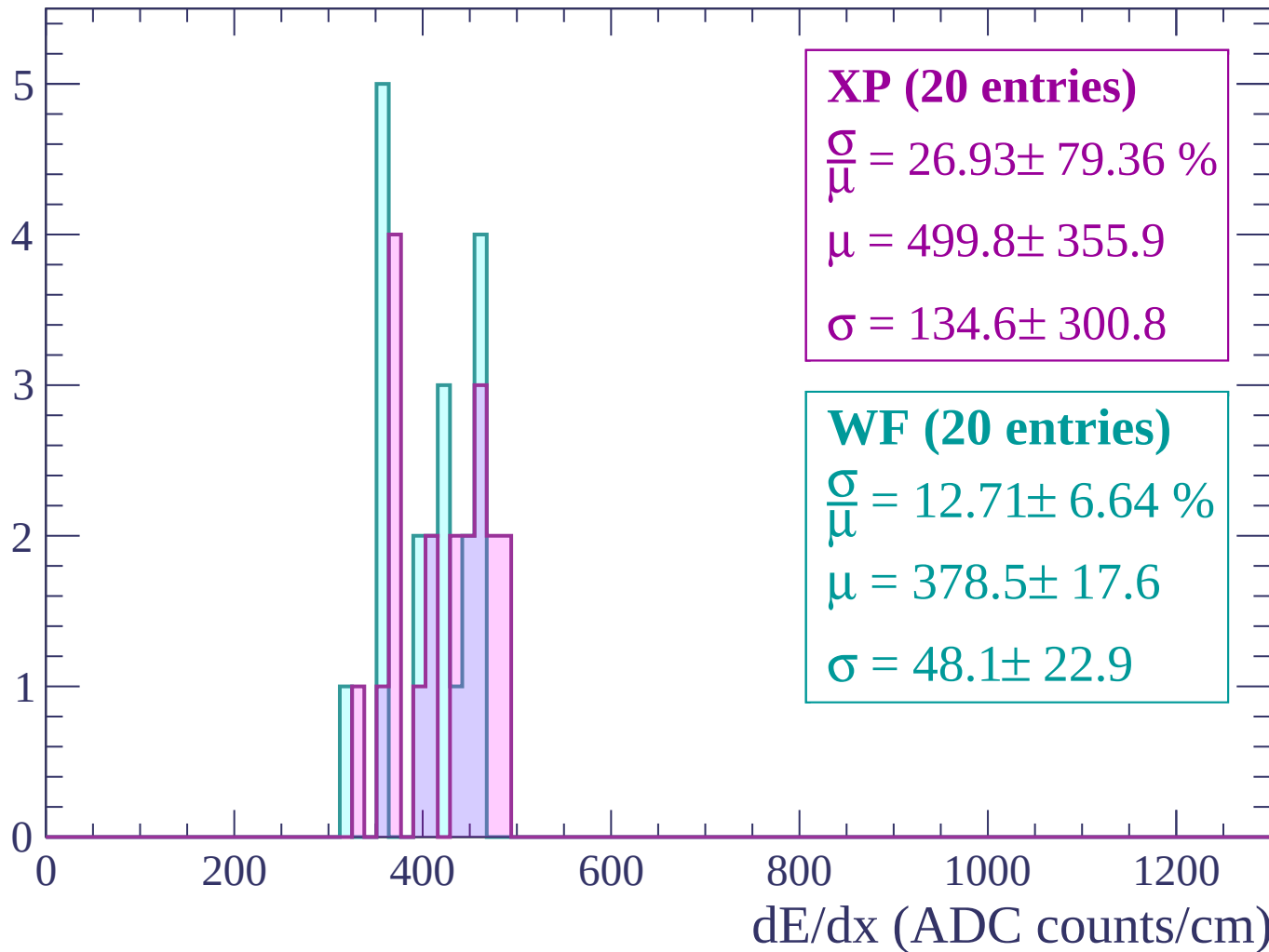
Energy loss | $2820 < p < 2880$

Count



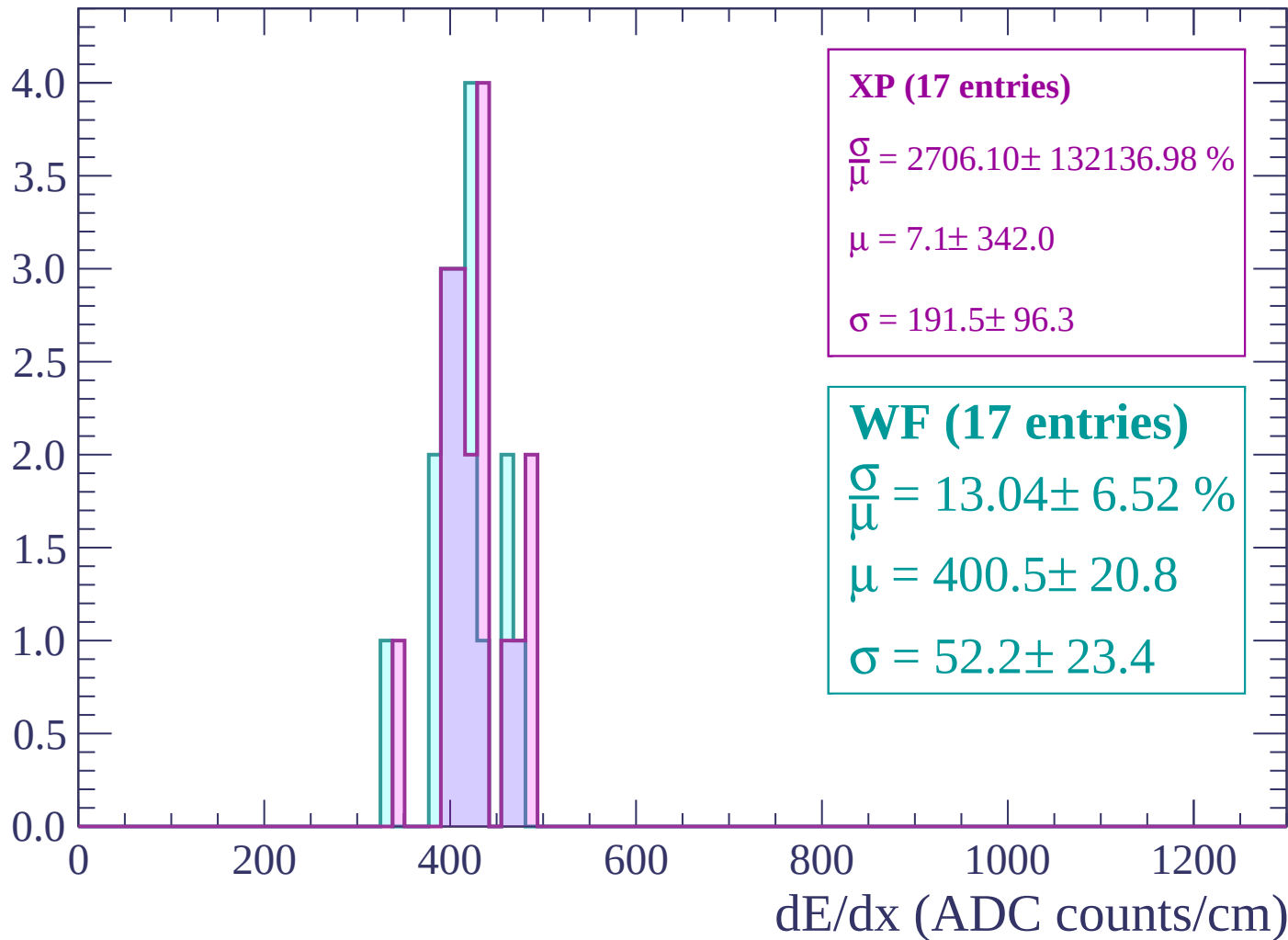
Energy loss | $2880 < p < 2940$

Count



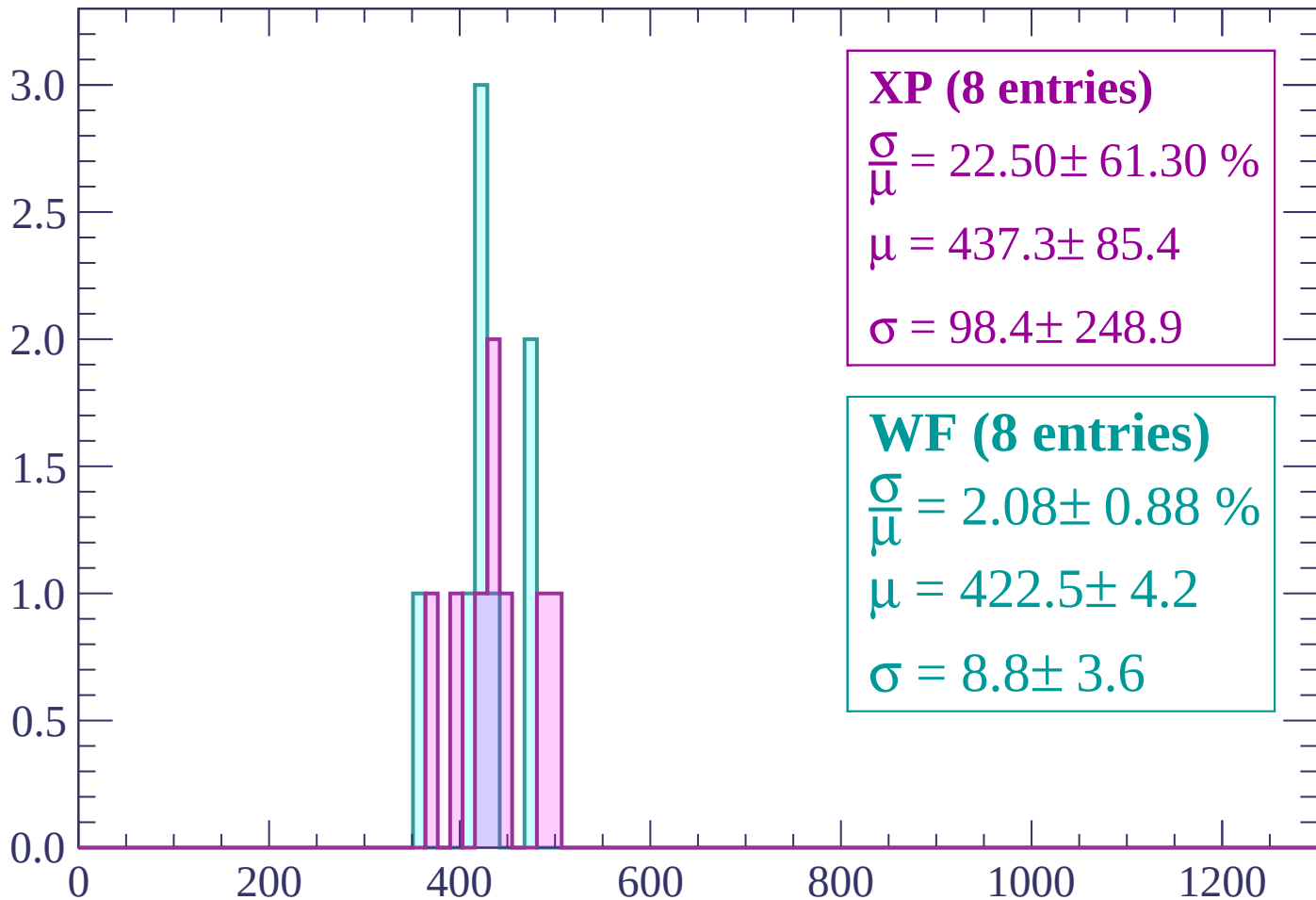
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



XP (8 entries)

$$\frac{\sigma}{\mu} = 22.50 \pm 61.30 \%$$

$$\mu = 437.3 \pm 85.4$$

$$\sigma = 98.4 \pm 248.9$$

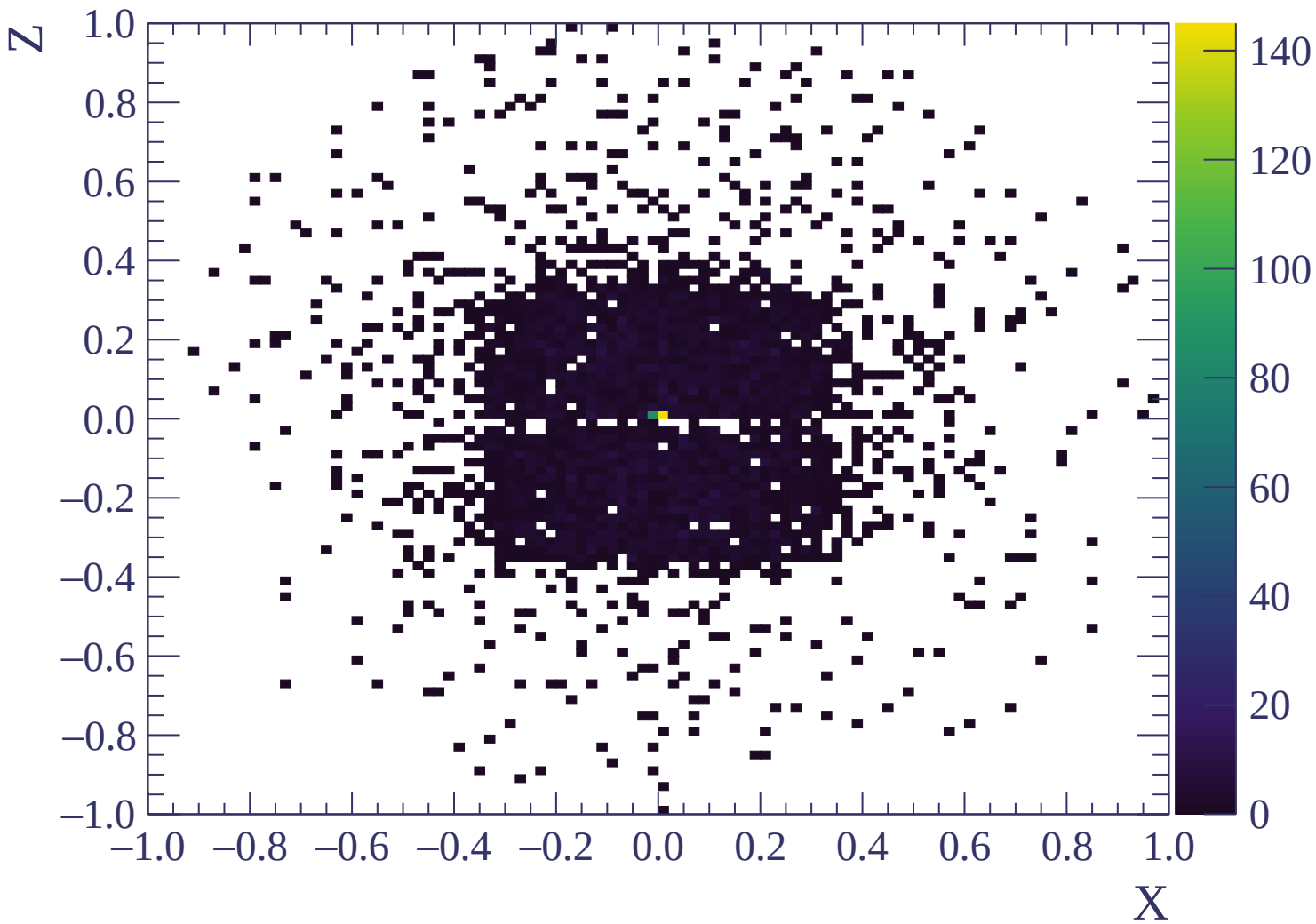
WF (8 entries)

$$\frac{\sigma}{\mu} = 2.08 \pm 0.88 \%$$

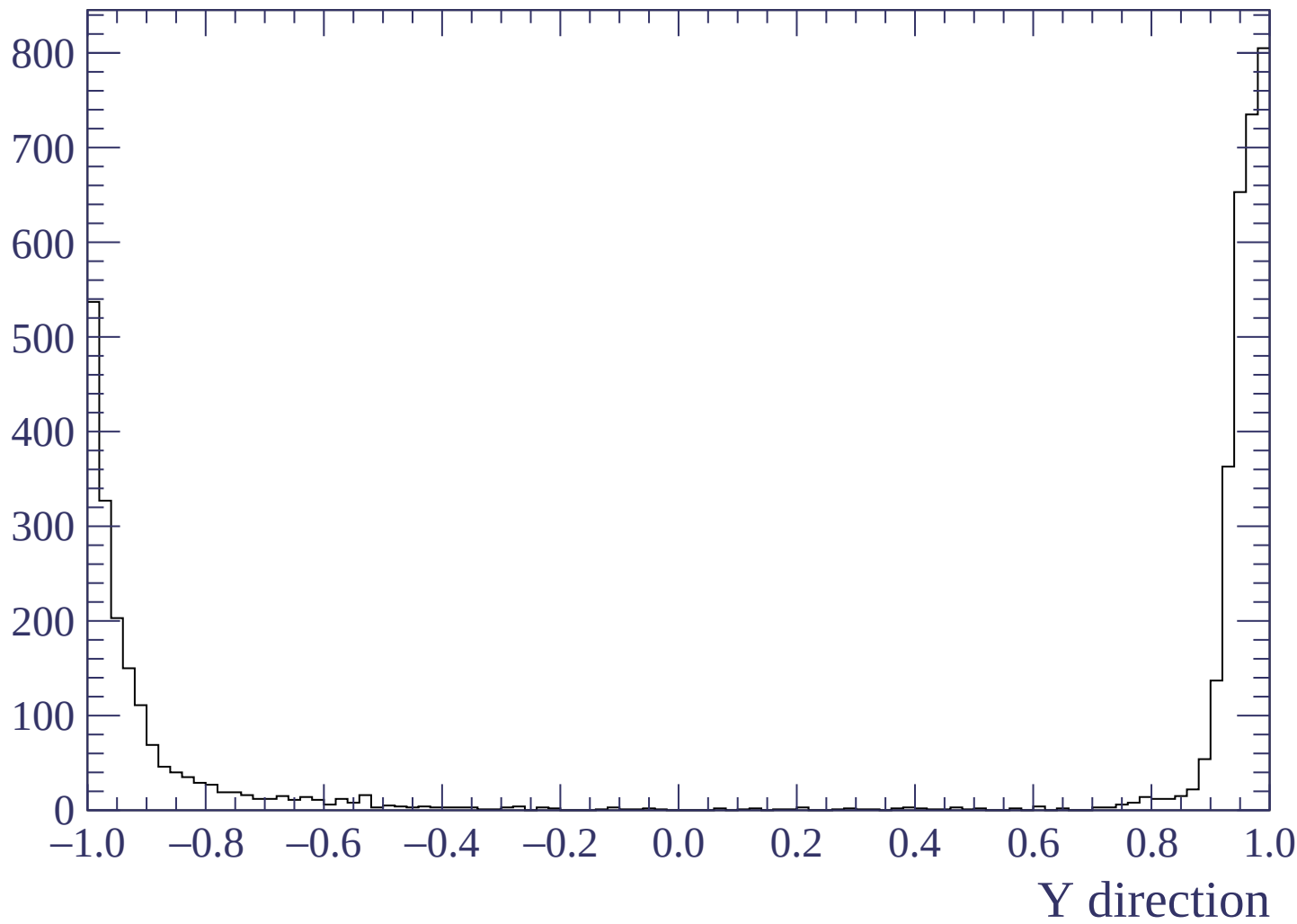
$$\mu = 422.5 \pm 4.2$$

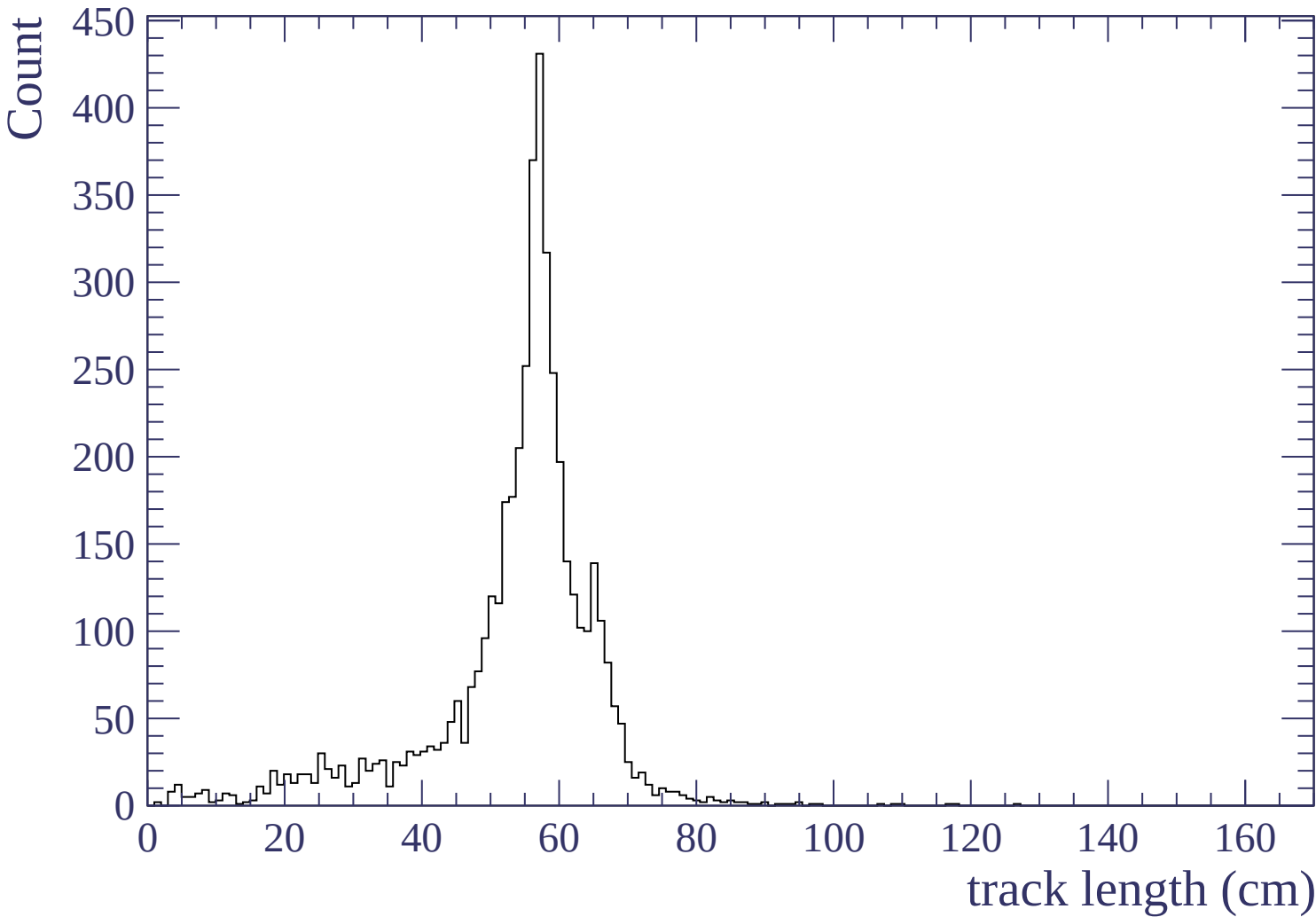
$$\sigma = 8.8 \pm 3.6$$

dE/dx (ADC counts/cm)

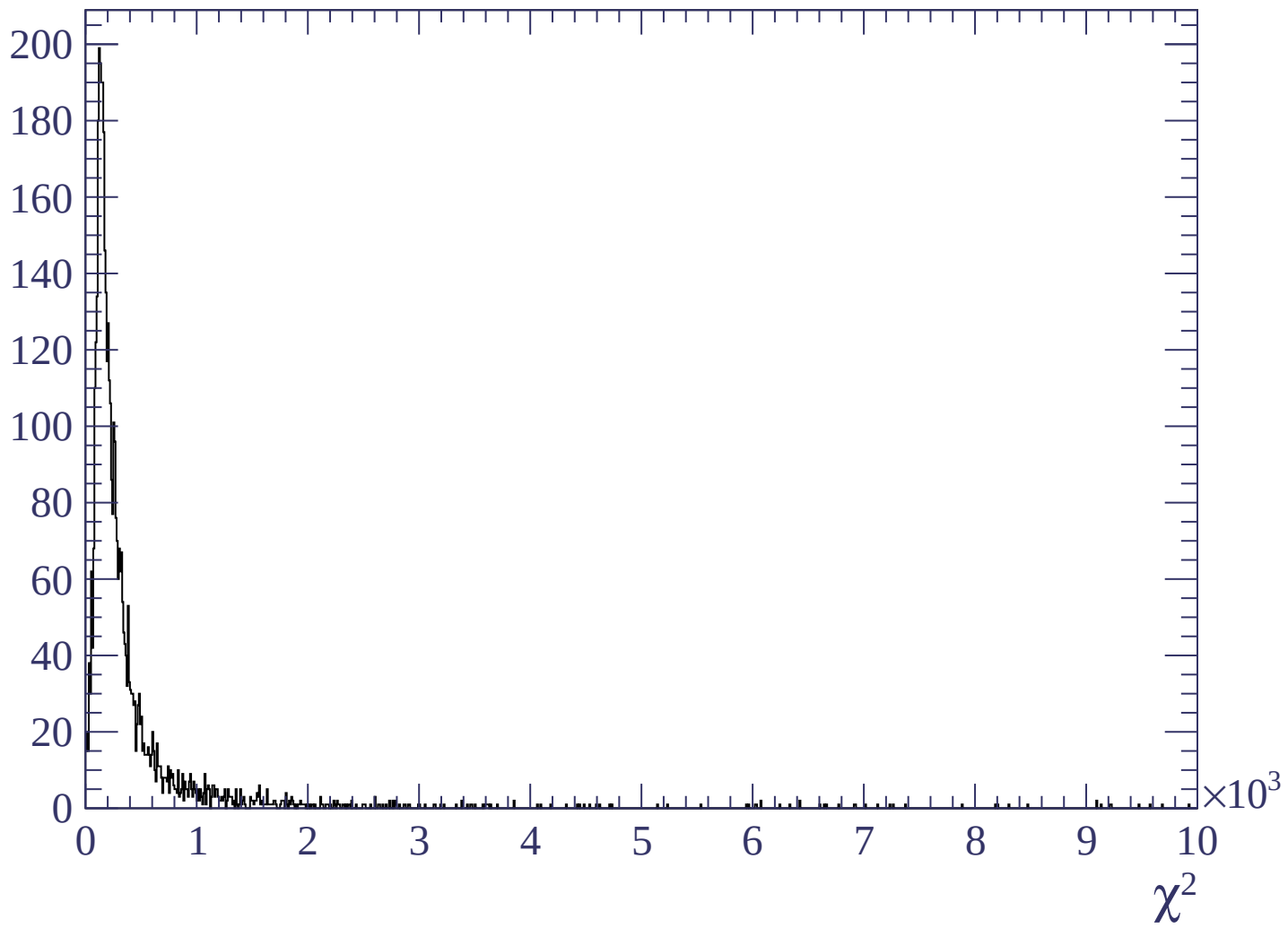


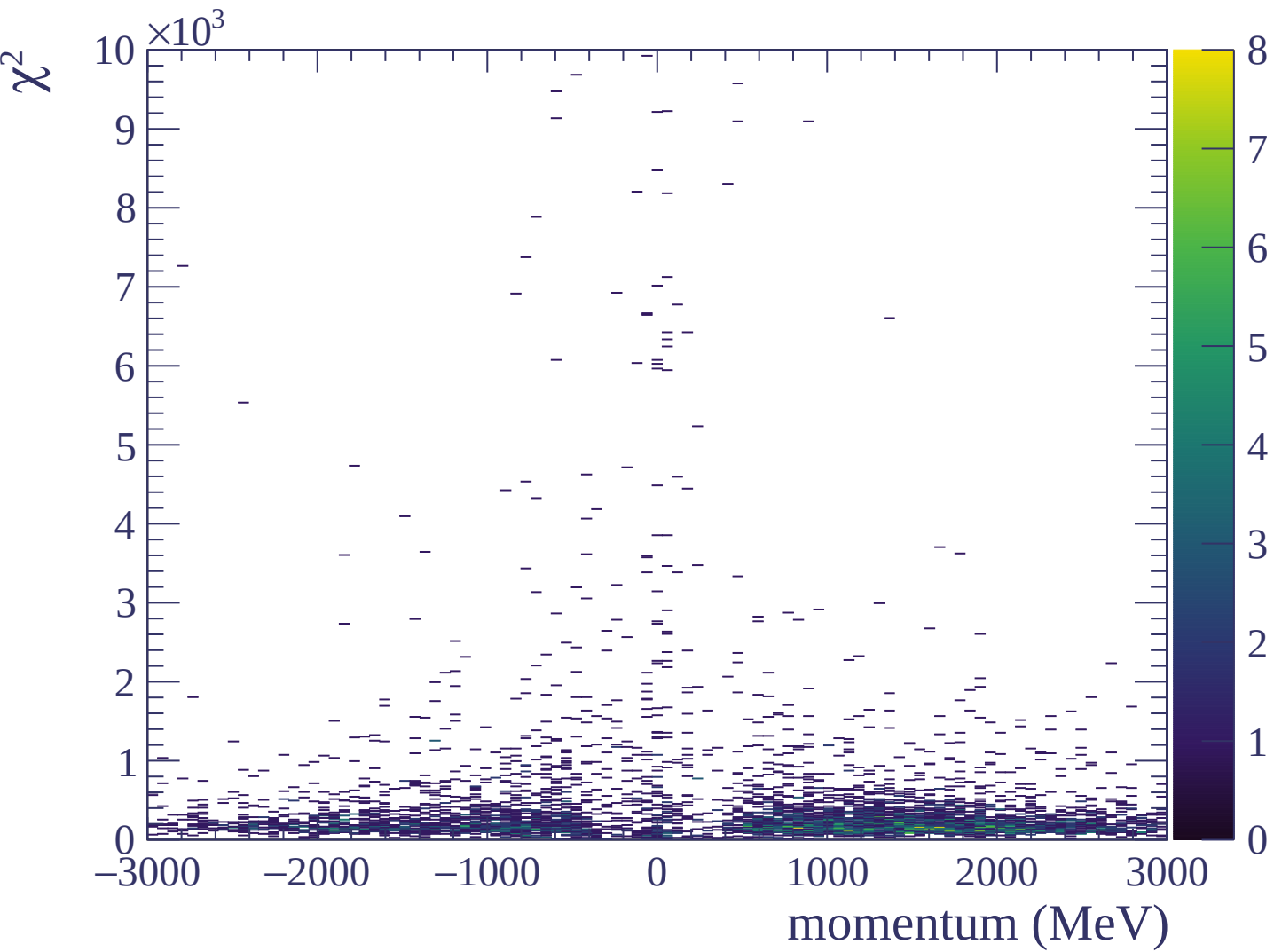
Count



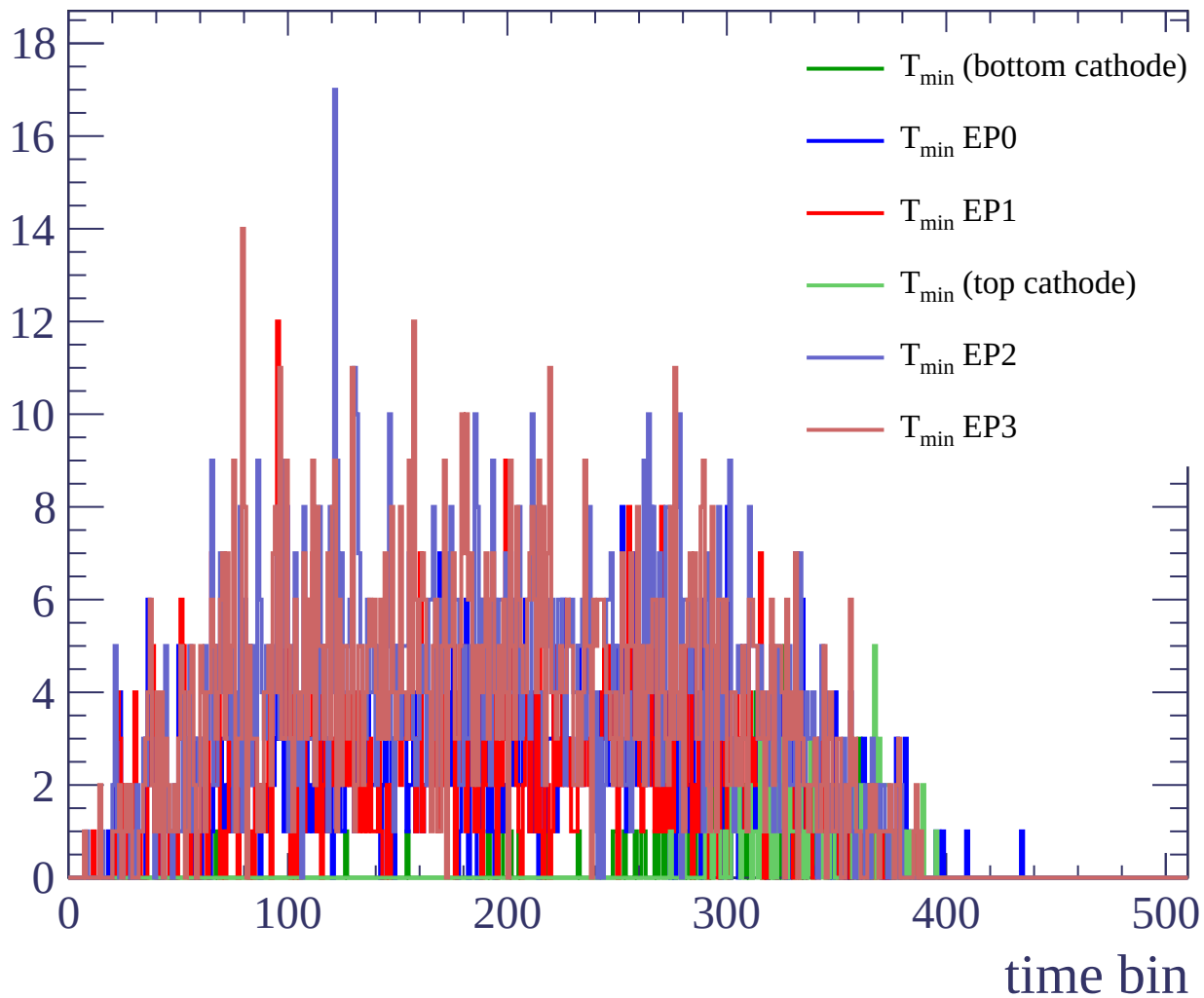


Count





Count



Count

